Statistical Climatology
MSc/PhD Course

Abdel Hannachi

Department of Meteorology, Stockholm University
106 91 Stockholm, Sweden

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Contents

Chapter 1

Basic Statistics and Probability Concepts

1.1 Exploratory basic statistics

1.1.1 Conventional parameters

A sample of data x_t, $t = 1, \dots, n$, is normally referred to as a *time series* when t is time. Figures 1.1 and 1.2 show two examples of time series. The first example (Fig. 1.1) represents the January North Atlantic Oscillation (NAO) time series from 1960 to 2004. The data come from the Climate Prediction Center (CPC) website[1]. For comparison, figure 1.2 shows a time series generated randomly. For a given time series, one can define a number of basic statistics:

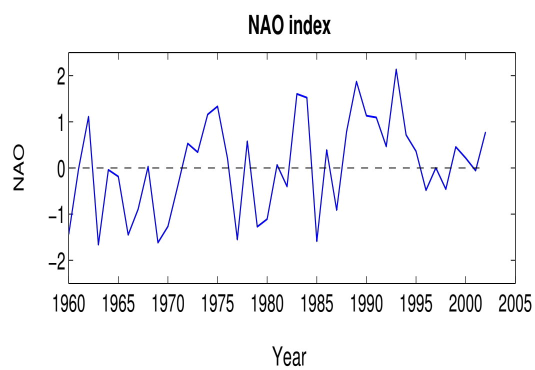

Figure 1.1: The January NAO time series

• Sample mean

$$\overline{x} = \frac{1}{n} \sum_{t=1}^{n} x_t \tag{1.1}$$

We normally denote by $x'_t = x_t - \overline{x}$, $t = 1, \dots, n$, as the anomaly time series, i.e., the zero-mean time series defined as the departure from the mean.

• Sample variance

[1]http://www.cpc.ncep.noaa.gov/data/teledoc/nao.shtml

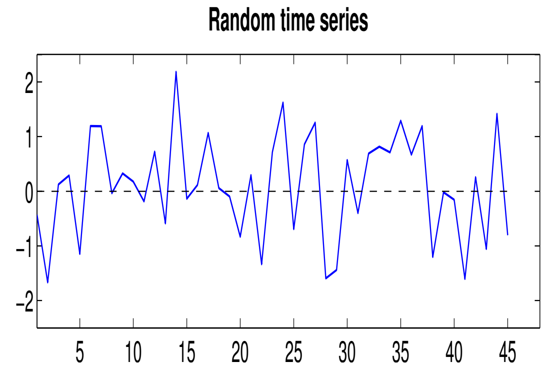

Figure 1.2: An example of a random time series

$$s^2 = \frac{1}{n-1}\sum_{t=1}^{n}(x_t - \overline{x})^2 = \frac{1}{n-1}\sum_{t=1}^{n}(x'_t)^2 \tag{1.2}$$

Exercise

Show that $\frac{n-1}{n}s^2 = \overline{x^2} - \overline{x}^2$

Explain the use of $n-1$ instead of n in eq (1.2).

• Standard deviation

$$s = \sqrt{s^2} \tag{1.3}$$

The mean and the variance represent the first two moments. One can also define other moments as shown below.

• High-order moments

The r'*th*, $r \geq 3$, high-order moment is defined by:

$$s_r = \frac{1}{n}\sum_{t=1}^{n}(x_t - \overline{x})^r = \frac{1}{n}\sum_{t=1}^{n}(x'_t)^r \tag{1.4}$$

Sometimes we also define the scaled high-order moment by scaling the above expression by s^r to yield s_r/s^r. The moments s_3 and s_4 are the *skewness* and *kurtosis* respectively[2].

An analysis of skewness and kurtosis in atmospheric low-frequency variability can be found in Sura and Hannachi (2015) and Hannachi (2010), and also Pires and Hannachi (2021). For example, Hannachi et al. (2003) discuss the skewness of sea surface temperature whereas Christiansen (2009) discuss the kurtosis of 20-hPa and 500-hPa geopotential heights.

1.1.2 Robust parameters

It is possible to define similar statistics but using instead robust measures. These measures are based on order statistics. We start first by ranking the sample to yield $x_{(1)} \leq x_{(2)} \leq \ldots \leq x_{(n)}$, and then compute the parameter of interest.

[2]Note that usually the kurtosis is defined with respect to that of a Gaussian, that is $s_4/s^4 - 3$

An example of a robust estimator of the mean is the median m, which is given by:

$$m = x_{(p)} \tag{1.5}$$

where p is the middle point of the sequence $1, 2, \ldots, n$

Comparison between the median and the mean

The mean and the median have different properties. Perhaps the most important is that the mean $\overline{x}$, for example, minimizes the mean square error (MSE) $f(a)$:

$$f(a) = \frac{1}{n-1}\sum_{t=1}^{n}(x_t - a)^2 \tag{1.6}$$

In fact we have

$$f(\overline{x}) = s^2 \leq f(a) \qquad \text{for all } a.$$

On the other hand the median m minimizes the mean absolute error (MAE)

$$g(a) = \frac{1}{n}\sum_{t=1}^{n}|x_t - a| \tag{1.7}$$

and we have in fact:

$$g(m) \leq g(a) \qquad \text{for all } a.$$

1.1.3 Histograms

The histogram of a time series is a chart showing the frequency as a function of data bins. To construct a histogram we first start by choosing a bin size h. The interval $[\min(x_t), \max(x_t)]$ is then divided into subintervals or bins of width h. A practical way to get the bin size is to divide the data range by the square root of the sample size, i.e. $(\max(x_t) - \min(x_t))/\sqrt{n}$. As a rule of thumb the number of bins can be chosen to be between 5 and 20. The histogram is then obtained by plotting the number of data points falling in each bin. Sometimes we scale these numbers by the total sample size.

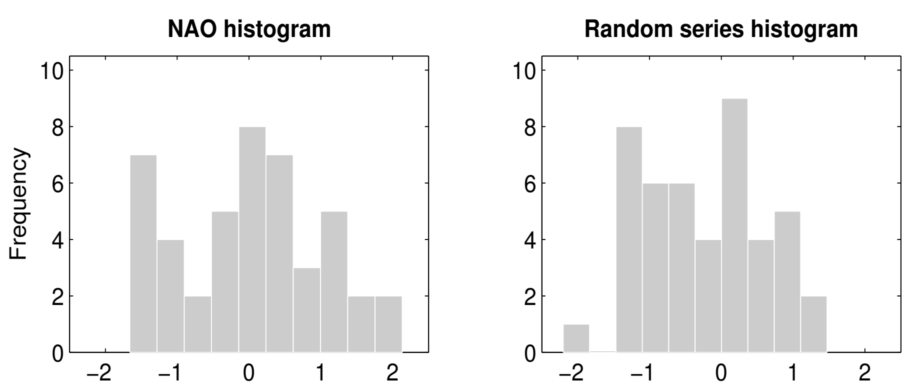

Figure 1.3: Histograms of the random time series and the NAO index shown in figures 1.1 and 1.2

Figure 1.3 shows the histograms of the time series shown in figures 1.1 and 1.2. The NAO time series suggests two peaks, that is a bimodal histograms. But most importantly the NAO histogram looks asymmetric with a longer tail on the positive NAO side compared to the other side. This is an example of a positive skewness.

1.2 Probability concepts

1.2.1 Random variable

Probability is used in our daily life, directly or indirectly, to make decisions when the outcome is not certain. Direct use of probability can be found for example in weather forecasting, health care, insurance and sport strategies. Indirect use of probability at a personal level can be in the form of subjective probability to make judgment. Many textbooks exist about probability, and I refer the reader to the book by DeGroot and Schervish (2012), which presents a comprehensible introduction to the subject. It is easily accessible with examples alongside, but many other books also provide similar style.

A real random variable X is a real-valued function defined on a sample space $\mathcal{S}$ representing the outcome of the random experiment. The random variable X assigns a value x to each outcome from $\mathcal{S}$, known also as events. This may seem mambo-jambo but the following example explains this in real terms.

Example

Coin-tossing experiment. In this experiment the outcomes are H and T for head and tail respectively. Hence $\mathcal{S} = \{H, T\}$. We can assign values 0 and 1 to H and T respectively so that $X = \{0, 1\}$.

1.2.2 Probability

A probability $Pr(.)$ is a positive additive function defined over the subsets of $\mathcal{S}$ taking values inside $[0, 1]$. It satisfies the following properties:

(1) $0 \leq Pr(A) \leq 1$ for any event A

(2) $Pr(\mathcal{S}) = 1$

(3) $Pr(A_1 U A_2 U \ldots U A_n) = \sum_{i=1}^{n} Pr(A_i)$ for disjoint events $A_1, A_2, \ldots, A_n$

There are discrete and continuous random variables. The outcomes of discrete random variables are discrete and the (potential) sample space can be infinite[3] and finite whereas for continuous variables one gets values with density over continuous sets such as whole intervals. Each random variable is defined by its probability density function (PDF) and cumulative distribution function (CDF). The table below shows the difference between both variables.

	Discrete random variable	Continuous random variable
PDF	$f(x) = Pr(X = x)$	$f(x)dx = Pr(x \leq X \leq x + dx)$
CDF	$F(x) = Pr(X \leq x) = \sum_{x_i \leq x} f(x_i)$	$F(x) = Pr(X \leq x) = \int_{-\infty}^{x} f(u)du$

Table 1. Definition of discrete and continuous random variables.

[3] as, for example, the case of the Poisson distribution.

We have the following properties (of density functions) in the continuous case:

- $f(x) \geq 0$ for all x for which $f(.)$ is defined, and $\int f(x)dx = 1$
- $F(.)$ is monotonically increasing, and $f(x) = \frac{dF(x)}{dx}$
- $F(-\infty) = 0$, $F(+\infty) = 1$, and $F(.)$ is left continuous. Figure 1.4 illustrates an example of

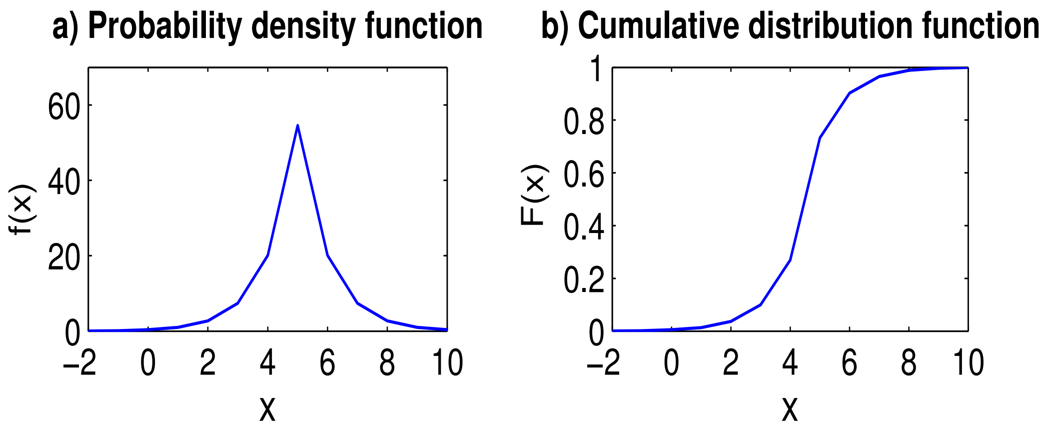

Figure 1.4: An example of a probability density (a) and cumulative distribution (b) functions. Note that the PDF $f(x)$ has been multiplied by 120.

a PDF (1.4a) and CDF (1.4b). Notice in particular, the monotonicity of the CDF. The area between the x-axis and the PDF curve (Fig. 1.4a) is one. On the figure this area is not one because the PDF was multplied by 120.

Exercise
Let X be a continuous random variable defined over an interval $[a, b]$. Compute $Pr(X = x_0)$, where x_0 is any point inside $[a, b]$.

Remark

> If X is a discrete RV with $Pr(X = x_k) = p_k$, $k = 1, \ldots n$, then X can be regarded as continuous variable with PDF given by:
>
> $$f(x) = \sum_{k=1}^{n} p_k \delta_{x_k}(x)$$
>
> where $\delta_{x_k}(.)$ is the Dirac delta function, which satisfies $\delta_{x_k}(x) = 1$ if $x = x_k$ and zero otherwise.

Exercise
We consider a continuous random variable defined over the set of real numbers, with the following PDF

$$f(x) = a \exp[-\frac{1}{2s^2}(x - m)^2] \tag{1.8}$$

with a and s being positive numbers, and m a real number. Derive the expression of a as a function of s and m and show that it is independent of m.

1.2.3 Expectation and moments

Let X be a real random variable. We have already given a definition of the moments above. Here we give the corresponding definitions in terms of the PDF $f(.)$ of X.

- *Expectation*

The expectation (or mean) of X is defined by:

$$E(X) = \int x f(x) dx = \mu,$$

and more generally $E(h(X)) = \int h(x) f(x) dx$, for any given (reasonably well behaving) function $h()$.

Exercise

Derive $E(X)$ for a discrete RV.

- *Moments*

The k'th, $k \geq 2$, order moment μ_k of X is:

$$\mu_k = E\left(X - \mu\right)^k = \int (x - \mu)^k f(x) dx$$

For example, for $k = 2$ we obtain the variance, i.e.

$$\mu_2 = \mathrm{var}(X) = E(X - \mu)^2 = \sigma^2$$

Exercise

Derive the mean and the variance of the RV with PDF given by eq (1.8).

- *Moment generating function*

The moment generating function $M_X()$ of a random variable X is a function defined on the real numbers and is given by:

$$M_X(t) = E(e^{tX}) = \int e^{tx} f(x) dx$$

The moment generating function can be used to compute the different moments of a random variable.

Three basic examples of random variables are given below: the Bernoulli, Poisson and normal distributions.

Example 1: Bernoulli distribution

The Bernoulli random variable X is discrete and takes only two values, say 0 and 1, with respective probabilities p and q, i.e. $Pr(X = 0) = p$, and $Pr(X = 1) = q = 1 - p$. The pdf is simply $f(x) = (1 - p)^x p^{1-x}$, with $x = 0$, or 1. An example is the coin tossing.

Exercise

Derive the mean and the variance of the Bernoulli random variable.

Example 2: Bernoulli distribution B(n,p)

The binomial distribution $B(n,p)$ is the distribution resulting from the number of successes obtained from n independent Bernoulli distributions with success probability p. Let $X = B(n,p)$, the probability of getting exactly k successes is:

$$Pr(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

where $\binom{n}{k} = \frac{n!}{k!(n-k)!}$. We also have $E(X) = np$, and $var(X) = np(1-p)$.

Example 3: Poisson distribution

The Poisson random variable with parameter $\lambda > 0$ is defined on the integer numbers by:

$$Pr(X = k) = \frac{\lambda^k}{k!} e^{-\lambda},$$

for $k = 0, 1, 2, \ldots$. This random variable is used to model counts, such as the number of customers visiting a bank in a given day of the year, or the number of fish caught by a fisherman.
Exercise
Derive the mean and the variance of the Poisson distribution.
Answer λ

Example 4: The uniform distribution

The uniform distribution $U_{[0,1]}$ on the interval $[0,1]$ has the pdf $f(x) = 1_{[0,1]}$ and cdf $F(x) = x 1_{[0,1]}$. The probability that $U_{[0,1]}$ falls in the interval $[a,b]$, $0 \leq a \leq b \leq 1$, is $b - a$.

Example 5: Normal distribution

Eq (1.8) is known as the normal or Gaussian probability density function with variance σ^2 and mean μ, i.e., $a = \frac{1}{\sigma\sqrt{2\pi}}$, $s^2 = \sigma^2$, and $m = \mu$. The standard Gaussian PDF corresponds to $\mu = 0$ and $\sigma = 1$, i.e.,

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

and the CDF, usually noted by $\Phi(.)$ is:

$$\Phi(x) = \int_{-\infty}^{x} f(x)dx.$$

Figure 1.5 shows the PDF of the standard Gaussian variable $\mathcal{N}(0,1)$. The shape of the normal PDF is sometimes known as the "bell" shape. The PDF decays exponentially hence concentrating most of the "mass" around the mean, which is zero here.

The Gaussian random variable has a symmetric probability distribution (about the mean), that is a zero-skewness. In addition, it is mesokurtic, meaning that it has zero excess kurtosis. Skewness is a measure of symmetry and kurtosis is a measure of peakedness. Positive excess kurtosis reflects a peaked (or leptokurtic) distribution. Negative excess kurtosis, on the

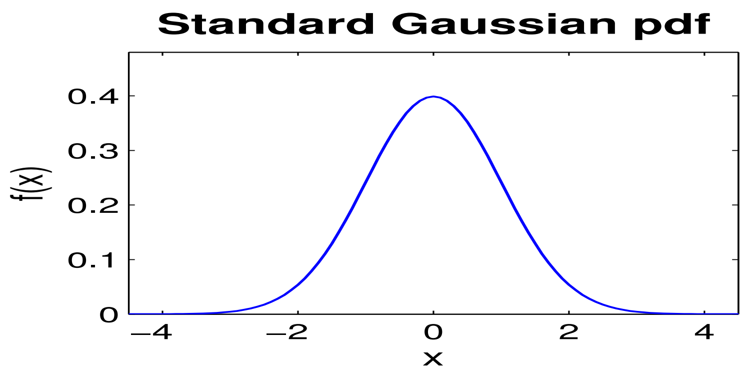

Figure 1.5: The probability density function of the standard normal $\mathcal{N}(0,1)$

other hand, yields a platykurtic distribution and reflects flatness of the probability density dunction. Note that a leptokurtic distribution yields fatter tails compared to the Gaussian, and yields higher probability of extremes whereas a platykurtic distribution has thinner tails than the Gaussian.

Remark: Relation between the normal and binomial distributions

Let $X = B(n,p)$, then for large n, $\frac{X-np}{\sqrt{np(1-p)}} \sim N(0,1)$. The proof of this result can be obtained using the moment generating function $M_X(t) = E(e^{tX})$ of X, i.e. $M_X(t) = (q+pe^t)^n$, where $q = 1-p$, and use a Taylor expansion of the moment generating function of the centred and scaled randon variable $\frac{X-np}{\sqrt{np(1-p)}}$.

1.3 Multivariate random variable

A vector $\mathbf{x} = (X_1, \ldots, X_p)^T$ of p univariate RV is a p-dimensional vector RV. The PDF of the vector RV $\mathbf{x}$, $f(x_1, \ldots, x_p)$, is a positive function of the multivariate vector $(x_1, \ldots, x_p)$ with unit total integral. The cumulative distribution function (CDF), $F(x_1, \ldots, x_p)$, is given by:

$$F(x_1, \ldots, x_p) = \int_{-\infty}^{x_1} \cdots \int_{-\infty}^{x_p} f(u_1, \ldots, u_p) du_1 \ldots du_p \tag{1.9}$$

The function $f(.)$ is known as the *joint* PDF. The univariate PDF $f_{X_k}(.)$ of any X_k, $1 \leq k \leq p$ is the *marginal* PDF of X_k, and is obtained by integrating the joint PDF over the remaining variables. The expectation of $\mathbf{x}$ is the vector

$$\boldsymbol{\mu} = (\mu_1, \ldots, \mu_p)^T = \int \mathbf{u} f(\mathbf{u}) d\mathbf{u} = \int \mathbf{u} f(u_1, \ldots, u_p) du_1 \ldots du_p$$

Componentwise, we have:

$$\mu_k = \int u_k f_{X_k}(u_k) du_k = \int \ldots \int u_k f(u_1, \ldots, u_p) du_1 \ldots du_p.$$

The covariance matrix $\boldsymbol{\Sigma}$ of the multivariate (or vector) random variable $\mathbf{x}$ is given by:

$$\boldsymbol{\Sigma} = E\left[(\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^T\right] = (\sigma_{ij}) \tag{1.10}$$

where σ_{ij} is the covariance between the random variables X_i and X_j, that is:

$$\sigma_{ij} = \mathrm{cov}(X_i, X_j) = E\left[(X_i - \mu_i)(X_j - \mu_j)\right] \tag{1.11}$$

The correlation coefficient between X_i and X_j is defined by:

$$\rho_{ij} = \frac{\sigma_{ij}}{\sqrt{\sigma_{ii}\sigma_{jj}}} = \frac{\mathrm{cov}(X_i, X_j)}{\sqrt{\mathrm{var}(X_i)\mathrm{var}(X_j)}} \tag{1.12}$$

Remarks

- Two RV X_1 and X_2 are said to be uncorrelated when $\mathrm{cov}(X_i, X_j) = 0$, or similarly $\rho_{ij} = 0$.
- Two random variables X_1 and X_2 with joint PDF $f(x_1, x_2)$ and marginals $f_{X_1}(.)$ and $f_{X_2}(.)$ are said to be independent when $f(x_1, x_2) = f_{X_1}(x_1)f_{X_2}(x_2)$.
- Independence always implies uncorrelatedness. The converse, however, is not true in general. That is, if X and Y are uncorrelated then, in general they are *not* independent, see a counter example below.

Exercise
Let X be the standard normal distribution $\mathcal{N}(0, 1)$, and let $Y = X^2$. Calculate $\mathrm{cov}(X, Y)$, the covariance between X and Y. Conclude.
Answer – Since $E(X^2) = \mathrm{var}(X) = 1$, $\mathrm{cov}(X, Y) = E(X(X^2 - 1)) = E(X^3) - E(X) = 0$. So the variables X and Y are uncorrelated, but obviously not independent.

Conditional distribution

Let X and Y be two random variables with joint PDF $f(.)$ and marginals $f_X(.)$ and $f_Y(.)$. When the random variable X is fixed to a certain value x_0, i.e. $X = x_0$, then one talks about *conditional* random variable $Y|X = x_0$. The conditional PDF of $Y|X = x_0$ is given by:

$$f_{Y|X=x_0}(y|x = x_0) = \frac{f(x_0, y)}{f_X(x_0)} \tag{1.13}$$

Note that the denominator serves to ensure that the integral of the PDF is one, i.e. $\int f_{Y|X=x_0}(y|x = x_0)dy = 1$, see section 1.2.2.

Example: The multivariate normal distribution

The multivariate normal random variable $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$ has the following probability density function:

$$f(\mathbf{x}) = \frac{1}{(2\pi)^{\frac{p}{2}}\sqrt{|\boldsymbol{\Sigma}|}} \exp\left[-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right] \tag{1.14}$$

The normal distribution is very important in weather and climate and many other disciplines.

The standard bi-dimensional Gaussian PDF is shown in Fig. 1.6. The thrre-dimensional plot of the PDF is shown by the bell shape in Fig. 1.6a. A two-dimensional contour plot of the PDF is shown in Fig. 1.6b.

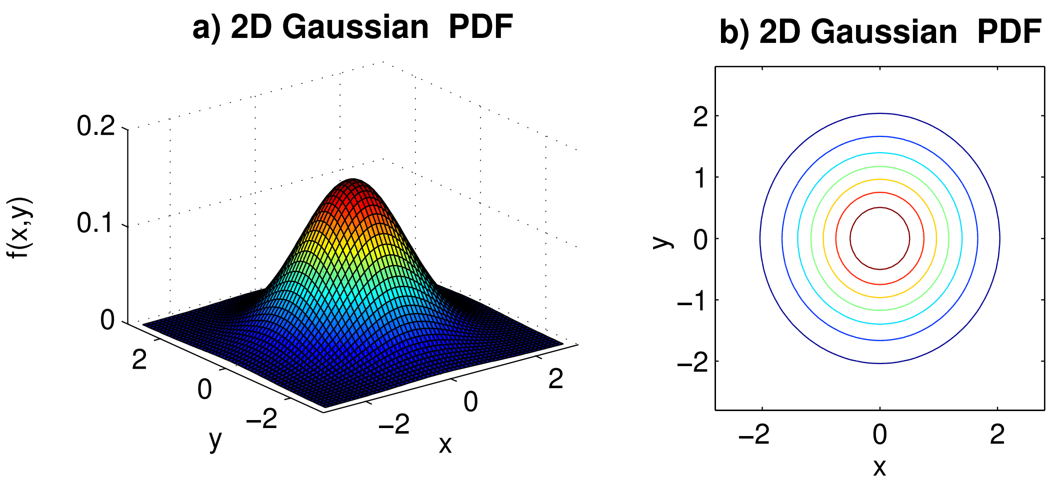

Figure 1.6: The probability density function of the standard bivariate Gaussian

Some basic properties of the normal distribution

The normal distribution has various nice properties. The following properties are perhaps amongst the most basic and also important:

• Independence and non-correlation are two equivalent properties in the space of normal random variables. That is, if X and Y are two normal random variables, then if X and Y are uncorrelated then they are independent and vice versa.
• A linear combination of uncorrelated normals is also normal. Precisely, let X and Y two uncorrelated normal random variables with respective means μ_X and μ_Y and variances σ_X^2 and σ_Y^2, i.e., $X \sim \mathcal{N}(\mu_X, \sigma_X^2)$ and $Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$, then $aX + bY \sim \mathcal{N}(a\mu_X + b\mu_Y, a^2\sigma_X^2 + b^2\sigma_Y^2)$.
• The conditional distribution of a Gaussian vector is also Gaussian.

Remark

If X and Y are two correlated random variables, with correlation coefficient ρ, then the variance of $aX + bY$ takes the form:

$$\sigma_{aX+bY}^2 = a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\ \mathrm{cov}(X,Y) = a^2\sigma_X^2 + b^2\sigma_Y^2 + 2\rho ab\sigma_X\sigma_Y$$

Exercise
Derive the above identity.

Exercise
Let X and Y be two normal random variables with zero-mean and respective variances σ_X^2 and σ_Y^2, and with correlation ρ. Find the expression of the joint PDF of the vector random variable (X, Y).
Answer – Refer to Eq (1.14).

Let Z be the standard normal distribution, then we have the following results:

• $Pr(-1 \leq Z \leq 1) \approx 68\%$ – That is a random value is *likely* to be within the interval $[-1, 1]$.
• $Pr(-2 \leq Z \leq 2) \approx 95\%$ – A random value is *very likely* to be within the interval $[-2, 2]$.
• $Pr(-2.6 \leq Z \leq 2.6) \approx 99\%$ – Any chosen value is *almost certainly* to be within the interval $[-2.6, 2.6]$.

Exercise: The bivariate normal
Let (X, Y) be a bivariate normal distribution, and let $\rho = corr(X, Y)$ (correlation coefficient between X and Y), $\sigma_x^2 = var(X)$, and $\sigma_y^2 = var(Y)$. Denote by $\mathbf{\Sigma}$ the covariance matrix of (X, Y).
Calculate the joint distribution $f(x, y)$ of (X, Y).
What is the nature of the marginal distributions $f_X(x)$ and $f_Y(y)$ of X and Y respectively.
Derive the conditional probability distribution of $Y|X = x_0$.
Conclude.
Answer. The covariance matrix is given by:

$$\mathbf{\Sigma} = \begin{pmatrix} \sigma_x^2 & \rho\sigma_x\sigma_y \\ \rho\sigma_x\sigma_y & \sigma_y^2 \end{pmatrix}$$

Its inverse is :

$$\mathbf{\Sigma}^{-1} = \frac{1}{1-\rho^2} \begin{pmatrix} \frac{1}{\sigma_x^2} & -\frac{\rho}{\sigma_x\sigma_y} \\ -\frac{\rho}{\sigma_x\sigma_y} & \frac{1}{\sigma_y^2} \end{pmatrix}$$

The determinant of the covariance marix is $|\mathbf{\Sigma}| = \sigma_x^2\sigma_y^2(1-\rho^2)$. Also, the term $(x\ y)\mathbf{\Sigma}^{-1}(x\ y)^T$ is equal to $\frac{1}{1-\rho^2}(\frac{x^2}{\sigma_x^2} - 2\frac{\rho xy}{\sigma_x\sigma_y} + \frac{y^2}{\sigma_y^2})$, and the joind pdf is

$$f(x, y) = \frac{1}{2\pi\sigma_x\sigma_y\sqrt{1-\rho^2}} \exp\left[-\frac{1}{2(1-\rho^2)}\left(\frac{x^2}{\sigma_x^2} - 2\frac{\rho xy}{\sigma_x\sigma_y} + \frac{y^2}{\sigma_y^2}\right)\right].$$

The marginal distribution of, e.g. X is obtained by integrating $f(x, y)$ with respect to y, i.e. $f_X(x) = \int f(x, y)dy$ to obtain $f_X(x) = \frac{1}{\sigma_x\sqrt{2\pi}}\exp(-x^2/2\sigma_x^2)$. Using the Bayes law $f(Y|X = x_0) = f(x_0, y)/f_X(x_0)$ yields

$$f(y|X = x_0) = \frac{1}{\sigma_y\sqrt{2\pi}\sqrt{1-\rho^2}} \exp\left[-\frac{1}{2}\frac{1}{\sigma_y^2(1-\rho^2)}(y - \rho\frac{\sigma_y}{\sigma_x}x_0)\right]$$

that is, $Y|X = x_0 \sim N(\rho\frac{\sigma_y}{\sigma_x}x_0, \sigma_y^2(1-\rho^2))$.

Exercise: The bivariate normal in polar coordinates
Consider again the bivariate normal distribution of (X, Y), with pdf $f()$, and with $\rho = 0$. We want to find the pdf in the polar coordinates. Let $x = r\cos\theta$, $y = r\sin\theta$, and recall the rule of variable change, i.e.,

$$E[h(X, Y)] = \int\int f(x, y)h(x, y)dxdy = \int\int f(r\cos\theta, r\sin\theta)h(r\cos\theta, r\sin\theta)|J|drd\theta.$$

where $|J|$ is the determinant of the Jacobian of the transformation, i.e., $|J| = det(\frac{\partial(x,y)}{\partial(r,\theta)}) = r$.
Derive the pdf $g(r, \theta)$ of (X, Y) in polar coordinates.

Derive the marginal pdf in the radial direction.

Answer. From the expression of $E[h(X,Y)]$, we get

$$g(r,\theta) = rf(r\cos\theta, r\sin\theta) = \frac{r}{2\pi\sigma_x\sigma_y} e^{-r^2(\sigma_x^{-2}\cos^2\theta + \sigma_y^{-2}\sin^2\theta)/2}.$$

Using the relations $\cos^2\theta = (1+\cos 2\theta)/2$ and $\sin^2\theta = (1-\cos 2\theta)/2$, the marginal distribution becomes:

$$\begin{aligned} g_r(r) &= \tfrac{1}{2\pi\sigma_x\sigma_y}\int_0^{2\pi} re^{-r^2(\sigma_x^{-2}(1+\cos 2\theta)+\sigma_y^{-2}(1-\cos 2\theta))/4}d\theta \\ &= \tfrac{1}{4\pi\sigma_x\sigma_y} re^{-r^2(\sigma_x^{-2}+\sigma_y^{-2})/4}\int_0^{\pi} e^{-r^2\cos\theta(\sigma_x^{-2}-\sigma_y^{-2})/4}d\theta \\ &= \tfrac{1}{4\pi\sigma_x\sigma_y} re^{-r^2(\sigma_x^{-2}+\sigma_y^{-2})/4} I_0(-r^2(\sigma_x^{-2}-\sigma_y^{-2})/4) \end{aligned}$$

where $I_0()$ is the modified Bessel function $I_n()$ of order $n=0$ ($I_n(z) = \frac{1}{\pi}\int_0^{\pi} e^{z\cos\theta}\cos n\theta d\theta$).

1.4 Further examples of probability distributions

We mentioned in the previous section the example of the normal distribution as the most widely used distribution. It has very useful properties discussed above. We mentioned, in particular, that uncorrelatedness does not imply, in general, independence. The only general exception is actually the normal distribution, that is when X and Y are Gaussian only then independence and uncorrelatedness are equivalent. In climate and other branches of science we also encounter other important distributions, which also involve the normal distribution directly or indirectly. These include mostly the chi-square, the t- and the F-distributions.

1.4.1 The chi-square distribution

The *chi-square* distribution is basically defined as the sum of squared independent normal distributions. Precisely, let $X_1, \ldots, X_k$ be a set of independent and identically distributed (IID) standard normal random variables, i.e., distributed as $\mathcal{N}(0,1)$ and independent, then the expression

$$\chi_k^2 = X_1^2 + \ldots X_k^2$$

is known as the Chi-square distribution with k degrees of freedom (d.o.f). The importance of the chi-square distribution is shown later in chapter 3. The PDF of χ_k^2 involves the gamma function $\Gamma(.)$, and takes the form:

$$f(x) = \frac{1}{2^{\frac{2}{k}}\Gamma(\frac{2}{k})} x^{\frac{2}{k}-1} e^{-\frac{x}{2}} 1_{x\geq 0} \tag{1.15}$$

where the gamma function, $\Gamma(.)$, defined over positive numbers is:

$$\Gamma(x) = \int_0^{\infty} u^{x-1}e^{-u}du$$

Note that for positive integers we get $\Gamma(n) = (n-1)!$. The expectation and variance of a chi-square distribution χ_n^2 are respectively $E(\chi_n^2) = n$ and $\mathrm{var}(\chi_n^2) = 2n$.

1.4.2 The t-distribution

Let $Z \sim \mathcal{N}(0,1)$ and $Y \sim \chi^2_n$ then the random variable:

$$X = \frac{Z}{\sqrt{\frac{Y}{n}}}$$

is known as the (student) *t-distribution* with n d.o.f, T_n, whose PDF is given by:

$$g(x) = \frac{\Gamma\left(\frac{n+1}{2}\right)}{\sqrt{n\pi}\Gamma\left(\frac{n}{2}\right)} \left(1 + \frac{x^2}{n}\right)^{-\frac{n+1}{2}}. \tag{1.16}$$

The mean of T_n distribution is $E(T_n) = 0$ for $n > 1$, but is undefined otherwise. Note also that the skewness of T_n is zero for $n > 3$ (undefined otherwise), hence the t-distribution is symmetrical (for $n > 3$). The t-distribution is fundamental in hypothesis testing, as it will be discussed in chapters 3 and 5.

1.4.3 The F-distribution

Let X and Y be two independent chi-square random variables with respective d.o.f n and m, i.e., $X \sim \chi^2_n$ and $Y \sim \chi^2_m$, then the random variable

$$F_{n,m} = \frac{X/n}{Y/m} \tag{1.17}$$

is known as *F-distribution*, $F_{n,m}$, with n and m d.o.f. The PDF of $F_{m,n}$ involves the beta function and takes the form:

$$f(x;n,m) = \frac{(m/n)^{m/2}}{B(\frac{m}{2},\frac{n}{2})} x^{\frac{n}{2}-1} \left(1 + \frac{n}{m}x\right)^{\frac{-n+m}{2}}$$

where $B(.,.)$ is the beta function defined by:

$$B(x,y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)} = \int_0^1 u^{x-1}(1-u)^{y-1}du.$$

1.4.4 The gamma distribution

The pdf of the gamma distribution with parameters $\lambda > 0$ and $\beta > 0$ is given by:

$$f(x) = \begin{cases} \frac{\lambda^\beta}{\Gamma(\beta)} x^{\beta-1} e^{-\lambda x} & \text{if} \quad x > 0 \\ 0 & \text{otherwise} \end{cases} \tag{1.18}$$

where $\Gamma()$ is the gamma function. If the parameter $\beta < 0$ the distribution is known as Erlang distribution. The parameter β is the shape parameter, and λ is the rate, or similarly $\theta = \lambda^{-1}$ is the scale parameter. So in terms of shape (β) and scale (θ) parameters the above expression (for $x > 0$) becomes $f(x) = \frac{1}{\theta^\beta \Gamma(\beta)} x^{\beta-1} e^{-x/\theta}$.

Exercise

Show that for the above gamma distribution $E(X) = \beta/\lambda$, and $var(X) = \beta/\lambda^2$. Show that $\phi(t) = (1 - it/\lambda)^{-\beta}$

1.4.5 Quantiles

Let X be a random variable with $f(.)$ and $F(.)$ as PDF and CDF respectively. For any $0 \leq \alpha \leq 1$, the equation:

$$F(x) = \alpha \tag{1.19}$$

has a unique solution given by:

$$x_\alpha = F^{-1}(\alpha) \tag{1.20}$$

The number x_α given in eq (1.20) is known as α'*th* quantile of the random variable X.

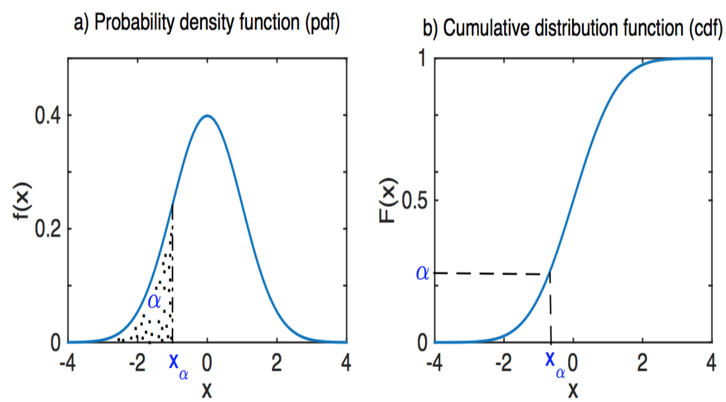

Figure 1.7: Representation of the α'*th* quantile x_α in the pdf (a) and the cdf (b) of a random variable with pdf $f()$ and cdf $F()$.

Note that other notations exist in the literature for quantiles. Figure 1.7 illustrates the α'*th* quantile in the pdf (1.7a) and the cdf (1.7b). In general if X has a standard notation of its CDF, such as $\Phi(.)$ for the standard normal, then F_α can also refer to the α'*th* quantile, which we will be using interchangeably in the sequel for the Gaussian distribution. For other distributions, mainly the chi-square (χ^2_n) and F-distribution ($F_{n,m}$) for our case we will use $\chi^2_n(\alpha)$ and $F_{n,m}(\alpha)$ for the α'*th* quantile for the chi-square and F-distribution respectively.

1.5 Advanced results in probability and statistics

1.5.1 Bayes law

One of the fundamental laws in probability is Bayes law. It describes the probability of an event conditional on observing another event. Bayesian law is mostly used in Bayesian estimation as in Monte Carlo simulations (chapter 9). Given two events A and B, with $Pr(B) \neq 0$, then the conditional probability of A, conditional on observing B can be obtained from the probability of observing, B conditional on A and the marginal probabilities of A and B. The basic concept of conditional probability, already mentioned earlier, is given by:

$$Pr(A|B)Pr(B) = Pr(A \text{ and } B). \tag{1.21}$$

Note that if A and B are independent $Pr(A|B) = Pr(A)$. Eq (21) yields the joint probability of A and B, i.e., $Pr(A, B) = Pr(A|B)Pr(B) = P(B|A)Pr(A)$. The above identities can be put together to yield Bayes law (or theorem):

Bayes law

The probability of event A, given that event B has been observed is

$$Pr(A|B) = \frac{Pr(A)Pr(B|A)}{Pr(A)Pr(B|A) + Pr(\overline{A})Pr(B|\overline{A})}, \tag{1.22}$$

where $\overline{A}$ stands for non A. Note that the denominator of Eq (1.22) is $Pr(A \wedge B) + Pr(\overline{A} \wedge B)$, i.e. $Pr(B)$.

Bayes law can also be given in terms of the probability distribution. If $f()$ is the joint pdf of random variables X and Y, then we have the conditional probability densities presented earlier, i.e.,

$$f_X(x|Y = y) = \frac{f_{X,Y}(x, y)}{f_Y(y)} = \frac{f_Y(y|X = x)}{f_Y(y)} f_X(x) \tag{1.23}$$

where $f_X()$ and $f_Y()$ are, respectively, the marginal pdfs of X and Y. Another important result in probability and statistics is the central limit theorem (CLT), presented next.

1.5.2 Central limit theorem

Assuming a sequence of idependent and identically distributed (iid) random variables $X_1, X_2, \ldots, X_n$ with mean μ and finite variance, $\sigma^2 = var(X_k) < \infty$, then the mean $\overline{X}_n = \frac{1}{n}\sum_{t=1}^{n} X_t$, properly scaled, converges in distribution towards the normal distribution. Precisely,

$$Z_n = \frac{\overline{X}_n - E(X_1)}{\sqrt{var(X_1)/n}} = \frac{\overline{X}_n - \mu}{\sqrt{\sigma^2/n}} \to N(0, 1) \text{ as } n \to \infty, \tag{1.24}$$

that is the probability function $F_n()$ of Z_n converges to the Gaussian probability distribution. Eq. (24), can also be written in a different form using the sequence $a_n = nE(X_1)$, $b_n = \sqrt{n}\sqrt{var(X_1)}$,

$$\frac{S_n - a_n}{b_n} \to N(0, 1)$$

where $S_n = \sum_{t=1}^{n} X_t$. This will be useful in extreme value theory (chapter 9).

Exercise

Assume that X_k, $k = 1, 2, \ldots$, are iid standard normal, i.e., $X_k \sim N(0, 1)$, find the pdf of $\overline{X}_n$ and Z_n.

Hint – Consider the mean and variance of $\overline{X}_n$.

1.5.3 Law of large numbers

There is also another related result, namely, the *law of large numbers*. Letting $X_1, \ldots X_n$ be a sequence of iid random variables with pdf $f()$ and cdf $F()$, and letting the mean $\mu = \int xf(x)dx = \int xdF(x)$, then the sample mean $\overline{X}_n = \frac{1}{n}\sum_{t=1}^{n} X_t$ converges to the mean μ, i.e. $\overline{X}_n \to \mu$ as $n \to \infty$. The weak law of large numbers (or Kintchin's law) is based on probabilistic convergence, that is, for any (small) $\varepsilon > 0$,

$$lim_{n\to\infty}\ Pr\left(|\overline{X}_n - \mu| < \varepsilon\right) = 1.$$

The strong (or Kolmogorov's) law is based on a almost certain convergence, i.e.,

$$Pr\left(lim_{n\to\infty}\,\overline{X}_n = \mu\right) = 1.$$

The law of large numbers can be extended to any integrable function $\phi()$, i.e. $\frac{1}{n}\sum_{t=1}^{n}\phi(X_t)$ converges to $E(\phi(X_1))$; $\frac{1}{n}\sum_{t=1}^{n}\phi(X_t) \to \int \phi(x)f(x)dx$.

Remark

> The law of large numbers can also be extended to dependent sequences, as is the case for Markov chains, discussed in later chapters.

1.5.4 Markov chains

Definition

A sequenceof random variables $X_1, X_2, \ldots$ is a (first-order) Markov chain if

$$Pr(X_{n+1} = x_{n+1}|X_n = x_n, \ldots, X_1 = x_1) = Pr(X_{n+1} = x_{n+1}|X_n = x_n). \tag{1.25}$$

This means that the future state of the chain, given the past and present, depends only on the present state. We assume here that the chain is homogeneous, i.e., Eq (25) does not depend on n (it can be written using a different time index, say m). The above definition means that the chain is completely determined by the initial distribution of X_0, and the transition probability, i.e., $Pr(X_{n+1}|X_n)$.

Examples of Markov chains

1. A sequence of iid is a trivial Markov chain.

2. The red-noise discussed in chapter 2, or first-order autoregressive model is a Markov (process), in which the time index is continuous.

3. Random walk – It is defined by the sequence, $X_0, X_1, \ldots$, with $X_0 = 0$ (with probability 1), and for $n \geq 1$, $X_n = X_{n-1} + U_n$, with $U_1, U_2, \ldots$ being a binary iid sequence, with $Pr(U_1 = 1) = p$, and $Pr(U_1 = -1) = q = 1 - p$. So, given X_{n-1}, the transition probability is given by the rule:

$$X_n = \begin{cases} X_{n-1} + 1 \text{ with probability } p \\ X_{n-1} - 1 \text{ with probability } 1 - p. \end{cases}$$

Figure 1.8 shows an example of a random walk. Note that the random walk has variance proportional to t. This process is obtained by running the following small Matlab code.

```
>> N=10000; X = zeros (N,1);
>> for t= 1:N-1
>> u = rand(1);
>> if (u > 0.5)
>> X(t+1) = X(t) + 1;
>> else
>> X(t+1) = X(t) - 1;
```

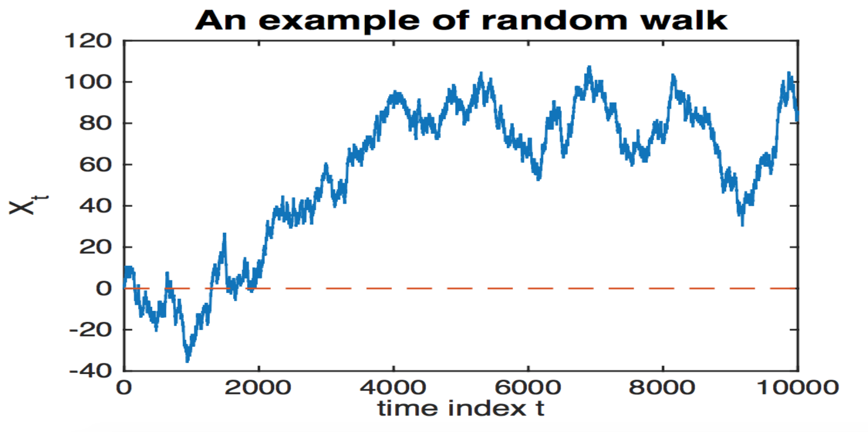

Figure 1.8: Example of random walk.

```
>> end
>> end
```

The Python commands are:

```
>>> import numpy as np
>>> N, t = 10000, 1
>>> X = N*[0]
>>> while t <= N-2:
>>>     u = np.random.random()
>>>     if (u > 0.5):
>>>         X[t+1] = X[t] + 1
>>>     else:
>>>         X[t+1] = X[t] - 1
>>>     t = t+1
```

Markov chains have states. A state of a Markov chain $X_1, X_2, \dots$ at time t is the value of X_t at time t, and the state space $\mathcal{S}$ of the chain is the set of values of the chain. The state space $\mathcal{S}$ can be discrete, finite or infinite, or continuous. For a discrete chain, with state $\mathcal{S}$, the set of probabilities $Pr(x_t|x_{t-1})$ is an element of $\mathbb{R}^{\mathcal{S}\times\mathcal{S}}$, i.e., a $d \times d$ matrix, where[4] d represents the size of $\mathcal{S}$, i.e. $d = |\mathcal{S}|$.

An example illustrating a Markov chain with 4 states S_1, S_2, S_2 amd S_4 is shown in Figure 1.9. The transition probability matrix $\mathbf{P} = (p_{ij})$, of the example shown in Figure 9.1 is given below.

$$\mathbf{P} = \begin{pmatrix} 0 & p_{12} & 0 & p_{14} \\ 0 & 0 & p_{23} & 0 \\ p_{31} & p_{32} & p_{33} & 0 \\ p_{41} & 0 & 0 & p_{44} \end{pmatrix}$$

Note that the sum of the terms in each row of $\mathbf{P}$ is 1. That is, for each i, $i = 1, \dots |\mathcal{S}|$,

[4] In set theory, $X^{\mathcal{S}}$ is the set of all functions from $\mathcal{S}$ to X.

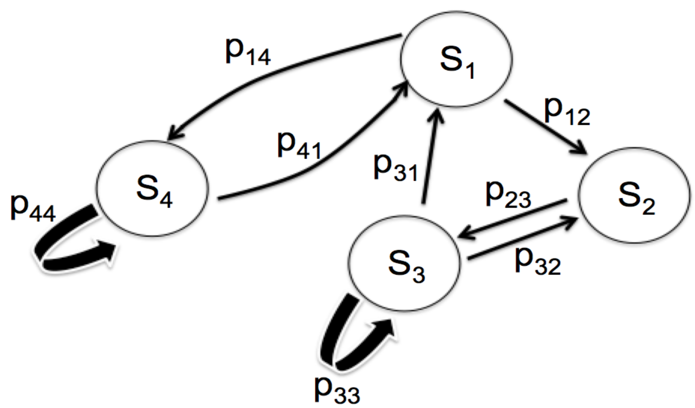

Figure 1.9: Illustration of a simple Markov chain with 4 states, S_1, S_2, S_3 and S_4.

$\sum_j p_{ij} = 1$. This also means that $\lambda = 1$, is an eigenvalue of $\mathbf{P}$ with eigenvector $\mathbf{1}$, i.e., $\mathbf{P1} = \mathbf{1}$.

Remark

Given the transion probability matrix $\mathbf{P}$, and the initial probabilities $\mathbf{q}_0 = (q_0^{(1)}, \ldots, q_0^{(d)})$, the transition probability matrix at step m is $\mathbf{P}^m$. That is the transition probability between S_i and S_j at step m is $(\mathbf{P}^m)_{ij} = Pr(X_{m+n} = S_i|X_n = S_j) = Pr(X_m = S_i|X_0 = S_j)$. The probability distribution at step m, $\mathbf{q}^{(m)} = (q_m^{(1)}, \ldots, q_m^{(d)})$, with $q_m^{(k)} = Pr(X_m = S_k)$, $k = 1, \ldots d$, given the initial probability distribution $\mathbf{q}_0$ is given by $\mathbf{q}^{(m)} = \mathbf{q}_0\mathbf{P}^m$.

Markov chains and ergodicity

Recurrence
A state S of a Markov chain is recurrent if the chain, having started in S, will eventually return to it with probability 1, i.e. $Pr(X_k = S|X_0 = S) = 1$ for some finite $k > 0$. If, on the other hand, $Pr(X_k = S|X_0 = S) < 1$ then the state is transient. If all the states are recurrent, then the chain is recurrent.
Irreducibility
A chain is irreducible if it can reach any state from any other state, i.e., $P^m(x, y) = P^m(x|y) = 1$, for some integer $m > 0$, where $P^m(x|y)$ is the transition probability from y to x at step m, i.e., $P^m(x|y) = Pr(X_m = x|X_0 = y)$.
Periodicity
A chain is aperiodic if, starting from a state S, there is no time constraint on its return to the same state. That is, $GCD\{m; P^m(x, x) > 0\} = 1$, where GCD is the greatest common divisor.

Remark

If the GCD is greater than 1, say for example 3, this means that the chain will return to the state x in cycles of multiples of 3, i.e., 3, 6, 9, ...

Ergodicity

An irreducible and aperiodic Markov chain is called ergodic. That is, there exists an integer $m > 0$, such that $P^m > 0$.

Stationary distribution of Markov chains

Stationary time series is addressed in more details in chapter 2, but we give here a brief description of stationarity of probability distributions in relation to Markov chains. A probability density function $f()$ is a stationary (or invariant) distribution of a Markov chain if $X_0 \sim f$ implies $X_n \sim f$ for all n, i.e.

$$f(y) = \sum_{x \in \mathcal{S}} P(y|x) f(x). \tag{1.26}$$

An ergodic Markov chain has at most one stationary distribution. In particular, a finite ergodic Markov chain, with transition probability matrix $\mathbf{P}$, has one stationary distribution, $\mathbf{f}$, satisfying $\mathbf{f} = \mathbf{fP}$.

Exercise

Find the stationary probability distribution of the Markov chaing of Fig. 9.1.

Note that a sufficient condition for the existence of a stationary distribution is the reversibility, also referred to as detailed balance, i.e. $P(y|x)f(x) = P(x|y)f(y)$. In fact, if this condition is satisfied, then

$$\sum_{x \in \mathcal{S}} P(y|x) f(x) = \sum_{x \in \mathcal{S}} P(x|y) f(y) = \left(\sum_{x \in \mathcal{S}} P(x|y) \right) f(y) = f(y)$$

For an ergodic chain, we have, for all A, B and m:

$$\lim_{n \to \infty} Pr\left(X_{n+m} \in B | X_m \in A\right) = \int_B f(x) dx \tag{1.27}$$

Furthermore, if $\phi()$ is integrable, then we have (the ergodic theorem):

$$\frac{1}{n} \sum_{t=1}^{n} \phi(X_t) \to \int \phi(x) f(x) dx \tag{1.28}$$

Remark

> Let the sequence $X_1, X_2, \ldots$ denote a Markov chain with stationary distribution $f()$. For a function $g()$ satisfying $var(g(X_1)) < \infty$, and letting $\overline{X}_n = \frac{1}{n} \sum_{t=1}^{n} g(X_t)$, we have the Markov chain central limit theorem (convergence in distribution):
>
> $$\frac{\overline{X}_n - \mu}{\sigma / \sqrt{n}} \to N(0, 1)$$
>
> where $\mu = E(g(X_1))$, and $\sigma^2 = var(g(X_1)) + 2 \sum_{t \geq 1} cov(g(X_1), g(X_{1+k}))$.

Chapter 2

Stationary Time Series

2.1 Time series and autocorrelation

2.1.1 Stationarity and autocorrelation

Let x_t, $t = 1, 2, \ldots n$ be a (real) time series. Each value x_t of the series is considered as a realisation of a random variable. We also speak of a realisation of a stochastic process.

Definition
The time series is said to be (second-order) stationary if the mean is constant and the covariance between x_t and x_s is a function of $(t - s)$, i.e.,
(1) $E(x_t) = \text{const}$
(2) $\text{cov}(x_t, x_s) = \gamma(t - s)$.

For a stationary time series x_t, $t = 1, 2, \ldots n$ the function

$$\gamma(\tau) = \text{cov}(x_{t+\tau}, x_t) = E\left[(x_{t+\tau} - \mu)(x_t - \mu)\right]$$

is known as the *autocovariance* function of the time series.

Remark

Note that $\gamma(0) = \sigma^2$.

Similarly, the autocorrelation function of the times series is given by

$$\rho(\tau) = \frac{\gamma(\tau)}{\sigma^2}$$

2.1.2 Properties of the auocovariance function

The autocovariance function has a number of useful properties:

- The autocovariance is bounded: $|\gamma(\tau)| \leq \gamma(0)$
- The autocovariance is even: $\gamma(-\tau) = \gamma(\tau)$

• The autocovariance is positive semi-definite[1] : For any p integers $\tau_1, \dots \tau_p$ and real numbers $a_1, \dots a_p$, then:

$$\sum_{i.j=1}^{p} \gamma(\tau_i - \tau_j) a_i a_j \geq 0$$

Exercise

Derive the above properties

Indication – Use the fact that for any real number λ, $\text{var}(\lambda x_t + x_{t+\tau}) \geq 0$ and that $\text{var}(\sum_{i=1}^{p} a_i X_{\tau_i)} \geq 0$.

2.1.3 Estimation of the autocovariance

Let x_t, $t = 1, \dots, n$ be a stationary time series. The mean is estimated by average $\overline{x}$, and the sample autocovariance $\hat{\gamma}(\tau)$ can be estimated by:

$$\hat{\gamma}(\tau) = \frac{1}{n-\tau} \sum_{k=1}^{n-\tau} (x_k - \overline{x})(x_{k+\tau} - \overline{x}) \tag{2.1}$$

In various textbooks the factor $\frac{1}{n}$ is used instead of $\frac{1}{n-\tau}$ in eq. (2.1) above. For large sample size and small lags the two estimators, however, are approximately the same.

Exercise

Assume that the stationary time series $x_1, \dots x_n$ is zero-mean and consider the estimator

$$\hat{\gamma}(\tau) = \frac{1}{n} \sum_{k=1}^{n-\tau} x_k x_{k+\tau}$$

Show that $E(\hat{\gamma}(\tau)) = \gamma(\tau) + O(\frac{1}{n})$. What about the estimator (2.1)?

An alternative expression to the sample autocovariance (2.1) is given as an integral and holds when the time series is continuous and defined over a finite interval $[0, T]$. This expression is given by Jenkins and Watts (1968, p 174):

$$\hat{\gamma}(\tau) = \begin{cases} \frac{1}{T} \int_0^{T-|\tau|} x(t) x(t+\tau) dt & \text{for } 0 \leq |\tau| \leq T \\ 0 & \text{otherwise} \end{cases} \tag{2.2}$$

Remark

> When the time series is periodic then the autocovariance function is also periodic. When the time series is not periodic, however, we get: $\hat{\gamma}(\tau) \to 0$ as $\tau \to \infty$.

2.1.4 Partial autocorrelation function

The autocorrelation function $\rho(k)$ includes implicitly all the effects at lags $1, 2, \dots, k$. Here we set the task of computing the correlation between x_t and x_{t-k}, but after allowing for the effect of the variables in between, i.e. when $x_{t-1}, x_{t-2}, \dots x_{t-k+1}$ are given, i.e.,

$$corr(x_t, x_{t-k} | x_{t-1}, \dots x_{t-k+1}) \quad (k \geq 2). \tag{2.3}$$

[1] or non-negative definite

Expression (2.3), considered as a function of the lag k, is by definition the partial autocorrelation function (pacf). The partial autocorrelation can be seen clearly within the autoregressive model framework as discussed in the next section.

2.2 Time series models

There is a whole plethora of time series models in the literature. Here I only mention the two most basic models.

2.2.1 White noise

A white noise is a sequence of IID random variables, normally taken to be Gaussian, ε_t, $t = 1, 2, \ldots$, with $E(\varepsilon_t) = 0$ and $E(\varepsilon_t^2) = \sigma_\varepsilon^2$. The autocorrelation of a white noise is the Dirac δ-function, i.e.

$$\rho(\tau) = \delta(\tau) = \begin{cases} 1 & \text{if } \tau = 0 \\ 0 & \text{otherwise} \end{cases} \tag{2.4}$$

2.2.2 Red-noise

The red-noise model is also known in the literature as the first-order autoregressive $AR(1)$ model. It is defined by:

$$x_t = \alpha x_{t-1} + \varepsilon_t \tag{2.5}$$

where ε_t is a white noise, in general Gaussian, with variance σ_ε^2.

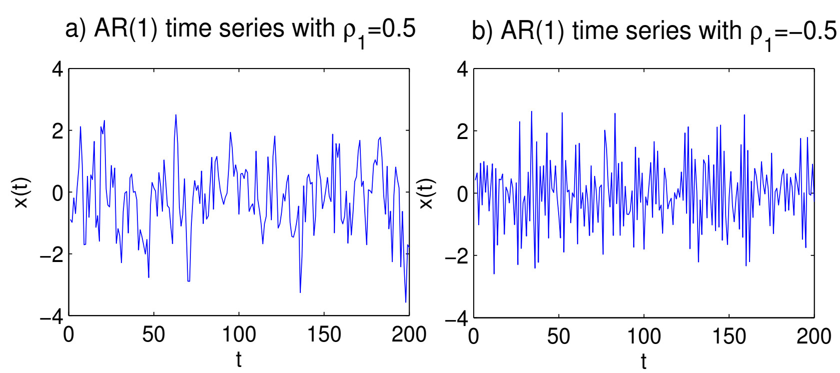

Figure 2.1: Two AR(1) time series with $\rho_1 = 0.5$ (a) and $\rho_1 = -0.5$ (b)

Figure 2.1 shows two examples of a AR(1) time series, one with $\rho_1 = 0.5$ (Fig. 2.1a) and the other with $\rho_1 = -0.5$ (Fig. 2.1b). Note the slow variation in Fig. 2.1a compared to Fig. 2.1b.

Exercise

Consider the red-noise model (2.5).

(1) Calculate $\gamma(\tau + 1)$ vs $\gamma(\tau)$

(2) Derive the expression of $\gamma(\tau)$ as a function of τ and $\gamma(1)$.

Answer – (1+2) The autocorrelation satisfies $\rho(\tau) = \alpha\rho(\tau - 1)$ or $\rho(\tau) = \alpha^\tau = (\rho(1))^\tau$. When $\alpha > 0$ we can get an alternative expression by letting $T_0^{-1} = -\log\alpha$. The autocorrelation becomes $\rho(\tau) = e^{-\frac{\tau}{T_0}}$. The number T_0 is known as the *e-folding* time of $\rho(.)$ Note that $\rho_1 = \alpha$ and $\sigma^2(1-\alpha^2) = \sigma_\varepsilon^2$. An example of autocorelation functions of AR(1) models

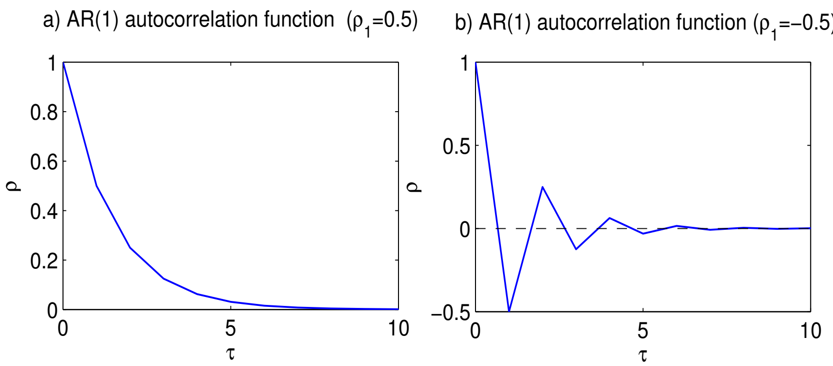

Figure 2.2: The autocorrelation function of a AR(1) model with $\rho_1 = 0.5$ (a) and $\rho_1 = -0.5$ (b)

is shown in Fig. 2.2, one $\rho_1 = 0.5$ (Fig. 2.2a) and the other with $\rho_1 = -0.5$ (Fig. 2.2b). The autocorrelation is characterised by a monotonic and fast decay when $\rho_1 > 0$ (Fig. 2.2a). When $\rho_1 < 0$ the autocorrelation is quite different (Fig. 2.1b). The autocorrelation decays again fast but in zig-zag fashion around the zero-autorrelation axis. Notice also the high frequency nature in the latter case compared to the former case. This was also noted in Fig.2.1. It is perhaps worth mentioning that this example is not common in atmospheric science. An artificial example is constructed in the remark below.

Remark

Consider a (continuous) sine wave $x_t = \cos(\pi t)$, and define a noise by $\varepsilon_{k+1} = \beta\varepsilon_k + \eta_k$, $k = 1, 2, \ldots$, with η_k, $k = 1, 2, \ldots$, are iid $N(0,1)$. Let the process y_k, $k = 1, 2\ldots$, defined by: $y_k = \alpha x_k \varepsilon_k$, with $0 < \alpha, \beta < 1$, then $\rho_1 < 0$

Exercise
Calculate the autocorrelation function $\rho(\tau)$ of y_k, $k = 1, 2, \ldots$.
Answer – $\rho_1 = -\alpha^2\beta$, and $\rho_2 = \alpha^2\beta^2$.

Exercise
Let μ and σ^2 be the mean and the variance of a (stationary) time series x_t, $t = 1, \ldots, n$. We wish to fit an $AR(1)$ model to the time series. Calculate α and σ_ε^2 in terms of μ, σ^2 and $\gamma(1)$.
Answer – Take the mean out and work with anomalies, and fit the same model to the anomalies, i.e. $x_t = \alpha x_{t-1} + \varepsilon_t$, where now x_t stands for the anomaly. Now the mean of the noise is zero, i.e. $E(\varepsilon_t) = 0$. Squaring both sides, and taking expectation we get $\sigma^2 = \alpha^2\sigma^2 + \sigma_\varepsilon^2$. Next we compute the lag-1 autocovariance by multiplying the left and right hand sides by x_{t-1} and taking expectation. We get $\gamma(1) = \sigma^2\alpha$, or similarly $\alpha = \rho(1)$, and therefore $\sigma_\varepsilon^2 = (1-\alpha^2)\sigma^2$. Note that $\alpha = 1$ does not yield AR(1), but a random walk, which is not stationary (σ^2 is not constant).

2.2.3 Autoregressive model of order p – AR(p)

The Autoregressive model of order p, $AR(p)$, is an extension of the $AR(1)$ model taking into account lagged autocorrelations up to lag p. The model reads:

$$x_t = \phi_1 x_{t-1} + \phi_2 x_{t-2} + \ldots + \phi_p x_{t-p} + \varepsilon_t \tag{2.6}$$

with ε_t, $t = 1, 2, \ldots$ is a sequence of IID with zero-mean and variance σ_ε^2. Note that in this model ε_t is correlated only with x_s for $s \geq t$.

Exercise

Consider a $AR(2)$ time series model

$$x_t = \phi_1 x_{t-1} + \phi_2 x_{t-2} + \varepsilon_t$$

(1) Derive the relationship satisfied by the autocorrelation function $\rho(\tau)$

(2) Given $\rho(0)$ and $\rho(1)$ how does $\rho(\tau)$ look like?

Answer – (1). $\rho(\tau + 1) = \phi_1 \rho(\tau) + \phi_2 \rho(\tau - 1)$. Note that the solution to this recurrence equation is given by $\rho(\tau) = a z_1^\tau + b z_2^\tau$ where z_1 and z_2 are the roots of $z^2 - \phi_1 z - \phi_2 = 0$ (assumed distinct), and a and b are obtained from the initial conditions $\rho(0)$ and $\rho(1)$.

The concept of partial autocorrelation introduced earlier fits in naturally here. Consider now the AR(p) model that we write as:

$$x_t = \phi_{p,1} x_{t-1} + \ldots \phi_{p,p} x_{t-p} + \varepsilon_t \tag{2.7}$$

The last coefficient $\phi_{p,p}$, that we denote $\pi(p)$, measures the effect of x_{t-p} on x_t but excluding the effect of the intervening variables $x_{t-1}, \ldots x_{t-p+1}$. It measures the correlation between x_t and x_{t-p} not accounted for by $x_{t-1}, \ldots x_{t-p+1}$, and provides the partial autocorrelation function at lag p for the model. Another way to understand the partial autocorrelation function can be obtained after noting that the coefficient $\phi_{p,p} = \pi(p)$ measures the excess correlation at lag p that is not accounted for by the $AR(p-1)$ model. That is, if we fit a $AR(p-1)$ model:

$$x_t = \phi_{p-1,1} x_{t-1} + \ldots \phi_{p-1,p-1} x_{t-p+1} + \varepsilon_t$$

then the autocorrelation function at lag $p-1$ becomes $\pi(p-1) = \phi_{p-1,p-1}$. In practical terms, the partial autocorrelation function $\pi(p)$ is obtained first by removing the effect of $x_{t-1}, \ldots x_{t-p+1}$ from x_t and x_{t-p} by regression, then computing the correlation between the residuals.

An application of AR models to the North Atlantic Oscillation (NAO) can be found in Onskog et al. (2017). They fitted AR(3) and AR(5) models to daily NAO time series. But because the noise was not Gaussian, the authors fitted leptokurtic distributions to fit the noise. Onskog et al. (2020) improved the modelling by fitting nonlinear AR models to the same daily NAO time series. These models were found to reproduce excellently the characteristic features of the time series, including skewness, distribution tail and the time scales of the positive and negative NAO phases.

2.2.4 Moving average model – MA(q)

The moving average of order q, MA(q), is defined by

$$x_t = \varepsilon_t + \theta_1 \varepsilon_{t-1} + \dots \theta_q \varepsilon_{t-q} \tag{2.8}$$

where $\theta_1, \dots \theta_q$ are constants, and ε_t is a white noise with zero-meand and variance σ_ε^2.

Exercise

Show that the MA(q) is 2nd-order stationary for all values of the parameters $\theta_1, \dots \theta_q$.

To get the autocovariance it is straightforward to compute the $E(x_t x_s)$, and we obtain:

$$\gamma(\tau) = \sigma_\varepsilon^2 \sum_{k=0}^{q-|\tau|} \theta_k \theta_{k+\tau} 1_{|\tau| \le q} \tag{2.9}$$

where $1_{|\tau| \le q}$ is one for all integer τ satisfying $|\tau| \le q$ and zero otherwise.

2.2.5 Autoregressive moving average model – ARMA(p,q)

In the AR(p) model the noise part in the right hand side of the equation is uncorrelated. An extension to this model is to include an autocorrelation to the noise. In this case the obtained model becomes an autoregressive moving average model of order p and q, $ARMA(p,q)$, and reads:

$$x_t - \phi_1 x_{t-1} - \dots - \phi_p x_{t-p} = \varepsilon_t + \theta_1 \varepsilon_{t-1} + \dots \theta_q \varepsilon_{t-q} \tag{2.10}$$

Basically the ARMA(p,q) is a combination of a AR(p) and MA(q).

Remark

> The above models are linear models. There are of course many other models including nonlinear ones, but these are not discussed here, see e.g., Kantz and Schreiber (2003) and Tong (2011).

2.3 Cross-covariance and cross-correlation

In this section we will consider two time series x_t and y_t, $t = 1, 2, \dots$, with respective autocorrelation functions $\gamma_x(.)$ and $\gamma_y(.)$. The cross-covariance function is $\gamma_{xy}(\tau) = \text{ cov } (x_t, y_{t+\tau})$. The cross-correlation is $\rho_{xy}(\tau) = \frac{\gamma_{xy}(\tau)}{\sigma_x \sigma_y}$. We now assume that we have finite samples of x_t and y_t, $t = 0, 1, \dots n-1$ supposed to be zero-mean. The sample cross-covariance is estimated by:

$$\hat{\gamma}_{xy}(\tau) = \frac{1}{n} \sum_{k=0}^{n-\tau-1} x_k y_{k+\tau}, \qquad 0 \le \tau \le n-2 \tag{2.11}$$

Since the autocovariance function is not symmetric, the above estimate can be separated into two parts depending on the sign of the lag τ to yield an alternative estimate: For a given $\tau \ge 0$, we have:

$$\begin{cases} \hat{\gamma}_{xy}(\tau) = \frac{1}{n} \sum_{t=0}^{n-\tau-1} (x_t - \overline{x})(y_{t+\tau} - \overline{y}) \\ \hat{\gamma}_{xy}(-\tau) = \frac{1}{n} \sum_{t=\tau}^{n-1} (x_t - \overline{x})(y_{t-\tau} - \overline{y}) \end{cases} \tag{2.12}$$

As for the autocovariance function we can also see here that

$$E\left(\hat{\gamma}_{xy}(\tau)\right) = (1 - \frac{\tau}{n})\gamma_{xy}(\tau) = \gamma_{xy}(\tau) + O(\frac{1}{n})$$

The estimator (2.11) is therefore asymptotically unbiased. If we use $(n-\tau)^{-1}$ instead of n^{-1} then we obtain an unbiased estimator.

2.4 ARMA autocovariance, stationarity and identification

2.4.1 ARMA autocovariance and stationarity

Delay operator

A convenient way to express ARMA models is to use the backward shift or delay operator B:

$$Bx_t = x_{t-1} \tag{2.13}$$

Exercise

1. Show that for $m \geq 1$, $B^m x_t = x_{t-m}$.
2. Can we define the inverse operator B^{-1}.
3. Show that formally we have: $(1-\alpha B)^{-1}x_t = (1 + \alpha B + \alpha^2 B^2 + \ldots) = x_t + \alpha x_{t-1} + \ldots$.

Hint – B^{-1} is the forward operator.

Exercise

Consider the mean $\overline{x}$ of a finite time series x_t, $t = 1, 2, \ldots n$. Find a relationship between $\overline{x}$ and x_n.

Answer – $\overline{x} = \frac{1}{n}\frac{1-B^n}{1-B}x_n$.

The continuous analogue of the delay operator is the continuous shift operatot and is defined for a continuous time series by:

$$B^u x(t) = x(t-u)$$

Exercise

Denoting by $D = \frac{d}{dt}$, show that formally $B^u = e^{-uD}$.

Answer – Use a Taylor expansion of $x(t-u)$.

ARMA autocovariance

The use of the operator B helps simplify the notation considerably. For instance, the (zero-mean) $AR(p)$ is written as:

$$P(B)x_t = (1 - \phi_1 B - \phi_2 B^2 - \ldots - \phi_p B^p)x_t = \varepsilon_t \tag{2.14}$$

where $P(.)$ is the polynomial:

$$P(z) = 1 - \phi_1 z - \phi_2 z^2 - \ldots \phi_p z^p. \tag{2.15}$$

We let $z_1, \ldots, z_p$ be the roots of the polynomial $z^p P(z^{-1}) = z^p - \phi_1 z^{p-1} \ldots - \phi_p$. By multiplying (2.6) by x_{t-k}, $k > 0$, taking expectation and dividing by the variance of (x_t), one gets the recurrence relation:

$$\rho_k - \phi_1 \rho_{k-1} - \ldots - \phi_p \rho_{k-p} = 0. \tag{2.16}$$

The general solution of the above equation is:

$$\rho_m = \sum_{k=1}^{s} \sum_{l=0}^{m_k - 1} a_{kl} |m|^l z_k^{|m|} \tag{2.17}$$

where s is the number of distinct roots and m_k is the multiplicity of z_k.

Remark

> Note that when all the roots are distinct we get $\rho_m = \sum_{k=1}^{p} a_k z_k^{|m|}$. The coefficients are determined from the initial conditions, i.e., $\rho_0 = 1, \rho_1, \ldots, \rho_{p-1}$.

It is clear that to have stationarity all the roots should be inside the unit circle, i.e., $|z_k| < 1$, $k = 1, \ldots p$. The autocorrelation is then a mixture of damped exponentials and sine/cosine waves.

Exercise

Consider the AR(2) model $P(B)x_t = \varepsilon_t$, where $P(z) = 1 - \phi_1 z - \phi_2 z^2$. Let z_1 and z_2 be the (distinct) roots of $z^2 P(z^{-1})$.

1. Show that $P(B)x_t = (1 - z_1 B)(1 - z_2 B)x_t = \varepsilon_t$.

2. Find a general solution of this difference equation, and derive the condition for (asymptotic) stationarity (use the result $(1-x)^{-1} = 1 + x + x^2 + \ldots$.)

Answer – 1. $P(z) = z^2(z^{-1} - z_1)(z^{-1} - z_2) = (1 - zz_1)(1 - zz_2)$

2. The general solution is obtained by (formally) inverting the above equation then adding a particular solution to $\phi(B)x_t = 0$. The particular solution takes the form $\alpha z_1^t + \beta z_2^t$ where α and β depend on x_0 and x_1. The other solution is of the form $(1 - z_1 B)^{-1}(1 - z_2 B)^{-1}\varepsilon_t$. Next, we consider the power expansion of $(1-x)^{-1}(1-y)^{-1}$ where we regroup all the terms with the same power, i.e. $1 + (x+y) + (x^2 + xy + y^2) + \ldots + (x^k + yx^{k-1} + \ldots + y^k) + \ldots$. Now each of these terms is of the form $\frac{x^{k+1} - y^{k+1}}{x - y}$, $k \geq 0$. This yields $x_t = \alpha z_1^t + \beta z_2^t + \sum_{k \geq 0} \frac{z_1^{k+1} - z_2^{k+1}}{z_1 - z_2} \varepsilon_{t-k}$. Asymptotic stationarity requires therefore the roots of $P(.)$ to be *outside* the unit circle, or similarly z_1 and z_2 to be *inside* the unit circle.

For an ARMA(p,q) the condition of stationarity is the same as that of AR(p). The autocovariance function satisfies a similar recurrence equation to that of the AR(p) except that an extra term is added to account for the covariance between x_t and ε_s, $s \leq t$.

2.4.2 Model identification

In section 2.2.2 we have seen a simple example to determine the parameters of a AR(1) model. That procedure can be extended to higher order models. The identification procedure is based on the recurrence relationship satisfied by the autocorrelation function. Let us consider, for example, the case of a AR(p) model (2.6) . The model is basically similar to a regression problem (see chapter 5 for more details), with unknowns $\phi_1, \ldots \phi_p$ and the noise variance. The parameters are obtained using a least-square estimates by solving the following minimization problem:

$$\min_{\phi_1,\ldots\phi_p} \sum_{t=p+1}^{n} \left[x_t - \sum_{k=1}^{p} \phi_k x_{t-k} \right]^2 . \tag{2.18}$$

After differentiating wrt to the parameters one obtains:

$$\hat{\gamma}_k = \sum_{l=1}^{p} \hat{\phi}_l \hat{\gamma}_{|k-l|}, \qquad k = 1, 2, \ldots p. \tag{2.19}$$

This system of equations is known as the *Yule-Walker* equations and can be written in a matrix form:

$$\hat{\mathbf{\Gamma}}_p \hat{\boldsymbol{\phi}} = \hat{\boldsymbol{\gamma}} \tag{2.20}$$

where $\hat{\mathbf{\Gamma}}_p = (\hat{\gamma}_{(k-l)})$, $k, l = 1, \ldots, p$, $\hat{\boldsymbol{\gamma}} = (\hat{\gamma}_1, \ldots, \hat{\gamma}_p)$, and $\hat{\boldsymbol{\phi}} = (\hat{\phi}_1, \ldots, \hat{\phi}_p)$. Systems (2.19) or (2.20) can be solved iteratively using the Durbin algorithm (Durbin 1960). For the MA or ARMA models the function to be minimized is nonlinear and non-quadratic and is to be solved iteratively using any descent algorithm, e.g., quasi-Newton (Box et al. 1994; Mukhtar 1977).

The Durbin algorithm is an iterative procedure with the following steps:

1. Set $k = 0$
2. Then $k = 1 + k$
   $\hat{\phi}_{k,k} = \left(\hat{\gamma}_k - \sum_{j=1}^{k-1} \hat{\phi}_{j,k-1} \hat{\gamma}_{k-j} \right) / \sigma^2_{k-1}, \quad j = 1, \ldots, k-1$
   $\hat{\phi}_{j,k} = \hat{\phi}_{j,k-1} - \hat{\phi}_{k,k} \hat{\phi}_{k-j,k-1}, \quad j = 1, \ldots, k-1$
   $\sigma^2_k = \sigma^2_{k-1}(1 - \hat{\phi}^2_{k,k})$
3. Stop if $|\hat{\phi}_{k,k}| < \varepsilon$, otherwise go to 2.

In the above algorithm ε is a small threshold. The numbers $\hat{\phi}_{k,k} = \hat{\pi}(k)$, $k = 1, \ldots p$, represent the sample partial autocorrelation function $\hat{\pi}(k)$, $k = 1, \ldots p$.

2.5 Independence and effective sample size

2.5.1 Background

Intuitively, it is reasonable to think that any real time series x_t, $t = 1, 2, \ldots$, has a non-trivial autocorrelation function, If there is a lag τ_0 such that $\rho(\tau)$, for $\tau \geq \tau_0$, is small enough, i.e., not 'too' different from zero, then it is reasonable to accept that any sub-sample with a sampling interval $\tau \geq \tau_0$ can be regarded as an independent subsample. In other words, the subsample x_t, $t = 1, \tau + 1, 2\tau + 1, \ldots$, for example, is an 'independent' sample.

Example

This is possible, in principle, when for instance the autocorrelation function decreases exponentially, which corresponds to a red-noise process.

2.5.2 Number of d.o.f

The number of d.o.f of a time series is the number n^* of independent samples x^*. Consider, for example, a time series x_t, $t = 1, 2, \ldots, n$ where we assume that the sampling interval corresponds to one unit, i.e., $\Delta t = 1$. If this time series is assumed to be a realisation of a $AR(1)$ process then Leith (1973) suggests:

$$n^* = \frac{n}{2T_0} = -\frac{n}{2} \log \rho(1) \tag{2.21}$$

where T_0 is the e-folding time of the autocorrelation function (see section 2.2.2 above). The idea behind Leith (1973) is that if x_t, $t = 1, 2, \ldots n$ is a IID time series with variance σ^2 then the mean $\overline{x} = n^{-1} \sum_{t=1}^{n} x_t$ has variance $\sigma_{\overline{x}}^2 = n^{-1}\sigma^2$.

Now, consider a continuous time series $x(t)$ defined for all values of t with lagged auto-covariance $\gamma(\tau) = E\left[x(t)x(t+\tau)\right] = \sigma^2 \rho(\tau)$. An estimate of the mean $\overline{x}_t$ for a finite time interval $[0, T]$ is:

$$\overline{x}_t = \frac{1}{T} \int_{t-\frac{T}{2}}^{t-\frac{T}{2}} x(u)du. \tag{2.22}$$

The variance of estimate (2.22) can be easily computed and yields:

$$\sigma_T^2 = \mathrm{var}(\overline{x}_t) = \frac{2\sigma^2}{T} \int_0^T \left(1 - \frac{u}{T}\right) \rho(u) du. \tag{2.23}$$

Exercise

1. Derive (2.23)
2. Compute σ_T^2 for a red-noise time series.
3. Conclude.

Answer – From (2.22) we have:

$$\begin{aligned} T^2 \sigma_T^2 &= E\left[\int_{t-\frac{T}{2}}^{t+\frac{T}{2}} \int_{t-\frac{T}{2}}^{t+\frac{T}{2}} x(s_1)x(s_2)ds_1 ds_2\right] = E\left[\int_{-\frac{T}{2}}^{\frac{T}{2}} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(t+s_1)x(t+s_2)ds_1 ds_2\right] \\ &= \sigma^2 \int_{-\frac{T}{2}}^{\frac{T}{2}} \int_{-\frac{T}{2}}^{\frac{T}{2}} \rho(s_2 - s_1) ds_1 ds_2 \end{aligned}$$

This integral can be computed by a change of variable $u = s_1$ and $v = s_2 - s_1$, which transforms the square $[-\frac{T}{2}, \frac{T}{2}]^2$ into a parallelogram $\mathcal{R}$ (see Fig. 2.3), i.e.,

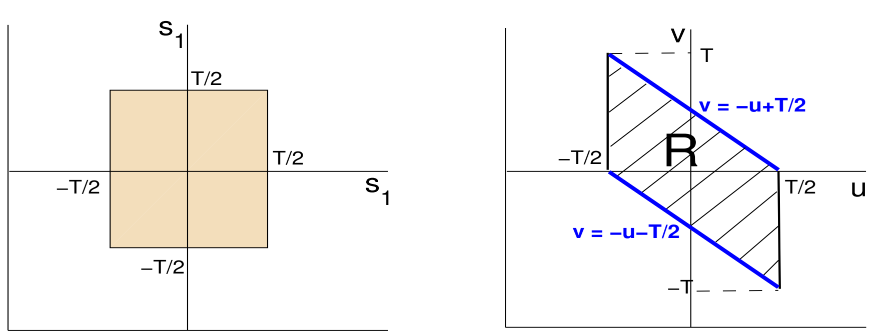

Figure 2.3: Domain change

$$T^2\frac{\sigma_T^2}{\sigma^2} = \int\int_{\mathcal{R}} \rho(u)dudv = \int_{-T}^{0} dv \int \rho(v)dudv + \int_0^T dv \int \rho(v)dudv$$
$$= \int_{-T}^{0} dv \int_{-v-\frac{T}{2}}^{\frac{T}{2}} \rho(v)dv + \int_0^T dv \int_{-\frac{T}{2}}^{-v+\frac{T}{2}} du = 2T\int_0^T (1-\tfrac{v}{T})\rho(v)dv$$

Hence $\frac{\sigma_T^2}{\sigma^2} = \frac{2}{T}\int_0^T (1-\frac{v}{T})\rho(v)dv$.

2. For a red-noise where $\rho(\tau) = e^{-\tau/T_0}$ one gets

$$\frac{\sigma_T^2}{\sigma^2} = 2\frac{T_0}{T}\left[1 - \frac{T_0}{T}(1-e^{-\frac{T}{T_0}})\right] \sim \frac{1}{\frac{T}{2T_0}}$$

for large values of T.

Remark

Note that here the e-folding time T_0 represents the integral of the autocorrelation function:

$$T_0 = \int_0^\infty \rho(\tau)d\tau.$$

In the above formulation, the time interval was assumed to be unity. If the time series is sampled every Δt then as $T_0 = -\frac{\Delta t}{\log\rho(\Delta t)}$, and then one gets:

$$n^* = -\frac{n}{2}\log\rho(\Delta t).$$

Bretherton et al. (1999) provides an improvement:

$$n^* = n\frac{1-\rho_1^2}{1+\rho_1^2} \tag{2.24}$$

where $\rho_1 = \rho(1)$.

Chapter 3

Statistical Significance and Hypothesis Testing

3.1 Background

The general objective of statistical significance is to test for the mean of a given sample of atmospheric (or any other type of) data. Let $x_1, x_2, \dots x_n$ be a sample of data (from the population), supposed to be independent, with variance σ^2. The variance of the mean $\overline{x}$ is given by:

$$\sigma_{\overline{x}}^2 = \frac{\sigma^2}{n} \tag{3.1}$$

and its square root is the *standard error* of the sample mean, and its sample value, i.e. $s/\sqrt{n}$ is usually used as an estimate of $\sigma/\sqrt{n}$, where s is the sample standard deviation, see Eq (1.2). In order to carry out the testing procedure we will assume that the sample comes from a normal distribution with mean μ and variance σ^2, $\mathcal{N}(\mu, \sigma^2)$.

Based on this assumption and that σ^2 is known, the z-statistic:

$$Z = \frac{\overline{x} - \mu}{\sigma_{\overline{x}}} = \frac{\overline{x} - \mu}{\sigma/\sqrt{n}} \tag{3.2}$$

is standard normal, i.e., $\mathcal{N}(0, 1)$. The z-statistic can be used to test for the mean μ of the population by providing a kind of margin error.

3.2 Confidence interval for the mean and the variance

3.2.1 Confidence interval for the mean

Broadly speaking the confidence interval for the mean is a kind of error margin. Suppose we want to construct the mean value of, say, the pressure in Stockholm in winter time from a sample $x_1, x_2, \dots, x_n$ of daily values. In order to achieve this we need some assumptions regarding the population from which the sample is drawn. In particular, we need some information about the variance σ^2 of the population. We have two cases:

a. Case when σ is known

We assume that our sample of pressure values are drawn from a normal distribution $\mathcal{N}(\mu, \sigma^2)$, and we suppose that σ is known. The objective is then to estimate the confidence interval for the population mean μ. We know from the $z-$*statistic* in eq (3.2) above that $Z = \frac{\overline{x}-\mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0,1)$. Consequently, we have

$$Pr\left(-1.96 \le Z \le 1.96\right) = Pr\left(-1.96 \le \frac{\overline{x}-\mu}{\sigma/\sqrt{n}} \le 1.96\right) \approx 95\%$$

that is:

$$Pr\left[\overline{x} - 1.96\frac{\sigma}{\sqrt{n}} \le \mu \le \overline{x} + 1.96\frac{\sigma}{\sqrt{n}}\right] \approx 0.95. \tag{3.3}$$

In this case we say that $[\overline{x} - 1.96\frac{\sigma}{\sqrt{n}},\ \overline{x} + 1.96\frac{\sigma}{\sqrt{n}}]$ is the 95% confidence interval of the population mean μ. Let us now denote by $1-\alpha = 95\%$, that is $\alpha = 5\%$. Then, letting $\phi(.)$

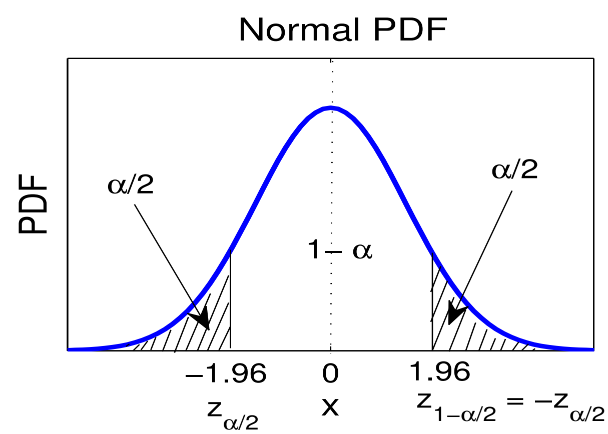

Figure 3.1: Critical levels in the Gaussian case, i.e., when σ is known

be the PDF of the standard normal $\mathcal{N}(0,1)$, i.e., $f(x) = (2\pi)^{-1/2}e^{-x^2/2}$, we have (see Fig. 3.1)

$$\int_{-1.96}^{1.96} \phi(z)dz = 1-\alpha.$$

To proceed, we let $z_{\alpha/2}$ be the number satisfying:

$$\int_{-\infty}^{z_{\alpha/2}} \phi(z)dz = \frac{\alpha}{2}$$

Let us now designate by $\Phi(.)$ the cumulative distribution function of $\mathcal{N}(0,1)$, i.e., $\Phi(x) = \int_{-\infty}^{x} \phi(u)du$, then the above equation means:

$$z_{\alpha/2} = \Phi^{-1}(\frac{\alpha}{2}) \tag{3.4}$$

and is known as the *z-critical level*. The notation $\Phi^{-1}(.)$ in eq. (3.4) refers to the inverse of $\Phi(.)$, and $z_{\alpha/2}$ is the $\alpha/2$'*th* quantile of $\mathcal{N}(0,1)$. We have $z_{1-\frac{\alpha}{2}} = -z_{\frac{\alpha}{2}}$.

In Matlab the quantiles from the normal distribution are obtained using the following commands:

```
>> x=-5:.1:5;
>> mu = 0.; sigma = 1.;
>> y = normpdf(x,mu,sigma) % PDF
>> plot(x,y)
>> P = normcdf(x,mu,sigma) % CDF
>> Q = norminv(P,mu,sigma) % Quantiles
>> R = normrnd(mu,sigma) % random number
>> [norminv(2.5/100) normcdf(-1.96)]
```

The corresponding Python commands are:

```
>>> import numpy as np
>>> import scipy.stats as sps
>>> x = np.arange(-5., 5., .1)
>>> mu, sig = 0., 1
>>> NN = sps.norm (mu, sig)
>>> y = NN.pdf (x)
>>> P = NN.cdf (x)
>>> Q = NN.ppf (x)
>>> R = np.random.normal(mu, sig, 1) # generates 1 random number
>>> a, b = NN.ppf(2.5/100.), NN.cdf(-1.96)
```

Remark

> The right hand side of eq. (3.4) is the $(\alpha/2)$'*th quantile* of the standard normal distribution (see section 1.4.4, chapter 1). Then we see, for example, that $z_{0.025} = \Phi^{-1}(0.025) \approx -1.96$ and also $z_{0.005} \approx -2.6$. For a given α, $0 \le \alpha \le 1$, the interval $[\overline{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}},\ \overline{x} + z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}]$ provides the $100(1-\alpha)\%$ confidence interval of the population mean μ.

b. Case when σ is unknown

The case when σ is known is straighforward. However, when σ is unknown, we have to estimate it from the sample in order to accomplish the inference. Here again we can distinguish two cases depending on the sample size.

a. Large sample size

When the sample size n is large we can approximate σ by its estimate s, i.e.

$$\sigma^2 \approx s^2 = \frac{1}{n-1}\sum_{k=1}^{n}(x_k - \overline{x})^2$$

and then take $Z = \frac{\overline{x}-\mu}{s/\sqrt{n}}$ as being approximately equal to $\mathcal{N}(0,1)$. As an example, we can take $[\overline{x} - 1.96s/\sqrt{n}, \overline{x} + 1.96s/\sqrt{n}]$ as the approximate 95% confidence interval of the mean μ when n is large.

a. Small sample size

In this case the above approximation is no more valid as σ can no longer be approximated by the sample standard deviation s. We assume again that the sample $x_t \sim \mathcal{N}(\mu, \sigma^2)$, $t = 1, \ldots n$, are IID. Then we know again that $\frac{\overline{x}-\mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0,1)$. Now, using the result from the end of chapter 1 (section 1.4) we know, for example, that

$$\frac{1}{\sigma^2}\sum_{k=1}^{n}(x_k - \overline{x})^2 \sim \chi^2_{n-1}$$

Since σ is unknown we use its estimator s instead and consider the variable $T = \frac{\overline{x}-\mu}{s/\sqrt{n}}$, which can be written using the $z-$statistic, to obtain the $t-$statistic:

$$T = \frac{\frac{\overline{x}-\mu}{\sigma/\sqrt{n}}}{s/\sigma} = \frac{Z}{\sqrt{\frac{U}{n-1}}}$$

where $U = (n-1)s^2/\sigma^2 = \frac{1}{\sigma^2}\sum_{k=1}^{n}(x_i - \overline{x})^2$. Since $Z \sim \mathcal{N}(0,1)$ and $U \sim \chi^2_{n-1}$, we have seen (see end of chapter 1) that

$$T \sim T_{n-1}$$

that is the t-distribution with $n-1$ d.o.f. Hence, if $t_{n-1}(\alpha/2)$ is the critical level (or value) for the *significance* level α, that is the $\alpha/2'th$ quantile of the t-distribution, $t_{n-1}(\alpha/2) = F^{-1}_{n-1}(\alpha/2)$ (if we designate by $F_{n-1}(.)$ the CDF of the t-distribution), then (see Fig. 3.2) $[\overline{x}+t_{n-1}(\alpha/2)\frac{s}{\sqrt{n}},\ \overline{x}-t_{n-1}(\alpha/2)\frac{s}{\sqrt{n}}]$ is the $100(1-\alpha)\%$ confidence interval for the population mean μ (Fig. 3.2). In Matlab the quantiles from the t-distribution are obtained using the

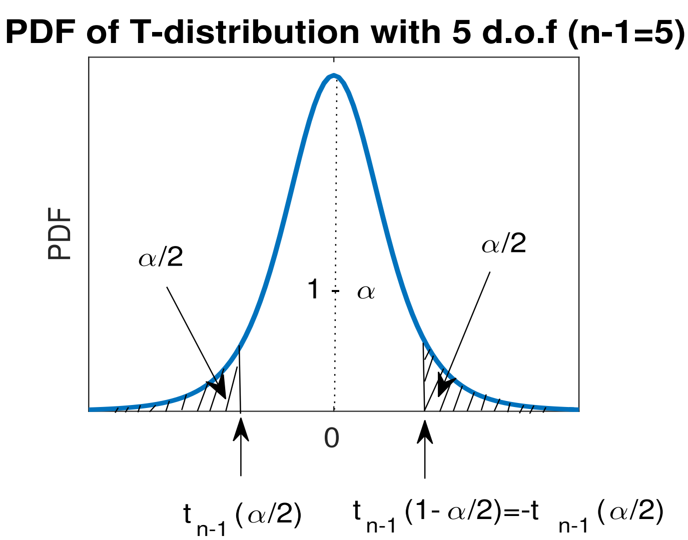

Figure 3.2: Critical levels for the mean with the t-distribution, i.e., when σ is not known

following commands:

```
>> x = -5:.1:5;
>> nu=4;
>> y = tpdf(x,nu);
```

```
>> P = tcdf(x,nu);
>> Q = tinv(P,nu);
>> tinv(2.5/100, [3 5 10 100 1000]);
```

The corresponding Python commands are:

```
>>> import numpy as np
>>> import scipy.stats as sps
>>> x = np.arange(-5., 5., .1)
>>> nu = 4
>>> TT = sps.t (nu)
>>> y = TT.pdf (x)
>>> P = TT.cdf (x)
>>> Q = TT.ppf (x)
```

3.2.2 Confidence interval for the variance

We assume again that we have a sample $x_1, x_2, \dots x_n$ from $\mathcal{N}(\mu, \sigma^2)$, and we estimate σ^2 using the sample variance s^2. Then we know that $\sum_1^n (x_i - \overline{x})^2 \sim \sigma^2 \chi^2_{n-1}$, that is

$$(n-1)\frac{s^2}{\sigma^2} \sim \chi^2_{n-1}. \tag{3.5}$$

For a given α (e.g., $\alpha = 5\%$) the $100(1-\alpha)\%$ confidence interval of σ^2 is then given by (see Fig. 3.3)

$$\frac{(n-1)s^2}{\chi^2_{n-1}(1-\alpha/2)} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi^2_{n-1}(\alpha/2)}. \tag{3.6}$$

where $\chi^2_{n-1}(\alpha/2)$ is the number that satisfies $\int_0^{\chi^2_{n-1}(\alpha/2)} f(u)du = \alpha/2$, where $f(.)$ is the PDF of χ^2_{n-1}. That is, if $F(.)$ denotes the CDF of χ^2_{n-1}, $\chi^2_{n-1}(\alpha/2) = F^{-1}(\alpha/2)$ and represents the $(\alpha/2)$'th quantile of χ^2_{n-1}. As an example, for $\alpha = 5\%$, the 95% confidence interval of σ^2 is given by $[\frac{(n-1)s^2}{\chi^2_{n-1}(0.975)}, \frac{(n-1)s^2}{\chi^2_{n-1}(0.025)}]$.

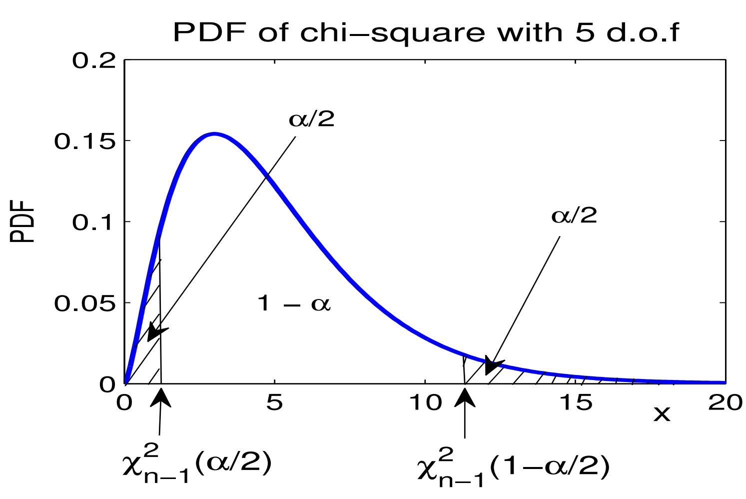

Figure 3.3: Critical levels for the variance with the chi-square distribution

In Matlab the quantiles form the ch-square distribution are obtained using the following commands:

```
>> x = 0:.1:5;
>> nu=5;
>> y = chi2pdf(x,nu);
>> P = chi2cdf(x,nu);
>> Q = chi2inv(P,nu);
```

The corresponding Python commands are:

```
>>> import numpy as np
>>> import scipy.stats as sps
>>> x = np.arange(0, 5, .1)
>>> nu = 5
>>> chi = sps.chi2(nu)
>>> y = chi.pdf (x)
>>> P = chi.cdf (x)
>>> Q = chi.ppf (x)
```

The number α is normally referred to as the *significance level* and $(1-\alpha)$ the confidence level. A consequence of this is that if, for example, the $100(1-\alpha)\%$ confidence interval of the mean μ does not contain zero then the mean is said to be significantly different from zero at the $100\alpha\%$ significance level.

3.3 Hypothesis testing

Hypothesis testing deals with drawing conclusions or making judgements about a given claim. The issue is to find out whether an observed difference is simply due to chance or is systematic.

Examples

1. We want to know for example whether the average winter temperature in Stockholm is the same as that of Uppsala.
2. We want to know whether the average height for men is the same as that for women.

In order to proceed one need to formulate the problem within its statistical framework. This requires three main steps, which can be decomposed further into four or even five steps.

• Null and alternative hypothesis

→ Null hypothesis H_0: "there is no difference"

→ Alternative hypothesis H_1: "there is a difference"

The statement H_0 is in general a hypothesis that the researcher hopes to reject in favor of H_1. Hypothesis testing can also be generalised to instances like saying that a certain parameter

θ, e.g. the mean, is in a certain interval or set Ω_0 versus θ is in Ω_1, etc. If the set Ω_0 contains a single value then H_0 is said to be simple otherwise it is composite.

Remark

> Note that the t-test situation corresponds to a composite H_0 because there is an uknown σ^2, i.e. the H_0 parameters consist of μ_0 and σ^2.

• Test statistic

The test statistic is the statistic used for the judgement. The two statistics presented in the previous section, i.e., $z-$ and $t-$statistics, are examples.

• Significance level

This is the level of error that the researcher is willing to accept. It represents the probability of rejecting H_0, at a given significance level (e.g., $\alpha = 1\%$ or 5%) when it is true.

Once the statistic, significance level and hypotheses are chosen on can then go to the decision. Note that in hypothesis testing the significance level α is normally chosen before computing the observed statistic.

3.3.1 Power of the test

When a decision is taken regarding any claim there is always the possibility that the decision is not the right one. For example, when we reject the hypothesis H_0 when it is true we commit an error. That error is known as *type I* error. However there is also another type of error and that it is when we fail to reject H_0, even though it is false[1]. That type of error is known as *type II* error, and the probability of committing this error is β. These error types are summarised in the contingency table (table 2). The number $1-\beta$ is referred to as the *power of the test*. So, the power of the test is simply the probability of rejecting the null hypothesis when it is in fact false, that is,

$$1-\beta = Pr(\text{reject } H_0|H_0 \text{ is false}).$$

These error types are summarised in the following contingency table (table 2).

	Null hypothesis H_0	
	T	F
Fail to reject	$1-\alpha$	β
	Correct decision	Type 2 error
Reject	α	$1-\beta$
	Type I error	Power

Table 2. Type I and II errors and the power a statistical test

[1]Note that in statistical tests terminology we do not say "we accept the alernative hypothesis H_1, even though it is false. The terminology is of the form "reject" or "fail to reject".

Ideally, of course, one would like to have $\beta = 0$. In practice, for a given significance level α the power depends of the statistic at hand and the distance between the the sample parameter estimates and the truth. One could choose actually a test statistic that has the strongest power. In general, however, for a given test if α is reduced then β is increased and vice versa.

3.3.2 The p-value

When a null hypothesis is rejected with a significance level α, it is in general still possible to reject it at a smaller value than α. Instead of fixing the significance level α it is in general better to report the smallest α such that H_0 is rejected with the observed data. This is known as the p-value.

When the p-value is small we have tendency towards rejection, and when it is large we have tendency to non-rejection. In the sequel we examine a few examples dealing with the mean and variance of a sample and also the cross-correlation and partial autocorrelation function of a stationary time series.

Remark

> It is worth pointing out that the issue of hypothesis testing and significance level has some association with philosophical arguments and has been subject to discussion in the literature regarding the valididty of published research findings by skeptics, e.g. Ioannidis (2005) and OSC (2015). For example, one issue is reproducibility, which can be quite difficult to proove, particularly in psychological science (OSC 2015). The study of Ioannidis (2005) suggests that it is quite difficult to demonstrate the validity of many current publication. In climate science, however, we believe that substantial amount of published research findings can be reproduced as we have good large amounts of data. In addition, most published weather and climate research findings provide detailed description of the data and methodologies. Nevertheless the above references should boost our efforts to do proper research with high level of reproducibility, and expose the results as they are.

3.4 Examples of tests applied to sample means and variances

3.4.1 Example 1: Testing the population mean

We suppose that we have a sample $x_1, \ldots x_n$ from a normal distribution $\mathcal{N}(\mu, \sigma^2)$ where μ is unknown but σ^2 is known and we want to check whether the mean μ equals a given value μ_0. Our hypotheses then are:

H_0: $\mu = \mu_0$
H_1: $\mu \neq \mu_0$

Suppose that we chose for our significance level the value $\alpha = \alpha_0$. Intuitively speaking the hypothesis H_0 will be rejected when $|\overline{x} - \mu_0|$ is large. We can consider the z-statistic:

$$Z = \frac{\overline{x}_n - \mu_0}{\sigma/\sqrt{n}}$$

The hypothesis will be rejected (see Fig. 3.4) when the z-statistic is in the critical region, i.e. $|Z| \geq -z_{\alpha_0/2} = z_{1-\alpha_0/2}$ that is:

$$|Z| \geq \Phi^{-1}(1 - \frac{\alpha_0}{2})$$

where $\Phi(.)$ is the CDF of the standard normal distribution $\mathcal{N}(0,1)$. Note that $\Phi(.)$ is related to what is known as the error function[2] erf(.) through $\Phi(x) = \frac{1}{2}(1 + erf(x))$.

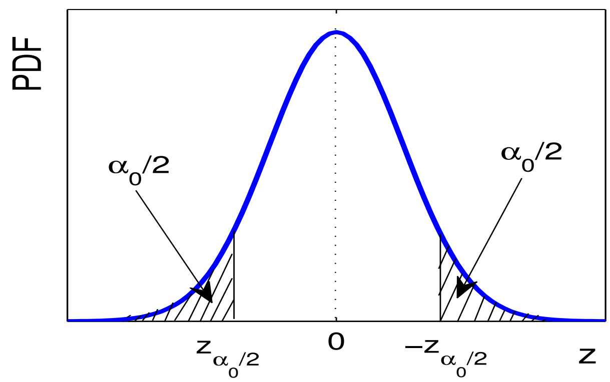

Figure 3.4: Critical levels for the mean with z-statistic

Remark

> Note that any observed statistic Z' such that $Z' \geq \Phi^{-1}(1 - \frac{\alpha_0}{2})$ will also reject H_0. The rejection/acceptance of a null hypothesis depends on the statistic, of course, and most importantly on the significance level α. A better method, however, to get away with the significance level is to report the p-value for the observed statistic. In this case the hypothesis H_0 is rejected for all α satisfying:
>
> $$\alpha \geq p = 2\,(1 - \Phi(|Z|))\,.$$
>
> So the p-value in this case is simply $2\,(1 - \Phi(|Z|))$.

3.4.2 Example 2: Comparison of two population means

Assume that we have two samples $x_1, \ldots x_{n_1}$ and $y_1, \ldots y_{n_2}$ from two normal distributions $\mathcal{N}(\mu_1, \sigma_1^2)$ and $\mathcal{N}(\mu_2, \sigma_2^2)$. Let us first assume that σ_1 and σ_2 are known but that μ_1 and μ_2 are not known. We consider the hypotheses:

H_0: $\quad \mu_1 = \mu_2$

[2] $erf(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2/2} dt$

H_1: $\mu_1 \neq \mu_2$.

The statistic is

$$Z = \frac{\overline{x}_{n_1} - \overline{y}_{n_2}}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

which is $\mathcal{N}(0,1)$ under the null hypothesis H_0. The remaining part of the procedure is then identical to the example 1 above.

Exercise

let x_t, $t = 1, 2, \ldots, n_1$, and y_t, $t = 1, 2, \ldots, n_2$ be two IID samples from two time series, with respective population variances σ_1^2 and σ_2^2. Compute the standard error of the difference $\overline{x}_{n_1} - \overline{y}_{n_2}$.

Answer – $\sigma^2_{\overline{x}-\overline{y}} = \sigma_1^2/n_1 + \sigma_2^2/n_2$

Assume now that $\sigma_1 = \sigma_2$ but unknown, and let s_1^2 and s_2^2 be the two sample variances. We the consider the statistic:

$$T = \frac{\overline{x}_{n1} - \overline{y}_{n2} - (\mu_1 - \mu_2)}{s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}.$$

This statistic is a t-student distribution with $\nu = n_1 + n_2 - 2$ d.o.f. and s is the pooled variance:

$$s = \sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}}$$

Exercise

Show that the statistic T defined above is t-distributed with $n_1 + n_2 - 2$ d.o.f.

Hint. Just divide the numerator and denominator by $\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2} = \sigma\sqrt{n_1^{-1} + n_2^{-1}}$. The numerator is then standard normal. The square of the denominator is simply $s^2/\sigma^2 = (n_1 + n_2 - 2)^{-1}(n_1 s_1^2/\sigma^2 + n_2 s_2^2/\sigma^2)$, and recalling the expression of s_1^2 and s_2^2, the denominator is distributed (under the null hypothesis) as $\sqrt{\chi^2_{n_1+n_2-2}/(n_1 + n_2 - 2)}$. Note that in the above expression of T, one can equally use for the pooled variance the expression $s^2 = ((n_1 - 1)s_1^2 + (n_2 - 1)s_2^2)/(n_1 + n_2 - 2)$, and the difference is negligible.

Remark

When both variances σ_1 and σ_2 are unknown, we can still use the two-sample t-statistic

$$T = \frac{\overline{x}_{n_1} - \overline{y}_{n_2} - (\mu_1 - \mu_2)}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}.$$

This statistic can be reasonably approximated by a t-distribution with $\min(n_1 - 1, n_2 - 1)$ d.o.f. An alternative, for reasonaly large values of n_1 and n_2, is to use an approximation of the number of d.o.f given by the following expression: $\frac{(n_1-1)(n_2-1)(s_1^2/n_1+s_2^2/n_2)^2}{(n_2-1)(s_1^2/n_1)^2+(n_1-1)(s_2^2/n_2)^2}$.

In Python the t-test can be applied using the built-in function *stats.ttest_ind*. In Matlab the *test* and *ttest2* are the scripts of t-test for one- and two-samples respectively.

3.4.3 Example 3: Comparison of two population variances

We assume that we have two samples $x_1, \dots x_{n_1}$ and $y_1, \dots y_{n_2}$ from two normal distributions $\mathcal{N}(\mu_1, \sigma_1^2)$ and $\mathcal{N}(\mu_2, \sigma_2^2)$. We would like to check whether the two population variances σ_1^2 and σ_2^2 are equal. We take:

H_0: $\sigma_1 = \sigma_2$

H_1: $\sigma_1 \neq \sigma_2$

The test statistic is

$$F = \frac{s_1^2}{s_2^2}$$

where $s_1^2 = \sum_{k=1}^{n_1} (x_k - \overline{x}_{n1})^2/(n_1 - 1)$ and similarly for s_2 from the second sample. We know that the statistic F is F-distribution with $n_1 - 1$ and $n_2 - 1$ d.o.f, i.e. F_{n_1-1, n_2-1}. The remaining of the procedure is then similar to the above example.

Exercise

Explain why F follows the F-distribution.

Answer. Under H_0, i.e. $\sigma_1 = \sigma_2 = \sigma$, we have $\frac{s_1^2/\sigma^2}{s_2^2/\sigma^2} \sim F$.

3.4.4 Confidence on proportions and median

Case of proportions

Consider an experiment in which a survey is conducted on the proportion of Y or N on a given proposal. Let n be the sample size of the people involved in the survey, n_Y and n_N be the number of people who voted by Y and N respectively ($n_Y + n_N = n$). Let p designate the proportion of people voting Y, the corresponding sample value is

$$\hat{p} = \frac{n_Y}{n} \tag{3.7}$$

The sample distribution of $\hat{p}$ is approximaltely normal (for large sample) $N(p, \sigma^2 = p(1 - p)/n)$. The approximate $100(1-\alpha)\%$ confidence interval of p is $\hat{p} \pm z_{\alpha/2}\sqrt{\hat{p}(1-\hat{p})/n}$, where $z_{\alpha/2}$ is the $\alpha/2$'th quantile of the standard normal.

Case of the median

Consider a sample of data $x_1, \dots x_n$, drawn from a pdf $f()$, with population median m. We assume $n = 2k + 1$, the sampling distribution of the median is a particular case of the distribution of order statistics. The distribution of the sample median $\hat{m}$ is approximately normal for large sample size, and:

$$\hat{m} \sim N(m, \frac{1}{8kf(m)^2}). \tag{3.8}$$

As an example, consider the normal distribution $N(\mu,\sigma^2)$. The distribution is symmetric (and unimodal) so the mean is identical to the median (and the mode). So $m=\mu$, and therefore $f(m)=f(\mu)=\frac{1}{\sigma\sqrt{2\pi}}$. Consequently, for large n, we get approximately $\hat{m}\sim N(\mu,\frac{\pi\sigma^2}{4k})$.

3.5 Examples from stationary time series

3.5.1 Confidence on the auto- and cross-correlation function

In chapter 2 an expression was given for the auto-covariance (and also cross-covariance) estimates of stationary time series. let x_t, $t=1,2,\ldots,n$, be a stationary Gaussian (zero-mean) time series and consider again the following estimator for the autocovariance function $\gamma(\tau)$:

$$\hat{\gamma}(\tau)=\frac{1}{n}\sum_{k=1}^{n-\tau}x_k x_{k+\tau} \tag{3.9}$$

This estimator is asymptotically unbiased, as was shown in chapter 2. The variance of this estimator (e.g., Priestly 1981, p 326), for large values of n is approximately

$$\sigma_\tau^2=\mathrm{var}(\hat{\gamma}(\tau))=\frac{1}{n}\sum_m\left(\gamma^2(m)+\gamma(m+\tau)\gamma(m-\tau)\right) \tag{3.10}$$

The autocorrelation can be estimated by:

$$\hat{\rho}(\tau)=\frac{\hat{\gamma}(\tau)}{\hat{\sigma^2}}. \tag{3.11}$$

where $\hat{\sigma^2}$ is thesample variance s^2.

Finding confidence intervals for the autocorrelation function is not an easy problem. For an IID time series $\hat{\rho}(\tau)$ is asymptotically normal (e.g., Anderson 1942; Fuller 1996) with:

$$E\left(\hat{\rho}(\tau)\right)=-\tfrac{1}{n}\quad\text{and}\quad\mathrm{var}\left(\hat{\rho}(\tau)\right)=\tfrac{1}{n} \tag{3.12}$$

Therefore the interval $[-\frac{1}{n}\pm\frac{1.96}{\sqrt{n}}]$, for example, provides the 95% confidence interval for $\rho(\tau)$ and, as a rule of thumb, we compare the sample autocorrelation function to the two horizontal lines $\pm 2/\sqrt{n}$ to get the 95% confidence level.

The uncertainty (or variance) of the sample cross-correlation is given by:

$$s_{xy}^2=\sigma^2\left(\rho_{xy}(\tau)\right)\approx\frac{1}{n-\tau}\sum_t\rho_x(t)\rho_y(t).$$

For example, at 5% significance level, and in order for $\rho_{xy}(\tau)$ to be different from zero one should have $|\hat{\rho}_{xy}(\tau)|>2s_{xy}$

3.5.2 Confidence on the partial autocorrelation function

We have discussed the procedure of AR(p) model fitting in the end of the previous chapter based on the Durbin algorithm. The partial autocovariance function $\pi(.)$ behaves exactly like the autocorrelation function $\rho(.)$. That is, the sample partial autocorrelation function $\hat{\phi}_{k,k}=\hat{\pi}(k)$ is approximately normally distributed with variance n^{-1} (for an IID sample). So again and as a rule of thumb, the 95% confidence limits are provided by using the interval

$[-2/\sqrt{n}, 2/\sqrt{n}]$. Now, in theory if the process is truly $AR(p)$ then the population partial autocorrelation function, $\phi_{k,k} = \pi(k)$, vanishes for all $k > p$. Therefore, a simple diagnostic for a $AR(p)$ model is provided by looking at which lag the partial autocorrelation function becomes non-significant.

3.6 Other methods of significance testing

3.6.1 Non parametric methods

The previous method of significance testing involves what is known as parametric methods because the underlying distribution is assumed to take a parametric distribution, namely the Gaussian distribution. Of course the normality assumption is not always satisfied, precipitation or surface temperature are good examples. One way to address this is to transform the data to become approximately Gaussian. This transformation is in general nonlinear, and can lead to difficulty in interpretation or conclusion. Another simpler way is to use non parametric methods. For instance, a popular method to test linear trends is to use the Mann-Kendall test (Mann 1945, Kendall 1975, Kendall and Gibbons 1990). Given a time series x_t, $t = 1, \ldots n$, the statistic T in this case is based on the sum of signed differences between all pairs of the time series:

$$T = \sum_{t=1}^{n-1} \sum_{s=t+1}^{n} \text{sign}(x_s - x_t) \tag{3.13}$$

Eq. (3.13) says that if $T > 0$ later observations tend to be larger then earlier observations. The variance of T can be calculated and is given by:

$$var(T) = \frac{1}{18} \left[n(n-1)(2n+5) - \sum_{p=1}^{P} f_p(f_p - 1)(2f_p + 5) \right] \tag{3.14}$$

where P is the number of tied groups , and f_p the number of data values (i.e., frequency) in the p'th tied group. Here, a tied group is a sample of data having equal values. Then, the statistic Z:

$$Z = \frac{T - sign(T)}{\sqrt{var(T)}} \tag{3.15}$$

is approximately normal. Large values of Z, e.g., larger than 1.96 or smaller than -1.96, indicate significant trend.

Another more advanced non-parametric test is that based on the Kolmogorov-Smirnov (KS) test. There are two such tests; the one-sample and the two-sample tests. Given a time series $x_1, \ldots x_n$, the empirical distribution function $F_n()$ is given by:

$$F_n(x) = \frac{1}{n} \sum_{t=1}^{n} 1_{x_t \le x}. \tag{3.16}$$

The one-sample KS test attempts to test the null hypothesis H_0: the sample comes from a specific distribution $F()$. It is based on computing the distance between $F()$ and $F_n()$:

$$D_n = \sup_x |F_n(x) - F(x)|. \tag{3.17}$$

If $F()$ is continuous, under the null hypothesis H_0, $\sqrt{n}D_n$ converges to the Kolmogorov distribution $K()$:

$$K(x) = \frac{\sqrt{2\pi}}{x} \sum_{k\geq 1} e^{-(2k-1)^2\pi^2/(8x^2)} \tag{3.18}$$

The null hypothesis H_0 can be rejected at a given significance level α if

$$\sqrt{n}D_n \geq K_\alpha = K^{-1}(1-\alpha) \tag{3.19}$$

where K_α is the $(1-\alpha)$'th quantile of the distribution $K()$.

The two-sample KS test compares two empirical distributions $F_n()$ and $G_m()$ based on two sample time series $x_1, \ldots x_n$, and $y_1, \ldots y_m$. The Statistic is:

$$D_{nm} = \sup_x |F_n(x) - G_m(x)|, \tag{3.20}$$

and the null hypothesis is rejected at significance level α if $D_{nm} > \sqrt{\frac{n+m}{2nm}}\sqrt{-log(\alpha/2)}$.

3.6.2 Monte Carlo and bootstrap methods

Another method of significance testing is based on resampling. It is based on generating many sub-samples based on the available time series and computing the test statistic. In this framework we need perhaps to go back and recall what hypothesis testing is about. We assume we have a sample $(x_1, \ldots, x_n)$ generated from a pdf $f(x|\theta)$ with parameter θ, and we look for testing the hypothesis H_0 vs H_1:

$$H_0 : \theta = \theta_0$$
$$H_1 : \theta > \theta_0$$

Note this is one-sided test, but we can also have a two-sided test, e.g., $H_1 : \theta \neq \theta_0$. To proceed, we then select a test statistic $h()$, i.e., a function of the data, which is meant to measure the departure from the null hypothesis H_0. The observed statistic is $t_o = h(x_1, \ldots, x_n)$, which in principle will be extreme if H_0 is not true. The test statistic then compares the observed value $t_o = h(x_1, \ldots, x_n)$ to the distribution $T = h(X_1, \ldots, X_n)$ where $X_1, \ldots X_n$ are iid random variates specified by H_0. Now, if the distribution T is known then we can compute the p-value for the test. For a one-sided test, the p-value is given by:

$$Pr(T \geq t_o | H_0 \text{ true}) \text{ or } Pr(T \leq t_o | H_0 \text{ true})$$

For a two-sided test for which t_o may be extreme on both tails, the p-value is given by:

$$\text{p-value} = Pr(T \leq -|t_o| \cup T \geq |t_o| \; |H_0 \text{ true}).$$

In this latter case one may need to centre the test statistic.

Monte Carlo testing

In Monte Carlo testing the pdf $f(.|\theta)$ is known and is used to generate samples and obtain the p-value. The algorithm for one-sided test ($H_1 : \theta > \theta_0$) is:

for j in {1, ...N}

simulate $(z_1, \dots z_n)$ with $Z_i \sim f(.|\theta_0)$

compute $t_j = h(z_1, \dots z_n)$

endfor

$P(T \geq t|H_0 \text{ true}) \approx \frac{1}{N}\sum_{j=1}^{N} 1_{\{t_j \geq t_o\}}$

So we see here that we do not need to derive the distribution of the test statistic. As an example, given a sample $x_1, \dots x_n$, with sample mean $\overline{x}$ derived from a normal pdf $f(x|\mu, \sigma^2)$, with known σ, we wish to get the p-value for a two-sided test, $H_0 : \mu = \mu_0$ versus $H_1 : \mu \neq \mu_0$. We then simulate N samples of size n from a normal distribution $N(\mu_0, \sigma^2)$, compute their means $\overline{z}_j, j = 1, \dots N$, and get the p-value $Pr(T \geq t_o|H_0 \text{ true}) \approx \frac{1}{N}\sum_{j=1}^{N} 1_{\{\overline{z}_j \geq \overline{x}\}}$. For a two-sided test, with $T = \overline{Z} - \mu_0$ and $t_o = \overline{x} - \mu_0$, the p-value is $Pr(|\overline{Z} - \mu_0| \geq |t_o|) \approx \frac{1}{N}\sum_i 1_{\{|\overline{z}_i - \mu_0| \geq |t_o|\}}$

Bootstrap testing

As shown above, Monte Carlo testing avoids the need to identify the analytical sampling distribution of the test statistic under the null hypothesis H_0. Of course, the null distribution is still needed. The question we ask now is whether we can still evaluate the test without even having to derive the null distribution. This can be achieved using Bootstrap.

Assume we have a given time series $x_1, \dots x_n$, and we wish to test if the population mean μ equals a specified value, say μ_0. Then using bootstrapping, one randomly draws (with replacement) N subsamples (e.g., $N = 1000$) from the time series, and compute the corresponding means $\overline{z}_1, \dots, \overline{z}_N$. A histogram of $\overline{z}_1, \dots, \overline{z}_N$ is then constructed, and the null hypothesis $H_0 : \mu = \mu_0$ is rejected if $\overline{x}$ is outside the bulk of the distribution, e.g., below the 5'th or above the 95'th percentile of $\overline{z}_1, \dots, \overline{z}_N$. The p-value can also be compute and is approximated by $\frac{1}{N}\sum_{j=1}^{N} 1_{\{|\overline{z}_j - \mu_0| \geq |\overline{x} - \mu_0|\}}$.

To compare the population means μ_x and μ_y of two time series $x_1, \dots x_n$ and $y_1, \dots y_m$ one follows the same procedure. Precisely, one generates subsamples from the two time series $x_1, \dots x_n$ and $y_1, \dots y_m$ and compute their respective means $\overline{u}_1, \dots \overline{u}_N$ and $\overline{v}_1, \dots \overline{v}_N$. A histogram of the difference $\overline{u}_i - \overline{v}_i$, $i = 1, \dots N$ (for paired observations) is then computed and the null hypothesis $H_0 : \mu_x - \mu_y = 0$ is rejected if the bulk of the distribution does not contain the zero value. If the two samples $x_1, \dots x_n$ and $y_1, \dots, y_m$ have different sample size, e.g., n and m, an approximate p-value of the test statistic of equality of means can be obtained following the algorithm of Efron and Tibshirani (1993, p. 221). Denoting by $z_1, \dots z_{n+m}$ the combined sample of the two time series, the algorithm for a two-sided test runs as follows:

draw, with replacement, N samples of size $n + m$ from $z_1, \dots z_{n+m}$ and let u and v be the first n and remaining m observations.

compute $T_i = \overline{u} - \overline{v}$, $i = 1, \dots N$

p-value $\approx \frac{1}{N}\sum_{i=1}^{N} 1_{\{|T_i| \geq |t_o|\}}$

This p-value is referred to as 'achieved significance level'.

Chapter 4

Spectral Analysis of Stationary Time Series

4.1 Background

Given a time series x_t, $t = 1, 2 \ldots n$, assumed to be stationary, different ways exist to analyse it. Time series can be studied in the time domain by analysing the autocorrelation function for prediction purposes for example. Alternatively[1], the time series can also be analysed in the frequency domain. This is achieved using Fourier transform for continuous (non periodic) functions or Fourier series for discrete series or periodic functions. For finite time series discrete Fourier transform can be used. If we are interested in identifying localised structures then time-frequency methods can be used. These methods include, for example, windowed FT, wavelets, and other methods such as singular system (spectrum) analysis.

Fourier transform for a non-periodic function

For a non-periodic (integrable) function $f(.)$ the Fourier transform is defined by:

$$g(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt \tag{4.1}$$

where ω is frequency (rd/s). The function $f(.)$ is then given by the inverse Fourier transform (FT):

$$f(t) = \int_{-\infty}^{\infty} g(\omega) e^{i\omega t} d\omega \tag{4.2}$$

One of the main results in Fourier transform theory is Parceval identity, which is a (energy) conservation property:

Parceval identity

$$\int_{-\infty}^{\infty} |f(t)|^2 dt = \int_{-\infty}^{\infty} |g(\omega)|^2 d\omega \tag{4.3}$$

[1] Yet another alternative is to use graph theory

Fourier transform for a periodic signal

For a periodic function $f(.)$, e.g., periodic over the interval $[-\pi, \pi]$ then $f(.)$ can be decomposed into an infinite Fourier series:

$$f(t) = \sum_{k=0}^{\infty} a_k \cos kt + b_k \sin kt. \tag{4.4}$$

The number k is referred to as the wave number, and

$$a_k = \tfrac{1}{(1+\delta_{0k})\pi} \int_{-\pi}^{\pi} f(x) \cos kx dx \quad \text{and} \quad b_k = \tfrac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin kx dx \tag{4.5}$$

where the Cronecker symbol δ_{ij} is one when $i = j$ and zero otherwise.

Remark

Note that the sum in eq. (4.4) spans all non-negative integers.

Exercise

Find the Fourier transform of a periodic function over the interval $[a, b]$

For a finite (real) time series $f(.)$ defined over the interval $[0, T]$ we can compute the discrete Fourier transform (DFT) as above, assuming periodicity with period T. Because of the orthogonality property satisfied by the trigonometric functions the frequencies ω that are represented in the discrete Fourier transform satisfy:

$$\frac{\omega}{2\pi} = k\frac{1}{T}$$

for $k = 0, 1, \ldots$. That is, the frequencies are integer multiples of the fundamental frequency ω_0, i.e., $\omega = k\omega_0$ with $\omega_0 = 2\pi/T$. In this case the DFT is defined through the Fourier coefficients given by:

$$g_n = \tfrac{1}{T} \int_0^T f(t) e^{-in\omega_0 t} dt, \quad n = 0, \pm 1, \pm 2, \ldots \tag{4.6}$$

The inverse DFT is then given by the complex Fourier series:

$$f(t) = \sum_k g_k e^{ik\omega_0 t} = \sum_k g_k e^{2i\frac{\pi}{T}kt} \tag{4.7}$$

which is similar to the periodic case, and it can be re-written as:

$$f(t) = \frac{A_0}{2} + \sum_{k\geq 1} [A_k \cos(k\omega_0 t) + B_k \sin(k\omega_0 t)] \tag{4.8}$$

The Parceval identity in this case is given by:

$$\frac{1}{T} \int_0^T |f(t)|^2 dt = \frac{A_0^2}{4} + \frac{1}{2} \sum_{k\geq 1} (A_k^2 + B_k^2) \tag{4.9}$$

with the coefficients A_k and B_k for $k = 1, 2, \ldots$, given by:

$$A_k = \tfrac{2}{T} \int_0^T f(t) \cos(k\omega_0 t) dt, \quad \text{and} \quad B_k = \tfrac{2}{T} \int_0^T f(t) \sin(k\omega_0 t) dt \tag{4.10}$$

Exercise

Using the complex Fourier series (4.7) derive (4.8) and (4.10).

Discrete Fourier transform for a finite discrete time series

For a finite discrete time series $x_0, x_1, \ldots, x_{n-1}$, the DFT is the sequence

$$y_m = \sum_{k=0}^{n-1} x_k e^{-2i\pi\frac{m}{n}k}, \quad m = 0, 1, \ldots n-1, \tag{4.11}$$

and has a finite number of frequencies. We can also obtain the original time series through the inverse DFT as

$$x_k = \frac{1}{n}\sum_{m=0}^{n-1} y_m e^{2i\pi\frac{k}{n}m}.$$

In this case the property of energy (or amplitude) conservation takes the form:

$$\sum_{k=0}^{n-1} |x_k|^2 = \frac{1}{n}\sum_{k=0}^{n-1} |y_k|^2. \tag{4.12}$$

The computation of the above DFT requires $O(n^2)$ operations, which is quite large for large values of n. The fast Fourier transform (FFT) algorithm achieves this in $O(n \log n)$. The FFT algorithm is based on approximating the length of the time series by a power of 2.

4.2 Spectra of stationary time series

4.2.1 Definition of power spectra

The "basic" definition of power spectra of a stationary stochastic process $X(t)$, for t real, is normally defined by:

$$f(\omega) = \int \gamma(\tau) e^{-i\omega\tau} d\tau \tag{4.13}$$

or equivalently $\gamma(\tau) = \int f(\omega) e^{i\omega\tau} d\omega$. Any sample power spectra $\hat{f}(.)$ is an estimator of the above expression (4.13) using a finite sample of data. It can be calculated using an estimator of the autocorrelation function $\hat{\gamma}(.)$, i.e.,

$$\hat{f}(k) = \int_{-T}^{T} \hat{\gamma}(\tau) e^{-ik\omega_0\tau} d\tau \tag{4.14}$$

with ω_0 being the fundamental frequency ($\omega_0 = 2\pi/T$). The above estimate $\hat{f}(.)$ is referred to as the *correlogram.* An alternative to the above estimator is to use the DFT of the time series. That is, if F_k, $k = 0, \pm 1, \pm 2, \ldots$, are the DFT coefficients of $X(t)$, assumed to be defined over the interval $[0, T]$, then we get:

$$\hat{f}(k) = T|F_k|^2, \quad k = 0, \pm 1, \pm 2, \ldots \tag{4.15}$$

This estimator is known as the *periodogram*, see the exercise below.

Exercise

If x_t is defined over $[0, T]$, the DFT sequence is given by $F_k = \frac{1}{T}\int_0^T x_t e^{-ik\omega_0 t} dt$, for $k = 0, \pm 1, \pm 2, \ldots$, ($\omega_0 = 2\pi/T$), and we know that

$$x_t = \sum_k F_k e^{ik\omega_0 t}.$$

a. Show that $F_{-k} = F_k^*$, where * stands for the complex conjugate.

b. Noting $a_k = F_k + F_{-k}$ and $b_k = i(F_k - F_{-k})$, i.e., $2F_k = a_k - ib_k$, show that a_k and b_k are real and that:

$$x_t = \frac{a_0}{2} + \sum_{k\geq 1}\left[a_k \cos(k\omega_0 t) + b_k \sin(k\omega_0 t)\right]$$

(i.e., $a_k + ib_k = \frac{2}{T}\int_0^T x_t e^{ik\omega_0 t}dt$).

c. The periodogram (or power spectrum) at frequency[2] k, for $k \geq 1$, is then:

$$\hat{f}(k) = T|F_k|^2 = T\left[real(F_k)^2 + imag(F_k)^2\right] = T\frac{a_k^2 + b_k^2}{4}$$

d. The Parceval identity is again:

$$\frac{1}{T}\int_0^T |x_t|^2 dt = \frac{a_0^2}{4} + \frac{1}{2}\sum_{k\geq 1}(a_k^2 + b_k^2) = |F_0|^2 + 2\sum_{k\geq 1}|F_k|^2 = \sum_k |F_k|^2.$$

Exercise

The objective of this exercise is to link the estimators of the autocovariance function and the spectrum using the DFT of the time series over $[0, T]$, and the assumptions are as above. Show that

$$\hat{\gamma}(\tau) = \sum_k |F_k|^2 e^{ik\omega_0\tau}$$

Indication– From the definition of the autocovariance estimator and the DFT we get

$$\hat{\gamma}(\tau) = \frac{1}{T}\int_0^{T-|\tau|} x_t x_{t+\tau} dt = \sum_{m,n} F_m F_n e^{im\omega_0\tau}\left[\frac{1}{T}\int_0^{T-|\tau|} e^{i(m+n)\omega_0 t}dt\right].$$

The term between brackets is ≈ 0 ($= 0$ when $\tau = 0$) when $m + n \neq 0$. We are then left with the case $m + n = 0$.

The results from the above exercise also mean that $|F_k|^2 = \frac{1}{T}\int_0^T \hat{\gamma}(\tau)e^{-ik\omega_0\tau}$. Note that this can also be obtained from the expression of the DFT coefficients, i.e., F_k, $k = 0, \pm 1, \ldots$ (i.e., $T|F_k|^2 = TF_kF_{-k} = T\frac{1}{T^2}\int_{-T/2}^{T/2}\int_{-T/2}^{T/2} x_t x_s e^{-ik\omega_0(t-s)}dtds$, which can be transformed to yield $T|F_k|^2 = \int_{-T}^{T}\hat{\gamma}(\tau)e^{-ik\omega_0\tau}d\tau$). In particular, we have

$$\hat{\gamma}(0) = \sum_k |F_k|^2.$$

If now x_t is a stationary process at discrete times $t = 0, \pm 1, \pm 2, \ldots$, with $\gamma(\tau)$, for $\tau = 0, \pm 1, \pm 2, \ldots$, as its autocovariance function, then the (power) spectrum is:

$$f(\omega) = \sum_\tau \gamma(\tau)e^{-2i\pi\omega\tau} = \gamma(0) + 2\sum_{\tau\geq 1}\gamma(\tau)\cos(2\pi\omega\tau) \tag{4.16}$$

for ω within $[-1/2, 1/2]$. It can also be seen, through the inverse transform, that we get:

$$\gamma(\tau) = \int_{-\frac{1}{2}}^{\frac{1}{2}} f(\omega)e^{2i\pi\omega\tau}d\omega \tag{4.17}$$

Remark

[2] in fact wave number

We note from the above that $\sigma^2 = \gamma(0) = \int_{-\frac{1}{2}}^{\frac{1}{2}} f(\omega)d\omega = 2\int_0^{\frac{1}{2}} f(\omega)d\omega$

Exercise
What is the interpretation of the above equation (4.17) in linking the variance σ^2 to the power spectrum $f(.)$?
Answer – see, e.g. Pires and Hannachi (2021).

4.2.2 Examples

Spectra and the autocovariance function

White noise The white noise model is the simplest model in time series analysis. It is not very useful in practice as noises do not tend to be IID, but tend to be autocorrelated. Following eq (4.16), the spectrum of a white noise with variance σ^2 is given by:

$$f(\omega) = \sigma^2. \tag{4.18}$$

Therefore white noise time series have equal power at all frequencies making it *white*

Red-noise The next model we encountered above was the red-noise, or $AR(1)$ model, i.e., $x_t = \rho_1 x_{t-1} + \varepsilon_t$, where ρ_1 is the lag-1 autocorrelation and ε_t, $t = 1, 2, \ldots$ being IID with zero-mean and variance σ_ε^2 and uncorrelated with x_s, $s \leq t$..

The power spectrum is given by

$$f(\omega) = \frac{\sigma_\varepsilon^2}{|1 - \rho_1 e^{-2i\pi\omega}|^2} = \frac{\sigma_\varepsilon^2}{1 + \rho_1^2 - 2\rho_1 \cos(2\pi\omega)}. \tag{4.19}$$

Exercise

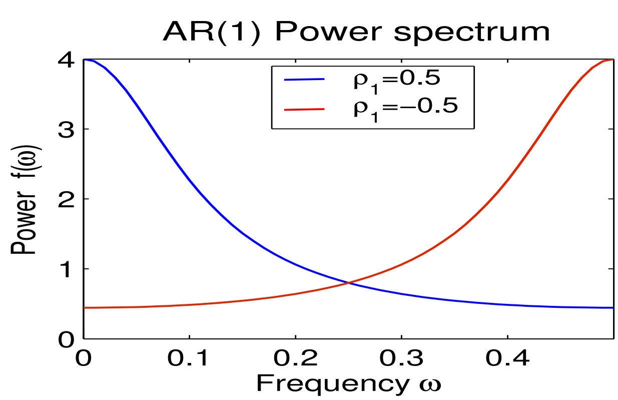

Figure 4.1: Power spectra of two AR(1) models with respective lag-1 autocorrelations 0.5 and -0.5

Derive eq. (4.19).
Answer – Keeping in mind that $\gamma(\tau) = \sigma^2\rho_\tau = \sigma^2\alpha^\tau$, with $\rho_1 = \alpha$, (and also $\sigma^2(1-\rho_1^2) = \sigma_\varepsilon^2$), eq. (4.16) yields

$$\frac{1}{\sigma^2} f(\omega) = \rho(0) + \sum_{\tau\geq 1} \rho(\tau)\left(e^{2i\pi\omega\tau} + e^{-2i\pi\omega\tau}\right) = 1 + \frac{\rho_1 e^{2i\pi\omega}}{1 - \rho_1 e^{2i\pi\omega}} + \frac{\rho_1 e^{-2i\pi\omega}}{1 - \rho_1 e^{-2i\pi\omega}}.$$

It is clear from the spectrum (4.19) that in general red-noises have more power at low frequencies compared to high frequencies. Of course the distribution of power in (4.19) depends on the lag-1 autocorrelation. If $\rho_1 < 0$ we get more power at high frequency as shown clearly in Fig. 4.1.

Exercise
Calculate the spectrum of the AR(2) model $x_t = \phi_1 x_{t-1} + \phi_2 x_{t-2} + \varepsilon_t$.
More generally, calculate the spectrum of a $AR(p)$ model, $x_t = \alpha_1 x_{t-1} + \ldots + \alpha_p x_{t-p} + \varepsilon_t$.

General expression of the ARMA spectra

A direct method to compute the spectra of ARMA processes is to make use of results from linear filtering. Filters are operators that operate on input time series to produce filtered series. Linear filters, in particular, are basically linear (discrete or continuous) combination of the inputs of time series and involve, for the discrete case, the backward shift operator B.

Exercise
Consider the delay operation $y_t = Bx_t$. Find the relation between the FT $y(\omega)$ and $x(\omega)$ of y_t and x_t respectively. Find a similar relationship when $y_t = \alpha x_t + \beta B x_t$.
Answer – $y(\omega) = (\alpha + \beta e^{i\omega}) x(\omega)$.

Let x_t, $t = 0, 1, 2, \ldots$ be a stationary time series and consider the filtering (or smoothing) equation:

$$y_t = \sum_{k=1}^{p} \alpha_k x_{t-k}.$$

Using the above exercise we get $y(\omega) = A(e^{i\omega}) x(\omega)$ where:

$$A(z) = \sum_{k=1}^{p} \alpha_k z^k$$

The function $a(\omega) = A(e^{i\omega})$ is refered to as the *transfer function* of the above linear filter. Now, the power spectrum of y_t is linked to that of x_t following:

$$f_y(\omega) = |a(\omega)|^2 f_x(\omega).$$

The application of this to the ARMA time series model (2.10)

$$\phi(B) x_t = \theta(B) \varepsilon_t$$

where $\phi(z) = 1 - \sum_{k=1}^{p} \phi_k z^k$ and $\theta(z) = 1 + \sum_{k=1}^{q} \theta_k z^k$ yields:

$$f_x(\omega) = \sigma_\varepsilon^2 |\frac{\theta(e^{i\omega})}{\phi(e^{i\omega})}|^2 \tag{4.20}$$

In the above equation it is of course assumed that the roots of $\phi(z)$ are outside unit circle (stationarity) and similarly for $\theta(z)$ (for invertibility, i.e., ε_t is written as a convergent power series in $x_t, x_{t-1}, \ldots$).

4.3 Estimation of power spectra and smoothing

4.3.1 Aliasing

Aliasing is a generic and ubiquitous phenomenon in signal analysis resulting from the transfer of power from high-frequency to lower frequency when the signal is sampled. Let us look at how this manifests practically then later we will show the theoretical root of this.

Assume we have a continuous signal $x(t)$. When the signal is discretised we can obtain say an infinite series $x_k = x(k\Delta t)$, $k = 0, 1, 2 \ldots$. Now, assume that we only observe a sample over a finite length of time, i.e., $T = n\Delta t$, i.e., we have x_k, $k = 1, 2, \ldots n$. This is referred to as *windowing*, i.e., looking at the signal through a finite window $T = n\Delta t$. The highest frequency that can be resolved is $\frac{1}{2\Delta t} = f_N$, that is the *Nyquist* frequency (1/2 sampling rate), whereas the lowest frequency we can measure is $f_l = \frac{1}{n\Delta t} = \frac{1}{T}$. Therefore the sampling frequency will have a frequency interval $\frac{1}{n\Delta t}$ and goes from f_l up to f_N, that is, we have $(\frac{1}{2\Delta t}))/(\frac{1}{n\Delta t}) = n/2$ frequencies.

A natural question that one might ask is the following; what is the effect, if any, of frequencies that are higher than f_N? It turns out that the power from these frequencies that corrupt the measured spectrum – hence *aliasing*. Let us look at an illustration with a sine wave

$$x(t) = a\cos\left(2\pi\frac{t}{\tau}\right)$$

with a period τ, e.g., $\tau = 0.02s$. Assume that the signal is windowed over $T = 0.1s = 5\tau$, with a sampling interval Δt, e.g., $\Delta t = \frac{T}{100} = 0.001s$ (see Fig. 4.2). For this particular signal $f_N = \frac{1}{2\Delta t} = 500(Hz)$.

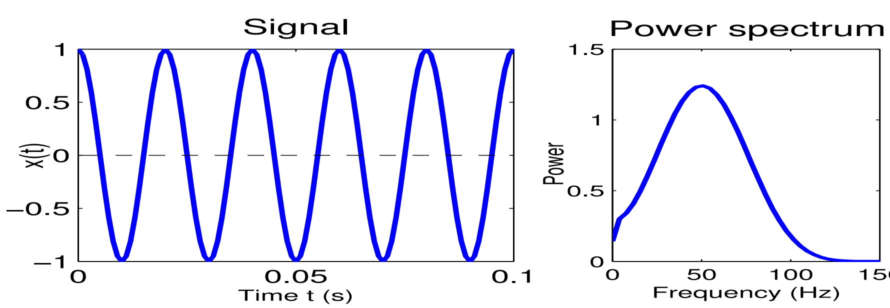

Figure 4.2: The signal $x(t) = \cos(2\pi t/\tau)$ with $\tau = 0.02s$ and a sampling interval $\Delta t = 0.001$ observed over a period $T = 0.1s$ (left), and the associated power spectrum based on Welch estimate showing a peak at $1/\tau = 50Hz$ (right).

Remark

The frequency of the signal is $\frac{1}{0.02} = 50(Hz) = f_s$.

The number of cycles or wavelengths is n, with $T = n\tau$, and in terms of frequency $f_s = n\frac{1}{T}$, or $f_s = nf_l$ ($\frac{1}{T}$ is the lowest frequency), i.e. $n = 5$ (Fig. 4.2). The spectrum of the sampled signal shows a peak at $f_s = 50Hz$!.

If now the period is smaller than the above value, i.e. the frequency f_s is getting larger, (and we decide to keep the same Δt), then we start to see changes in the spectrum.

For example, if $f_s = 200$ ($\tau = 0.005$) or $f_s = 400$ ($\tau = 0.0025$), see Fig. 4.3. This figure can

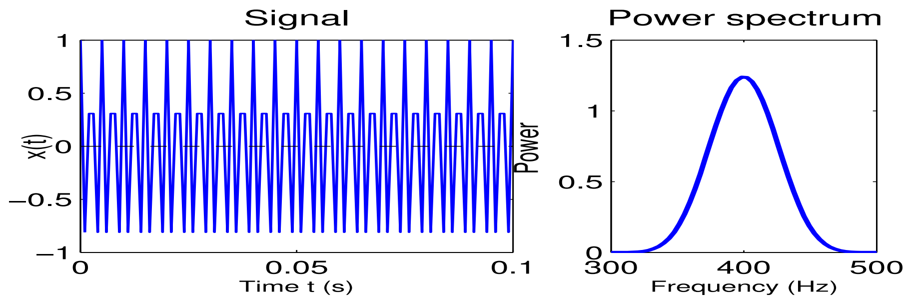

Figure 4.3: Same as figure 4.2 but with $\tau = 0.0025s$, i.e. $\tau^{-1} = 400Hz$

be generated using the following few lines of Matlab:

```
>> tau=0.0025; Dt = 0.001; T=0.1; t=0:Dt:T; x=cos(2*pi*t/tau);
>> subplot(2,2,1); plot (t,x);
>> [P F] = pwelch(x); subplot(2,2,2); plot(F/(2*pi*Dt), P);
```

The equivalent in Python is

```
>>> import numpy as np
>>> from math import *
>>> import matplotlib.pyplot as plt
>>> from scipy.signal import welch
>>> [tau, Dt, T] = [0.0025, 0.001, 0.1]
>>> t = np.arange (0, T, Dt)
>>> x = cos(2. * pi/tau)
>>> plt.plot(t, x)
>>> f, P = welch(x)
>>> plt.plot (f, P)
>>> plt.grid()
>>> plt.show ()
```

The spectrum still shows a peak at the right frequency − but note that the signal (when plotted in time domain) is not as 'nice' as for $\tau = 0.02$. This is simply because increasing f_s (i.e., decreasing τ), with same Δt we get fewer samples that are measured for each period of the wave. The power is still in the right frequency because $f_s < f_N$, i.e., $\tau > 2\Delta t$. As we increase f_s above f_N we see that the spike of the power spectrum shifts to lower frequency as shown in figure 4.4.

The reason is that as the period of the signal τ decreases (and becomes less than $\frac{1}{f_N} = 2\Delta t$) so the time Δt between samples becomes nearly equal to the period τ of the wave, and samples look like those from a low-frequency signal, i.e., the samples will show only slow variation (Fig. 4.5). In brief, aliasing occurs when the sampling is not fast enough, that is

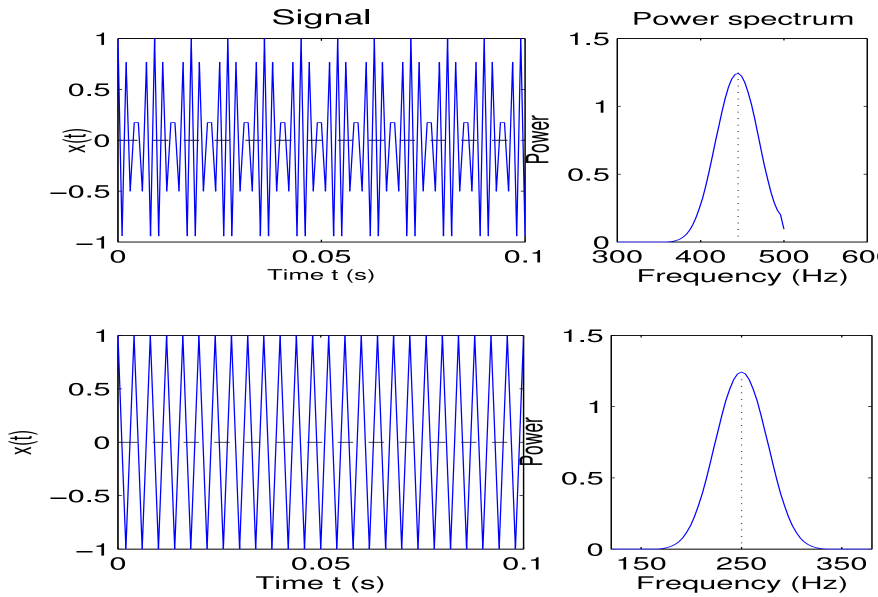

Figure 4.4: Same as figure ...but with $\tau = 0.0018$, i.e., $\tau^{-1} = 555.6Hz$ (top) and $\tau = 0.0008$, i.e. $\tau^{-1} = 1250Hz$ (bottom)

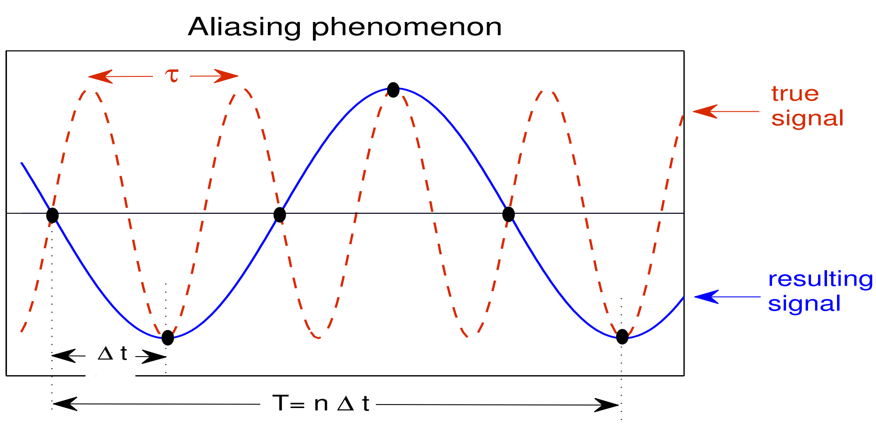

Figure 4.5: Aliasing resulting from sampling with a sampling rate smaller than twice the highest frequency of the signal, or similarly, the sampling interval is larger than half the period of the shortest wave in the signal ($2\Delta t > \tau$).

when the Nyquist time ($2\Delta t$) is larger than the period of the shortest wave in the signal. In fact, sampling must occur faster than twice the highest frequency in the signal in order to avoid aliasing.

Theoretically, this has to do with what is known as the *sampling theorem*. Assume that the continuous signal has $\gamma_c(.)$ and $f_c(.)$ as continuous autocorrelation function and power spectrum respectively, and $\gamma(.)$ and $f(.)$ those of the sampled signal, e.g., $\gamma_k = \gamma_c(k\Delta t)$. Then from the relationship between $\gamma_c(.)$ and $f_c(.)$ we get

$$f(\omega) = \sum_k f_c(\omega + \frac{2\pi k}{\Delta t}) = \sum_{k\geq 0} f_c(\omega + 2k\omega_N) + \sum_{k>1} f_c(\omega - 2k\omega_N)$$

for $-\omega_N < \omega < \omega_N$, and where ω is the radian frequency, (i.e., $\omega = 2\pi f$).

It is clear from the above equation that if the continuous signal $x(t)$ has no power for frequencies $|\omega| \geq \omega_N$ then clearly $f(\omega) = f_c(\omega)$, and hence no information is lost by sampling. In practice, however, this may not be the case as $f_c(.)$ may have power in the range $\omega \geq \omega_N$, and what we observe is that after sampling the obtained spectrum $f(\omega)$, ω in $[-\omega_N, \omega_N]$, results from the accumulation of power from the continuum spectrum at high frequencies $\omega \pm 2k\omega_N$, $k = 0, 1, \ldots$. These frequencies are said to be aliases of one another – i.e., every frequency outside $[-\omega_N, \omega_N]$ has an alias inside it.

Remark

Aliasing leads to :

1. Mistaking a high frequency signal for a low frequency one.
2. Can corrupt/affect the entire shape of the spectrum, particularly when there is a broad range of frequencies in the signal.

Hence, $f(\omega)$ will be obtained by 'folding' the continuum spectrum $f_c(.)$ at the folding frequencies $\pm(2k+1)\omega_N$, $k = 0, 1, 2, \ldots$, back onto the Nyquist interval $[-\omega_N, \omega_N]$.

The sampling theorem then says that if $x(t)$ has a band limited spectrum, i.e. $f_c(\omega) \approx 0$ for $|\omega| > \omega_0$, then the optimal choice of the sampling interval would be $\Delta t = \frac{\pi}{\omega_0}$. In practice, to avoid aliasing we can

- take small sampling intervals, ideally $\omega_N > \omega_s$, where ω_s is the highest frequency in the signal.
- filter the signal by removing all frequencies $\omega \geq \omega_N$, before sampling (using, e.g., analog electrical filters).

4.3.2 Estimation of power spectra

The DFT coefficients of a finite time series x_t, $t = 0, 1, \ldots n-1$ are given by:

$$F_k = \sum_{t=0}^{n-1} x_t e^{-2i\pi kt/n}, \qquad k = 0, 1, \ldots n-1. \tag{4.21}$$

The time series x_t, $t = 0, 1, \ldots n-1$, is then given by the inverse DFT:

$$x_t = \frac{1}{n} \sum_{k=0}^{n-1} F_k e^{2i\pi kt/n}. \tag{4.22}$$

As it was shown above (see exercise in section 4.2.1), it can be seen that $F_{n-k} = F_k^*$, $k = 1, 2, \ldots n-1$. Here we will suppose, for convenience, that n is odd. The time series can be decomposed into Fourier components (see exercise above):

$$x_t = \frac{a_0}{2} + \sum_{k=1}^{(n-1)/2} \left[a_k \cos(2\pi \frac{k}{n} t) + b_k \sin(2\pi \frac{k}{n} t) \right] \tag{4.23}$$

where the coefficients a_k and b_k, $k = 0, 1, \ldots (n-1)/2$, are given by:

$$a_k = \tfrac{2}{n} \textstyle\sum_{t=0}^{n-1} x_t \cos(2\pi \frac{k}{n} t) \quad \text{and} \quad b_k = \tfrac{2}{n} \textstyle\sum_{t=0}^{n-1} x_t \sin(2\pi \frac{k}{n} t) \tag{4.24}$$

Note that $a_k - ib_k = 2F_k^*/n$. As stated above the spectrum is then given by:

$$\hat{f}(k) = \frac{n}{4}(a_k^2 + b_k^2) = \frac{1}{n}|F_k|^2 = \frac{1}{n}|\sum_{t=0}^{n-1} x_t e^{-2i\pi \frac{k}{n}t}|^2 \tag{4.25}$$

As it can be noted this last expression is the raw periodogram

Remark

For any ω_k, $k = 1, \dots m$, with $m = (n-1)/2$, the sequence $e^{i\omega_k t}$, $t = 1, \dots n$, satisfies a number of constraints:

- $\sum_{t=1}^n (\cos(\omega_k t) + i\sin(\omega_k t)) = 0$
- $\sum_{t=1}^n \cos(\omega_k t)\sin(\omega_l t) = 0$
- $\sum_{t=1}^n \cos(\omega_k t)\cos(\omega_l t) = \sum_{t=1}^n \sin(\omega_k t)\sin(\omega_l t) = \frac{n}{2}\delta_{kl}$

The periodogram can also be obtained directly via a regression problem (see chapter 5). For a time series x_t, $t = 0, 1, \dots n-1$ (n is odd). The regression takes the form:

$$x_t = \alpha_0 + \sum_{k=1}^m \alpha_k \cos(\omega_k t) + \sum_{k=1}^m \beta_k \sin(\omega_k t) + \varepsilon_t \tag{4.26}$$

where $\omega_l = 2\pi l/n$, $l = 1, 2 \dots m$ and $m = (n-1)/2$. The regression problem (chapter 5) boils down to a linear system of equations:

$$\mathbf{X}^T\mathbf{X}\hat{\boldsymbol{\theta}} = \mathbf{X}^T\mathbf{x} \tag{4.27}$$

where $\mathbf{x} = (x_0, \dots, x_{n-1})^T$, $\mathbf{X}$ is the design matrix which contains the independent variables in (4.26), and $\hat{\boldsymbol{\theta}} = (\hat{\alpha_0}, \hat{\alpha_1}, \hat{\beta_1}, \dots, \hat{\alpha_m}, \hat{\beta_m})^T$. As it turns out (see chapter 5) the total variance can be split into two terms as:

$$\begin{aligned}\frac{1}{n}\sum_{t=0}^{n-1}(x_t - \overline{x})^2 &= \frac{1}{n}\sum_{k=1}^m \frac{2}{n}\left[\left(\sum_{t=0}^{n-1} x_t \cos\omega_k t\right)^2 + \left(\sum_{t=0}^{n-1} x_t \sin\omega_k t\right)^2\right] \\ &= \frac{1}{n}\sum_{t=0}^{n-1}\frac{2}{n}|\sum_{t=0}^{n-1}(x_t - \overline{x})e^{i\omega_k t}|^2\end{aligned}$$

The raw periodogram $I(\omega)$ is normally defined so that the sample variance matches the total area below $I(.)$. Basically, if the frequency interval is scaled to be $[0, T]$ and divided into equal bins of width $\frac{2\pi}{n}$ then the area corresponding to this bin is $\frac{2\pi}{n}I(\omega_k)$. Using the above equation this means:

$$I(\omega) = \frac{1}{n\pi}|\sum_{t=0}^{n-1}(x_t - \overline{x})e^{i\omega_k t}|^2 \tag{4.28}$$

Remark

Note that in eq. (4.25) the term $\frac{1}{\pi}$ is not included unlike the equation given above. This is more a matter of taste and is not universal.

Exercise

Show using the raw estimator of the periodogram in eq (4.25) that

$$\frac{1}{s^2}\hat{f}(k) = 1 + 2\sum_{\tau=1}^{n-1}\hat{\rho}(\tau)\cos(\frac{2\pi k}{n}\tau)$$

Answer – Consider $\hat{f}(k) = \frac{1}{n}\sum_t\sum_s x_t x_s e^{2i\pi\frac{k}{n}(t-s)}$, which is derived from the rhs of (4.25). This can be decomposed into a sum of square, $\frac{1}{n}\sum_{t=0}^{n-1} x_t^2$, and a sum of cross-terms, i.e., $\frac{1}{n}\sum_{\tau=1}^{n-1}\sum_{t-s=\tau} x_t x_s \left(e^{2i\pi\frac{k}{n}\tau} + e^{-2i\pi\frac{k}{n}\tau}\right)$ (the time series is assumed to be zero-mean). To see this, one simply writes the previous sum as $n^{-1}\mathbf{xAx}$, where $\mathbf{x} = (x_0, \ldots, x_{n-1})^T$, and $\mathbf{A} = (a_{lm})$ with $a_{lm} = e^{i\omega(l-m)}$ and $\omega = 2\pi k/n$. Because the matrix has constant diagonals, we can compute the sum diagonal-wise. The sum of the cross-terms along the τ'*th* diagonals below and above and the main diagonal is $n^{-1}(x_0 x_\tau + x_1 x_{\tau+1} + \ldots x_{n-\tau-1}x_{n-1})2\cos(\tau\omega)$, that is $2\hat{\gamma}(\tau)\cos(\tau\omega)$. By summing over τ one gets

$$\hat{f}(k) = s^2 + \frac{2}{n}\sum_{\tau=1}^{n-1}\left(\sum_{t-s=\tau} x_t x_s\right)\cos(2\pi\frac{k}{n}\tau) = s^2 + 2\sum_{\tau=1}^{n-1}\hat{\gamma}(\tau)\cos(2\pi\frac{k}{n}\tau).$$

We have seen in chapter 2 that the autocorrelation estimate used above is not unbiased (asymptotically unbiased). This bias is inherited by the spectrum estimate.

Exercise

Calculate the expectation of the spectrum.

Conclude.

Answer – Using the above expression one gets:

$$\begin{aligned} E\left(\hat{f}(k)\right) &= \sigma^2 + \tfrac{1}{n}\textstyle\sum_{\tau=1}^{n-1}\left[\sum_{|t-s|=\tau} E(x_t x_s)\right]\cos(2\pi\frac{k}{n}\tau) \\ &= \sigma^2 + \tfrac{2\sigma^2}{n}\textstyle\sum_{\tau=1}^{n-1}(n-\tau)\gamma(\tau)\cos(2\pi\frac{k}{n}\tau) \end{aligned}$$

which yields finally:

$$E\left(\hat{f}(k)\right) = \sigma^2\left(1 + 2\sum_{\tau=1}^{n-1}(1-\frac{\tau}{n})\rho(\tau)\cos(2\pi\frac{k}{n}\tau)\right)$$

Therefore the estimator is biased.

From the above exercise it is clear that the raw estimator of the power spectrum is always biased and therefore cannot be used as is. Let us have a look at what happens asymptotically. When the sample size n tends to infinity then the discrete frequencies $\omega_k = 2\pi\frac{k}{n}$ will tend towards pointwise values ω. Therefore by passing to the limit we get:

$$\lim_{n\to\infty} E\left(\hat{f}(\omega_k)\right) = \sigma^2\left(1 + 2\sum_{\tau=1}^{\infty}\rho(\tau)\cos(\tau\omega)\right) = f(\omega) \tag{4.29}$$

This shows that the sample spectrum is *asymptotically unbiased*, again exactly like the estimator of the autocovariance function.

Remark

An (unbiased) estimator whose variance goes to zero as $n\to\infty$ is called a *consistent* estimator. This is exactly like the sample mean. However, the sample autocovariance function is *asymptotically consistent*. The sample variance of the sample spectrum $\hat{f}(\omega)$ is

$$\mathrm{var}\left(\hat{f}(\omega)\right) \approx f^2(\omega)$$

and hence the sample spectrum is inconsistent.

Exercise
Let x_t, $t = 1, \ldots, n$ be an IID $\mathcal{N}(0, \sigma^2)$ with periodogram $I(\omega) = \frac{1}{n}(a^2(\omega) + b^2(\omega))$ with $a(\omega) + ib(\omega) = \sum_{t=1}^{n} x_t e^{i\omega}$ and $\omega = 2\pi k/n$.
1. What is the distribution of $a(\omega)$ and $b(\omega)$? Compute $\mathrm{var}(a(\omega))$, $\mathrm{var}(b(\omega))$, and $\mathrm{cov}(a(\omega), b(\omega))$.
2. Deduce the distribution of $I(\omega)$. Conclude.
Answer – $a(\omega)$ and $b(\omega)$ are normal, and $\mathrm{var}(a(\omega)) = \mathrm{var}(b(\omega)) = n\sigma^2/2$ and $\mathrm{cov}(a(\omega), b(\omega)) = 0$. Hence $I(\omega) \sim \sigma^2\chi_2^2/2$. Therefore this estimator is unbiased but inconsistent.

4.3.3 Smoothing of the spectrum

The results of the previous exercise can be extended to any Gaussian process with spectrum $f(.)$, i.e., $I(\omega_k) \sim f(\omega_k)\chi_2^2/2$, in addition to the fact that $f(\omega_k)$ and $f(\omega_l)$ are independent for $k \neq l$. The objective next is to find a better estimator, e.g., an estimator with asymptotic consistency. Now, assuming a smooth spectrum, let us consider the expression

$$\hat{f}_*(\omega_k) = \frac{1}{2p+1} \sum_{l=-p}^{p} I(\omega_{k+l}).$$

The estimator $\hat{f}_*(\omega_k)$ is distributed as $f(\omega_k)\chi^2_{2(2p+1)}/(4p+2)$. The variance is now $f^2(\omega_k)/(2p+1)$, hence smaller than that of the original estimator, that is $I(\omega_k)$.

When a time series is observed over a finite time interval then we know that we do not get an exact estimate of the spectrum, i.e. the sample spectrum is inconsistent. One way to obtain a consistent estimator is to divide the time series into chunks of equal size then take the average of the obtained individual sample spectra. This way of obtaining an estimate of the power spectra is known as *Bartlett smoothing*.

Now because the time series is basically obtained using a weighting function $w(t)$, normally $1_{[0,T]}$ or $1_{[-T/2,T/2]}$ known also as boxcar window, the obtained spectrum is simply a convolution of the true spectrum with a spectral 'window'. For example, for the boxcar window $1_{[-T/2,T/2]}$ the spectral window is $\frac{\sin(\omega T/2)}{\omega T/2}$, i.e. its Fourier transform.

The smoothed periodogram takes the form:

$$\hat{f}_s(\omega) = \int_{-\pi}^{\pi} \hat{f}(\theta)\Lambda(\omega - \theta)d\theta \tag{4.30}$$

where $\Lambda(.)$ is the spectral window (or smoothing kernel).

Remark

In the discrete form the smoothed spectrum takes the form:

$$\hat{f}_s(\omega) = \sum_k \hat{f}(\omega_k)\Lambda(\omega - \omega_k)$$

where $\omega_k = 2\pi \frac{k}{n}$, for $k = -\left[\frac{n+1}{2}\right], \ldots, \left[\frac{n}{2}\right]$, where the notation $[x]$ stands for the integer part of x.

The smoothed spectrum can also be written using the autocorrelation function:

$$\frac{1}{s^2}\hat{f}_s(\omega) = \sum_{k=-(n-1)}^{n-1} \lambda_k \hat{\rho}(k) \cos \omega k \tag{4.31}$$

where $\lambda_k = \int_{-\pi}^{\pi} \Lambda(\theta)\cos(k\theta)d\theta$, and represents the lag-window and can be regarded as the dual (or Fourier transform) of the spectral window. In general $\lambda_k = 0$ for $|k| > M$, yielding:

$$\frac{1}{s^2}\hat{f}_s(\omega) = \lambda_0 + 2\sum_{k=1}^{M} \lambda_k \hat{\rho}(k) \cos \omega k \tag{4.32}$$

Examples of windows

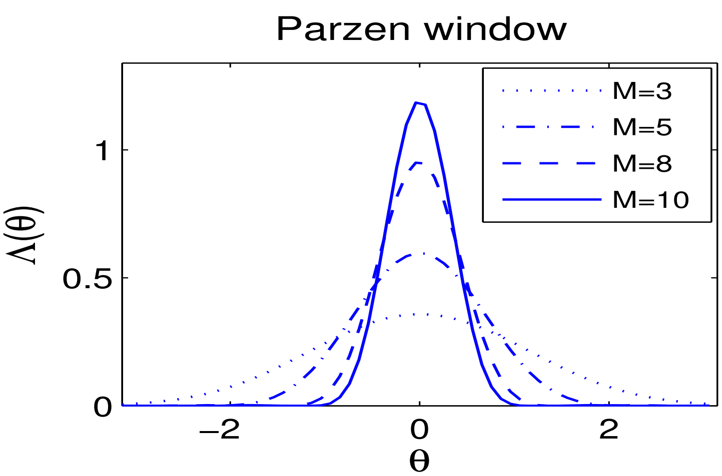

Figure 4.6: Parzen window for different values of the parameters M

- Parzen window – It has the following spectral window:

$$\Lambda(\theta) = \frac{6}{\pi M^3}\left[\frac{\sin(M\theta/4)}{\sin(\theta/2)}\right]^4 \left(1 - \frac{2}{3}\sin^2(\theta/2)\right)$$

- Bartlett window – This spectral window has the following lag-window:

$$\lambda_k = 1 - \frac{k}{M}$$

for $0 \leq k \leq M$. Figure 4.6 shows different Parzen windows for different values of M. Notice in particular that as M increases the lag window becomes narrower. As it has been discussed above, the smoothing decreases the variance. Since M can be regarded as a time resolution it is clear that the variance increases with M and vice versa.

Remark

Large M means that (4.32) is not much different from the non-smoothed periodogram, hence it has large variance. For small M, on the other hand, (4.32) becomes localised in time (since it only considers the first few lags and neglects the others) and broad in spectral domain ('like' white noise). This is like averaging over a large number of bins in spectral domain, yielding hence small variance.. Figure 4.7 shows the relation between the time and spectral windows for different shapes of the time windows.

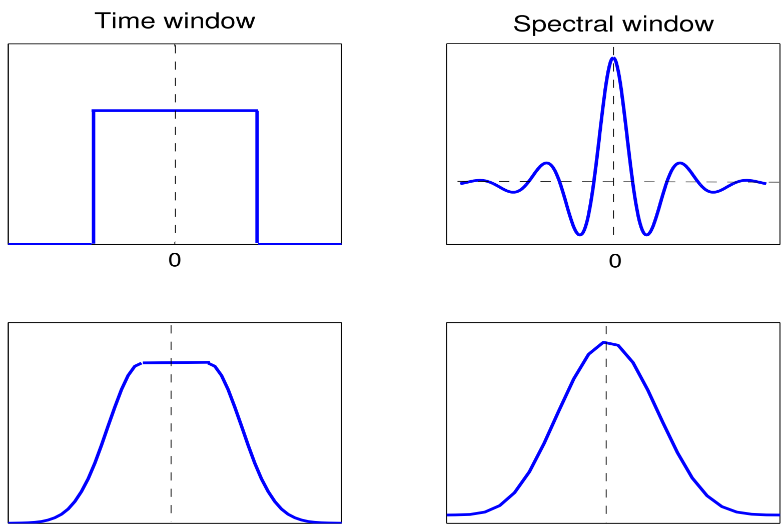

Figure 4.7: Relationship between the time and spectral windows

The degree of smoothing depends on T where small T yields broad spectral windows, i.e. strong smoothing and vice versa. For large T (i.e. large n) the spectral window is peaked, i.e. with good spectral resolution, but there are many lobes leading to what is known as *spectral leakage*. The bandwidth is like $1/T$. Of course, the ideal spectral window has a narrow central lobe and no side lobes. The particular choice of the window depends on the phenomenon of interest.

4.4 Statistical significance

As it was mentioned above, when a time series x_t, $t = 1, 2, \ldots$, is normally distributed then the periodogram is approximately distributed as χ_2^2. For the smoothed periodogram the distribution depends, in general, on the smoothing window. In general, the approximated distribution of $\hat{f}_s(\omega)$ is:

$$\frac{\hat{f}_s(\omega)}{f(\omega)} \sim \frac{\chi_\nu^2}{\nu} \tag{4.33}$$

where the number of d.o.f takes the form:

$$\nu = \frac{2n}{\sum_{k=-M}^{M} \lambda_k^2}. \tag{4.34}$$

Exercise

Find the number of d.o.f. ν for the smoothed periodogram $f_*(.)$ discussed earlier, i.e. $\hat{f}_*(\omega_k) = (2p+1)^{-1} \sum_{l=-p}^{p} I(\omega_{k+l})$

Answer – Use eq (4.34) with $\Lambda(\omega) = (2p+1)^{-1} \sum_{k=-p}^{p} \delta_{\omega_k}(\omega)$ plus a Fourier (cosine) transformation to yield $\lambda_k = (2p+1)^{-1} \sum_{j=-p}^{p} \cos(k\omega_j)$.

So the $100(1-\alpha)\%$ confidence interval for $f(\omega)$ is

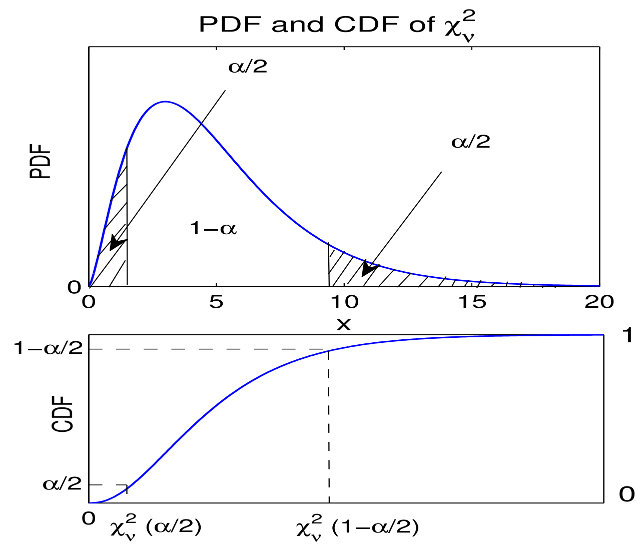

Figure 4.8: Critical levels for the sample spectrum using the chi-square distribution

$$Pr\left(\chi^2_\nu(\alpha/2) < \nu\frac{\hat{f}_s(\omega)}{f(\omega)} < \chi^2_\nu(1-\alpha/2)\right) = 1-\alpha$$

yielding the confidence interval $[\nu\frac{\hat{f}(\omega)}{\chi^2_\nu(1-\alpha/2)},\ \nu\frac{\hat{f}(\omega)}{\chi^2_\nu(\alpha/2)}]$ as shown in Fig. 4.8. Here again the notation $\chi^2_\nu(\alpha/2)$ is the $\alpha/2$'*th* quantile of the χ^2_ν.

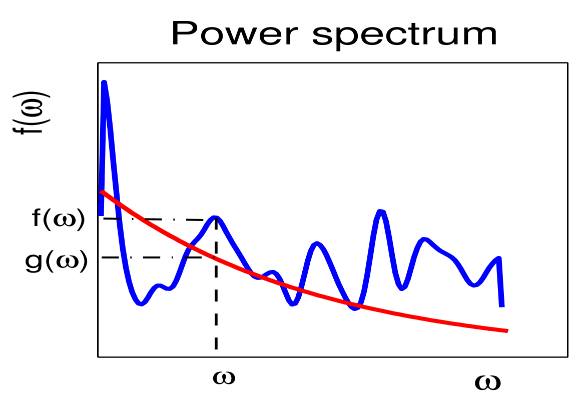

Figure 4.9: Comparison between the estimated power spectrum and the null power spectrum

So to test the spectrum we can use the null hypothesis
H_0: $f(\omega) = g(\omega)$ versus $f(\omega) \neq g(\omega)$,
where $g(\omega)$ is the spectrum of interest, e.g. red-noise, etc. (see Fig. 4.9). In this case we can use an estimate of $g(\omega)$ and then the expression $\frac{\hat{f}(\omega)}{\hat{g}(\omega)}$ behaves like a F-distribution.

4.5 Cross-spectra

The cross-spectrum of two time series (x_t) and (y_t) is defined using the cross-covariance by:

$$f_{xy}(\omega) = \sum_{\tau} \gamma_{xy}(\tau) e^{-2i\pi\omega\tau}. \tag{4.35}$$

Note that because the cross-covariance is not symmetric (unlike the autocovariance) the cross-spectrum is now complex. The real part of the cross-spectrum is referred to as the *co-spectrum* and the imaginary part is the *quadrature spectrum*. These are used to define the *coherence* and *phase*. Precisely, the coherence $C(\omega)$ and phase $\varphi(\omega)$ of the cross-spectrum $f_{xy}(.)$ are defined respectively by:

$$C(\omega) = \frac{|f_{xy}(\omega)|^2}{f_x(\omega) f_y(\omega)} \quad \text{and} \quad \varphi(\omega) = \tan^{-1} \frac{imag(f_{xy}(\omega))}{real(f_{xy}(\omega))}. \tag{4.36}$$

Note that the normalisation in $C(\omega)$ serves to isolate the influence of coupling. As a result we have:

$$0 \leq C(\omega) \leq 1.$$

The coherence gives a measure of the square of the correlation coefficient in the frequency domain whereas the phase reflects the lag between the two time series also in the frequency domain.

Remark

The co-spectrum is even and the quadrature spectrum is odd.

The sample cross-spectrum is given by:

$$\hat{f}_{xy}(k) = \sum_{\tau=-(n-1)}^{n-1} \hat{\gamma}_{xy}(\tau) e^{-2i\pi \frac{k}{n} \tau} \tag{4.37}$$

Alternatively, we can go back to the periodogram by using the DFT of the times series, and we get:

$$\hat{f}_{xy}(k) = \frac{a_k^x - ib_k^x}{2} \frac{a_k^y - ib_k^y}{2} = Co(k) - iQu(k) \tag{4.38}$$

Exercise

1. Using the expression of the raw periodogram show that

$$\hat{f}_{xy}(\omega) = I_{xy}(\omega) = \frac{1}{n} \sum_{t,s=0}^{n-1} x_t y_s e^{-i\omega(t-s)}$$

2. Show that $\hat{f}_{xy}(\omega) = (a_x(\omega) - ib_x(\omega))(a_y(\omega) - ib_y(\omega))/4$, and derive the expression of $a_x(\omega)$ and $b_x(\omega)$.

Answer – We can use a similar procedure as for the power spectrum of a single time series. The rhs of the above equality reads: $\hat{\gamma}(0) + n^{-1} \sum_{\tau=1}^{n-1} \sum_{k=1}^{n-\tau} \left[x_k y_{\tau+k} e^{-i\tau\omega} + y_k x_{\tau+k} e^{i\tau\omega} \right]$, which is precisely (4.37). We also have $a_x(\omega) - ib_x(\omega) = 2n^{-1} \sum_{t=0}^{n-1} x_t e^{-i\omega t}$, see also (4.24).

The coherence is then given by:

$$\hat{C}(\omega) = 2\frac{|\hat{f}_{xy}(\omega)}{\sqrt{|a_k^x + ib_k^x|^2 + |a_k^y + ib_k^y|^2}} \tag{4.39}$$

Exercise
From the sample estimate of the cross-spectrum (also bivariate periodogram) calculate $\hat{C}^2(\omega)$.
Answer – $\hat{C}(\omega) = 1$

From the above exercise we learn that the 'raw' sample coherence cannot be used as an estimator of the population coherence. One can estimate it, however, using the smoothing procedure presented earlier in connection with the periodogram.

Remark

The cross-spectrum is (exactly and theoretically) defined from the covariance $\gamma_{xy}(\tau) = \mathrm{cov}(x_t, y_{t-\tau})$, that is

$$f_{xy}(\omega) = \frac{1}{2\pi}\sum_{k=-\infty}^{\infty} \gamma_{xy}(\tau)e^{-iw\tau} = R_{xy}(\omega)e^{i\varphi(\omega)}$$

with $-\pi \leq \varphi_{xy}(.) \leq \pi$ (see Fig.) From these one can define the squared coherence by:

$$\rho_{xy}^2(\omega) = \frac{R_{xy}(\omega)}{f_x(\omega)f_y(\omega)}$$

Exercise
We consider the time series defined by:

$$y_t = \theta x_{t-\tau} + \varepsilon_t$$

with x_t and ε_t are two $\mathcal{N}(0,1)$ uncorrelated IID.
Compute $\gamma_{xy}(\omega)$, $f_{xy}(\omega)$, $f_{xx}(\omega)$ and $f_{yy}(\omega)$.
Compute ρ_{xy}^2
Hint. – $f_{xy}(\omega) = \frac{\theta}{2\pi}e^{i\omega k}$, with $-\pi \leq \varphi = \omega k \leq \pi$

Remark

From a finite sample one needs to estimate cross-spectrum. The cross-periodogram is the basic estimator. Now, the cross-spectrum is

$$I_{xy}(\omega) = \frac{1}{2\pi}\sum_{k=-(n-1)}^{n-1} \hat{\gamma}_{xy}(k)e^{-i\omega k}$$

where $\hat{\gamma}_{xy}(.)$ is the sample cross-correlation, see eqs (2.11)-(2.12). It is straightforward to see that

$$I_{xy}(\omega_k) = \frac{1}{2\pi}\frac{1}{n}\sum_{t=1}^{n} x_t e^{-i\omega_k t}\left(\sum_{t=1}^{n} y_t e^{-i\omega_k t}\right)^*,$$

after using the fact that $\sum_{t=1}^{n} e^{i\omega_k t} = 0$ for any non-zero Fourier frequency ω_k.

Exercise

Show that

$$\frac{|I_{xy}(\omega)|^2}{I_{xx}(\omega)I_{yy}(\omega)} = 1,$$

which implies that the (raw) periodogram cannot be used to estimate the squared coherence.

Chapter 5

Regression Analysis

5.1 Background

Regression and correlation concepts trace their origin back to the days of Sir Francis Galton, a cousin of Charles Darwin. Although the correlation coefficient bears Pearson's name[1] who studied rigorously the mathematical aspect of product moment correlation, the concept of regression/correlation was due to Galton. Note that Pearson was a colleague of Galton and his biographer after the latter's death.

In the context of regression analysis it is perhaps worth mentioning a bit of history on the regression terminology. The word "regression" literally means "going back", and is not literally linked to "correlation". Galton found what is known as "regression to the mean". This is a phenomenon observed in statistics in which if a variable is extreme on its current observation then it will tend to be (tend to have been) closer to the mean in the next (previous) measurement. Galton, in fact, observed this phenomeneon in the context of simple linear regression[2]. See, e.g., Stigler (1997) for more details on the history of regression to the mean.

The experiment with regression started in 1875 when Galton got interested in studying inheritence in the world of plants. He was interested precisely in studying the way genetic characteristics of one plant generation manifests in the next generation (Galton 1894). Galton plotted the weight of daughter seeds versus mothers' ones and realised that the weights of the former seeds, from a particular size of mother seed, follow approximately a straight line. This is reported in Galton's biography by Pearson (1930):"Thus he naturally reached a straight regression line, and the constant variability for all arrays of one character for a given character of a second. I was, perhaps, best for the progress of the correlation calculus that this simple special case should be promulgated first, it is so easily grasped by the beginner".

It was in 1896 that Pearson published his mathematical treatment of correlation/regression in the Philosophical Transaction of the Royal Society (Pearson 1896), where he showed, in particular, that the regression slope and correlation coefficient from a set of pairs of data (x_i, y_i) can be derived from the product-moment, i.e. $\sum(x_i - \overline{x})(y_i - \overline{y})/n$. We note, however, that Sir Gilbert Walker mentions Pierre Simon Laplace (1749-1827) and Giovanni Plana

[1]Pearson product moment correlation

[2]http://en.wikipedia.org/wiki/Regression_toward_the_mean

(1781-1864) as contributors to the introduction of "*product-term*" in the correlation, and Karl Friedrich Gauss (1777-1855) for contributing to the mathematical theory of correlation (Walker 1929), see also the discussion in Denis (2001).

5.2 Simple regression equation

5.2.1 Exploratory measures

In simple regression we assume that we have two time series x_i and y_i, $i = 1, 2, \ldots n$, and we think/believe that these two variables are related or one variable depends on the other. These two variables could be, for example, total annual precipitation over Europe and the annual NAO index. This points to the possibility that the time series can be realization of two random processes. Actually, in regression analysis the *dependent/response variable,* e.g. precipitation, is considered as randomly varying even after it has been described (to the extent possible) by the other, named *independent/predictor/explanatory* variable, which need not be. In practice, however, this requirement may be relaxed, that is the predictor and response variables can both be realization of two random variables. But when the explanatory variable X is random we normally resort to conditional expectations.

Before indulging into the regression equations the relationship between both time series is to be considered using exploratory tools, e.g., time series and scatter plots. Figure 5.1 shows

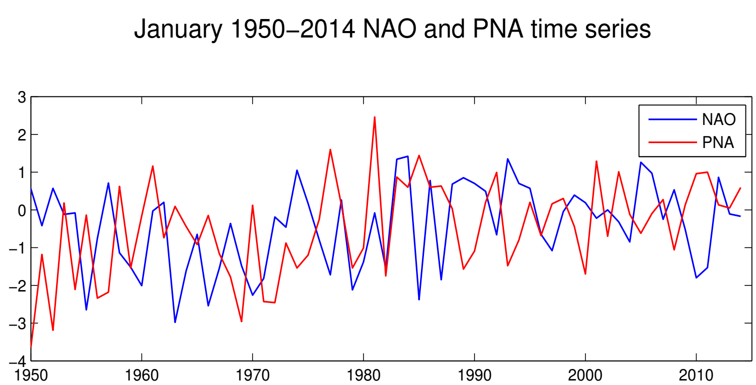

Figure 5.1: January NAO and PNA time series for the period 1950-2014.

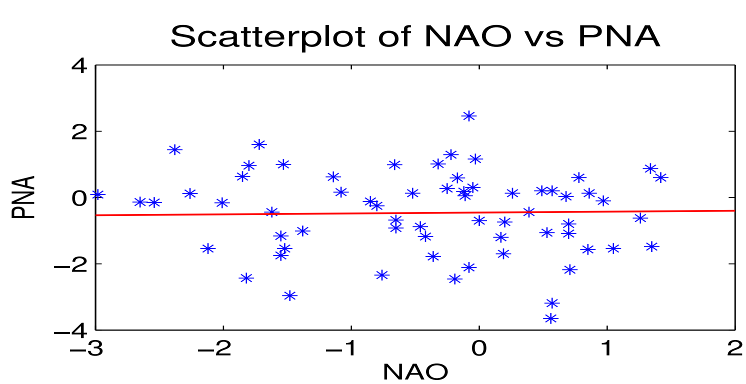

Figure 5.2: Scatterplot of January NAO and PNA time series for the period 1950-2014.

the time series for January North Atlantic oscillation (NAO), a prominent mode of variability ovet the North Atlantic region and the Pacific North America (PNA) pattern, a prominent mode of variability over the northern east Pacific and the North American continent. A scatter plot of NAO versus PNA is shown in Fig. 5.2. The 'best' line fit relating the PNA to the NAO is also shown. The equation of that line is $PNA = 0.027 * PNA - -0.454$. It can be seen that the slope is pretty small. The correlation between the time series is -0.119.

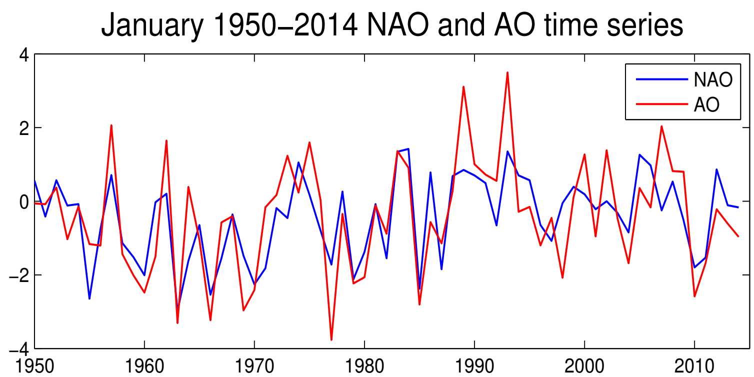

Figure 5.3: January AO and NAO time series for the period 1950-2014.

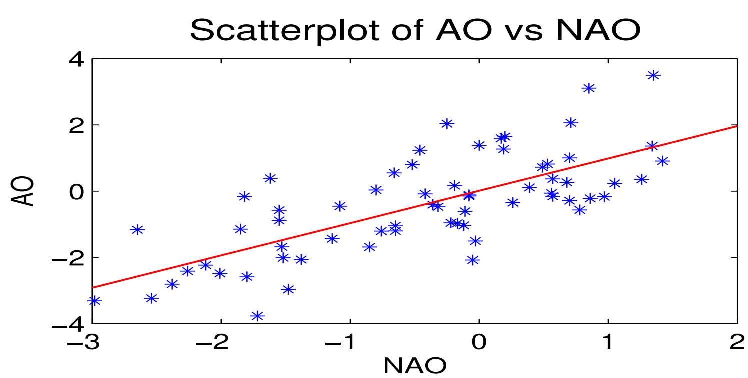

Figure 5.4: Scatterplot of January AO and NAO time series for the period 1950-2014.

Let us now look at the example of the Arctic Oscillation (AO) and its link to the NAO. The AO is also a prominent mode of variability in sea level pressure over the whole northern hemisphere. The AO and the NAO time series are plotted together for comparison in figure 5.3. As expected the two time series are highly correlated. A scatterplot is shown in figure 5.4. The best fit line is given by $AO = 0.975NAO + 0.009$. The correlation coefficient is now 0.72.

As we learned from the above examples, the next step could be to compute the Pearson product moment correlation[3]:

$$r_{xy} = \frac{\sum(x_i - \overline{x})(y_i - \overline{y})}{\sqrt{\sum(x_i - \overline{x})^2 \sum(y_i - \overline{y})^2}} \tag{5.1}$$

The Pearson correlation coefficient r_{xy} is *asymptotically* normally distributed as:

$$r_{xy} \sim \mathcal{N}\left(\rho_{xy}, \frac{1}{n}(1 - \rho_{xy}^2)^2\right). \tag{5.2}$$

[3] A more robust correlation coefficient is provided by Spearman's rank correlation (see chapter 9), but has no role in the regression topic discussed here

where ρ_{xy} is the population correlation coefficient. In practice, however, it is better to use the Fisher z-transform, which goes to the normal distribution as the sample size increases:

$$z = \frac{1}{2}\log\left(\frac{1+r_{xy}}{1-r_{xy}}\right) \to \mathcal{N}\left(\frac{1}{2}\log\frac{1+\rho_{xy}}{1-\rho_{xy}}, \frac{1}{n-3}\right) \tag{5.3}$$

assuming $\rho_{xy} \neq 0$. Using eq (5.3) we can draw the confidence limits for the correlation coefficient. For a given significance level α the confidence interval of ρ_{xy} is given by $[\tanh(z_-),\ \tanh(z_+)]$ where

$$z_\pm = z \pm \frac{1}{\sqrt{n-3}}\Phi_{1-\frac{\alpha}{2}} \tag{5.4}$$

where $\Phi_{1-\frac{\alpha}{2}}$ is the $(1-\frac{\alpha}{2})$'th quantile of the standard normal $\mathcal{N}(0,1)$, i.e.

$$\Phi_{1-\frac{\alpha}{2}} = \Phi^{-1}\left(1-\frac{\alpha}{2}\right)$$

Figure 5.5 shows the critical region along with the quantile.

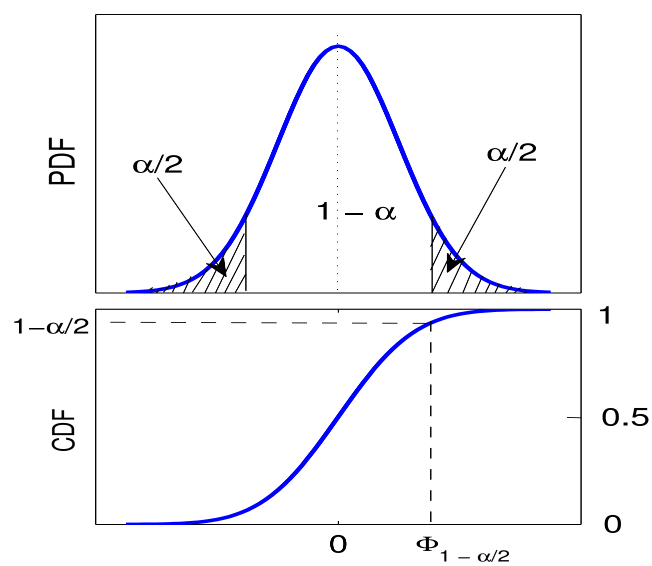

Figure 5.5: Critical region and $(1-\alpha/2)$'th quantile of the normal distribution

Remarks

- The above result is valid for large sample sizes.
- For small to moderate sample sizes one can use the t-distribution. For example, to test the null hypothesis $H_0 : \rho_{xy} = 0$ versus the alternative $H_1 : \rho_{xy} \neq 0$ one can use the statistic:

$$T = r_{xy}\sqrt{\frac{n-2}{1-r_{xy}^2}} \tag{5.5}$$

which follows a t-distribution with $n-2$ d.o.f.

5.2.2 The simple regression model

We assume that (x_i, y_i), $i = 1, 2, \ldots n$, are realisation of two random variables X_i and Y_i. The simple linear regression model is based on the assumption that the conditional mean of the variable $Y_i|X_i = x_i$ takes the form:

$$E\left(Y_i|X_i = x_i\right) = \alpha + \beta x_i. \tag{5.6}$$

This means that the conditional random variable $Y_i|X_i = x_i$ is normally distributed as:

$$Y_i|X_i = x_i \sim \mathcal{N}\left(\alpha + \beta x_i, \sigma_Y^2(1 - \rho_{xy}^2)\right).$$

In fact, the basic assumptions of the simple linear regression model are:

(i) linearity,

(ii) normality, and

(iii) constant variance.

Note, however, that the second assumption (ii), i.e. normality, is not needed for parameter estimation, only for confidence and tests. These assumptions are schematised in figure 5.6. Note that in this figure the conditional means are $\mu_k = E(Y|X = x_k)$, for $k = 1, \ldots n$. These means are linearly related to the explanatory variables x_k, $k = 1, \ldots n$. So, for each x_k, $k = 1, \ldots n$, the variable $Y|X = x_k$ is normal with mean μ_k and constant variance σ_Y^2,

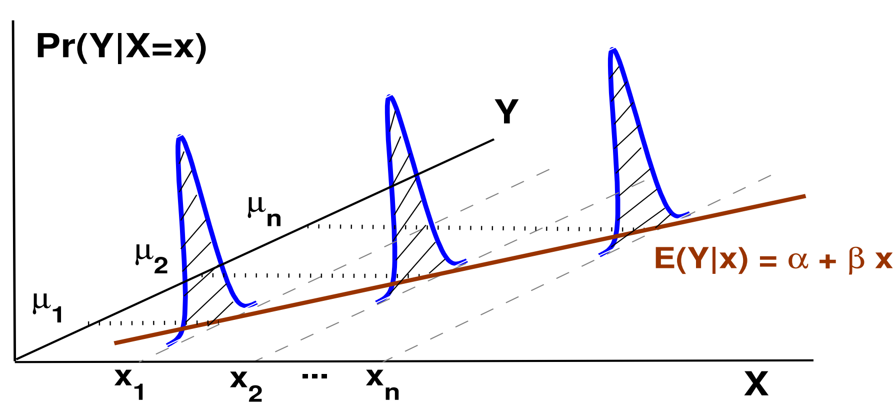

Figure 5.6: Fundamental assumptions for the linear regression model

Exercise

Derive the above equation, namely the variance of $Y_i|X_i = x_i$.

Hint – Use the expression of the conditional distribution (chap. 1).

These realisations then satisfy the model:

$$Y_i = \alpha + \beta x_i + \varepsilon_i \tag{5.7}$$

where ε_i are zero-mean Gaussian IID random variables. In the sequel, and for consistency with the lower case notation of the exaplanatory variable the response variable will be denoted also in lower case. Since α, β and ε_i, $i = 1, 2, \ldots n$ are unknown, they can be estimated by minimizing the sum of squared errors, SSE, i.e.

$$(\hat{\alpha}, \hat{\beta}) = argmin_{\alpha,\beta} \left(SSE = \sum_{i=1}^{n} \varepsilon_i^2 = \sum_{i=1}^{n} (y_i - \alpha - \beta x_i)^2 \right) \tag{5.8}$$

which yields the least square estimates:

$$\hat{\beta} = \frac{\sum x_i y_i - n\overline{x}\,\overline{y}}{\sum x_i^2 - n\overline{x}^2} \quad \text{and} \quad \hat{\alpha} = \overline{y} - \hat{\beta}\overline{x} \tag{5.9}$$

Exercise

Derive the above expressions of $\hat{\alpha}$ and $\hat{\beta}$. Note that $\hat{\beta}$ equivalently takes the form $\hat{\beta} = \sum(x_i - \overline{x})(y_i - \overline{y}) / \sum(x_i - \overline{x})^2$.

Notation

A set of terminology is often used in regression analysis, and it is useful to present them here. By refering again to the model (5.7) we have:

+ Predicted value: $\hat{y}_i = \hat{\alpha} + \hat{\beta} x_i$
+ Residual: $y_i - \hat{y}_i = y_i - \overline{y} - \hat{\beta}(x_i - \overline{x}) = \hat{\varepsilon}_i$
+ Sum of squares of x: $S_{xx} = \sum_i (x_i - \overline{x})^2$
+ Sum of squares due to regression (SSR) $SSR = \sum_i (\hat{y}_i - \overline{y})^2 = \hat{\beta} S_{xy}$

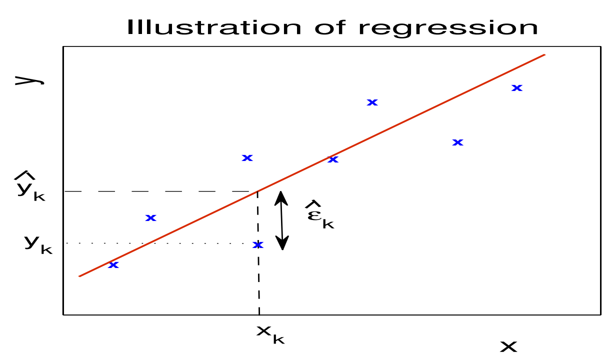

Figure 5.7: Illustration of a scatter plot showing the regression line $\hat{y}_k = \hat{\alpha} + \hat{\beta} x_k$ and the residuals $\hat{\varepsilon}_k$

Figure 5.7 illustrates an example of a scatter plot along with the regression line and the residuals.

Exercise

Assuming $\overline{x} = \overline{y} = 0$, show that $SSR = \hat{\beta} S_{xy}$
Hint. – since $\hat{y}_i = \hat{\beta} x_i$, one gets $\sum \hat{y}_i^2 = \hat{\beta}^2 \sum x_i^2 = \hat{\beta}(\sum x_i y_i / \sum x_i^2) \sum x_i^2$

+ Sum of cross-product (S_{xy}): $S_{xy} = \sum_i (x_i - \overline{x})(y_i - \overline{y})$
+ Total sum of square (SST) $S_{yy} = \sum_i (y_i - \overline{y})^2$
+ Sum of squared errors (SSE) or Residual sum of squares: $SSE = S_{yy} - \hat{\beta} S_{xy} = \sum_i (y_i - \hat{y}_i)^2 = \sum_i \hat{\varepsilon}_i{}^2$

Exercise

Derive the last identity of SSE.
Hint. – From $y_i - \overline{y} = \hat{\beta}(x_i - \overline{x}) + \hat{\varepsilon}_i$ (can take $\overline{x} = \overline{y} = 0$). Then $\hat{\varepsilon}_i{}^2 = y_i^2 + \hat{\beta}^2 x_i^2 - 2\hat{\beta} x_i y_i$ one gets $\sum \hat{\varepsilon}_i{}^2 = S_{yy} + \beta \left[(\sum x_i y_i)/(\sum x_i^2)\right] \sum x_i^2 - 2\beta S_{xy} = S_{yy} - \hat{\beta} S_{xy}$

The least square estimate gives us a partition of the total variability as:

$$SST = SSR + SSE \tag{5.10}$$

Another important quantity in linear regression is the *coefficient of determination* or R-square given by:

$$R^2 = \frac{SSR}{SST} \tag{5.11}$$

Exercise

Show that $R^2 = \frac{(\sum (x_i - \overline{x})(y_i - \overline{y}))^2}{\sum (x_i - \overline{x})^2 \sum (y_i - \overline{y})^2} = r_{xy}^2$
Hint. From the previous exercises, $SSR = \hat{\beta} S_{xy}$, and $SST = SSR + SSE = SSR + SS_{yy} - \hat{\beta} S_{xy} = S_{yy}$, and then $R^2 = \frac{\hat{\beta} S_{xy}}{S_{yy}} = \frac{S_{xy}^2}{SxxS_{yy}}$.

Remark

> Note that to avoid the confusion of the total sum of square with sea surface temperature (SST) we also use RegSS, ResSS and TotSS to mean respectively sum of square due to regression (SSR), sum of squared errors (SSE), and total sum of squares (SST).

5.2.3 Significance tests

Here we set out to investigate the significance of the slope β. Based on the above model (conditional on $X_i = x_i$) the coefficient $\hat{\beta}$ can be seen as a random variable and is seen as a linear combination of y_i, $i = 1, 2, \ldots n$

Exercise

Let Y_i, $i = 1, \ldots n$, be n random variables. Calculate $var(\sum_{i=1}^n \alpha_i Y_i)$.
Ans. $\sum_{i,j} \alpha_i \alpha_j cov(Y_i, Y_j)$.

Exercise

We assume (for simplicity, bt this does not affect the result) that $\overline{x} = \overline{y} = 0$,

1. Calculate $E(\hat{\beta})$
2. show that var$(\hat{\beta}) = \frac{\sigma^2}{\sum x_i^2} = \frac{\sigma^2}{S_{xx}}$ where $\sigma^2 = var(y_i^2) = var(\varepsilon_i)$.

Hint. 1. Use (5.9), recalling that $E(y_i) = \alpha + \beta x_i$ and $\overline{x} = 0$. This yields $E(\hat{\beta}) = \beta$.

2. $var(\hat{\beta}) = E(\hat{\beta} - \beta)^2$ and use Eq (5.9) and (5.7). Alternatively, can use the above exercise keeping in mind that $cov(Y_i, Y_j) = cov(\varepsilon_i, \varepsilon_j) = \sigma^2 \delta_{ij}$.

It is clear that

$$\hat{\beta} \sim \mathcal{N}(\beta, \frac{\sigma^2}{S_{xx}}) \tag{5.12}$$

Since σ^2 is unknown, we have to estimate it from the sample using the residual variance. The unbiased estimate of σ^2 is

$$\hat{\sigma}^2 = \frac{SSE}{n-2} = \frac{1}{n-2} \sum_{i=1}^{n} \varepsilon_i^2. \tag{5.13}$$

The confidence interval of the slope β is then obained using the t-distribution[4]. At the α_0-significance level the confidence interval of β is given by:

$$[\hat{\beta} - t_{n-2}(1 - \frac{\alpha_0}{2})\frac{\hat{\sigma}}{\sqrt{S_{xx}}},\ \hat{\beta} + t_{n-2}(1 - \frac{\alpha_0}{2})\frac{\hat{\sigma}}{\sqrt{S_{xx}}}] \tag{5.14}$$

where $t_{n-2}(1 - \frac{\alpha_0}{2})$ is the $(1 - \frac{\alpha_0}{2})$'th quantile of the t-distribution t_{n-2} with $n-2$ d.o.f. This confidence interval is illustrated in figure 5.8

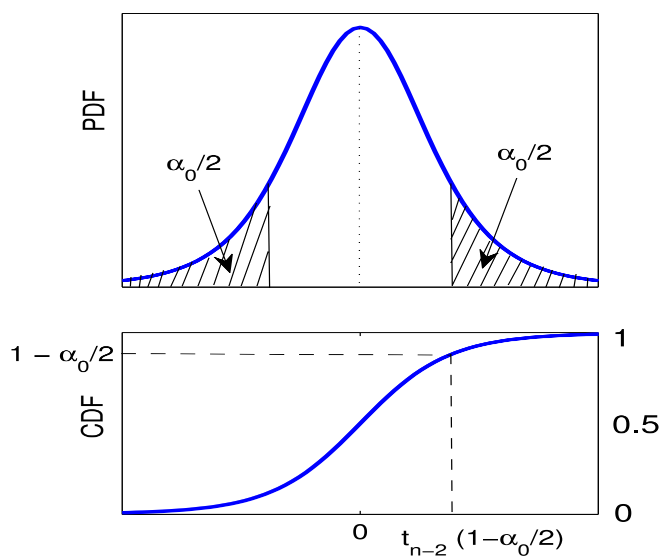

Figure 5.8: Critical levels using the t-distribution

Remark

The test statistic used for testing the regression slope β is $T = \frac{\hat{\beta}-\beta}{\sigma/\sqrt{S_{xx}}}$. Since we do not know σ, we approximate it with $\hat{\sigma}$, i.e. $T = \frac{\hat{\beta}-\beta}{\hat{\sigma}/\sqrt{S_{xx}}}$. By dividing the numerator and the denominator by σ and using eq (5.13) yields the following statistic $T = \frac{(\hat{\beta}-\beta)/(\sigma/\sqrt{S_{xx}})}{\sqrt{\frac{SSE/\sigma^2}{n-2}}}$, which is a student distribution with $n-2$ d.o.f.

[4]Why not the normal distribution? (Ans. because the variance is unknown and is estimated from the sample. This is particularly the case for small to moderate sample sizes.)

To test the hypothesis

$H_0 : \beta = 0$ versus $H_1 : \beta \neq 0$,

and the statistic to be used is

$$T = \frac{\hat{\beta}}{\hat{\sigma}/\sqrt{S_{xx}}} \sim t_{n-2}$$

Exercise

Confidence interval on the intercept $\hat{\alpha} = \overline{y} - \hat{\beta}\overline{x}$.

a. Show that $\mathrm{var}(\hat{\alpha}) = \sigma^2 \left(\frac{1}{n} + \frac{\overline{x}^2}{S_{xx}}\right) = \sigma^2 \frac{\sum x_i^2}{nS_{xx}}$ and therefore $\hat{\alpha} \sim \mathcal{N}\left(\alpha, \sigma^2 \frac{\sum x_i^2}{nS_{xx}}\right)$

b. For a given significance level α_0 calculate the $100(1-\alpha_0)$ confidence interval of the intercept α.

Answer – $[\hat{\alpha} \pm t_{n-2}(1 - \frac{\alpha_0}{2}) \frac{\hat{\sigma}}{\sqrt{nS_{xx}/\sum x_i^2}}]$.

Exercise

We want to compute the confidence interval on a predicted value $\hat{y}_k$.

a. From the equation $\hat{y}_k = \overline{y} + \hat{\beta}(x_k - \overline{x})$ show that $\mathrm{var}(\hat{y}_k) = \sigma^2 \left(\frac{1}{n} + \frac{(x_k - \overline{x})^2}{\sum (x_i - \overline{x})^2}\right)$

b. Find the confidence interval of $\hat{y}_k$ for a given α_0-significance level.

Answer – $[\hat{\alpha} + \hat{\beta}x_k \pm \hat{\sigma} \left(\frac{1}{n} + \frac{(x_k - \overline{x})^2}{\sum (x_i - \overline{x})^2}\right)^{1/2} t_{n-2}(1 - \frac{\alpha_0}{2})]$.

Post-analysis

Once the regression model is fitted some steps need to be checked for validation purposes. The model should be checked for consistency and adequacy.

• Most often one should look at the residuals, $\hat{\varepsilon}_i$, $i = 1, 2, \ldots n$. In fact, if the model were valid these residuals should be approximately normal around zero, with constant variance. This case of constant variance is referred to as *homoscedasticity.*

• To check for homoscedasticity one could use a scatter plot of $\hat{\varepsilon}_t$ vs x_t or $\hat{y}_t$. Now if there is a systematic pattern the residuals are said to *heteroscedastic*, in which case the model may be changed, through using, e.g., the logarithm or a power transformation such as $y_t = \alpha + \beta\sqrt{x}_t + \varepsilon_t$, etc.

• For independence, it is worth looking at the autocorrelation function of $\hat{\varepsilon}_t, t = 1, \ldots, n$ to check for any serial correlation. As an example, a regression model was fitted to the NAO and AO time series (Fig. 5.4). The model ($AO = \alpha + \beta NAO$) was given in section 5.2.1, i.e., $\alpha = .01$ and $\beta = 0.978$. The standard errors of α and β are 0.143 and 0.120 respectively. So the 95% confidence interval of β, for example, is [0.7391.218]. So we can safely reject the null hypothesis H_0: $\beta = 0$ at the 5% significance level.

Figure 5.9a shows the residuals from the regression model. The histogram of the residuals is shown in figure 5.9b. **Remark**

> In case of homoscedasticity, one can use alternative. One model is to look for nonlinear transformation of the data. Another robust model is the quantile regression, which is based on quantiles and includes extreme values. This is a convenient model especially to model extremes in weather and climate.

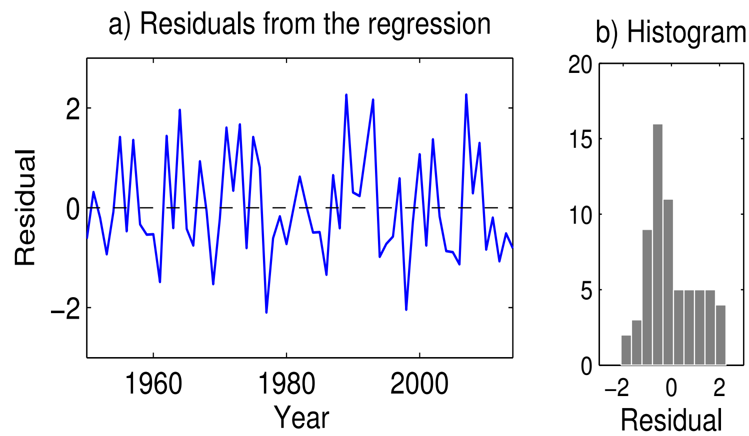

Figure 5.9: Residuals from the regression model between AO and NAO (a) and their histogram (b).

Quantile regression was applied by Finke and Hannachi (2022) to analyse surface temperature extremes and their link to the stratosphere.

5.3 Multiple regression

5.3.1 Background

The previous simple regression model can be re-written using matrix notation by introducing the so-called *design matrix* $\mathbf{X}$:

$$\mathbf{X} = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix} \tag{5.15}$$

In this case the model (5.7) becomes:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} \tag{5.16}$$

where $\mathbf{y} = (y_1, \ldots, y_n)^T$, $\boldsymbol{\beta} = (\alpha, \beta)^T$, and $\boldsymbol{\varepsilon} = (\varepsilon_1, \ldots, \varepsilon_n)^T$. The sum of squared error, SSE, is:

$$SSE = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) = \boldsymbol{\varepsilon}^T \boldsymbol{\varepsilon} \tag{5.17}$$

The minimisation of (5.17) yields the estimators:

$$\hat{\boldsymbol{\beta}} = \left(\mathbf{X}^T\mathbf{X}\right)^{-1} \mathbf{X}^T\mathbf{y} \tag{5.18}$$

Remark

From the above regression model and eq. (5.18) the predicted, or fitted, variable $\hat{\mathbf{y}}$ from eq. (5.16), i.e. $\mathbf{y} = \hat{\mathbf{y}} + \boldsymbol{\varepsilon}$, is written as

$$\hat{\mathbf{y}} = \mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1} \mathbf{X}^T\mathbf{y} = \boldsymbol{\mathcal{H}}\mathbf{y}$$

The matrix $\boldsymbol{\mathcal{H}}$ is known as the *hat* (or sometimes influence) matrix because it transforms $\mathbf{y}$ into $\hat{\mathbf{y}}$.

5.3.2 Multiple regression

The model presented previously is quite simple and one would expect that in practice we will have more than one explanatory variable. These variables can be of different nature. For example, for the case of European precipitation one can have, in addition to the NAO index, ENSO, Mediterranean SST and perhaps Asian monsoon. But we can also have one single explanatory variable, e.g. NAO index, but that the relationship is nonlinear such as a polynomial function:

$$precip = \alpha + \beta_1 NAO + \ldots + \beta_p (NAO)^p + \varepsilon.$$

In this case we deal with multiple regression, namely:

$$y_i = \alpha + \beta_1 x_{i,1} + \ldots + \beta_p x_{i,p} + \varepsilon_i, \tag{5.19}$$

which is similar to eq (5.16) except that now the design matrix is formed using the p factors $x_{k,i}$, $k = 1, \ldots p$ and $i = 1, \ldots n$, i.e.,

$$\mathbf{X} = \begin{pmatrix} 1 & x_{1,1} & \ldots & x_{1,p} \\ 1 & x_{2,1} & \ldots & x_{2,p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n,1} & \ldots & x_{n,p} \end{pmatrix} \quad \text{and} \quad \boldsymbol{\beta} = \begin{pmatrix} \alpha \\ \beta_1 \\ \vdots \\ \beta_p \end{pmatrix} \tag{5.20}$$

The residual variance is:

$$\sigma^2 = E(\boldsymbol{\varepsilon}^T \boldsymbol{\varepsilon}) = \|\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}\|^2 \tag{5.21}$$

We note again here that $\hat{\boldsymbol{\beta}}$ is an unbiased estimator of $\boldsymbol{\beta}$.

Exercise

Show that $E(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}$.

Hint. – Use (5.19) or (5.16) along with (5.18) to show that $E(\hat{\boldsymbol{\beta}}) = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T E(\mathbf{y}) = \boldsymbol{\beta}$.

The covariance matrix of the estimator $\hat{\boldsymbol{\beta}}$ is given by:

$$\mathrm{cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 \left(\mathbf{X}^T\mathbf{X}\right)^{-1} \tag{5.22}$$

Exercise

Derive the above identity.

Hint – Use the estimator (5.18) and the fact that the covariance of $\mathbf{y}$ is $\sigma^2\mathbf{I}$.

We therefore see that

$$\hat{\boldsymbol{\beta}} \sim \mathcal{N}\left(\boldsymbol{\beta}, \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}\right) \tag{5.23}$$

The residual sum of squares $\hat{\boldsymbol{\varepsilon}}^T\hat{\boldsymbol{\varepsilon}} = \mathbf{y}^T\mathbf{y} - \mathbf{y}^T\mathbf{X}\hat{\boldsymbol{\beta}}^T$ leads to an has an (unbiasedly) estimated variance:

$$\hat{\sigma}^2_{\boldsymbol{\varepsilon}} = \frac{1}{n-p-1}\hat{\boldsymbol{\varepsilon}}^T\hat{\boldsymbol{\varepsilon}} \tag{5.24}$$

Exercise

Show that $\hat{\boldsymbol{\varepsilon}}^T\hat{\boldsymbol{\varepsilon}} = \mathbf{y}^T\mathbf{y} - \mathbf{y}^T\mathbf{X}\hat{\boldsymbol{\beta}}$.
Answer – From $\hat{\boldsymbol{\varepsilon}} = \mathbf{y} - \hat{\mathbf{y}}$ one gets $\hat{\boldsymbol{\varepsilon}}^T\hat{\boldsymbol{\varepsilon}} = \mathbf{y}^T\mathbf{y} - 2\mathbf{y}^T\mathbf{X}\hat{\boldsymbol{\beta}} + \hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{X}\hat{\boldsymbol{\beta}}$, and use (5.18) for the last term.

Remark

Let $\mathbf{C} = (\mathbf{X}^T\mathbf{X})^{-1} = (c_{ij})$, and letting $\beta_0 = \alpha$, then for any $i, j = 0, \dots p$, $var(\hat{\beta}_j) = \sigma^2 c_{jj}$. Using the available sample, the standard error of $\hat{\boldsymbol{\beta}}$ is $s_e\mathbf{C}^{1/2}$, and consequently for any $j = 0, \dots p$, the standard error of $\hat{\beta}_j$ is

$$stde(\hat{\beta}_j) = s_e\sqrt{c_{jj}}$$

where s_e^2 is the estimator of σ^2, i.e. $s_e^2 = \frac{1}{n-p-1}\sum(y_i - \hat{y}_i)^2$, and $\hat{\beta}_j \sim N(\beta_j, \sigma^2 c_{jj})$. It follows that:

$$\frac{\hat{\beta}_j - \beta_j}{s_e\sqrt{c_{jj}}} \sim t_{n-p-1}$$

Note that, as in the simple regression case, see eq (5.10), the total sum of squares is:

$$TotSS = (\mathbf{y} - \overline{\mathbf{y}})^T(\mathbf{y} - \overline{\mathbf{y}}) = ResSS + RegSS \tag{5.25}$$

Also in a similar fashion to the simple linear regression, the estimator $\hat{\boldsymbol{\beta}}$ follows approximately a Chi-square, that is:

$$\frac{1}{\sigma^2}\left(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\right)^T\mathbf{X}^T\mathbf{X}\left(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\right) \sim \chi^2_{p+1} \tag{5.26}$$

The above equation contains σ^2, which is not known, and thums does not seem useful directly, but it will be shown later that it can be useful by using rations of two quantities containing σ^2.

Remarks

It is straightforward to show the following identities:
$TotSS = (\mathbf{y} - \overline{\mathbf{y}})^T(\mathbf{y} - \overline{\mathbf{y}}) = \mathbf{y}^T\left(\mathbf{I}_n - \frac{1}{n^2}\mathbf{N}^2\right)\mathbf{y} = SST$
$ResSS = (\mathbf{y} - \hat{\mathbf{y}})^T(\mathbf{y} - \hat{\mathbf{y}}) = \mathbf{y}^T\left(\mathbf{I}_n - \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\right)\mathbf{y} = SSE$
$RegSS = (\hat{\mathbf{y}} - \overline{\mathbf{y}})^T(\hat{\mathbf{y}} - \overline{\mathbf{y}}) = \mathbf{y}^T\left(\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T - \frac{1}{n^2}\mathbf{N}^2\right)\mathbf{y} = SSR$
where $\mathbf{N} = \mathbf{1}\mathbf{1}^T$ with $\mathbf{1}^T = (1, 1, \dots, 1)^T$ is the vector of ones with length n, i.e., $\mathbf{N}$ is the $n \times n$ matrix of ones ($\mathbf{N}^2 = n\mathbf{N}$.)

Similarly we have

$$\frac{1}{\sigma^2}TotSS = \frac{1}{\sigma^2}(\mathbf{y} - \overline{\mathbf{y}})^T(\mathbf{y} - \overline{\mathbf{y}}) \sim \chi^2_{n-1}, \tag{5.27}$$

and

$$\frac{1}{\sigma^2}SSE = \frac{1}{\sigma^2}\hat{\boldsymbol{\varepsilon}}^T\hat{\boldsymbol{\varepsilon}} \sim \chi^2_{n-p-1}. \tag{5.28}$$

Therefore $RegSS \sim \sigma^2\chi^2_p$, we recall here that $R^2 = SSR/TotSS$.
The Matlab code is :
>> [B Bint] = regress (y, X)
and with Python we can use:

```
>>> from sklearn.linear_model import LinearRegression
>>> regressor = LinearRegression()
>>> regressor.fit(X, y)
>>> # display coefficients
>>> print(regressor.coef_)
```

5.3.3 Confidence domain

Confidence on the slopes

In order to compute the confidence domain of the parameter $\boldsymbol{\beta}$ we consider again equations (5.26) and (5.28). Recalling that $(\chi_a^2/a)/(\chi_b^2/b) \sim F_{a,b}$, we get, by taking the ratio of (5.26) to (5.28):

$$\left(\boldsymbol{\beta}-\hat{\boldsymbol{\beta}}\right)^T \left(\mathbf{X}^T\mathbf{X}\right) \left(\boldsymbol{\beta}-\hat{\boldsymbol{\beta}}\right) \sim \frac{p+1}{n-p-1}\left(\hat{\boldsymbol{\varepsilon}}^T\hat{\boldsymbol{\varepsilon}}\right) F_{p+1,n-p-1} \tag{5.29}$$

The lhs of eq (5.29) is a quadratic function in the parameter space (i.e. $\boldsymbol{\beta}$) centered around around the estimate $\hat{\boldsymbol{\beta}}$. The confidence region can therefore be obtained using (5.29). In most cases we seek, however, to test the significance of the slope coefficients $\boldsymbol{\beta}_1 = (\beta_1, \dots, \beta_p)^T$. Let $\mathbf{X}_1$ be the design matrix obtained from $\mathbf{X}$ but without the column of ones, i.e. $\mathbf{X}_1 = (x_{i,j})$, $i = 1, \dots n$, $j = 1, \dots p$.

Remark

> Assuming that there is no constant term (i.e. $\alpha = 0$) the number of parameters in this case is p. But if we keep model (5.19) and test only $\boldsymbol{\beta}_1$ then we still have $p+1$ parameters.

We let $\mathbf{A} = (a_{ij})$, $i,j = 1, \dots p$, be the square sub-matrix of $(\mathbf{X}^T\mathbf{X})^{-1} = (x_{ij}^{-})$, $i,j = 1, \dots p+1$, corresponding to the slope coefficients in $\boldsymbol{\beta}_1$, i.e.

$$(\mathbf{X}^T\mathbf{X})^{-1} = \begin{pmatrix} x_{11}^{-} & x_{12}^{-} & \dots & x_{1,p+1}^{-} \\ x_{21}^{-} & x_{22}^{-} & \dots & x_{2,p+1}^{-} \\ \vdots & & \ddots & \\ x_{p+1,1}^{-} & x_{p+1,2}^{-} & \dots & a_{p+1,p+1} \end{pmatrix} = \begin{pmatrix} x_{11}^{-} & | & \dots & x_{1,p+1}^{-} \\ - & - & - & - \\ \vdots & | & \mathbf{A} & \\ x_{p+1,1}^{-} & | & & \end{pmatrix} \tag{5.30}$$

that is $\mathbf{A} = (x_{ij}^{-})$, $i,j = 2, \dots p+1$, then basically, eqs (5.26) and (5.28) respectively become

$$\frac{1}{\sigma^2}\left(\hat{\boldsymbol{\beta}}_1-\boldsymbol{\beta}_1\right)^T \mathbf{A}^{-1} \left(\hat{\boldsymbol{\beta}}_1-\boldsymbol{\beta}_1\right) \sim \chi_p^2 \tag{5.31}$$

$$\frac{1}{\sigma^2} SSE \sim \chi_{n-p-1}^2 \tag{5.32}$$

where $\mathbf{A}$ is the submatrix of $(\mathbf{X}^T\mathbf{X})^{-1}$ as shown in eq (5.30). Now as in (5.29), the $100(1-\alpha)\%$ region for $\boldsymbol{\beta}_1$ is given by the ellipsoid

$$\left(\boldsymbol{\beta}_1-\hat{\boldsymbol{\beta}}_1\right)^T \mathbf{A}^{-1} \left(\boldsymbol{\beta}_1-\hat{\boldsymbol{\beta}}_1\right) \leq \frac{p}{n-p-1}\left(\hat{\boldsymbol{\varepsilon}}^T\hat{\boldsymbol{\varepsilon}}\right) F_{p,n-p-1}(1-\alpha) \tag{5.33}$$

centered at $\hat{\boldsymbol{\beta}}_1$. In eq (5.33) $F_{p,n-p-1}(1-\alpha)$ is the $(1-\alpha)$'th quantile of the F-distribution with p and $n-p-1$ d.o.f.

Exercise

Derive eq. (5.33)

Answer – We look here for the $100(1-\alpha)\%$ confidence region for $\boldsymbol{\beta}_1$, i.e. we seek the region in the $\boldsymbol{\beta}_1$-parameter space containing $100(1-\alpha)\%$ of the whole mass of the distribution:

$$Pr\left[\frac{\left(\boldsymbol{\beta}_1-\hat{\boldsymbol{\beta}}_1\right)^T \mathbf{A}^{-1}\left(\boldsymbol{\beta}_1-\hat{\boldsymbol{\beta}}_1\right)/p}{\hat{\boldsymbol{\varepsilon}}^T\hat{\boldsymbol{\varepsilon}}/(n-p-1)} \leq x\right] = Pr\left(F_{p,n-p-1}\leq x\right) = 1-\alpha.$$

So clearly $x = F_{p,n-p-1}(1-\alpha)$, i.e. the $(1-\alpha)$'th quantile of $F_{p.n-p-1}$, or the inverse of the CDF of $F_{p.n-p-1}$ ($x = F^{-1}(1-\alpha)$ if $F(.)$ designates the CDF of $F_{p.n-p-1}$). This is illustrated in Fig. 5.10. In Matlab the quantiles from the F-distribution are obtained using the following

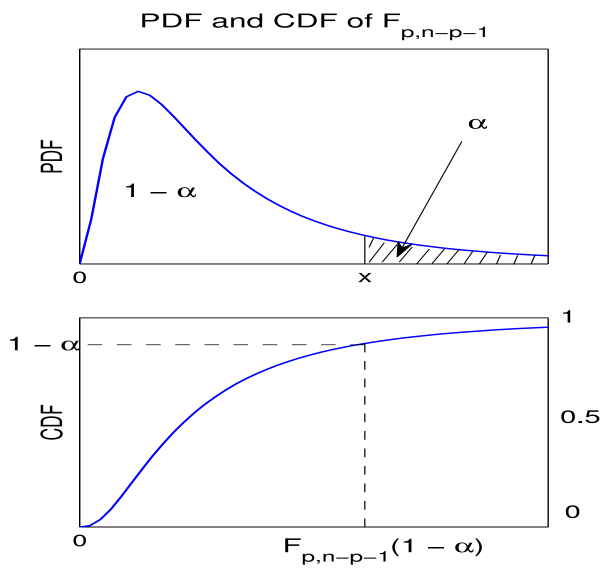

Figure 5.10: Critical region and $(1-\alpha)$'*th* quantile of the F-distribution

commands:

```
>> x = 0:.1:5;
>> m = 6; n = 6;
>> y = fpdf(x,m,n);
>> P = fcdf(x,m,n);
>> Q = finv(P,m,n);
```

The corresponding Python commands are:

```
>>> import numpy as np
>>> import scipy.stats as sps
>>> x = np.arange(0, 5, .1)
>>> m, n = 6, 6
>>> f = sps.f(m, n)
>>> y = f.pdf (x)
```

```
>>> P = f.cdf (x)
>>> Q = f.ppf (x)
```

The above confidence region is known as the *joint confidence domain.* Note the above test can also be applied by considering any fixed parameter $\boldsymbol{\beta}_1^{(0)}$ (not necessarily $\boldsymbol{\beta}_1$).
Note that in general $\mathbf{A}$ is not the inverse of $\mathbf{X}_1^T\mathbf{X}_1$. If, however, our model did not include α, i.e. we force α to be zero, then $\mathbf{A} = (\mathbf{X}_1^T\mathbf{X}_1)^{-1}$ and $n-p-1$ becomes $n-p$ in the above equation.

Remark

Sometimes we use the following notation, which is (adopted from) ANOVA (analysis of variance) table:

Source	Sum of square	d.o.f	Mean square (MS)	F
Regression	SSR	p+1-1	$MSR = SSR/p$	MSR/MSE
Residuals	SSE	$n-(p+1)$	$MSE = SSE/(n-p-1)$	
Total	SST or $TotSS$	$n-1$	$TotSS/(n-1)$	

Table 2. ANOVA table from the regression model (5.19)

If the model is forced to go through the origin then the number of d.o.f of SSE becomes $n-p$ and that of the total becomes n. Note also that as for the sum $SSR+SSE = TotSS$, we have the same equation for the d.o.f.

An equivalent way of expressing the test of the parameters $\boldsymbol{\phi}$, e.g. 5.32, is to use the ANOVA table to test the null hypothesis that all the slopes are zero, i.e., $H_0 : \boldsymbol{\beta}_1 = 0$. This is known as the *global* or *omnibus* null hypothesis. We can use the above ANOVA table 2 using the $F-$statistic (recall that $R^2 = RegSS/TotSS$),

$$F = \frac{RegSS/p}{ResSS/(n-p-1)} = \frac{n-p-1}{p}\frac{R^2}{1-R^2} \sim F_{p,n-p-1} \tag{5.34}$$

see also eq (5.33), and we can reject H_0 when F is sufficiently small. Precisely, for a given $100\alpha\%$ significance level we can reject H_0 when $F_{obs} > F_{p,n-p-1}(1-\alpha)$, where $F_{p,n-p-1}(1-\alpha)$ is the $(1-\alpha)$'th quantile of $F_{p,n-p-1}$.

Remark

The number of d.o.f of a statistic (which is different from that of dynamical systems) is basically the effective dimension of a hyperplane in which the system resides. A few examples are provided below

- Example 1. – For a set of n random variables $Y_1, \ldots Y_n$ the quantity $\sum(Y_i - \overline{Y})^2$ has $n-1$ d.o.f. because of the equality $\sum(Y_i - \overline{Y}) = 0$, which represents a hyperplane (note that one equation means take away one d.o.f).

- Example 2. – In a regression model $y_i = \alpha + \beta x_i + \varepsilon_i$ the residuals $\hat{\varepsilon}_i = y_i - \hat{\alpha} - \hat{\beta} x_i$ satisfy $\sum \hat{\varepsilon}_i = \sum x_i \hat{\varepsilon}_i = 0$, (intersection of two hyperplanes) hence $n-2$ d.o.f.

- Example 3. – Similarly $SSR = \sum_1^n (\hat{\alpha} + \hat{\beta} x_i - \overline{y})^2$ has $2-1=1$ d.o.f (two parameters $(\hat{\alpha}, \hat{\beta})$ and one equation $\sum (\hat{\alpha} + \hat{\beta} x_i - \overline{y}) = 0$).

Confidence on the mean as a function of x

As for the simple regression case, see eq. (5.6), one can compute the conditional mean $E(Y|X_1 = x_1, \ldots, X_p = x_p)$ of the variable Y given $X_k = x_k$, $k = 1, \ldots, p$. The mean of the regression model (5.19), for a given $\mathbf{x}$, is given by $\mu = E(Y|X_1 = x_1, \ldots, X_p = x_p) = \mathbf{x}^T \boldsymbol{\beta}$ where $\mathbf{x} = (1, x_1, \ldots, x_p)^T$, i.e. the vector containing one and the p independent variables. The estimate of μ is

$$\hat{\mu} = \mathbf{x}^T \hat{\boldsymbol{\beta}}. \tag{5.35}$$

Now, using (5.22) or (5.23) and keeping in mind that $var(\mathbf{x}^T \hat{\boldsymbol{\beta}}) = \mathbf{x}^T cov(\hat{\boldsymbol{\beta}}) \mathbf{x}$ one gets:

$$\hat{\mu} \sim \mathcal{N}\left(\mathbf{x}^T \boldsymbol{\beta}, \sigma^2 \mathbf{x}^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}\right), \tag{5.36}$$

and we can compute the confidence interval of the mean response, i.e. obtaining an interval estimate of $E(Y|X = \mathbf{x})$.

Exercise

Derive the above result.

Hint. Use the fact that for two vectors $\mathbf{a}$ and $\mathbf{b}$, $(\mathbf{a}^T \mathbf{b})^2 = \mathbf{a}^T (\mathbf{b} \mathbf{b}^T) \mathbf{a}$.

Remark

Recall that Eq (5.36) means that:

$$\frac{\hat{\mu} - \mathbf{x}^T \boldsymbol{\beta}}{\sigma \sqrt{\mathbf{x}^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}}} \sim N(0,1),$$

and using the estimator $\hat{\sigma}^2$ of σ^2, i.e. $s_e^2 = \frac{1}{n-p-1} \sum (y_i - \hat{y}_i)^2$, we get

$$\frac{\hat{\mu} - \mathbf{x}^T \boldsymbol{\beta}}{s_e \sqrt{\mathbf{x}^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}}} \sim T_{n-p-1}$$

We can therefore use the t-distribution to test the mean μ. So for a specific significance level α, the $100(1-\alpha)\%$ confidence interval of the mean response $E(Y|X = \mathbf{x}_0)$, for any given $\mathbf{x}_0$, is given by the interval:

$$[\mathbf{x}_0^T \hat{\boldsymbol{\beta}} \pm \hat{\sigma} \sqrt{\mathbf{x}_0^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_0} t_{n-p-1}(\frac{\alpha}{2})] \tag{5.37}$$

where again $t_{n-p-1}(\frac{\alpha}{2})$ is the $(\frac{\alpha}{2})$'th quantile of the t-distribution T_{n-p-1} with $n-p-1$ degrees-of-freedom.

Although (5.29) and (5.33) provide a more comprehensive confidence levels one can also compute the confidence level for a given regression coefficient used in eq (5.19). If we are interested in β_k, $k = 0, \ldots, p$ (and $\beta_0 = \alpha$) then we apply the above procedure (as for the mean μ) with $\mathbf{x}$ being the $(p+1)$-length vector of zeros except the k'th element which is 1. For example, for the intercept α we take $\mathbf{x} = (1, 0, 0, \ldots, 0)^T$, and for β_1 we have $\mathbf{x} = (0, 1, 0, \ldots, 0)^T$, etc. Of course the joint confidence level is more appropriate as the parameter estimates are not uncorrelated, i.e. the covariance matrix of $\boldsymbol{\beta}$ is not diagonal.

Remark

> The theory of regression analysis is based on the independence of the various estimators encountered. In fact, we have the following main results on independence:
>
> - SSE (or $ResSS$), i.e. $SSE = \sum(y_i - \hat{\alpha} - \hat{\beta}_1 x_{i,1} - \ldots)^2$ is independent of $(\hat{\alpha}, \hat{\beta}_1, \ldots, \hat{\beta}_p)$ or $\hat{\sigma}^2$.
> - Under the null hypothesis $H_0 : \beta_1 = \beta_2 = \ldots = \beta_p = 0$, SSE and SSR are independent.

Test summary

The test for a single parameter is different from the joint test. For a single parameter β_j the hypothesis of interest is:
$H_0 : \beta_j = 0$ vs $H_1 : \beta_j \neq 0$
and is based on the t-statistic: $t = \frac{\hat{\beta}_j - \beta_j}{stde(\hat{\beta}_j)}$. Under the null hypothesis $t \sim t_{n-p-1}$. The confidence interval, for a given significance level α, is $\hat{\beta}_j \pm t_{\alpha/2, n-p-1} s_e \sqrt{c_{jj}}$, with $s_e^2 = \frac{1}{n-p-1} \sum(y_i - \hat{y}_i)^2$, and $\mathbf{X}^T\mathbf{X}^{-1} = \mathbf{C} = (c_{ij})$.
The joint test of significance of the coefficients β_j, $j = 1, \ldots p$, also referred to as significance of regression, is:
$H_0 : \beta_1 = \beta_2 = \ldots \beta_p$ vs H_1 : At least one $\beta_j \neq 0$, $j = 1, \ldots p$.
The F-ratio test is:
$F = \frac{n-p-1}{p} \frac{\sum(\hat{y}_i - \overline{y})^2}{\sum(y_i - \hat{y}_i)^2} = \frac{n-p-1}{p} \frac{RegSS}{ResSS} \sim F_{p,n-p-1}$
and H_0 is rejected if $F > F_{p,n-p-1}(1-\alpha)$.

5.3.4 Nested model testing

It is also possible to test a submodel of the full model, eq. (5.19). In other words, the objective here is to test a smaller or null model, e.g.,

$$Y_i = \alpha + \beta_1 x_{i,1} + \ldots + \beta_q x_{i,q} \tag{5.38}$$

with $q < p$, against the full model , eq (5.19). So the null hypothesis to be tested is:

$$H_0 : \beta_{q+1} = \beta_{q+2} = \ldots = \beta_p = 0 \tag{5.39}$$

Let $\hat{y}_{i,f}$, and $y_{i,n}$, $i = 1, \ldots n$, be the fitted values based on the full (eq 5.19) and null (eq 5.38) models respectively. The corresponding ANOVA table is:

Source	Sum of square	d.o.f	Mean square	F
Difference	$SSD = \sum_1^n (\hat{y}_{f,i} - \hat{y}_{n,i})^2$	$p - q$	$SSD/(p-q)$	$\frac{n-p-1}{p-q}\frac{SSD}{SSE}$
Full	$SSE = \sum_1^n (y_i - \hat{y}_{f,i})^2$	$n-p-1$	$SSE/(n-p-1)$	
Null	$SST = \sum_1^n (y_i - \hat{y}_{n,i})^2$	$n-q-1$		

Table 2. ANOVA table from the nested regression model

The F-ratio is then given be

$$F = \frac{n-p-1}{p-q} \frac{\sum_{i=1}^n (\hat{y}_{f,i} - \hat{y}_{n,i})^2}{\sum_{i=1}^n y_i - \hat{y}_{f,i})^2},$$

and the null is rejected if $F > F_{p-q,n-p-1}(1-\alpha)$, for a significance level α.

5.4 The example of the periodogram

In chapter 4 we have briefly discussed how the periodogram is obtained from a regression analysis. Here we wish to present some details regarding that regression. We consider again a time series x_t, $t = 0, \dots n-1$ (n odd), and we seek the time series as a linear combination of $\cos\omega_k t$ and $\sin\omega_k t$, $k = 1, \dots, m$ where $m = (n-1)/2$. The regression equation was given in eq (4.26) and takes the form:

$$\mathbf{z} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} \tag{5.40}$$

where $\mathbf{z} = (z_0, \dots, z_{n-1})^T$, $\boldsymbol{\beta} = (\alpha_0, \alpha_1, \beta_1, \dots, \alpha_m, \beta_m)^T$, $\boldsymbol{\varepsilon} = (\varepsilon_0, \dots \varepsilon_{n-1})^T$, and

$$\mathbf{X} = \begin{pmatrix} 1 & \cos\omega_1 & \sin\omega_1 & \dots & \cos\omega_m & \sin\omega_m \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & \cos n\omega_1 & n\sin n\omega_1 & \dots & \cos n\omega_m & \sin n\omega_m \end{pmatrix}$$

The solution is given in eq (5.18), i.e. $\boldsymbol{\beta} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{z}$. Now, using the properties satisfied by the sequence $\cos\omega_k t$, and $\sin\omega_k t$, $t = 1, \dots, n$ (see the remark in section 4.4.1) the matrix $\mathbf{X}^T\mathbf{X}$ is simply diagonal, i.e.,

$$\mathbf{X}^T\mathbf{X} = diag(n, \frac{n}{2}, \dots, \frac{n}{2}).$$

the application of eq (5.10), keeping in mind that the above regression model is actually exact as the number of parameters equals the sample size, i.e. $\boldsymbol{\varepsilon} = \mathbf{0}$, yields:

$$\sum_{t=0}^{n-1} (\mathbf{z} - \mathbf{1}\overline{z})^T(\mathbf{z} - \mathbf{1}\overline{z}) = 2\sum_{k=1}^{m} \frac{1}{n}\left[\left(\sum_{t=1}^{n} z_{t-1}\cos\omega_k t\right)^2 + \left(\sum_{t=1}^{n} z_{t-1}\sin\omega_k t\right)^2\right] \tag{5.41}$$

Exercise

Derive the above equality.

Answer – We can write $\hat{\mathbf{z}} = (\mathbf{z} - \mathbf{1}\overline{z}) + \mathbf{1}\overline{z}$ and then $\hat{\mathbf{z}}^T\hat{\mathbf{z}} = (\mathbf{z} - \mathbf{1}\overline{z})^T(\mathbf{z} - \mathbf{1}\overline{z}) + n\overline{z}^2$. But since $\hat{\mathbf{z}}^T\hat{\mathbf{z}} = \mathbf{z}^T\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{z} = n\overline{z}^2 + 2n^{-1}\sum_{k=1}^m \left[(\sum_{t=1}^n z_{t-1}\cos\omega_k t)^2 + (\sum_{t=1}^n z_{t-1}\cos\omega_k t)^2\right]$. we obtain the required equation.

Now, in the rhs of the above equality, eq (5.41), one recognises the expression (between brackets) of the periodogram $I(\omega_k)$ and the equality reflects the fact that the variance of the time series equals the area below the periodogram, i.e.,

$$s_z^2 = \sum_{k=1}^{m} \frac{2}{n} I(\omega_k)$$

using bins of width $\frac{2}{n}$.

Remark

Note that if we have used $\frac{1}{n\pi}$ instead of $\frac{1}{n}$ in the expression of $I(\omega_k)$ then the bin width becomes $\frac{2\pi}{n}$.
Note also that here the objective is not to test the regression coefficients, rather it is the spectrum that we are after.

Chapter 6

Empirical Orthogonal Functions

6.1 Background and notation

The method of empirical orthogonal functions (EOFs) is one of the most widely used methods in atmospheric science. EOF analysis s also known as principal component analysis (PCA), and is based on performing an eigen analysis of the sample covariance matrix of the data. It is an exploratory technique for multivariate data, which aims at explaining and interpreting the variability in the data. A number of textbooks discuss EOF method in details, see, e.g., Jolliffe (2002), and more recently the text of Hannachi (2021), with more references therein. Some pre-requisites, notations and terminologies are needed before presenting the method.

6.1.1 Data matrix

In weather and climate we normally deal with fields, i.e. large gridded data sets. Our spatio-temporal fields are time series of two or three-dimensional variables such as gridded sea surface temperature or wind.

We assume that we have a time series of fields, i.e. a multivariate time series $\mathbf{x}_k = \mathbf{x}(t_k)$, $k = 1, \ldots, n$, where each $\mathbf{x}_k$ represents a two- or three-dimensional gridded variable at time t_k and gives the values of the variable at say p grid points. In the sequel we will consider $\mathbf{x}_k$ as consisting of p values x_{kj}, $j = 1, \ldots, p$ and the whole time series can be cast in terms of an array or data matrix $\mathbf{X}$ of the form:

$$\mathbf{X} = (\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n)^T = \begin{pmatrix} x_{1,1} & \ldots & x_{1,p} \\ \vdots & & \vdots \\ x_{n,1} & \ldots & x_{n,p} \end{pmatrix} \tag{6.1}$$

In (6.1) n is the sample size and p is the number of variables (or grid points), and represents the dimension of the problem. The above notation is quite useful and allows various operations to be accomplished. The mean field is noted as $\overline{\mathbf{x}}$ and is obtained by averaging $\mathbf{X}$ vertically, i.e.:

$$\overline{\mathbf{x}} = \frac{1}{n} \left(\sum_{i=1}^{n} x_{i1}, \ldots, \sum_{i=1}^{n} x_{ip} \right)^T .$$

When the mean (or climatology) is subtracted from the data matrix we are left with the anomaly data matrix $\mathbf{X}'$, i.e. $\mathbf{X}' = (\mathbf{x}_1 - \overline{\mathbf{x}}, \ldots, \mathbf{x}_n - \overline{\mathbf{x}})^T$. In the sequel it will be assumed that data matrix has been centered by subtracting the mean, i.e. $\mathbf{X}' = \mathbf{X}$. The (sample) covariance matrix is then:

$$\mathbf{S} = \frac{1}{n-1}\sum_{k=1}^{n}\left(\mathbf{x}_k - \overline{\mathbf{x}}\right)\left(\mathbf{x}_k - \overline{\mathbf{x}}\right)^T = \frac{1}{n-1}\mathbf{X}^T\mathbf{X} \tag{6.2}$$

Exercise

Derive the expression of the elements s_{ij} of $\mathbf{S}$. What are the two main properties of $\mathbf{S}$?

6.1.2 Singular value decomposition theorem

Before diving further it is helpful to present some theoretical background on the singular value decomposition (SVD) in linear algebra (e.g., Horn and Johnson 1985). Let $\mathbf{X}$ be a (non-trivial) $p \times p$ square matrix then there is always a complex number $\lambda \neq 0$ and a vector $\mathbf{v}$ such that

$$\mathbf{X}\mathbf{v} = \lambda\mathbf{v} \tag{6.3}$$

The number λ is known as and *eigenvalue* and $\mathbf{v}$ the associated eigenvector of $\mathbf{X}$.

Fundamental result: eigenvalue decomposition of a real square matrix

The matrix $\mathbf{X}$ has p eigenvalues $\lambda_1, \ldots, \lambda_p$ and associated eigenvectors $\mathbf{v}_1, \ldots, \mathbf{v}_p$. These eigenvalues can be complex, in which case they come in pairs of complex conjugate and similarly for the eigenvectors.

Eigenvalue decomposition of a symmetric matrix

Let $\mathbf{X}$ be a $p \times p$ symmetric matrix with eigenvalues $\lambda_1, \ldots, \lambda_p$ and associated eigenvectors $\mathbf{v}_1, \ldots, \mathbf{v}_p$ then we have the following decompositions:

$$\mathbf{X} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^T \tag{6.4}$$

where $\boldsymbol{\Lambda} = diag\left(\lambda_1, \ldots, \lambda_p\right)$ and $\mathbf{U}$ is an orthogonal matrix, i.e. $\mathbf{U}\mathbf{U}^T = \mathbf{I}_p$, and $\mathbf{I}_p$ is the identity matrix of order p. The columns of $\mathbf{U}$ are the normalized eigenvectors of $\mathbf{X}$. We say in this case that $\mathbf{X}$ is diagonalisable. For a complex matrix one has to substitute the notation $(^T)$ with $(^{*T})$ in (6.4), i.e., the transpose of the complex conjugate.

Remark

> When $\mathbf{X}$ is not symmetric, then $\mathbf{X}$ is diagonalisable when (6.4) holds but after substituting $\mathbf{U}^T$ with $\mathbf{U}^{-1}$.

Singular value decomposition (SVD) theorem

Any $n \times p$ matrix $\mathbf{X}$ can be decomposed as

$$\mathbf{X} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{V}^T \tag{6.5}$$

where $\mathbf{U}$ and $\mathbf{V}$ are orthogonal, i.e.

$$\mathbf{U}^T\mathbf{U} = \mathbf{V}^T\mathbf{V} = \mathbf{I}_r \tag{6.6}$$

where $\mathbf{I}_r$ is the identity matrix of order r, the rank of $\mathbf{X}$, and $\mathbf{\Lambda} = diag(\lambda_1, \ldots, \lambda_p)$. The diagonal matrix $\mathbf{\Lambda}$ contains the singular values $\lambda_k \geq 0$, $k = 1, \ldots, p$. The rank is the maximum number of independent columns (or rows) of $\mathbf{X}$. The column vectors of $\mathbf{U}$ and $\mathbf{V}$ are known, respectively, as the left and right singular vectors of $\mathbf{X}$ and $\lambda_1, \ldots, \lambda_p$ the associated singular values.

6.2 Empirical orthogonal functions

6.2.1 What are they?

The objective is to find a linear combination of the variables that explains maximum variance. In other words we seek vectors $\mathbf{a}$ such that the time series $\mathbf{Xa}$ has maximum variance. Mathematically, this is equivalent to:

$$\begin{aligned} &\max\left(\mathbf{a}^T\mathbf{X}^T\mathbf{X}\mathbf{a}\right) \\ &\text{subject to } \mathbf{a}^T\mathbf{a} = 1 \end{aligned} \tag{6.7}$$

Solutions to (6.7) are given by the following eigenvalue problem:

$$\mathbf{Sa} = \lambda^2\mathbf{a}. \tag{6.8}$$

Exercise
Derive eq (6.8).

Remark

> An alternative to the above eigenvalue problem formulation is to seek vectors $\mathbf{u}$ and $\mathbf{v}$ satisfying:
>
> $$\begin{aligned} &\max\left(\mathbf{u}^T\mathbf{X}\mathbf{v}\right) \\ &\text{subject to } \mathbf{u}^T\mathbf{u} = 1 = \mathbf{v}^T\mathbf{v} \end{aligned}$$
>
> leading to the singular value decomposition of the data matrix $\mathbf{X}$, see eqs (6.5-6.6).

The empirical orthogonal functions (EOFs) are therefore the eigenvectors of the sample covariance matrix arranged in decreasing order of the eigenvalues, i.e. $\lambda_1^2 \geq \lambda_2^2 \geq \ldots \geq \lambda_p^2$. The projection of the data onto the EOFs provides times series, referred to as principal components (PCs).

6.2.2 Computing EOFs/PCs

The EOFs can be computed from the sample covariance matrix through an eigenvalue decomposition routine. For example, in Matlab the routine *eig* provides this. If $\mathbf{X}$ is the (anomaly) $n \times p$ data matrix then the following few lines of code solves the problem:

```
>> load X
>> size(X)
```

```
>> S = cov (X);
>> [U, D] = eig (S);
```

and the eigenvalues and eigenvectors are stored respectively in D and U. Once the EOFs are obtained the PCs can be derived by projecting the data matrix onto the EOF loadings as:

```
>> PCs = X*U;
```

In Python, after loading the data matrix X we get:

```
>>> import numpy as np
>>> import scipy.linalg as la
>>> Xt = X.transpose() # can also use Xt = X.T
>>> Cov = np.cov (Xt)
>>> D, U = la.eig(Cov)
>>> PCs = X * U
```

Note that in np.cov(X), the variables are in rows and observations in columns of the object X. An alternative to the covariance matrix is to use the SVD theorem:

```
>> [V D U] = svd (X);
```

and in Python it is:

```
>>> V, D, U = la.svd (X)
```

The PCs are then stored in the columns of **V**, the EOF loadings are in the columns of **U**, and the singular values are in the diagonal matrix **D**.

6.3 Properties and interpretation

6.3.1 Properties

The EOFs have a number of properties that make them useful but at the same time these same properties can be associated with some problems particularly when it comes to interpretation. The following are some of the main properties:

- By construction the EOFs are orthogonal (see Fig. 6.1) and the PCs are uncorrelated.
- The leading EOF/PC explains maximum variance, and the next EOF/PC has the largest next explained variance and being orthogonal/uncorrelated to the previous one, etc.
- Because of the orthogonality property, the EOFs tend to be global, i.e. they span the whole domain as illustrated in Fig. 6.1. Using the eigenvalues of the covariance matrix[1] one can

[1] or the square of the singular values of the data matrix **X**

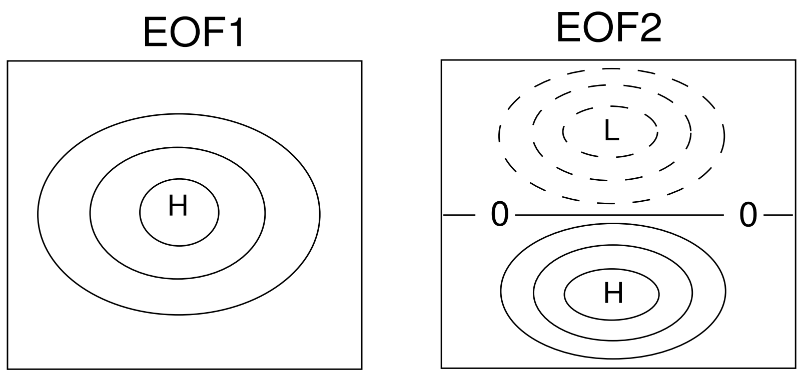

Figure 6.1: Illustration of a simple monople EOF1 and dipole EOF2.

compute the standard errors of the eigenvalues. If λ_k is the k'*th* eigenvalue the percentage Per_k of explained variance of the k'*th* EOF/PC is given by:

$$Per_k = 100\frac{\lambda_k}{\sum_{i=1}^{r}\lambda_i}\%. \tag{6.9}$$

The eigenvalues are subject to uncertainties due to sampling. As a rule of thumb, the standard error of λ_k, due to sampling variability, is of the order

$$\delta\hat{\lambda}_k \approx O\left(\hat{\lambda}_k\sqrt{\frac{2}{n}}\right) \tag{6.10}$$

A confidence interval can be constructed. For example the standard 95% confidence interval is approximated by the interval:

$$[\hat{\lambda}_k \pm 1.96\delta\hat{\lambda}_k] \tag{6.11}$$

Based on the above uncertainty, EOFs may have nearly identical eigenvalues, in which case the EOFs (or the eigenvalues) are said to be degenerate. This has consequences on the interpretation as it is discussed in the next section. EOFs have also other properties that can affect their physical interpretation:

• Domain-dependence – EOFs do not have, in general, an intrinsic and independent structure. For example, if we extend the domain or change its geometry we do not expect to get the same patterns.
• Global feature – the EOFs tend in general to span the whole domain. This means that EOFs are in general unable to capture localised features.
• Mixing – EOF analysis tends to mix signals to achieve optimality, which is again a handicap to physical interpretability.

6.3.2 EOF Interpretation

Because of the geometric constraints satisfied by the EOFs and the PCs attaching a physical significance/interpretation/meaning can be misleading, and should in general be avoided, unless there is a strong reason for doing so. If, for example, EOF1 has a constant sign over a domain then EOF2 will necessarily have a dipolar structure, in addition to the global feature mentioned above (see Fig. 6.1). So in some sense EOFs are 'predictable'.

"Physical" modes are not expected to be necessarily tagged to the amount of variance explained. For example, although the leading EOFs have the largest variability, there is the possibility that some physical modes may not be represented by these EOFs, or that they could be associated with EOFs of lower ranks. Perhaps the most popular example in atmospheric science literature is the North Atlantic Oscillation (NAO)/the Arctic Oscillation (AO) dichotomy.

Scaling is another issue encountered in EOF analysis. This problem occurs when we compute EOFs of combined variables with different units, e.g., mb/hPa and ^{o}K. This problem is related to the fact that the PCs depend critically on the scale used to measure the variables. This problem can be solved by standardising the variables, i.e. using the correlation matrix instead.

6.4 Rotation

6.4.1 What is it and why

Because of the constraints satisfied by EOFs interpretation can be difficult as discussed above. In many cases we are interested in local structures because of the interest in isolating patterns that can ease physical interpretation. This can be achieved by what is known as EOF *rotation*. Literally one attempts to rotate a number of EOFs (or PCs) by maximizing certain criteria. This rotation helps to focus the loading coefficients in a particular region.

6.4.2 The rotation procedure

Rotation attempts mainly to reduce or increase the loading coefficients of EOFs to the nearest whole number so to make them *simple*. There are two types of rotation namely, orthogonal and oblique, but orthogonal rotation tends to be more widespread. In orthogonal rotation a rotation matrix $\mathbf{R}$ is applied to a set of EOFs (or PCs). The rotated EOFs (REOFs) are defined by:

$$\mathbf{B}_m = \mathbf{U}_m \mathbf{R} = (\mathbf{b}_1, \ldots, \mathbf{b}_m) \tag{6.12}$$

where $\mathbf{U}_m = (\mathbf{u}_1, \ldots, \mathbf{u}_m)$ is a set of m EOFs to be rotated. These REOFs are then required to maximize a "simplicity criterion". One of the most popular criteria is the VARIMAX:

$$\max \left(\sum_{k=1}^{m} \left[p \sum_{j=1}^{p} b_{jk}^4 - \left(\sum_{j=1}^{p} b_{jk}^2 \right)^2 \right] \right) \tag{6.13}$$

Note that (6.13) represents the "spatial" variance of the squared elements of $\mathbf{B}_m$. This operation boils down to tending the loading coefficients towards 0 or ± 1.

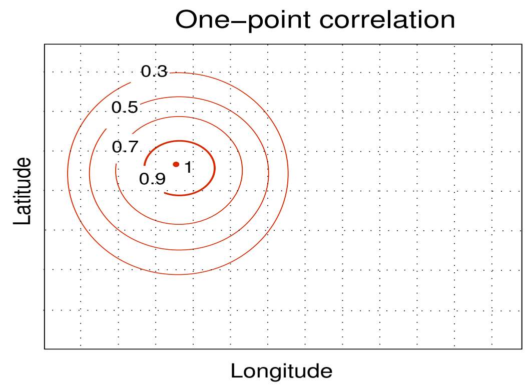

Figure 6.2: An example of one correlation map without teleconnection

6.5 Teleconnection and eigen empirical teleconnection

Teleconnection maps are obtained using one point correlation. Such a map is simply a row or column of the correlation matrix $\mathbf{C} = (c_{ij})$. The basic structure of this correlation map has a correlation one at the base point and decreases outward (Fig. 6.2).

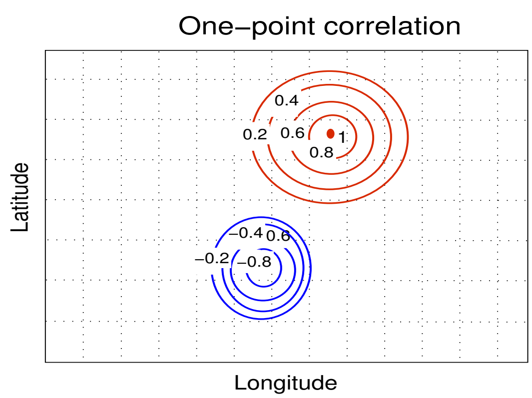

Figure 6.3: An example of one correlation map with teleconnectionas as indicated by the second negative centre

Sometimes, however, this structure can change radically where the pattern gets augmented by a second centre with opposite sign and remote from the base point (Fig. 6.3). It is this centre that make the difference. This represents a teleconnection pattern. Teleconnection patterns exist in large scale atmosphere. Examples include the NAO, the Pacific North America (PNA) pattern, and the Scandinavian pattern. The teleconnectivity T_i at the i'*th* grid point is defined by:

$$T_i = -\min_j c_{ij} \tag{6.14}$$

This teleconnectivity map provides a way to locate regions that are significantly inter-related.

An extension of teleconnectivity is possible and can be obtained by looking for patterns derived from regression coefficients between the selected base points and all other points. This operation is performed iteratively, and yields the eigen empirical teleconnection (EOT). Once an EOT is found the next one is obtained in a similar fashion after removing, through regression, the effect of the previous ones (e.g. van den Dool et al., 2000).

Chapter 7

Extended and Complex EOFs and Principal Oscillation Patterns

7.1 Background

EOFs attempt to decompose any field into ”stationary” patterns and varying amplitudes in time. EOFs do take spacial covariance between grid points into consideration, but ignore any temporal auto- and cross-correlation. This means that EOFs are, in general, unable to capture propagating signals.

One remedy to this is to take the temporal dimension into consideration. Various ways exist to achieve this. Perhaps the simplest is to extend the data matrix by including the temporal dimension. This leads to extended EOFs (EEOFs). Another way is to *complexify* the field so to obtain a complex time series then apply EOFs. This leads to complex EOFs (CEOFs).

7.2 Extended EOFs

In extended EOFs the data matrix is extended by introducing 'extra' variables. If $\mathbf{x}_t = (x_{t1}, \ldots, x_{tp})$ is the field at time t, then we consider a window of length M and then look through the time series using this sliding window. The new data matrix will look like:

$$\boldsymbol{\mathcal{X}} = \begin{pmatrix} \mathbf{x}_1 & \mathbf{x}_2 & \ldots & \mathbf{x}_M \\ \mathbf{x}_2 & \mathbf{x}_3 & \ldots & \mathbf{x}_{M+1} \\ \vdots & \vdots & \ldots & \vdots \\ \mathbf{x}_{n-M+1} & \mathbf{x}_{n-M+2} & \ldots & \mathbf{x}_n \end{pmatrix} \tag{7.1}$$

This data matrix is now $(n-M+1)\times pM$, and it is the new matrix that will go into the EOF machinery. A SVD decomposition of eq (7.1) leads to the extended EOFs and associated extended PCs. As it can be seen from eq (7.1) the number of dimensions now is pM. This means that an extended EOF has pM dimensions, that is M patterns. These patterns should reflect the propagating structure of the EEOF. Apart from this the same basic drawbacks and constraints encountered in EOFs are basically inherited by the extended EOFs. EEOFs can be used to identify propagating structures and to filter data. In one dimension extended

EOF analysis is also known as singular system[1] analysis (SSA) and has been used to identify periodic or quasi-periodic signals from time series. More details can be found in Hannachi et al. (2007).

7.3 Complex or Hilbert EOFs

Hilbert transform starts first by *complexifying* the field. The reason behind this transformation is that waves are easily and conveniently expressed using complex representation. We assume we have a real scalar field $\mathbf{x}_t = (x_{t1}, \ldots, x_{tp})$, which is represented by its Fourier transform:

$$\mathbf{x}_t = \sum_{\omega} \left(\mathbf{a}(\omega) \cos(\omega t) + \mathbf{b}(\omega) \sin(\omega t) \right) \tag{7.2}$$

Eq (7.2) can be transformed into a complex field by introducing a complex representation whose real part is precisely given by eq (7.2). The complexified field is given by:

$$\mathbf{z}_t = (z_{t1}, \ldots, z_{tp}) = \sum_{\omega} \mathbf{c}(\omega) e^{-i\omega t} \tag{7.3}$$

where the complex field $\mathbf{c}(\omega)$ is then given by:

$$\mathbf{c}(\omega) = \mathbf{a}(\omega) + i\mathbf{b}(\omega). \tag{7.4}$$

The Hilbert transform of eq (7.2) is given by:

$$\mathcal{H}(\mathbf{x}_t) = \sum_{\omega} \left(\mathbf{a}(\omega) \cos(\omega t) - \mathbf{b}(\omega) \sin(\omega t) \right) \tag{7.5}$$

Going back to eq (7.3) and taking into account the Hilbert transform in eq (7.5) we can see that the complexified field becomes:

$$\mathbf{z}_t = \mathbf{x}_t - i\mathcal{H}(\mathbf{x}_t) \tag{7.6}$$

In Matlab this is obtained using the following command:

```
>> Xr = randn(50,1);
>> X = hilbert(Xr);
```

In Python the above code is:

```
import numpy as np
Xr = np.random.randn(50, 1)
import scipy.signal.signaltools as sigtool
X = sigtool.hilbert(Xr)
```

The Hilbert EOFs (HEOFs) are then obtained from the eigenanalysis of the Hermitian covariance matrix:

$$\mathbf{S} = \frac{1}{n-1} \sum_{t=1}^{n} \mathbf{z}_t \mathbf{z}_t^{*T} \tag{7.7}$$

[1] also singular spectrum analysis

or via SVD applied to the data matrix $\mathbf{Z} = (z_{ij})$. The HEOFs, being complex, provide other measures such as the amplitude and phase functions, see, e.g., Hannachi et al. (2007) for further details.

7.4 Principal oscillation patterns

A simple dynamical model for the atmosphere is the first-order (vector) autoregressive model:

$$\mathbf{x}_t = \mathbf{A}\mathbf{x}_{t-1} + \boldsymbol{\varepsilon}_t \tag{7.8}$$

where $\boldsymbol{\varepsilon}_t$ is independent in time, with covariance matrix $\mathbf{Q}$, and uncorrelated with $\mathbf{x}_k$, $k \leq t-1$. The matrix $\mathbf{A}$ can be estimated from eq (7.8) by:

$$\mathbf{A} = \boldsymbol{\Sigma}_1 \boldsymbol{\Sigma}_0^{-1} \tag{7.9}$$

where $\boldsymbol{\Sigma}_0$ is the covariance matrix of the time series $\mathbf{x}_t$, $t = 1, 2, \ldots$ and $\boldsymbol{\Sigma}_1$ is the lag-1 autocovariance matrix of $\mathbf{x}_t$.

In order for (7.8) to be stable we require that all eigenvalues of $\mathbf{A}$ satisfy $|\lambda_k| < 1$. We assume that $\mathbf{A}$ is diagonalisable, with eigenvalues $\lambda_1, \ldots, \lambda_p$, and associated eigenvalues $\mathbf{u}_1, \ldots, \mathbf{u}_p$, i.e.,

$$\mathbf{A} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^{-1} \tag{7.10}$$

The vectors $\mathbf{u}_1, \ldots, \mathbf{u}_p$ are known as principal oscillation patterns (POPs). Now, by decomposing the state vectors following the POPs, the dynamics boil down to a set of p first-order autoregressive $AR(1)$ models:

$$z_k(t) = \lambda_k z_k(t-1) + \varepsilon_k(t) \tag{7.11}$$

with $\lambda_k = |\lambda_k| e^{i\omega_k}$. The evolution of the expected values of eq (7.11) yields a damped oscillator with period $T_k = \frac{2\pi}{\omega_k}$ and damping or e-folding time $\tau_k = -\frac{1}{\log|\lambda_k|}$.

Remark

> Note that, unlike the case of the covariance matrix, the matrix $\mathbf{U}$ in the above decomposition is not orthogonal in general. So the POPs are in general neither orthogonal (in space) nor uncorrelated (in time).

Let $\mathbf{u}_k$ be the k'*th* POP with the decomposition into real and imaginary parts

$$\mathbf{u}_k = \mathbf{u}_k^r + i\mathbf{u}_k^i$$

The evolution of the (noise-free or expected) k'*th* POP yields:

$$z_k(t)\mathbf{u}_k + z_k^*(t)\mathbf{u}_k^* = \lambda_k^t \mathbf{u}_k + \lambda_k^{*t}\mathbf{u}_k^* = 2|\lambda_k|^t \left(\mathbf{u}_k^r \cos(\omega_k t) - \mathbf{u}_k^i \sin(\omega_k t)\right) \tag{7.12}$$

Hence within the 2D plane of $\mathbf{u}_k = (\mathbf{u}_k^r, \mathbf{u}_k^i)$ the evolution of the system can be described by a succession of patterns:

$$\mathbf{u}_k^r \longrightarrow -\mathbf{u}_k^i \longrightarrow -\mathbf{u}_k^r \longrightarrow \mathbf{u}_k^i \longrightarrow \mathbf{u}_k^r$$

ocuppied at successive times $t_m = \frac{m\pi}{2\omega_k}$, $m = 0, 1, 2, \ldots$ (Fig. 7.1).

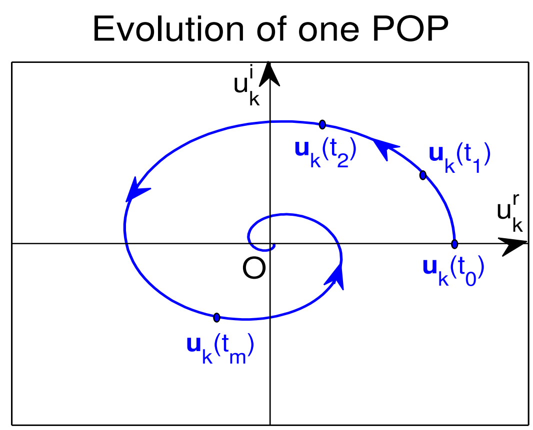

Figure 7.1: The evoluion of the k'*th* POP within the plane (u_k^r, u_k^i), showing the positions of $\mathbf{u}_k(t)$ at successive times $t_0, t_1, \ldots, t_m$

7.5 Methods for coupled patterns

Although EOFs can be applied to combined fields, after taking care of the variance of the fields, the leading patterns explain maximum variance of the combined fields. In many cases the objective is to find fields having strongest association, such as the association between, e.g., SST and SLP.

We assume we have two fields (i.e., two data matrices) $\mathbf{X}$ and $\mathbf{Y}$, with respective sizes $n \times p$ and $n \times q$. To find the patterns from $\mathbf{X}$ and $\mathbf{Y}$ that are most associated we can use two methods: maximum covariance analysis (MCA) and canonical correlation analysis (CCA).

7.5.1 Maximum covariance analysis

Here we start by forming the $p \times q$ covariance matrix:

$$\mathbf{C}_{\mathbf{XY}} = \frac{1}{n}\mathbf{X}^T\mathbf{Y} \tag{7.13}$$

which can be decomposed using SVD:

$$\mathbf{C}_{\mathbf{XY}} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T \tag{7.14}$$

where $\mathbf{U} = (\mathbf{u}_1, \ldots, \mathbf{u}_r)$, $\mathbf{V} = (\mathbf{v}_1, \ldots, \mathbf{v}_r)$, and $\mathbf{\Sigma} = diag(\sigma_1^2, \ldots, \sigma_r^2)$ and where the singular values have been arranged in decreasing order $\sigma_1^2 \geq \sigma_2^2 \geq \ldots \geq \sigma_r^2$. The patterns $\mathbf{u}_1$ and $\mathbf{v}_1$ are the leading patterns of $\mathbf{X}$ and $\mathbf{Y}$ respectively that have the largest covariance, and $(\mathbf{u}_2, \mathbf{v}_2)$ are the next patterns with the next largest covariance and being orthogonal to $\mathbf{u}_1$ and $\mathbf{v}_1$ respectively. This method is also known as singular value analysis.

7.5.2 Canonical correlation analysis

When the fields are highly correlated one can apply EOF analysis of both fields in a first step. One then selects a number of leading EOFs or PCs from each field and apply MCA between the obtained PCs. CCA can also be defined alternatively as the patterns whose time series have maximum correlation instead of covariance. In this case the canonical correlation patterns are obtained by solving the eigenvalue problems:

$$\begin{aligned}\mathbf{C_{YX}u} - \mu\mathbf{C_{YY}v} &= 0 \\ \mathbf{C_{XY}v} - \lambda\mathbf{C_{XX}u} &= 0\end{aligned} \tag{7.15}$$

This system can be transformed to yielding a single eigenvalue problem:

$$\mathbf{C_{XX}^{-1}C_{XY}C_{YY}^{-1}C_{YX}u} = \lambda\mu\mathbf{u} \tag{7.16}$$

or similarly

$$\mathbf{C_{YY}^{-1}C_{YX}C_{XX}^{-1}C_{XY}v} = \lambda\mu\mathbf{v} \tag{7.17}$$

Exercise

Show that $\mu = \lambda$ in the above eqs (7.16)-(7.17).

Answer – Use eq (7.15) to show that $\lambda = \mu = \mathbf{u}^T\mathbf{C_{XY}v}$ (recall that $\mathbf{u}^T\mathbf{C_{XX}^{-1}v} = 1 = \mathbf{v}^T\mathbf{C_{YY}^{-1}}$).

Remark

Note that now eq (7.15) can be written as a generalised eigenvalue problem:

$$\mathbf{Aw} = \nu\mathbf{Bw}$$

where the matrices $\mathbf{A}$, $\mathbf{B}$ and the vector $\mathbf{w}$ are given by:

$$\mathbf{A} = \begin{pmatrix} \mathbf{C_{YX}} & \mathbf{O} \\ \mathbf{O} & \mathbf{C_{XY}} \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} \mathbf{O} & \mathbf{C_{YY}} \\ \mathbf{C_{XX}} & \mathbf{O} \end{pmatrix} \quad \text{and} \quad \mathbf{w} = \begin{pmatrix} \mathbf{u} \\ \mathbf{v} \end{pmatrix}$$

Once these eigenvectors and eigenvalues are obtained and ordered in decreasing order the pairs $(\mathbf{u}_1, \mathbf{v}_1), (\mathbf{u}_2, \mathbf{v}_2), \ldots$ provide the patterns with maximum correlation. For example, $\mathbf{u}_1$ and $\mathbf{v}_1$ represent the patterns from $\mathbf{X}$ and $\mathbf{Y}$, respectively, that are the most correlated.

7.6 Coupled patterns and multivariate regression

7.6.1 Background

PCA is normally applied to a single field to find out the modes of variability of the data. It can also be applied to more than one field, such as wind field at different levels. The objective is always to identify directions that maximize variance but not to look for relationships between the different data sets. This task belongs to other methods such as maximum covariance analysis (MCA), and canonical correlation analysis (CCA), which use two data sets and aim at finding projections maximizing covariance and correlation respectively. MCA and CCA use both data sets in a symmetric way in the sense that we are not seeking to explain one data set as a function of the other.

Methods that fit in the regression framework exist and are based on considering one data set as predictor and the other as predictand. Three main methods attempt to achieve this, namely, redundancy analysis (RDA), principal predictors analysis (PPA) and principal regression analysis (PRA). RDA (von Storch and Zwiers 1999, Wang and Zwiers 1999) aims at selecting predictors that maximize the explained variance. PPA (Thacker 1999) seeks to select predictors that maximize the sum of squared correlations. PRA (Yu et al., 1997), as its name indicates, fits regression models between the principal components of the predictor data and each of the predictand elements individually.

Tippett et al. (2008) discusses the connection between the different methods of finding coupled patterns and multivariate regression through a SVD of the regression matrix.

7.6.2 Coupled patterns and multivariate regression

In multivariate regression one attempts to explain the variability of the predictand in terms of a linear model of the predictor variables:

$$\mathbf{y} = \mathbf{A}\mathbf{x} + \boldsymbol{\varepsilon}' \tag{7.18}$$

The regression matrix $\mathbf{A}$ is given by:

$$\mathbf{A} = <\mathbf{y}\mathbf{x}^T><\mathbf{x}\mathbf{x}^T>^{-1} \tag{7.19}$$

where the bracket stands for the expectation operator, i.e. $< . >= E(.)$. The least-squares fitted value of the predictand is given by:

$$\hat{\mathbf{y}} = \mathbf{A}\mathbf{x} = <\mathbf{y}\mathbf{x}^T><\mathbf{x}\mathbf{x}^T>^{-1}\mathbf{x}. \tag{7.20}$$

Now, given a $p \times n$ data matrix $\mathbf{X}$ of predictors and a $q \times n$ matrix $\mathbf{Y}$ of predictand or dependent variables, then the least-squares regression matrix is givne by:

$$\mathbf{A} = \mathbf{Y}\mathbf{X}^T\left(\mathbf{X}\mathbf{X}^T\right)^{-1}. \tag{7.21}$$

Note that in (21) $\mathbf{X}$ and $\mathbf{Y}$ are assumed to be scaled by $\sqrt{n-1}$. In many instances one usually transforms one or both data sets using a linear transformation for various reasons purposes, such as reducing the number of variables, etc., and one would like to identify the regression matrix of the transformed variables. If, in model (18), $\mathbf{x}$ and $\mathbf{y}$ are replaced respectively by $\mathbf{x}' = \mathbf{L}\mathbf{x}$ and $\mathbf{y}' = \mathbf{M}\mathbf{y}$, where $\mathbf{L}$ and $\mathbf{M}$ are two matrices, then the model becomes:

$$\mathbf{y}' = \mathbf{A}'\mathbf{x}' + \boldsymbol{\varepsilon}'. \tag{7.22}$$

Using the data matrices $\mathbf{X}' = \mathbf{L}\mathbf{X}$ and $\mathbf{Y}' = \mathbf{M}\mathbf{Y}$ the new regression matrix is:

$$\mathbf{A}' = \mathbf{Y}'\mathbf{X}'^T\left(\mathbf{X}'\mathbf{X}'^T\right)^{-1} = \mathbf{M}\left(\mathbf{Y}\mathbf{X}^T\right)\mathbf{L}^T\left[\mathbf{L}\mathbf{X}\mathbf{X}^T\mathbf{L}^T\right]^{-1}. \tag{7.23}$$

In the particular case where $\mathbf{L}$ is invertible one gets:

$$\mathbf{A}' = \mathbf{M}\mathbf{A}\mathbf{L}^{-1} \tag{7.24}$$

Remarks

When $\mathbf{L}$ is invertible one gets a number of useful properties:

- The predicted value of the transformed variables is directly linked to the predicted value of the transformed variables as:

$$\hat{\mathbf{y}'} = \mathbf{A}'\mathbf{x}' = \mathbf{M}\hat{\mathbf{y}} \tag{7.25}$$

- We also get, via eqs (18) and (22), a relationship between the sum of squared errors:

$$\boldsymbol{\varepsilon'}^T\boldsymbol{\varepsilon'} = (\mathbf{y}' - \hat{\mathbf{y}'})^T(\mathbf{y}' - \hat{\mathbf{y}'}) = (\mathbf{y} - \hat{\mathbf{y}})^T\mathbf{M}\mathbf{M}^T(\mathbf{y} - \hat{\mathbf{y}}) \tag{7.26}$$

The last equality in (26) means, in particular, that both the squared error $(\mathbf{y} - \hat{\mathbf{y}})^T(\mathbf{y} - \hat{\mathbf{y}})$ and the positive semi-definite quadratic functions of the error $(\mathbf{y} - \hat{\mathbf{y}})^T\mathbf{M}\mathbf{M}^T(\mathbf{y} - \hat{\mathbf{y}})$ are minimized.

From the above remarks it can be seen, by choosing particular forms for the matrix $\mathbf{M}$, such as rows of the identity matrix, that the error covariance of each variable separately is also minimized. Consequently the full regression model also embeds the individual models for the predictand variables (Tippett et al. 2008). By using the SVD of the regression matrix $\mathbf{A}$ the regression coefficients can be interpreted in terms of correlation, explained variance, standardized expalined variance and covariance (Tippett et al. 2008). For example, by using a whitening transformation, i.e., using the scaled PCs of both variables, then (as in the unidimensional case) the regression matrix $\mathbf{A}'$ simply becomes the correlation matrix between the (scaled) predictand and predictors.

We now consider the SVD of the matrix $\mathbf{A}'$, i.e., $\mathbf{A}' = \mathbf{U}\mathbf{S}\mathbf{V}^T$, then $\mathbf{U}^T\mathbf{Y}'$ and $\mathbf{V}^T\mathbf{X}'$ decomposes the (pre-whitened) data into time series that are maximally correlated, and uncorrelated with subsequent ones. In terms of the original variables the weight vectors satisfy respectively $\mathbf{Q}_x^T\mathbf{X} = \mathbf{V}^T\mathbf{X}'$ and $\mathbf{Q}_y^T\mathbf{Y} = \mathbf{V}^T\mathbf{Y}'$ and are given by:

$$\mathbf{Q}_x = \left(\mathbf{X}\mathbf{X}^T\right)^{-1/2}\mathbf{V} \quad \text{and} \quad \mathbf{Q}_y = \left(\mathbf{Y}\mathbf{Y}^T\right)^{-1/2}\mathbf{U}. \tag{7.27}$$

Associated with these weight vectors are the pattern vectors

$$\mathbf{P}_x = \left(\mathbf{X}\mathbf{X}^T\right)^{1/2}\mathbf{V} \quad \text{and} \quad \mathbf{P}_y = \left(\mathbf{Y}\mathbf{Y}^T\right)^{1/2}\mathbf{U} \tag{7.28}$$

Remark

The pattern vectors are similar to EOFs (associated with PCs) and satisfy $\mathbf{P}_x\mathbf{Q}_x^T\mathbf{X} = \mathbf{X}$, and similarly for $\mathbf{Y}$. These equations are solved in a least square sense, see, e.g., Tippett et al. (2008). Note that the above condition leads to $\mathbf{P}_x^T\mathbf{Q}_x = \mathbf{I}$, and similarly for the predictand variables.

Hence the data are decomposed into patterns with maximally correlated time series and uncorrelated with subsequent predictor and predictand ones. The regression matrix is also decomposed as:

$$\mathbf{A} = \mathbf{P}_y\mathbf{S}\mathbf{Q}_x^T$$

and hence CCA diagonalises the regression (Tippett et al. 2008).

As for the univariate case, when only the predictor variables are pre-whitened the regression matrix[2] represents the explained variance of the corresponding (individual) regression. This is what RDA attempts to achieve. If in addition the predictand variables are scaled by the corresponding standard deviation, then the regression matrix is precisely what PPA is about.

When the predictor variables are scaled by the covariance matrix, i.e., $\mathbf{X}' = \left(\mathbf{X}\mathbf{X}^T\right)^{-1}\mathbf{X}$, then the regression matrix becomes:

$$\mathbf{A}' = \mathbf{Y}\mathbf{X}^T,$$

which represents the covariances between predictand and predictors, hence MCA. Tippett et al. (2008) applied different transformation (or filtering) to a statistical downscaling problem of precipitation over Brazil using a GCM. They found that CCA provided the best overall results based on correlation as a measure of skill.

Remark

MCA, CCA and RDA can be brought together in a unified-like approach through the scaled SVD (Swenson 2014). If $\mathbf{X} = \mathbf{U}_x\mathbf{S}_x\mathbf{V}_x^T$ is a SVD of the data matrix $\mathbf{X}$, then the data are scaled as

$$\mathbf{X}^* = \left(\mathbf{U}_x\mathbf{S}_x^{\alpha-1}\mathbf{V}_x^T\right)\mathbf{X} = \mathbf{U}_x\mathbf{S}_x^{\alpha}\mathbf{V}_x^T$$

and similarly for $\mathbf{Y}$. Scaling SVD is then obtained by applying the SVD

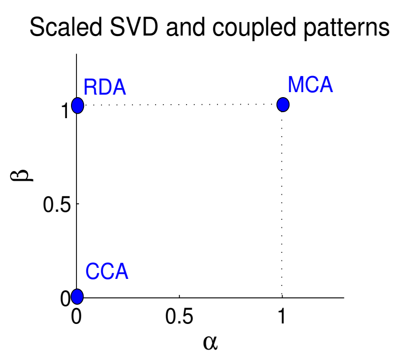

Figure 7.2: CCA, RDA and MCA within the $\alpha - \beta$ plane of scaled SVD

to the cross-covariance matrix $\mathbf{X}^*\mathbf{Y}^{*T}$. It can be seen that CCA, MCA and RDA can be recovered from scaled SVD by using respectively $\alpha = \beta = 1$, $\alpha = \beta = 0$, and $\alpha = 0$, $\beta = 1$ (see Fig. 7.2). Swenson (2014) points out that other intermediate values of $0 \leq \alpha, \beta \leq 1$ can isolate coupled signals better.

[2] in fact, the absolute value of the elements of the matrix

Chapter 8

Analysis of Variance - ANOVA

8.1 Background

Chapter 3 discusses statistical significance with particular focus on the comparison of two population means, based on two samples. The test statistic was based on the student t-distribution. What happens if we want to compare the means of more than two groups, or more than two normal distributions? the answer to this question is achieved by what is known as *analysis of variance* (ANOVA).

8.2 ANOVA setup

ANOVA can be placed in the general context of linear regression. It is, in fact, a special case of regression in which the explanatory (or independent) variables are categorical, but the response variable is continuous. ANOVA deals with continuous responses obtained from an experiment having one or more categorical explanatory variables. These are referred to as *factors*. If we have one factor, then we talk about *one-way ANOVA*, and if we have two (or more) factors, we talk about *two-way* (or multi-way) ANOVA. Each factor has more than one level, which can be interpreted as different "values" of the factor. For example, with one factor and two levels, we get the one-way ANOVA with two levels. This results in the two-sample t-test.

Examples

1) Sugar level of patients practising sport activity at different levels, e.g. once, twice or three times a week. This represents a case of one factor: sport activity, with three levels.
2) Blood pressure reaction of patients to different cocentrations of a given drug. Here again we deal with one factor: drug, with different levels (concentration).
3) Reaction (e.g., blood pressure) of males and females to different concentrations of a drug. Here we deal with two factors namely, gender, with two levels (male and female), and drug (with different concentration levels).

Given a set of measurements or observations from an experiment involving one or more factors, with different levels, the objective is to compare the (population) means of the different groups under consideration. This is achieved by analysing a number of variance-like quantities. In fact, ANOVA is based on splitting the total variability into different parts or sources as is presented below.

8.3 One-way ANOVA

8.3.1 Problem description and notation

In the theoretical layout of one-way ANOVA we assume that we have random samples from p different normal distributions with the same variance σ^2. We then seek to compare the means based on the observed values from the samples. For example, measuring the acidity of four types of orange constitutes a one-way example.

Specifically, let p designate the number of groups or levels/values of the explanatory variable or factor. For $i = 1, \dots p$, let n_i, $\overline{x}_i$, and s_i designate, respectively, the sample size, mean and standard deviation of the i'*th* group. So, if x_{ij} represents the j'*th* response from the i'*th* group/level, then

$$\begin{aligned} \overline{x}_i &= \tfrac{1}{n_i} \textstyle\sum_{j=1}^{n_i} x_{ij} \\ \overline{s}_i^2 &= \tfrac{1}{n_i - 1} \textstyle\sum_{j=1}^{n_i} (x_{ij} - \overline{x}_i)^2 \end{aligned} \tag{8.1}$$

The total sample size is $n = \sum_{i=1}^{p} n_i$, and the overall average is $\overline{x} = \frac{1}{n} \sum_{ij} x_{ij}$.

Exericse
Let SST designate the sum of squares total: $SST = \sum_{i=1}^{p} \sum_{j=1}^{n_i} (x_{ij} - \overline{x})^2$, show that $SST = \sum_{i=1}^{p} n_i (\overline{x}_i - \overline{x})^2 + \sum_{i=1}^{p} \sum_{j=1}^{n_i} (x_{ij} - \overline{x}_i)^2$
Hint – Consider for example only observations in the i'*th* group: $\sum_{j=1}^{n_i} (x_{ij} - \overline{x})^2 = \sum_{j=1}^{n_i} (x_{ij} - \overline{x}_i)^2 + n_i (\overline{x}_i - \overline{x})^2$, then take a sum over the index i.

The above exercise shows that the SST can be decomposed into the sum of two terms, the variability between sample or SSB:

$$SSB = \sum_{i=1}^{p} n_i (\overline{x}_i - \overline{x})^2, \tag{8.2}$$

and the variability within sample (or error) SSW (or SSE):

$$SSW = \sum_{i=1}^{p} \sum_{j=1}^{n_i} (x_{ij} - \overline{x}_i)^2. \tag{8.3}$$

It is clear that SST measures the total variation of the sample around the overall mean $\overline{x}$, whereas SSB measures the variation of the group means around $\overline{x}$, and SSW (or SSE) is the residual and measures the variation of observations around the group means, and we have (see exercise above):

$$SST = SSW + SSB.$$

Remark

> In some textbooks, e.g., DeGroot and Schervish (2002), SST, SSB and SSW (or SSE) are noted, respectively, as S^2_{TOT}, S^2_{Betw}, and S^2_{resid}.

Given the data, following the one-way layout, the next step is to investigate if different levels of the factor affect the observed data in the same way. Precisely, the comparison of population means μ_i, $i = 1, \dots p$, is achieved by testing the null hypothesis H_0 versus an alternative hypothesis H_1:

$$\begin{aligned} H_0 :\ \mu_i &= \mu \qquad \text{for all } i = 1, \dots p \\ H_1 :\ \mu_i &\neq \mu \qquad \text{for some } i = 1, \dots p \end{aligned} \tag{8.4}$$

To complete the above test, a number of assumptions are in place.

8.3.2 Assumptions and model formulation

Three main assumptions are used when applying ANOVA, namely:

(i) The data are assumed to be obtained independently and randomly.

(ii) Normality assumption within each group (or each factor level.)

(iii) The variance σ^2 is the same for all normal distributions (or all groups).

It follows from the above assumptions that $E(x_{ik}) = \mu_i$, and $var(x_{ik}) = \sigma^2$, for $k = 1, \dots n_i$, and $i = 1, \dots p$. The implied model is then:

$$x_{ik} = \mu_i + \varepsilon_{ik} \tag{8.5}$$

with $x_{ik} \sim N(\mu_i, \sigma^2)$, and $\varepsilon_{ik} \sim N(0, \sigma^2)$.

Note that (8.5) can also be written as $x_{ik} = \mu + \alpha_i + \varepsilon_{ik}$ and the hypothesis H_0 becomes $\alpha_i = 0$ for all $i = 1, \dots p$.

8.3.3 Testing hypotheses and ANOVA table

The above decomposition of the total sum of squares SST (or S^2_{TOT}) implies, in particular, that if for instance SSB is greater than SSW, then this would suggest that the population means are significantly different. The test is based essentially on this observation, i.e., a comparison between SSB and SSE.

Remark

> As was mentioned in the background section, the one-way ANOVA layout can be cast into the general linear model (multiple linear regression), in which the $n \times p$ design matrix contains only 0 and 1. The i'th $n_i \times p$ bloc matrix (submatrix) contains only ones in the i'th column, and zero elsewhere (Degroot and Schervish 2002).

Furthermore, for each $i = 1, \dots p$, we have $\sigma^{-2} \sum_{k=1}^{n_i} (x_{ik} - \overline{x}_i)^2 \sim \chi^2_{n_i - 1}$, and therefore:

$$\frac{1}{\sigma^2} SSW = \frac{1}{\sigma^2} S^2_{Resid} \sim \chi^2_{n-p}. \tag{8.6}$$

It can also be shown that SSW and SSB are independent. Now, under the null hypothesis H_0, we have $\frac{1}{\sigma^2}SSB \sim \chi^2_{p-1}$ (DeGroot and Schervish 2002). This can also be obtained from the dentity $SST = SSW + SSB$ plus the fact that $\sigma^{-2}SST \sim \chi^2_{n-1}$. Consequently, under the null hypothesis the ratio F of the between group sum of squares to the within sum of squares (error term) follows a F distribution, i.e.,

$$F = \frac{SSB/(p-1)}{SSW/(n-p)} = \frac{SSB/(p-1)}{SSE/(n-p)} \sim F_{p-1,n-p} \tag{8.7}$$

The null would then be rejected, at a significance level α, if the observed value F is larger than a critical value F_{cr}, i.e., $F > F_{cr} = F^{-1}_{p-1,n-p}(1-\alpha)$, where $F^{-1}_{p-1,n-p}$ is the quantile function of $F_{p-1,n-p}$ (Fig. 8.1). The above calculations are normally summarized in a table,

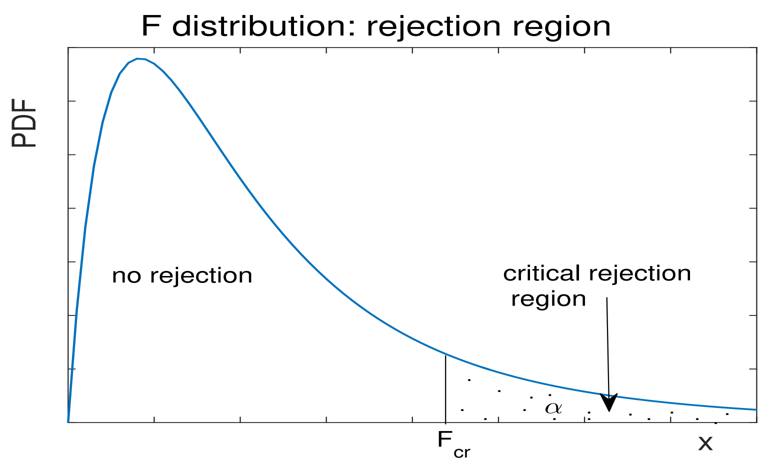

Figure 8.1: Rejection region for a F distribution.

referred to as the ANOVA table as shown in Table 8.1

Table 8.1: One-way ANOVA table

Source of variation	Sum of squares	Degrees of freedom	Mean square	F-ratio
Between sample	SSB	p-1	$MSSB = \frac{SSB}{p-1}$	$\frac{MSSB}{MSSE}$
Within sample	SSW (or SSE)	n-p	$MSSW = \frac{SSW}{n-p}$	
Total	SST	n-1		

Example

The following table (Table 8.2) shows the acidity, measured in mg/ml, of 4 types of oranges, namely (i) naval orange, (ii) blood orange, (iii) mandarin, and (iv) clementine.

Here we have one factor (orange) with 4 levels with $p = 4$, $n_1 = 3$, $n_2 = 4$, $n_3 = 4$, $n_4 = 3$, and the total sample size is $n = 14$. The means are $\overline{x}_1 = 9.6$, $\overline{x}_2 = 10.6$, $\overline{x}_3 = 8.9$, $\overline{x}_4 = 10.3$, and $\overline{x} = 9.8$. We want to test the null hypothesis H_0: the means are identical

Table 8.2: Orange acidity

Naval	blood	mandarin	clementine
8.9	10	9.6	10.2
10	12	8.9	11
9.8	9.5	9	9.6
	11	8	

Table 8.3: ANOVA table for orange acidity data

Source	SS	d.o.f.	mean square	F-ratio
Between samples	7	3	2.3	3.3
Within samples	6.7	10	0.7	
Total	13.6	13		

against the alternative H_1: there is difference, at a significance level $\alpha = 5\%$. The corresponding ANOVA table is given in Table 8.3

The critical value can be found using Matlab:

$>> n_1 = 3; n_2 = 10;$
$>> Fcr = Finv(1 - 0.05, 3, 10)$

The result is $F_{cr} = 3.71 > 3.3$. Therefore, there is no evidence to reject the null hypothesis.

8.4 Two-way ANOVA

In the two-way ANOVA one deals with the effects of two factors, e.g., A and B, each with a number of levels, say n_A and n_B. And the objective is to investigate whether there is an effect of each factor. For example, the effect of (1) age as one factor, with say 3 levels (young, adult, and old) and (2) fat consumption as the second factor, with say 2 levels (high and low) on the cholesterol level in the blood.

The data now can be presented in the form of a table with two entries; rows (or factor A) and columns (or factor B) with their respective levels as shown in Table 8.4. The table has $n_A \times n_B$ entries, and at the i'th row, $i = 1, \ldots n_A$, and j'th column, $j = 1, \ldots n_B$, the observation is x_{ij} when there are no replicates. If there are replicates then we will have more than one observation at each entry of the table, i.e. x_{ijk}, with k being the index of the replicates. It is also possible to assess the interaction between the two factors when more than one measurement is available at each combination of factors' levels, i.e. $k > 1$. When there is significant interaction between the two factors the model is referred to as interaction model, otherwise it is additive.

Table 8.4: Structure of a two-way ANOVA table with several levels at each factor, with no replicates, i.e. one observation in each cell.

	Factor B			
Factor A	1	2	$\dots$	n_B
1	x_{11}	x_{12}	$\dots$	x_{1n_B}
2	x_{21}	x_{22}	$\dots$	x_{2n_B}
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
n_A	$x_{n_A 1}$	$x_{n_A 2}$	$\dots$	$x_{n_A n_B}$

Table 8.5: Two-way ANOVA table

Source	SS	d.o.f.	MS	F-ratio
factor A	SSA	$n_A - 1$	MSSA	MSSA/MSSE
factor B	SSB	$n_B - 1$	MSSB	MSSB/MSSE
A×B	SSAB	$(n_A - 1)(n_B - 1)$	MSSAB	MSSAB/MSSE
Error	SSE	$n - n_A n_B$	MSSE	
Total	SST	n-1		

The ssumptions in two-way ANOVA are similar to those of one-way. One also assumes that a given factor has same effect at all levels of the other factor. Let x_{ijk} designate the k'*th* observation in the (i,j)'*th* cell, the model of the two-way ANOVA is:

$$x_{ijk} = \mu_{ij} + \varepsilon_{ijk} \tag{8.8}$$

with $\varepsilon_{ijk} \sim N(0, \sigma^2)$, for $i = 1, \dots n_A$, $j = 1, \dots n_B$, and $k = 1, \dots n_{ij}$, where n_{ij} is the number of observations at the (i,j)'*th* cell. The total number of observations is $n = \sum_{i=1}^{n_A} \sum_{j=1}^{n_B} n_{ij}$. Note that in Eq. (8.8) the mean μ_{ij} can be decomposed as: $\mu_{ij} = \mu + \alpha_i + \beta_j + \gamma_{ij}$, where α_i and β_j represent the additive main effect from level i of the first factor and level j of the second factor, and γ_{ij} is the interaction effect.

The results of the two-way ANOVA can also be summarized in a two-way ANOVA table shown in Table 8.5. It is similar to the one-way ANOVA table with two more lines, one for the second factor and the other for the interaction (for more than one replicates). The F-ratios are F-distributed, for example $\frac{MSSA}{MSSE} \sim F_{n_A - 1, n - n_A n_B}$, and the test statistics are then based on comparing the F-ratios in the last column of the above ANOVA table to the corresponding critical values, and decide on the effect of each factor or their interaction.

In the above table, we have the decomposition of the total sum of squares, $SST = SSA + SSB + SSAB + SSE$. Let use denote $n = \sum_{i=1}^{n_A} \sum_{j=1}^{n_B} n_{ij}$, $n_{i\bullet} = \sum_{j=1}^{n_B} n_{ij}$, $n_{\bullet j} = \sum_{i=1}^{n_A} n_{ij}$. We also denote:

$\overline{x}_{\bullet\bullet\bullet} = \frac{1}{n} \sum_{i=1}^{n_A} \sum_{j=1}^{n_B} \sum_{k=1}^{n_{ij}} x_{ijk}$,
$\overline{x}_{i\bullet\bullet} = \frac{1}{n_{i\bullet}} \sum_{j=1}^{n_B} \sum_{k=1}^{n_{ij}} x_{ijk}$,

Table 8.6: Example of the structure of a two-way ANOVA table with more than one replicate (3 here). The two factors here are grains and grass hay

Factors		Grains	
Grass hay	No grains	once/day	twice/day
once/day	x_{111}	x_{121}	x_{131}
	x_{112}	x_{122}	x_{132}
	x_{113}	x_{123}	x_{133}
twice/day	x_{211}	x_{221}	x_{231}
	x_{212}	x_{222}	x_{232}
	x_{213}	x_{223}	x_{233}

$\overline{x}_{\bullet j \bullet} = \frac{1}{n_{\bullet j}} \sum_{i=1}^{n_A} \sum_{k=1}^{n_{ij}} x_{ijk}$,
$\overline{x}_{ij\bullet} = \frac{1}{n_{ij}} \sum_{k=1}^{n_{ij}} x_{ijk}$.

Note that if all the cells have the same number of observations (or replicates), r, then $n_{ij} = r$ for all $i = 1, \dots n_A$, $j = 1, \dots n_B$, and then $n_{i\bullet} = rn_B$, $n_{\bullet j} = rn_A$, and $n = rn_A n_B$. The different sum of squares are then given by:

$$
\begin{aligned}
SSA &= \textstyle\sum_{i=1}^{n_A} n_{i\bullet}(\overline{x}_{i\bullet\bullet} - \overline{x}_{\bullet\bullet\bullet})^2 \\
SSB &= \textstyle\sum_{j=1}^{n_B} n_{\bullet j}(\overline{x}_{\bullet j\bullet} - \overline{x}_{\bullet\bullet\bullet})^2 \\
SSAB &= \textstyle\sum_{i=1}^{n_A} \sum_{j=1}^{n_B} n_{ij}(\overline{x}_{ij\bullet} - \overline{x}_{i\bullet\bullet} - \overline{x}_{\bullet j\bullet} + \overline{x}_{\bullet\bullet\bullet})^2 \\
SSE &= \textstyle\sum_{i=1}^{n_A} \sum_{j=1}^{n_B} \sum_{k=1}^{n_{ij}} (x_{ijk} - \overline{x}_{ij\bullet})^2 \\
SST &= \textstyle\sum_{i=1}^{n_A} \sum_{j=1}^{n_B} \sum_{k=1}^{n_{ij}} (x_{ijk} - \overline{x}_{\bullet\bullet\bullet})^2.
\end{aligned}
$$

In this model, one would be interested, for example, in analysing the effect of the two factors and check whether these effects are additive. We can rewrite Eq (8.8) as

$$\mu_{ij} = \mu + \alpha_i + \beta_j + \gamma_{ij} \tag{8.9}$$

for $i = 1, \dots n_A$, $j = 1, \dots n_B$, with $\sum \alpha_i = \sum \beta_j = 0$, and $\sum_i \gamma_{ij} = \sum_j \gamma_{ij} = 0$. The α_i and β_j, $i = 1, \dots n_A$, and $j = 1, \dots n_B$ are the main effects of factors A and B respectively. To test the additivity of factors A and B, we use the hypotheses:

$$
\begin{aligned}
&H_0 : \gamma_{ij} = 0, \; i = 1, \dots n_A, j = 1, \dots n_B, \\
&H_1 : \gamma_{ij} \neq 0, \; \text{for at least one pair} (i_0, j_0).
\end{aligned}
$$

The hypothesis H_0 would then be rejected at significance level α if MSSAB/MSSE $> F^{-1}_{(n_A-1)(n_B-1), n-n_A n_B}(1-\alpha)$ in case of no replicates ($n_{ij} = 1$). When there are m observations in the cells, i.e. $n_{ij} = m$, then the number of d.o.f of SSE becomes $n_A n_B(m-1)$.

If we reject H_0 this means there is no significant interaction between the two factors, and that the effect is additive. If in addition factor A, for example, is not significant, that is the p-value corresponding to $MSSA/MSSE$ is large, then this means that factor B is the only factor that has significant effect on the measured variable under investigation.

For example, let us assume we are interested in studying the growth rate of farm animals when they are fed with two different kinds of food: (1) grains and (2) grass hay (see Table 8.6). The farmer can pick up say 18 calves at random (divided into 6 groups of three), and feed them with grass hay either twice or three times a day (i.e., two levels) and grains either twice/day, three times/day or none (i.e., three levels). The farmer then measures their weight after say two months. Depending on the results of the ANOVA table we can get different conclusions. If, for example, the interaction effect and the grain factor are not significant then we can conclude that only grass hay is the only significant effect on growth rate of calves.

Chapter 9

Introduction to Machine Learning

9.1 Background

Learning is the process of improving, at some task, with experience. The performance is normally evaluated using some measure, which can represent the magnitude of the fitting error. It is also labelled sometimes energy, costfunction or loss function.

Simple example

As an illustrative example, imagine I provide you with a few pictures: goat (G), rabbit (R) and a kangarou (K). Next, I give you a picture of a wallaby, and ask you to put it in one of these three groups. What is your answer? Then I give you a picture of a bird and ask you the same question. What will be your answer?

The above is a simple example of classification problem. Of course this example does not need computer power. However, if I give you thousands of pictures, then the problem becomes intractable visually and can only be solved using computers. Data mining is a term used when dealing with high-dimensional data and refers to digging deep into the data in order to gain patterns and identify hidden relationships. This yields knowledge, which can be used in prediction and gaining wisdom.

Unlike human learning, which involves memory and consciousness (i.e. knowledge), machine learning (ML) is mostly linked to performance. ML, a branch of artificial intelligence, refers to procedure of using computers to 'learn' from a given large data but with no explicit human intervention of programming. The term is believed to originate from the work of Turing (1950), see e.g., Hsieh (2023). Expressed in a simple wording, ML is a way of programming by example (Lindholm et al. 2023). A related definition is provided by Wikipedia, namely that ML is about the development and use of algorithms that can learn from data and extends to unseen data and performs tasks without explicit instruction. ML is divided mainly into two or three categories:
(1) supervised learning (regression, time series prediction, classification)
(2) unsupervised learning (clustering, dimension reduction, PDF estimation)
(3) reinforcement learning, which involves training software to make decisions in an optimal

way (gaming, robotics, traffic control, etc).

In supervised learning, one relies on the existence of a test set, or training sample, for which we know the input and output, also referred to as tagged/labelled example.

Example of regression problem

Assume we have precipitation data R_t along with temperature T_t and pressure P_t data for $t = 1, \ldots N$. We also assume that we are given T_{N+1} and P_{N+1}, and we like to estimate R_{N+1}. One simple way is to fit a simple model linking R_t to T_t and P_t, such as a linear regression model (see chapter 5):

$$R_t = \alpha P_t + \beta T_t + \varepsilon_t. \tag{9.1}$$

The training sample, corresponding to $t = 1, \ldots N$, is used to obtain estimates of the regression parameters α, β and the noise variance. The fitted values are given by $\hat{R}_i = \alpha P_i + \beta T_i$, $i = 1, \ldots N$, and the predicted value of R_{N+1} is $\hat{R}_{N+1} = \alpha P_{N+1} + \beta T_{N+1}$.

A typical example of supervised learning is classification. It is the process of finding a model that identifies the 'class' or 'category' of a new observation, given that we know the categories from the training sample.

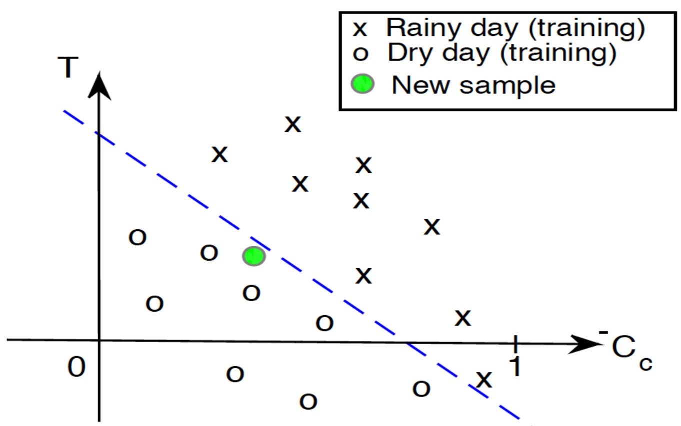

Figure 9.1: A schematic example of rainy and dry days in the temperature-cloud cover plane. The dashed line refers to a separating surface based on the crosses and empty circles (training sample) and the filled circle represents a new observation.

Examples of typical classification problems

1. *Rainy and dry days* – Given a set of cloud cover C_c (between 0 and 1) and surface temperature T_s sample for which we know whether it is rainy or dry (Fig 9.1). Next we get a new observation of cloud cover and surface temperature, and the task is to categorise it (Fig 9.1).

2. *Weights and heights of males and females* – We are given a training set of weights (W) and heights (H) of a group of males and females. A new measure of W and H was then obtained, and set out to identify the gender of this new observation (Fig. 9.2).

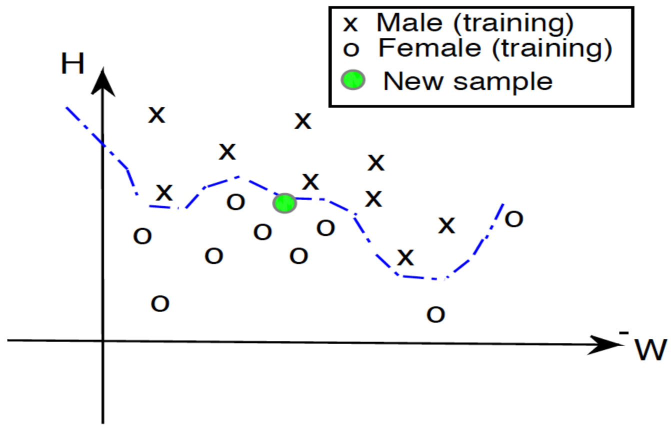

Figure 9.2: As in figure 9.1, but for the weight and height of males and females. This an illustrative case of a nonlinear separating surface.

Supervised learning is based on model building and training. Model building involves learning, which yields the model construction. The model is then used to predict the class label and also testing the model. In model training the data are divided into a training (or test) set and a validation set. The former is used to train the model and the latter is used for model performance. The term data mining mentioned previously contains two main branches, (i) predictive, and (ii) descriptive (Fig 9.3). In summary the classification steps

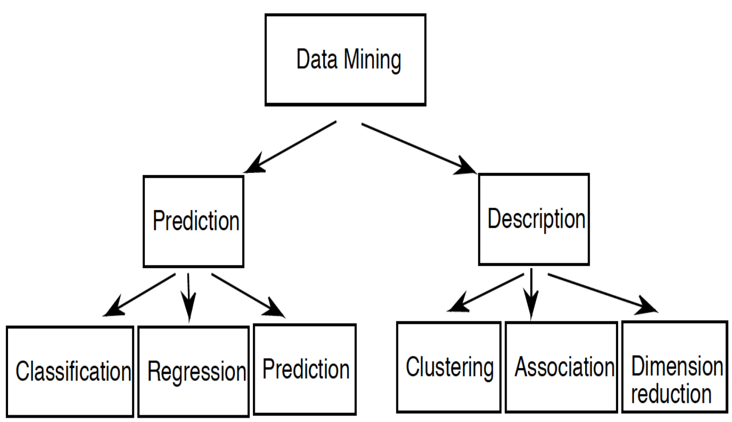

Figure 9.3: Two main branches of data mining: prediction (supervised learning) and description (unsupervised learning).

are, (1) data collection and processing, (2) feature selection (e.g. picking the most relevant attributes), and model selection, training and evaluation.

Remark

> The types od data used in ML can be binary and nominal (ordinal, continuous or discrete).

There are different types of classifiers including logistic regression and decision trees discussed later in the chapter, and more advanced methods, namely neural networks and support vector machines discussed in the following chapters. ML is applied in various branches of science, e.g., market analysis, IT, weather forecasting, patterns recognition such as security, education.

Before exploring more advanced ML methods, we are going to discuss the simple case of linear classification.

9.2 Linear discriminant analysis-LDA

9.2.1 Gaussian LDA

Let us discuss the example of males/females weight and height assuming linearly separating surface (Fig. 9.4). We normally assume that our data are centered. Here we have two classes

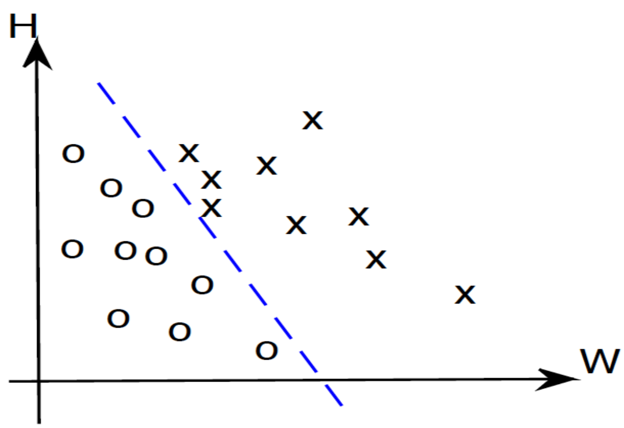

Figure 9.4: Illustration of the case of a linearly separating surface between the two classes of males and females weight/height.

that we label "0" and "1" with respective means $\boldsymbol{\mu}_0$ and $\boldsymbol{\mu}_1$ and covariances $\boldsymbol{\Sigma}_0$ and $\boldsymbol{\Sigma}_1$:
$->$ *The problem* - identify a good prediction for the class $Y = \{0, 1\}$ of a new observation $\mathbf{x}$.
$->$ *Assumption* - we assume here that the probability distribution of $\mathbf{x}$, given the class, is normal, i.e., $Pr(\mathbf{x}|Y = 0) \sim N(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0$ and $Pr(\mathbf{x}|Y = 1) \sim N(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1)$. Using the log-likelihood, $\mathbf{x} \in \{Y = 1\}$ if

$$(\mathbf{x}-\boldsymbol{\mu}_0)^T\boldsymbol{\Sigma}_0^{-1}(\mathbf{x}-\boldsymbol{\mu}_0) - \frac{1}{2}\log|\boldsymbol{\Sigma}_0| - (\mathbf{x}-\boldsymbol{\mu}_1)^T\boldsymbol{\Sigma}_1^{-1}(\mathbf{x}-\boldsymbol{\mu}_1) + \frac{1}{2}\log|\boldsymbol{\Sigma}_1| > \tau. \tag{9.2}$$

where τ is some threshold.

The assumption in LDA is that the covariance matrices are identical, $\boldsymbol{\Sigma}_0 = \boldsymbol{\Sigma}_1 = \boldsymbol{\Sigma}$, and full rank. The log-likelihood yields, with $\tau = 0$, the condition:

$$2(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)^T\boldsymbol{\Sigma}^{-1}\mathbf{x} > (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)^T\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 + \boldsymbol{\mu}_0). \tag{9.3}$$

Figure 9.5 shows the interpretation of Eq (3). The dashed line shown in the figure provides a separating hyperplane between both classes. This hyperplane is orthogonal to $\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)$.

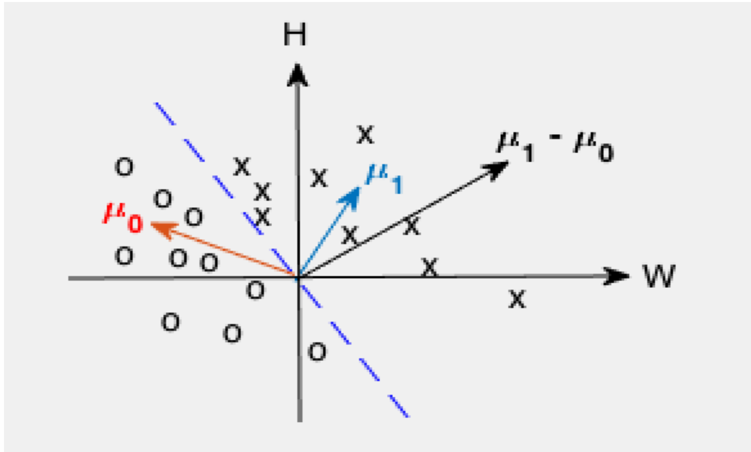

Figure 9.5: As in figure 9.4 (with centred data) showing the means of the two groups $\boldsymbol{\mu}_0$ and $\boldsymbol{\mu}_1$ and the separating surface.

Remark

In the basic LDA the posterior probability of an observation $\mathbf{x}$ is Gaussian, i.e.

$$Pr(\mathbf{x}|S_i) = \frac{1}{(2\pi)^{\frac{p}{2}} |\mathbf{S}_i|^{\frac{1}{2}}} \exp\left[-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu}_i)^T \mathbf{S}_i^{-1}(\mathbf{x}-\boldsymbol{\mu}_i)\right] \tag{9.4}$$

and the assumption is $\mathbf{S}_i = \mathbf{S}$, $i = 1, \ldots K$, for K classes. So an observation $\mathbf{x}$ is in class S_i if the odds satisfy:

$$\frac{Pr(S_i|\mathbf{x})}{Pr(S_j|\mathbf{x})} > 1, \qquad \text{for } j \neq i \tag{9.5}$$

or that $log(odds) > 0$, that is

$$log(\frac{p_i}{p_j}) - \frac{1}{2}\boldsymbol{\mu}_i^T \mathbf{S}^{-1}\boldsymbol{\mu}_i + \frac{1}{2}\boldsymbol{\mu}_j^T \mathbf{S}^{-1}\boldsymbol{\mu}_j + \mathbf{x}^T \mathbf{S}^{-1}(\boldsymbol{\mu}_i - \boldsymbol{\mu}_j) > 0 \tag{9.6}$$

and that the hyperplane

$$log(\frac{p_i}{p_j}) - \frac{1}{2}\boldsymbol{\mu}_i^T \mathbf{S}^{-1}\boldsymbol{\mu}_i + \frac{1}{2}\boldsymbol{\mu}_j^T \mathbf{S}^{-1}\boldsymbol{\mu}_j + \mathbf{x}^T \mathbf{S}^{-1}(\boldsymbol{\mu}_i - \boldsymbol{\mu}_j) = 0, \tag{9.7}$$

is the decision boundary (p_i is the prior probability of class S_i). If the assumption $\mathbf{S}_i = \mathbf{S}$, $i = 1 \ldots K$, is dropped then one gets the quadratic discriminant analysis (Hsieh 2023; Wilks 2011). Note that Eq (6) can be split to yield $y_i(\mathbf{x}) > y_j(\mathbf{x})$, where $y_i(\mathbf{x})$ is a linear discriminant function.

9.2.2 Fisher LDA

In Fisher LDA no assumption, such as Gaussianity, is used. Let $\mathbf{a}$ be a vector and $\mathbf{x}$ an observation, and consider the quantity $\mathbf{a}^T\mathbf{x}$. Given the data sample the quantity $\mathbf{a}^T\mathbf{x}$ yields two one-dimensional distributions. Fisher considered the ratio of the between-groups variance to the within-group variance:

$$\rho = \frac{(\mathbf{a}^T\boldsymbol{\mu}_1 - \mathbf{a}^T\boldsymbol{\mu}_0)^2}{\mathbf{a}^T\boldsymbol{\Sigma}_1\mathbf{a} + \mathbf{a}^T\boldsymbol{\Sigma}_0\mathbf{a}} \tag{9.8}$$

Maximizing Eq (8) with respect to $\mathbf{a}$ yields $\mathbf{a} = \alpha(\boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_0)^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)$.

Exercise

Derive the expression of $\mathbf{a}$ maximizing Eq (8).

The vector $\mathbf{a}$ is orthogonal to the hyperplane separating both classes. And the problem can be transformed into a one-dimensional problem by projecting the data onto the vector $\mathbf{a}$, and the separation is made based on the obtained one-dimensional distributions.

The above expression can be written using the between-class and within-class scatter matrices $\mathbf{S}_B = (\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)^T$ and $\mathbf{S}_W = \mathbf{S}_1 + \mathbf{S}_2$, respectively, yielding the maximization of the Raleigh quotient R:

$$\mathbf{u}^* = \underset{\mathbf{u}}{\text{argmax}} R = \frac{\mathbf{u}^T \mathbf{S}_B \mathbf{u}}{\mathbf{u}^T \mathbf{S}_W \mathbf{u}}. \tag{9.9}$$

The solution is given by the generalized eigenvalue problem:

$$\mathbf{S}_B \mathbf{u} = \lambda \mathbf{S}_W \mathbf{u} \tag{9.10}$$

or similarly:

$$\mathbf{S}_W^{-1} \mathbf{S}_B \mathbf{u} = \lambda \mathbf{u} \tag{9.11}$$

where $\mathbf{S}_W^{-1}$ is the inverse of $\mathbf{S}_W$ if it exists or its generalized inverse if it is singular. The leading eigenvector $\mathbf{u}_1$ associated with the largest eigenvalue λ_1 is used for the discriminant analysis. This is the known *Fisher discriminant analysis*. The data are then projected subsequently onto $\mathbf{u}_1$ and classification is obtained. A data point $\mathbf{x}$ is assigned to class C_1 if it is closer to $\boldsymbol{\mu}_1$ than to $\boldsymbol{\mu}_2$, i.e.,

$$|\mathbf{u}_1^T(\mathbf{x} - \boldsymbol{\mu}_1)| \leq |\mathbf{u}_1^T(\mathbf{x} - \boldsymbol{\mu}_2)| \tag{9.12}$$

This can be generalized to more than the leading eigenvector, see, e.g. Hsieh (2023) and Wilks (2011).

Remark

> Since $\mathbf{S}_B = (\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)^T$ the leading eigenvector is given by $\mathbf{u}_1 = \mathbf{S}_W^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$

Exercise

Derive the expression of $\mathbf{u}_1$ in the above remark.

Hint. Just notice that $R = \frac{(\mathbf{u}^T(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2))^2}{\mathbf{u}^T \mathbf{S}_W \mathbf{u}}$ then use $\nabla_{\mathbf{u}} R = \mathbf{0}$ to get $\mathbf{S}_W \mathbf{u}_1 = \alpha(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$. Alternatively, simply expand $\mathbf{S}_W^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^T \left(\mathbf{S}_W^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)\right)$, which is proportional to $\mathbf{S}_W^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$.

Eq. (12) can be rewritten using the mean $\boldsymbol{\mu} = 0.5\,(\boldsymbol{\mu}_1 + \boldsymbol{\mu}_2)$ (Wilks 2011), so that $\mathbf{x}$ is in group 1 if $\mathbf{u}_1^T \boldsymbol{\mu} \leq \mathbf{u}_1^T \mathbf{x}$, and in group 2 otherwise.

Exercise

Show that the total variance can be decomposed into a sum of the between and within group variance. Refer to section chapter 8 (section 3) for the derivation.

Fisher discriminant analysis can be extended to multiclass (K classes, $K > 2$) case. This can be achieved by using the pooled within-groups covariance matrix and the (overall) between-groups covariance matrix, and finding the eigenvalues/eigenvectors as in Eq (11). Consequently, data point $\mathbf{x}$ is then assigned to group S_i if (Wilks 2011):

$$\sum_{k=1}^{K}\left(\mathbf{u}_k^T(\mathbf{x}-\boldsymbol{\mu}_i)\right)^2 \leq \sum_{k=1}^{K}\left(\mathbf{u}_k^T(\mathbf{x}-\boldsymbol{\mu}_j)\right)^2, \text{ for all } j \neq i \tag{9.13}$$

9.2.3 Multiple LDA

The above derivation in the Gaussian LDA is based on a 2-class case. What about multiple classes? When there are $c > 2$ classes the procedure is different from the above, and various methods can be applied. The following two methods, mentioned above, are most used in practice, namely;
(1) pairwise classification using all $c(c-1)/2$ classifiers, and
(2) one against the rest, i.e., selecting one class on its own and the remaining in another class, and then doing the same for the latter, etc.

Remark

> The last method, and to some extent also the former method yields to some kind of ambiguity, where some regions in the input space will be ambiguously classified.

In the pairwise classification, with $c(c-1)/2$ binary discriminant functions, each point is classified to its class based on the majority of votes amongst the different classifiers. This also will lead to an ambiguously classified region. These ambiguities can be overcome by considering one single c-class classifier, and then decide on the correct class based on the value of the discriminant function (Bishop 2006).

Remark

> It is useful to mention that in addition to classification, LDA can be used as a dimension reduction method.

9.3 Classification and logistic regression

Logistic regression was introduced in chapter 11 as an example of generalized linear model. Logistic (or logit) regression can also be used for classification for both; the two- and multi-class cases. For the binary case with two outcomes (Yes/No, 0/1, -1/1, etc.) the outcome is expressed using the logistic function $g(z) = (1 + e^{-z})^{-1}$ having values in $[0, 1]$, that is

$$Pr(Y = y|\mathbf{x}; \theta) = \left[g(h_{\boldsymbol{\theta}}(\mathbf{x}))\right]^y \left[1 - g(h_{\boldsymbol{\theta}}(\mathbf{x}))\right]^{1-y} \tag{9.14}$$

with $Y = \{0, 1\}$, and $h_{\boldsymbol{\theta}}()$ is the predictor function parametrized by $\boldsymbol{\theta}$ (e.g., $h_{\boldsymbol{\theta}}(\mathbf{x}) = \boldsymbol{\theta}^T\mathbf{x}$). The parameter(s) can be obtained through maximum likelihood estimates (MLE) or similarly:

$$\hat{\boldsymbol{\theta}} = \underset{\boldsymbol{\theta}}{\operatorname{argmin}} \left[-l(\boldsymbol{\theta}) = -log(L(\boldsymbol{\theta}))\right] \tag{9.15}$$

where $L()$ is the likelihood function

$$L(\boldsymbol{\theta}) = \prod_{t=1}^{n} \left[g(h_{\boldsymbol{\theta}}(\mathbf{x}_t))\right]^{y_t} \left[1 - g(h_{\boldsymbol{\theta}}(\mathbf{x}_t))\right]^{1-y_t}$$

Using the fact that $\frac{d}{dz}g(z) = g(z)(1 - g(z))$, the gradient of the log-likelihood $l()$ yields:

$$\nabla l(\boldsymbol{\theta}) = \sum_{t=1}^{n} \left(y_t - g(h_{\boldsymbol{\theta}}(\mathbf{x}_t))\right) \nabla_{\boldsymbol{\theta}} h_{\boldsymbol{\theta}}(\mathbf{x}_t) \tag{9.16}$$

where y_t is the observed outcome corresponding to $\mathbf{x}_t$, $t = 1, \ldots n$. A gradient descent type algorithm can then be used to learn the parameter $\boldsymbol{\theta}$. The predicted class of a future observation $\mathbf{x}_0$ will be 0 if $g(h_{\boldsymbol{\theta}}(\mathbf{x}_0)) < 0.5$ and 1 otherwise.

For the multiclass case, the multinomial logistic (or softmax) regression can be used. With K classes one gets:

$$g(h_{\boldsymbol{\theta}}(\mathbf{x})) = \left(Pr(Y = 1|\mathbf{x}; \boldsymbol{\theta}), \ldots, Pr(Y = K|\mathbf{x}; \boldsymbol{\theta})\right)^T = \frac{1}{\sum_{k=1}^{K} \exp(\boldsymbol{\theta}_k^T \mathbf{x})} \left(e^{\boldsymbol{\theta}_1^T \mathbf{x}}, \ldots, e^{\boldsymbol{\theta}_K^T \mathbf{x}}\right)^T \tag{9.17}$$

If $h_{\boldsymbol{\theta}}(\mathbf{x})$ is a linear function of $\mathbf{x}$ (which is generally the case), with parameters $(\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_K)$, a $m \times K$ matrix is obtained, where m is the dimension of the explanatory variable $\mathbf{x}$. The costfunction to be minimized is:

$$-l(\boldsymbol{\theta}) = -\sum_{t=1}^{n} \sum_{k=1}^{K} 1_{Y_t = k} log\left(\frac{e^{\boldsymbol{\theta}_k^T \mathbf{x}_t}}{\sum_{i=1}^{K} e^{\boldsymbol{\theta}_i^T \mathbf{x}_t}}\right), \tag{9.18}$$

where $1_X()$ is the indicator function taking 1 if X is true and 0 otherwise. The assignment of a future observation $\mathbf{x}_0$ will be the class having the largest value of $e^{\boldsymbol{\theta}_k^T \mathbf{x}_0}$, $k = 1, \ldots K$.

9.4 Random Forest

In machine learning, supervised learning refers to applications in which the training set is composed of pairs of input data and their corresponding target output. Classification in which the target (or classes) are composed of finite number of discrete categories, is a good example in which the classes of the training data input are known. When the target data are composed of continuous variables one gets regression.

Random forest (RF) is a supervised learning algorithm for classification and regression problems (Breiman 2001), based on what is known as *decision trees*, which are briefly described below.

9.4.1 Decision trees

What are they?

Decision trees are the building blocks of random forests. They aim, based on a training set, at predicting the output of any data from the input space. A decision tree is based on a sequence of binary selections, and looks like a (reversed) tree (Bishop 2006). Precisely, the input sample is (sequentially) split into two or more homogeneous sets based on the main features of the input variables. The following simple example illustrates the basic concept. Consider the set $\{-5, -3, 1, 2, 6\}$, which is to be separated using the main features (or variables), namely sign (+/-) and parity (even/odd). Starting with the sign the binary splitting yields the two sets $\{1, 2, 6\}$ and $\{-3, -5\}$. The last set is homogeneous, i.e. a class of negative odds. The other set is not, and we use the parity feature to yield $\{1\}$ and $\{2, 6\}$. This can be summarized by a decision tree shown in Figure 9.6. Each decision tree is composed of:

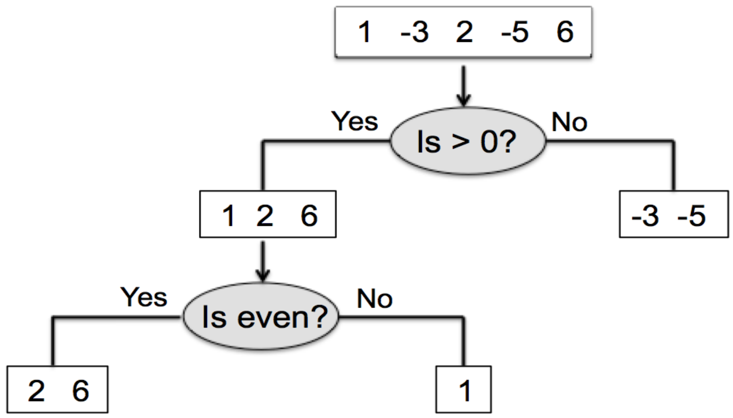

Figure 9.6: Decision tree of the simple dataset shown in the top box.

- Root node – containing the entire sample
- Splitting node
- Decision node – a subnode that splits into further subnodes
- Terminal node or leaf – a node that cannot split further
- Branch – a subtree of the whole tree
- Parent and child nodes.

In the above illustrative example, the root node is the whole sample. The set $\{1, 2, 6\}$, for example, is a decision node whereas $\{2, 6\}$ is a terminal node (or leaf). Also, the last subset, i.e. $\{2, 6\}$, is a child node of the parent node $\{1, 2, 6\}$. The splitting rule in the above example is quite simple, but for real problems more criteria are used depending on the type of problem at hand, namely regression or classification, which are discussed next.

Classification and regression trees

As is mentioned above, decision trees attempt to partition the input (or predictor) space using a sequence of binary splitting. This splitting is chosen to optimize a splitting criterion, which depends on the nature of the predictor, e.g. discrete/categorical versus continuous, and the type of problem at hand, e.g., classification versus regression. In the remaining part of this chapter we assume that our training set is composed of n observations $(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_n, y_n)$, where, for each k, $k = 1, \ldots n$, $\mathbf{x}_k = (x_{k1}, \ldots, x_{kd})^T$ and contains the d variables (or features), and y_k is the response variable. The splitting proceeds recursively with each unsplit node by looking for the 'best' binary split.

Case of categorical predictors

We discuss here the case of categorical predictors. A given unsplit (parent) node, containing a subset $\mathbf{x}_1, \ldots, \mathbf{x}_m$ of the training set, is split into two child nodes: a left (L) node and a right (R) node. For regression, a common criterion F is the mean square residual, or variance of the response variable. For this sub-sample in the parent node the costfunction (variance) is:

$$F = \frac{1}{m} \sum_{i=1}^{m} (y_i - \overline{y})^2 \tag{9.19}$$

where $\overline{y}$ is the average response of this subset. Now, if F_L and F_R are the corresponding variance for the (left and right) child nodes, then the best split is based on the variable (or feature) yielding maximum variance reduction, i.e. $max|F - F_{LR}|$, or similarly minimizing F_{LR}, i.e. $\min(F_{LR})$, where $F_{LR} = m^{-1}(m_L F_L + m_R F_R)$, with m_L and m_R being the sample sizes of the left and right (child) nodes respectively. Note again that to obtain the best result, every possible split based on every variable (among the d variables) is considered.

The splitting criterion for classification, with say K classes, is different. Let p_k designate the proportion of the k'*th* class in a given (unsplit) node, i.e. $p_k = m^{-1} \sum_{i=1}^{m} 1_{\{y_i = k\}}$, where 1_A is the indicator of set A, i.e. 1 if A is true and 0 otherwise. Then the most commonly used criterion is the Gini index (e.g. Cutler et al, 2012):

$$F = \sum_{k<k'}^{K} p_k p_{k'} = 1 - \sum_{k=1}^{K} p_k^2, \tag{9.20}$$

and the 'best' split corresponds again to the one maximizing $|F - F_{LR}|$ or minimizing F_{LR}.

Case of continuous predictors

In the case of continuous predictor variables, the splitting involves sorting the values of the predictor subset in the node of interest and consider all splits between consecutive pairs (Cutler et al. 2012). Precisely, denote again by $\mathbf{x}_1, \ldots, \mathbf{x}_m$ the continuous d-dimensional subset (of the training set) in the node at hand. For each variable j, $j = 1, \ldots, d$, the observations are sorted as $x_{1,j} \leq x_{2,j} \leq \ldots \leq x_{m,j}$, and a set is chosen of thresholds $\theta_{0,j}, \theta_{1,j}, \ldots \theta_{n+1,j}$, with $\theta_{0,j} \leq x_{1,j}$; $x_{n+1,j} \leq \theta_{n,j}$ and $x_{t,j} \leq \theta_{t,j} \leq x_{t+1,j}$, for $t = 1, \ldots n-1$. Note that any choice of the θ parameters satisfying the above inequalities yields the same results. For example, a common choice (after sorting) is chosen as the midpoint of any two consecutive values, i.e. $\theta_{t,j} = 0.5(x_{t,j} + x_{t+1,j})$ for $t = 1, \ldots n-1$. Next, for each t and j, $t = 1, \ldots n$, and $j = 1, \ldots d$, a binary split is defined based on the condition $1_{\{x_{.j} \leq \theta_{t,j}\}}$, where $x_{.j}$ stands for the j'*th* variable of any observation. In other words, for any choice of θ (among the $(n+1)d$ choices of $\theta_{t,j}$) and any choice of variable, a split is performed on the node under consideration. This requires around $O(m^2 d)$ operations. For each of these splits, the costfunction F_{LR} is computed, and the best split is again chosen based on minimizing F_{LR}. Using a fast algorithm, the number of operations can be reduced to $O(dm\log(m))$. We remind again that each split is based on one variable or feature.

In summary, the above steps can be summarized in the following algorithm (e.g. Cutler at al. 2012). Given a training set $(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)$, where $\mathbf{x}_t = (x_{t1}, \dots, x_{td})^T$ and y_t, $t = 1, \dots n$, denote respectively the d predictors and response, the decision tree algorithm proceeds as follows:

(1) *Starting* – Begin with the training set in the root node.

(2) *Tree construction* – Each unsplit node is split into two child nodes based on the best split, as detailed above by finding the predictor variable among $1, \dots d$, that minimizes F_{LR}.

(3) *Prediction* – When the tree is obtained, a new input variable $\mathbf{x}$ (not from the training set) is passed through the tree until it falls in a terminal node (or leaf), denoted l, with responses $y_{l_1}, \dots, y_{l_p}$ (obtained from the training set). The prediction for $\mathbf{x}$ is then given by $y_{\mathbf{x}} = p^{-1}\sum_{t=1}^{p} y_{l_t}$ for regression , and $y_{\mathbf{x}} = \underset{y}{\operatorname{argmax}} \sum_{t=1}^{p} 1_{\{y_{l_t}=y\}}$ for classification, where y refers to anyone class.

As an illustration, if a new observation, e.g., $x = 10$, is introduced in Fig. 9.6, then the final leaf is the one containing $\{2, 6\}$, with corresponding classes $y_{x=2}$ and $y_{x=6}$.

Remarks

- Besides the two optimization criteria mentioned above, there is a number of other criteria that are used in decision trees. These include Chi-square, entropy and gain ratio.
- Decision trees have a number of advantages but also disadvantages. The following are the main advantages:

  – > it is a non-parametric method

  – > it is easy to understand and implement

  – > it can use any data type (categorical or continuous).
- Although decision trees can be applied to continuous variables, they are not ideal because of categorization. But the main disadvantage is that of overfitting, related to the fact that the tree can grow unnecessarily – e.g. ending with one leaf for each single observation.

One way to overcome the main downside mentioned above, i.e. overfitting, is to apply what is known as *pruning*. Pruning consists of trimming off unnecessary branches starting from the leaf nodes in such a way that accuracy is not much reduced. This can be achieved, for example, by dividing the training set into two subsets, fitting the (trimmed) tree with one subset and validating it with the other subset. The best trimmed tree corresponds to optimizing the accuracy of the validation set. There is another more attractive way to overcome overfitting, namely *random forest* discussed next.

9.4.2 Random forest: definition and algorithm

Random forest is an algorithm based on an ensemble of large number of individual decision trees aimed to obtain better prediction skill. The random element in the random forest comes from the bootstrap sampling, which is based on two key rules:

- Tree building is based on random sub-samples of the training set.
- Splitting nodes is based on random subsets of the variables (or features).

Note that no prunning is done in random forest. The final outcome is then obtained by averaging the predictions of the individual trees from continuous response (regression) or by combined voting for categorical response (classification). The algorithm for RF is given below:

Random forest algorithm

(1) Begin with the training set $(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_n, y_n)$, and decide a number I for the bootstrap samples and a number N of predictor variables to be chosen among the original d variables.

(2) $i = 1$

Decide a bootstrap subsample B_i placed in a root node

- Randomly select N predictors
- Identify the best binary split based on the N selected variables, and split the parent node accordingly.
- Continue until the i'*th* decision tree is obtained
- If $i < I$ then $i = 1 + i$ then go to (2), otherwise go to (3).

(3) For a new data $\mathbf{x}$ the prediction is given by $\hat{y}_{\mathbf{x}} = I^{-1} \sum_{i=1}^{I} y_{\mathbf{x},i}$ for regression, and $\hat{y}_{\mathbf{x}} = \underset{y}{\operatorname{argmax}} \sum_{i=1}^{I} 1_{\{y_{\mathbf{x},i}=y\}}$ for classification, where as before $y_{\mathbf{x},i}$ is the predicted response at $\mathbf{x}$ from the i'*th* tree.

9.4.3 Out-of-bag data, generalization error and tuning

Out-of-bag data and generalization error

A convenient and robust way to get an independent estimate of the prediction error is to use, for each tree in the random forest, the observations that do not get selected in the bootstrap; a tool similar to cross-validation. These data are referred to as *out-of-bag* (oob) data. It is precisely these data which are ued to estimate the performance of the algorithm, and works as follows. For each input data $(\mathbf{x}_t, y_t)$ from the training set, find the set T_t of those (N_t) trees that did not include this observation. Then, the oob prediction at $\mathbf{x}_t$ is given by $y_{oob,t}$, i.e. $N_t^{-1} \sum_{T_t} y_{\mathbf{x}_t,i}$ for regression and $\underset{y}{\operatorname{argmax}} \sum_{T_t} 1_{\{y_{\mathbf{x}_t,i}\}}$ for classification, and where $y_{\mathbf{x}_t,i}$ is the prediction of $\mathbf{x}_t$ based on the i'*th* tree of the random forest. These predictions are then used to calculate the oob error, ε_{oob}, given by (e.g. Cutler et al. 2012), $\varepsilon_{oob} = n^{-1} \sum_{t=1}^{n} (y_t - y_{oob,t})^2$ for regression or $n^{-1} \sum_{t=1}^{n} 1_{\{y_t \neq y_{oob,t}\}}$ for classification.

Parameter selection

To get the best out of the random forest, experience shows that the number of trees in the forest can be chosen to be large enough (Breiman 2001), e.g. several hundreds. Typical values of the selected number of variables in the random forest depends on the type of the problem at hand. For classification, a standard value of N is $\sqrt{d}$, whereas for regression it is of the order $n/3$.

Remarks

1. Random forest algorithm has a number of advantages inherited from decision trees. The main advantages are: accuracy, robustness (to outliers) in addition to being easy to use and reasonably fast. The algorithm can also handle missing values in the predictors, and scale well with large samples (Hastie et al. 2009).

2. The algorithm, however, has few main disadvantages. Random forest tends to be difficult to interpret, in addition to being not very good at capturing relationships involving linear combinations of predictor variables (Cutler et al. 2012).

Clustering is a pettern recognition technique belonging to unsupervised learning problems. Its objective is to identify groups of similar features/characteristics. Several methods exist in machine learning to conduct clustering. We discuss below two particular examples, namely K-means and hierarchical clustering. Note that no target values exist in clustering.

9.5 K-means clustering

We assume we have a dataset $\mathbf{x}_t$, $t = 1, \dots n$ within $\mathbb{R}^d$. Given the number K, the goal is to partition the data into K clusters/groups. Intuitively, a cluster is a group of data whose inter-point distances are less than distances to points outside the cluster. Let us designate by $\boldsymbol{\mu}_k$, $k = 1, \dots K$, the centres of the clusters. We also let

$$r_{ik} = \{ \begin{array}{l} 1 \text{ if } \mathbf{x}_i \text{ is in cluster } k, C_k \\ 0 \text{ otherwise,} \end{array} \tag{9.21}$$

that is $r_{ik} = 1_{C_k}(\mathbf{x}_i)$, and define the distortion measure $J = \sum_{i=1}^n \sum_{k=1}^K r_{ik} \|\mathbf{x}_i - \boldsymbol{\mu}_k\|^2$. Then we have:

$$\{r_{ik}, \boldsymbol{\mu}_k\} = Argmin\ J \tag{9.22}$$

Then, the procedure of clustering is to assign $\mathbf{x}_i$ to the closest centre, i.e.,

$$r_{ik} = \{ \begin{array}{l} 1 \text{ if } k = \underset{j}{\operatorname{argmin}} \|\mathbf{x}_i - \boldsymbol{\mu}_j\|^2 \\ 0 \text{ otherwise,} \end{array} \tag{9.23}$$

then we optimize a quadratic function with respect to $\boldsymbol{\mu}_k$, and get:

$$\boldsymbol{\mu}_k = \frac{\sum_i r_{ik} \mathbf{x}_i}{\sum_i r_{ik}} \tag{9.24}$$

Note that in the above equation $\sum_i r_{ik}$ is the number of points in the k'*th* cluster. The centrs $\boldsymbol{\mu}_k$ represent the mean of data points $\mathbf{x}_i$ assigned to cluster k, that is the k-means algorithm. In fact, the algorithm is based on re-assigning data points to clusters an re-computing the cluster means. This procedure is repeated until no further change in the assignment is observed, or until a maximum number of iterations is reached (MacQueen 1967). In summary, the centres $\mathcal{Z} = \{\boldsymbol{\mu}_1, \dots \boldsymbol{\mu}_K\}$ of the clusters are solution to the minimization problem

$$min \sum_i d(\mathbf{x}_i, \mathcal{Z}) \tag{9.25}$$

where $d(\mathbf{x}_i, \mathcal{Z})$ is the distance between $\mathbf{x}$ and $\mathcal{Z}$, i.e., $d(\mathbf{x}_i, \mathcal{Z}) = \underset{\mathbf{q} \in \mathcal{Z}}{min} \|\mathbf{x} - \mathbf{q}\|^2$. Now given our dataset $\mathbf{x}_1, \dots \mathbf{x}_n$ and K, the algorithm reads:

K-means algorithm

Initialise randomly K data centres as initial seeds
While converegnce do
  Assign each data point to the closest centre
  Compute the new means of the clusters
end while

9.6 Hierarchical clustering

Hierarchical clustering is an alternative to K-means. It has two main advantages over K-means:

a. It does not require the number of clusters
b. It results in a tree-like diagram, known as *dendrogram*.

This does not reduce the value of K-means algorithm. The K-means is in fact computationally efficient, and is easy to understand and implement. However, because of the number of clusters, which is a difficult parameter to get, K-means is less flexible than hierarchical clustering. In addition, the K-means algorithm may be unstable and sensitive to outliers. Nonetheless bith methods cannot handle categorical data directly, and may have difficulties with non continuous data or with large variance. Finally, it should be mentioned that both methods are exploratory methods, i.e., are not model-based.

9.6.1 Main types of hierarchical clustering

Hierarchical clustering has two main types, namely, agglomerative and divisive.

Agglomerative clustering

The tree-like of the agglomerative clustering (AGNES: agglomerative nesting), we start with the individual patterns, that is each element constitutes a single-element cluster (*leaf*). Then at each every two 'similar' clusters are merged into one cluster (*node*). This procedure is than continued until all elements are combined into one cluster, i.e. *the root* (Fig. 9.7).

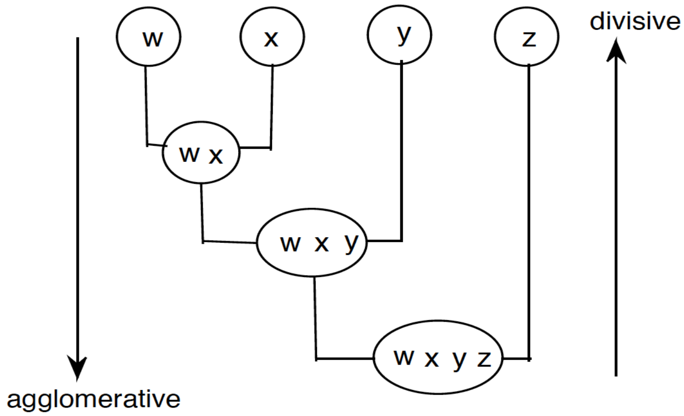

Figure 9.7: An illustrative example of the agglomerative and divisive algorithms.

Divisive clustering

This clustering method (DIANA: divisive analysis) is the inverse of AGNES. It starts with the whole data, and at each step the most heterogeneous cluster is divided into two clusters (Fig. 9.7).

9.6.2 Dissimilarity measures

There are different dissimilarity measures used in hierarchical clustering. The following are the main measures used:

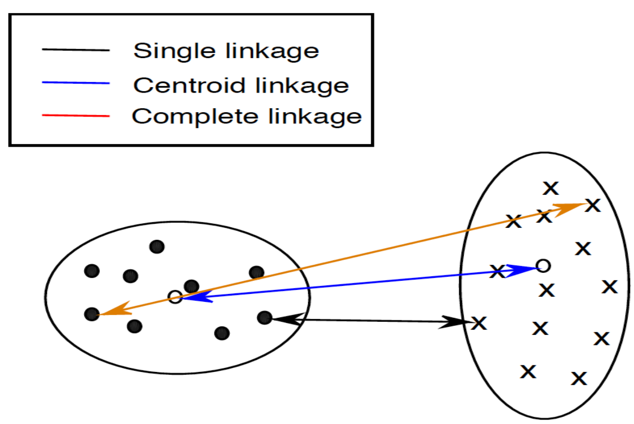

Figure 9.8: Illustration of three measures of similarity between two classes.

- Complete linkage – Here the distance between two clusters as the maximum amongst all pairwise distances between points from the two clusters (Fig. 9.8). Complete linkage yields more compact clusters.
- Single linkage – This is based on the minimum distance between the elements of the two clusters. The single linkage yields in general long and loose clusters.
- Average linkage – This is based on the average of the pairwise distances between the lelements of both clusters

- Centroid linkage – It merges clusters with minimum distance between their centroids.
- Ward's method – Here clusters with minimum distance between them are merged. Wards's method tends to minimize the total within-cluster variance.

9.6.3 Main steps of hierarchical clustering

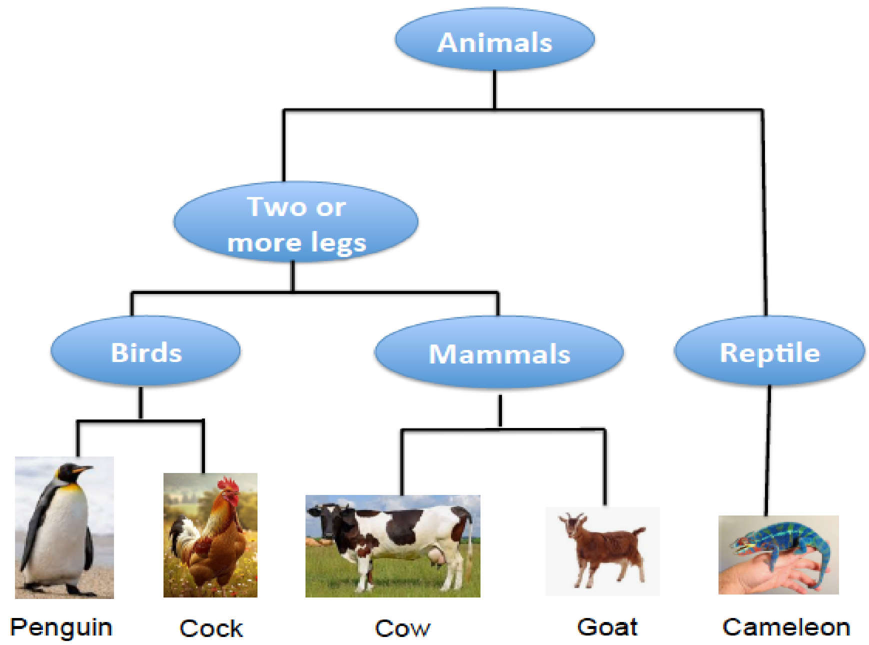

Figure 9.9: Example of dendrogram of a set of 5 animals.

Several steps are followed when we apply hierarchical clustering to a dataset. For example, for agglomerative clustering we have the following steps:

1) Compute the distance matrix between pairs of elements.
2) Merge closest clusters using one of the above dissimilarity measures.
3) Update the distance matrix (between the newly obtained clusters).
4) Repeat steps 1), 2) and 3) above until one cluster is obtained.

Figure 9.9 shows a simple example of dendrogram for clustering animals. Going from top to bottom, for example, represents the divisive algorithm. If we are interested in two clusters as an example, then we cut the dendrogram just after the first (from the top) split and we obtain the walking group (with 4 elements) and the crawling group (with one element).

In Matlab k-means clustering is obtained using the script *kmeans*. Given a data matrix X of size $N \times p$ and a number of clusters K, the clusters are obtained as:

```
>> classes = kmeans (X, K);
```

In Python, the script is *KMeans* and can be use, e.g. as:

```
>>> from sklearn.cluster import KMeans
>>> K=2
```

```
>>> classes = KMeans (n_clust=K, n_init=10).fit(X)
```

The classes are found in *classes.labels_*, given by integers from 0 to $K-1$. The centres of the classes are found in *classes.cluster_centers_*.

Hierarchical clustering is also provided in Matlab and Python and other softwares. In matlab, the script *cluster* can be used to apply clustering to the data matrix, and the *dendrogram* can be called to get the dendrogram.

In Python, the following few lines can be used to cluster the data matrix X into for example K clusters using ward linkage:

```
>>> from scipy.cluster.hierarchy import linkage, dendrogram, fcluster
>>> linkage_matrix = linkage (X, method='ward')
>>> dendrogram (linkage.matrix)
>>> cluster_labels = fcluster (linkage_matrix, K, criterion='maxclust')
```

Chapter 10

Artificial Neural Networks

10.1 Background

To provide an easy introduction to artificial neural network (ANN), let us go back to the classification problem discussed in the previous chapter. We assume we have two classes as shown in figure 10.1. The separating hyperplane is orthogonal to the vector $\mathbf{a} = (a_0, a_1, \ldots a_p)^T$. The hyperplane equation is:

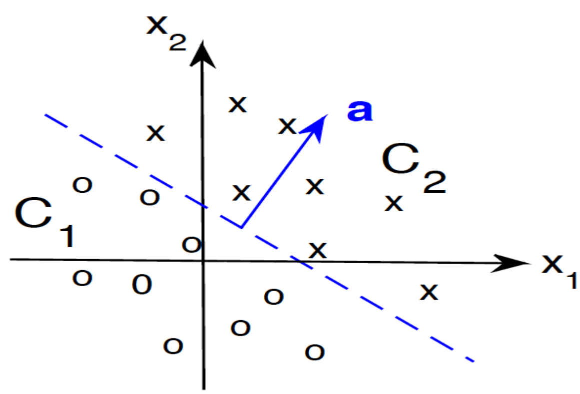

Figure 10.1: Illustration of a separating surface between two classes C_1 and C_2

$$a_0 + a_1 x_1 + \ldots a_p x_p = (1\ x_1 \ldots x_p)\mathbf{a} = \mathbf{a}^T\mathbf{x} = 0, \tag{10.1}$$

and a new pattern $\mathbf{x}$ is assigned to C_1 if $\mathbf{a}^T\mathbf{x} < 0$, and to C_2 if $\mathbf{a}^T\mathbf{x} > 0$.

10.1.1 Rosenblatt's perceptron

Now let us express the above formulation into the following chart (Fig. 10.2). In this chart the first box on the left constructs the product $\mathbf{a}^T\mathbf{x}$, which is then fed into the activation (the next box), which is a binary step function ($1_{x>0}$, i.e., 1 if $x > 0$ and 0 otherwise) to produce the output o/y, that is:

$$o/y = 1_{x>0}(\mathbf{a}^T\mathbf{x}) = \{ \begin{array}{l} 1 \text{ if } \mathbf{a}^T\mathbf{x} = \sum_{i=0}^{p} a_i x_i > 0 \\ 0 \text{ otherwise} \end{array} \tag{10.2}$$

The above flow chart representation is known as *Rosenblatt's perceptron* (e.g., Vapnik 2000,

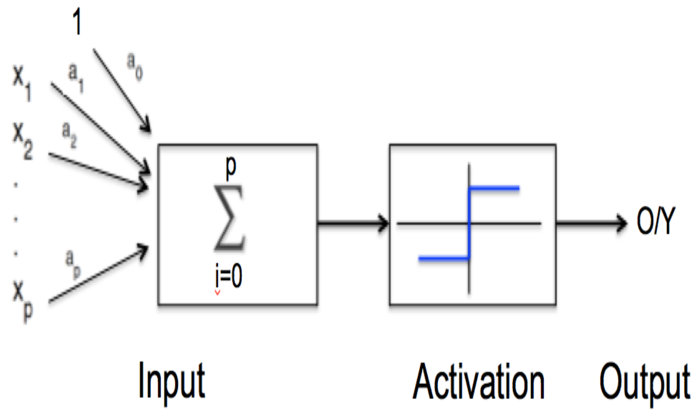

Figure 10.2: Basic Rosenblatt's perceptron with hard delimiter

Bishop 2006) with hard delimiter (the step function). It is clear that the (step function) perceptron achieves the same classification as in Fig. 10.1 for linearly separable problems. Rosenblatt's perceptron is associated with the neural network model proposed earlier by McCulloch and Pitts (1943).

The perceptron attempts to mimic the human neuron (Fig. 10.3). In neurobiology, and following our current knowledge, the neuron or nerve cell (Fig. 10.3) receives signals from other neurons through dendrites, which are fine and branched structures. If the stimulus exceeds certain threshold the neuron becomes activated, and the signal gets propagated through a thin long strand, labelled axon, to the the synapse where the signal can be picked up by neighbouring neurons.

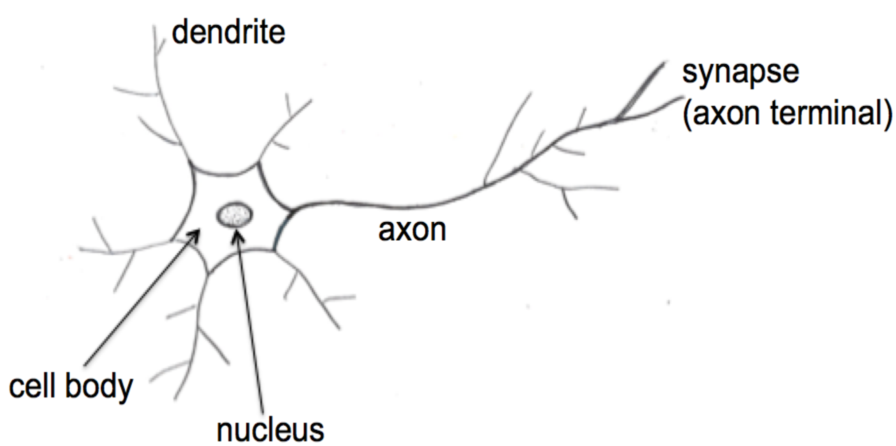

Figure 10.3: Simplified model of a biological nerve cell or neuron.

The perceptron model (Fig. 10.2) is a basic mathematical simplification of the complex structure of the nerve cell (Fig. 10.3). This perceptron is important in binary classification expressed by the following theorem.

Perceptron convergence theorem

If the two classes are linearly separable (Fig. 10.4) and the training set presented to the perceptron then the perceptron converges after a finite number of steps.

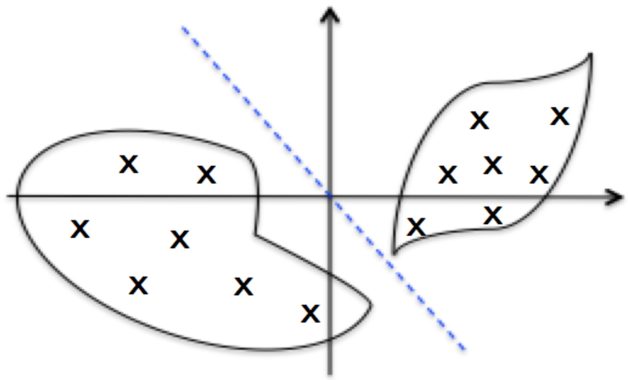

Figure 10.4: Example of linearly separable classes

The above perceptron can be extended by using a different activation function yielding the single neuron model.

Application to the Boolean function

The above perceptron convergence theorem (see for example Hertz et al. (1991) and Bishop (2006)) shows that linearly separable (binary) classification problems can be solved with the perceptron having a step activation function given by $sign(\mathbf{a}^T\mathbf{x})$, see Eqs (13.1-13.2). Basic simple examples of linearly separable problems include several Boolean functions[1]. Consider the Boolean OR function, with truth table 0111 (0OR0=0, 0OR1=1, etc.), which is illustrated in Fig. 10.5 in the plane (x_1, x_2) showing four patterns with two (binary) classes (True or 1, and False or 0). For example the class of the origin (0,0) is 0 (0OR0=0). It is clear that the OR function is linearly separable, which is also the case for the Boolean AND function.

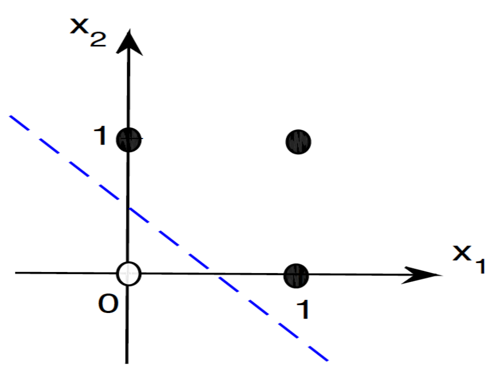

Figure 10.5: Four patterns of the Boolean OR function showing linear separability of the two classes shown respectively by filled and empty circles.

Exercise

Show that the Boolean OR function is linearly separable.

Indication. Is it possible to find $\mathbf{w} = (w_1, w_2)^T$ and b such that $\mathbf{w}^T\mathbf{x} + b = 0$ separates the

[1] A Boolean function is a function that takes binary inputs and produces one binary output.

two classes of Fig. 10.5, that is $\mathbf{w}^T\mathbf{x} + b$ is positive (negative) for the first (second) class? For example we get the conditions: $b < 0$, $w_1 + b > 0$, $w_2 + b > 0$, and $w_1 + w_2 + b > 0$, for which $w_1 = w_2 = 1$ and $b = -0.5$ is a solution.

10.1.2 The single neuron model

Rosenblatt's perceptron is the first and basic artificial neural net. In this perceptron one can use a different activation function $g()$ to yield the single neuron model (Fig. 10.6). Note that for the output unit we use interchangeably y or o, i.e. y/o. In the single neuron net the

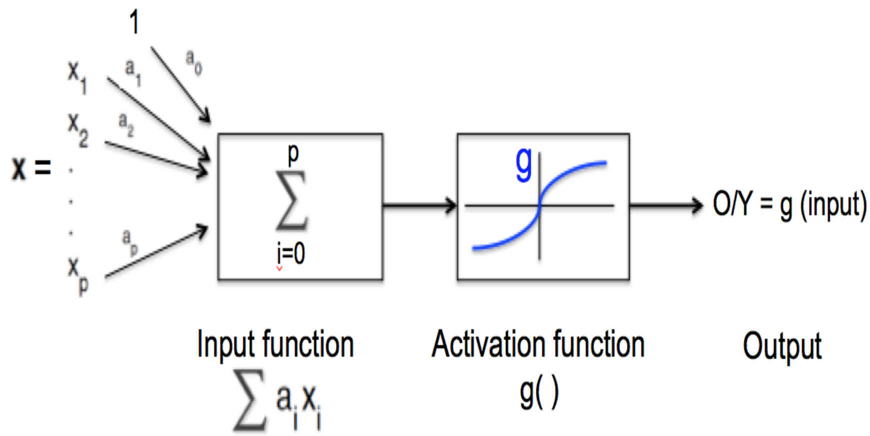

Figure 10.6: Basic Rosenblatt's perceptron with activation function $g()$

activation function can be linear or nonlinear, and both have different properties discussed below.

10.2 Learning with single neuron model

10.2.1 Linear activation function

Here we assume that the output is a linear function of the input parameter (Fig. 10.7). In

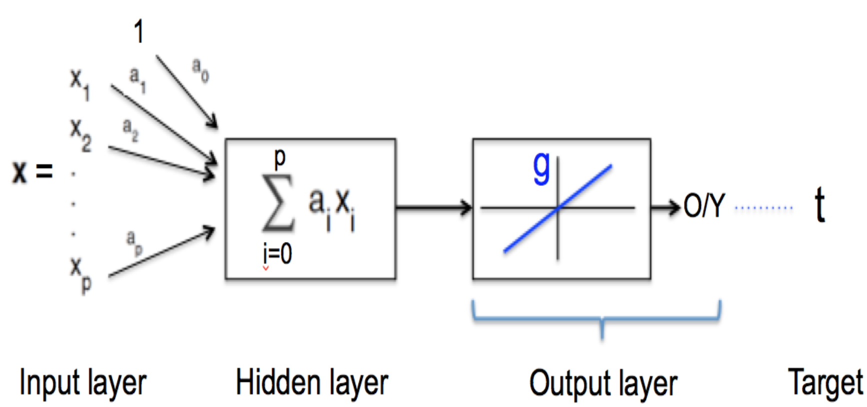

Figure 10.7: Single neuron network with linear activation

this setup we assume we have a training $\mathbf{x}_1, \ldots \mathbf{x}_n$ (n is the sample size of the training set), along with the target set $t_1, \ldots t_n$. The output is linear, i.e., $y/o = \mathbf{a}^T\mathbf{x}$ for an input pattern $\mathbf{x}$, and the objective is to train this net by minimizing the residual/error:

$$E = \frac{1}{2}\sum_{i=1}^{n}(o_i - t_i)^2 = E(\mathbf{a}). \tag{10.3}$$

with $o_i = \mathbf{a}^T\mathbf{x}_i$, $i = 1, \ldots n$, and $\mathbf{x}_i = (1, x_{i1}, \ldots x_{ip})^T$. The identification of the weights $a_0, a_1, \ldots a_p$ is referred to as the *learning process*. Various algorithms are used to learn the network. One of the main algorithms is the gradient descent (Fig. 10.8), which performs the optimisation by computing the gradient ∇E and moving downgradient:

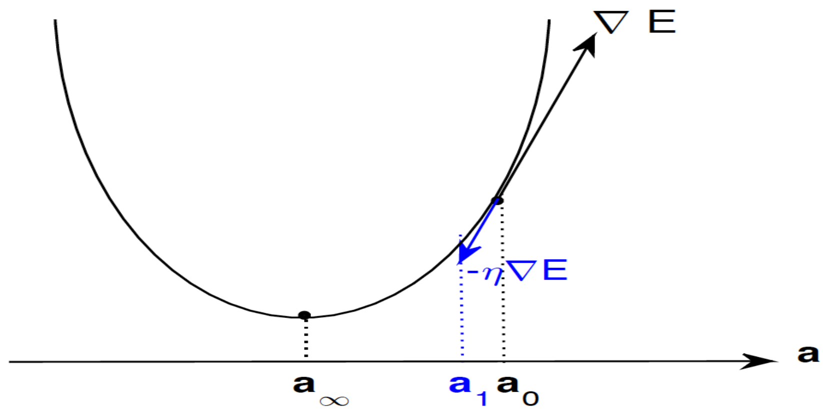

Figure 10.8: Illustration of the gradient descent algorithm starting at $\mathbf{a}_0$ and converging to the minimum $\mathbf{a}_\infty$

$$a_i \leftarrow a_i + \Delta a_i; \text{ with } \Delta a_i = -\eta\frac{\partial E}{\partial a_i}. \tag{10.4}$$

where $0 < \eta \leq 1$ is the learning rate, and

$$\frac{\partial E}{\partial a_i} = \sum_{d=1}^{n}(t_d - y_d)(-x_{id}). \tag{10.5}$$

The learning algorithm reads:

Algorithm 0

* Initialization: choose small random numbers for $\mathbf{a} = (a_0, a1, \ldots a_p)^T$

* Iteration: Do until convergence

Initialise $\Delta a_i = 0$

For each training input $\mathbf{x}$ compute the output y/o

$\Delta\mathbf{a} \leftarrow \Delta\mathbf{a} + \eta(t - y)\mathbf{x}$

$\mathbf{a} \leftarrow \mathbf{a} + \Delta\mathbf{a}$

enddo

One of the main results is that the algorithm converges in a finite number of iterations. Other algorithms are used particularly for the nonlinear case. One particular algorithm is the stochastic gradient descent, in which at each iteration one randomly chosen input data is used to calculate the gradient instead of using the whole sample.

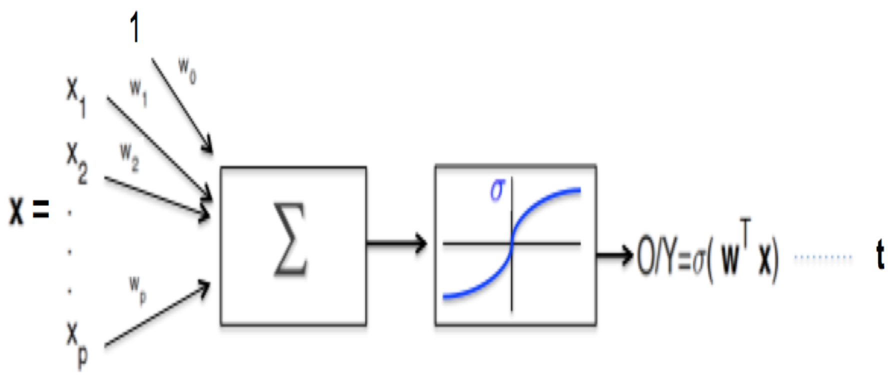

Figure 10.9: Single neuron network with nonlinear sigmoid

10.2.2 The nonlinear case

This case corresponds to a (nonlinear) sigmoid (Fig. 10.9) with output:

$$o/y = \sigma(\mathbf{w}^T\mathbf{x}), \tag{10.6}$$

where $\sigma()$ is a sigmoid function, such as $\sigma(x) = (1 + e^{-x})^{-1}$. Here, and for convenience, we use the letter "w" for the weights in accordance with the NN literature in general. A nice useful property of $\sigma()$ is related to its derivative, namely, $\frac{d\sigma}{dx} = \sigma(1-\sigma)$, which is used during minimization. The same gradient descent algorithm can be used as before but the gradient is now computed based on the new sigmoid function. The previous single neuron model can do binary classification with half-plane as decision surface, but cannot do more complicated tasks such as multiclass classification, which need more complex networks.

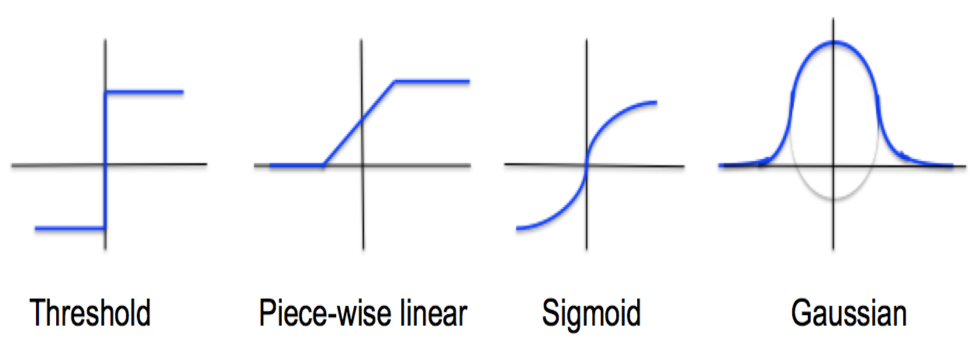

Figure 10.10: Example of various activation functions

Figure 10.10 shows various forms of activation functions, namely, binary step function $g() = 1_{x>0}()$, piece-wise linear such as the rectified linear unit (ReLU) $g() = x1_{x>0}()$, the sigmoid function $g() = \sigma()$, and the Gaussian $g(x) = e^{-x^2/2}$. Other activation functions also exist in the literature. These activation functions have advantages and disadvantages. For example, the peice-wise linear functions are simple but they have discontinuous derivatives.

10.3 Multilayer perceptron

10.3.1 Description of the perceptron

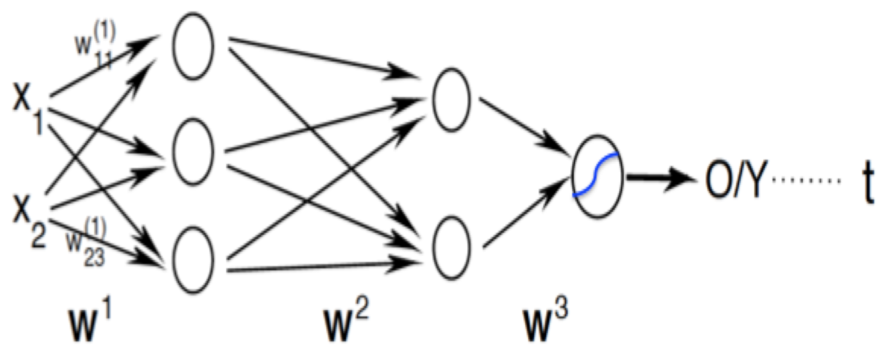

Figure 10.11: Example of multilayer perceptron comprising an input layer, two hidden layers and an output layer. The target is shown as "t"

Multilayer perceptron (MLP) networks can have several hidden layers (Fig. 10.11), also referred to as deep neural network. Each neuron in the inner and output layers is a sigmoid function. In general, but not always, the sigmoid functions for the different layers are identical. In addition, each layer has its own weights. For example, in Fig. 10.11 the weights of the first layer are $\mathbf{W}^{(1)} = (w_{ij}^{(1)})$ with $i = 1, 2$ and $j = 1 : 3$. Perceptrons with hidden layers can solve many nonlinear problems, such as nonlinear regression and nonlinear classification problems (Mehlig 2022). The main learning procedure used in MLP is the backpropagation algorithm discussed below.

10.3.2 The backpropagation algorithm

Consider the composition function $h(x) = f_1(f_2(x))$. The derivative of $h()$ is obtained using the chain rule of differentiation, i.e. $\frac{dh(x)}{dx} = \frac{df_1}{dx}(f_2(x))\frac{df_2(x)}{dx}$. The chain rule of derivatives can extend to the derivative of $f_1(f_2 \ldots f_m(x))\ldots)$. Now consider the following deep neural net with just one node in each layer and target t (Fig. 10.12). To find the gradient of the

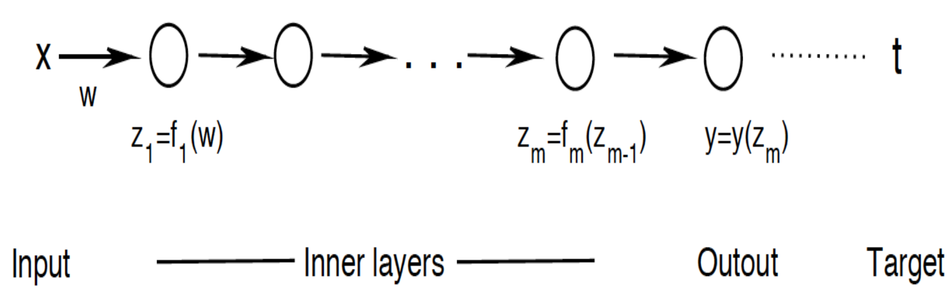

Figure 10.12: Illustration of the backpropagation algorithm in a deep neural network

costfunction $E = \frac{1}{2}(t-y)^2$ we extend the above chain rule to get:

$$\frac{dE}{dw} = -(t-y)\frac{dy}{dw} = (y-t)\frac{df_m}{dz_m}\cdots\frac{df_1}{dw} = -(t-y)\frac{dy}{dz_m}\cdots\frac{dz_2}{dz_1}\frac{dz_1}{dw} \tag{10.7}$$

In the above equation, one can see the backward cascade of derivatives starting from the end of the net. This rule can then be extended to a MLP. The update rule of the weight reads:

$$w_{k+1} = w_k + \eta \frac{\partial E}{\partial w_k} \tag{10.8}$$

where k refers to the iteration step and η is the *learning rate*, which is chosen apriori, but can change along the iteration process.

Example with one sigmoid unit

Consider a simple one sigmoid unit (e.g., Fig. 10.9). The gradient of E with respect to w_i, the i'*th* component of $\mathbf{w}$, with input pattern $\mathbf{x}$ (x_i being the i'*th* component of $\mathbf{x}$), e.g., Eq. (2.6):

$$\frac{\partial E}{\partial w_i} = - \sum_{\text{training set}} (t-y)y(1-y)x_i \tag{10.9}$$

The above update rule with the expression of the gradient yields the following updating algorithm for a given weight w_{ij}:

Algorithm 1

Initialization of the weights w_{ij} to small random numbers
Iteration: Do until convergence
Use the training data and get outputs
For output unit k, y_k
$\delta_k \leftarrow y_k(1-y_k)(t_k - y_k)$
For hidden unit h
$\delta_h \leftarrow y_h(1-y_h)\sum_{\text{output}} w_{hk}\delta_k$
$w_{ij} \leftarrow w_{ij} + \Delta w_{ij} = w_{ij} + \eta\delta_j x_{ij}$
enddo

where x_{ij} is the i'*th* input into the j'*th* unit. The above algorithm can be easily extended to MLP by introducing the layers $l = 1, \dots L$, with $l = 1$ and $l = L$ representing, respectively, the first hidden and the output layers. After initialization the backpropagation loop of the algorithm reads:

Algorithm 2

Do until convergence
Choose an input training pattern $\mathbf{x}$ (with output/target $\mathbf{o}/\mathbf{t}$)
$\delta_i^L = g'(h_i^L)(t_i - y_i^L)$
for l=L-1,1 do
$\delta_i^l = g'(h_i^l)\sum_j w_{ij}^{l+1}\delta_j^{l+1}$
$\Delta w_{ij}^l = \eta\delta_i^l y_j^{l-1}$
enddo
enddo

In the above algorithm h_i^l is the input to the i'*unit* of the l'*th* layer, and the same activation function $g()$, with derivative $g'()$, is used in the net. Note also that y_i^0 corresponds to the input x_i.

Remarks

1. In NN the number of neurons (or units) in the input layer equals the input space dimension. But the number of neurons in the output layer depends on the problem at hand. For example, we need:
   $->$ 1 unit in regression and binary classification
   $->$ K units for multiclass classification with K classes
   $->$ In some unsupervised learning the number of output layer units equals the dimension of the input data (e.g., auto-encoder NN).
2. There is no rule for the choice of the number of hidden (and the number of their neurons), which depends in general on the accuracy of the network.

10.3.3 Types and architecture of NN

Large numbers of NN architectures have been developped in the literature of MLP, see for example Brunton and Kutz (2022), and references therein for the description of 18 NN architectures. But for brevity we discuss below a few types of neural nets mostly used in climate science:

- Feed forward nets, including single layer perceptron and MLP.
- Kohonen self organising maps (SOM).
- Radial basis networks.
- Autoencoder nets.
- Convolutional neural networks.
- Recurrent neural nets.

10.3.4 Example of output of a two-layer neural net

Figure 10.13 shows an example of a two-layer feedforward neural net with one hidden layer, with $h()$ and $\sigma()$ being the sigmoid or activation functions of the hidden and output layers respectively. We also let M and K units for the hidden and output layers respectively and with respective weights $\mathbf{W}^{(1)} = (w_{ij}^{(1)})$ and $\mathbf{W}^{(2)} = (w_{ij}^{(2)})$. Given an input pattern $\mathbf{x} = (x_1, \ldots x_p)^T$, the output of the j'$th$ unit of the output layer is:

$$\begin{aligned} y_j(\mathbf{x}, \mathbf{W}) &= \sigma\left[\sum_{i=1}^M w_{ij}^{(2)} z_i + w_0^{(2)}\right] \\ &= \sigma\left[\sum_{i=1}^M w_{ij}^{(2)} h\left(\sum_{l=1}^p w_{li}^{(1)} x_l + w_0^{(1)}\right) + w_0^{(2)}\right] \end{aligned} \tag{10.10}$$

Exercise

Consider Fig. 10.13 along with Eq. (13.10), assuming $p = 2$, $M = 2$ and $K = 1$. Calculate the output y and the gradient of the costfunction, assuming n training input patterns $\mathbf{x}_i$, and targets t_i, $i = 1, \ldots n$.

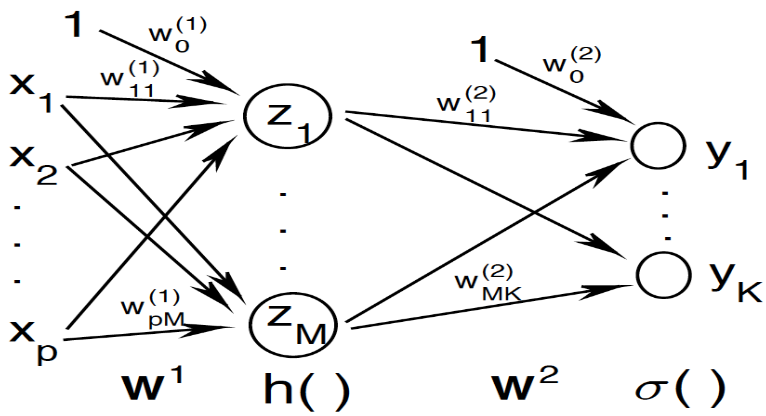

Figure 10.13: Example of a two-layer neural net with activation functions $y()$ and $\sigma()$

10.4 Kohonen Self-Organizing Maps

10.4.1 Background

Broadly speaking there are two main classes of training networks, namely supervised and unsupervised training. The former refers to the case in which a target output exists for each input pattern (or observation) and the network learns to produce the required outputs. When the pair of input and output patterns are not available, and we only have a set of input patterns then one deals with unsupervised training. In this case, the network learns to identify the relevant information from the available training sample. Clustering is a typical example of unsupervised learning. A particularly interesting type of unsupervised learning is based on what is known as competitive learning, in which the output network neurons compete among themselves for activation resulting, through *self-organization*, in only one activated neuron: the winning neuron, at any time. The obtained network is referred to as *self-organinzing map*, SOM, (Kohonen 2001). SOM or Kohonen Network (Rojas 1996) has two layers, the input and output layers. In SOM, the neurons are positioned at the nodes of a usually low-dimensional (one- or two-dimensional) lattice. The positions of the neurons follow the principle of neighborhood so that neurons dealing with closely related input patterns are kept close to each other following a meaningful coordinate system (Kohonen 2001, Haykin 2009). In this way, SOM maps the input patterns onto the spatial locations (or coordinates) of the neurons in the low-dimensional lattice. This kind of discrete maps is referred to as *topographic map*. SOM can be viewed as a nonlinear generalization of principal component analysis (Ritter 1995, Haykin 2009). SOM is often used for clustering and dimensionality reduction (or mapping).

10.4.2 SOM algorithm

Mapping identification and Kohonen network

Each input pattern $\mathbf{x} = (x_1, \ldots, x_d)^T$ from the usually high-dimensional input space is mapped onto $i(\mathbf{x})$ in the output space, i.e. neuron lattice (Fig. 10.14).

A particularly popular SOM model is the Kohonen network, which is a feed-forward

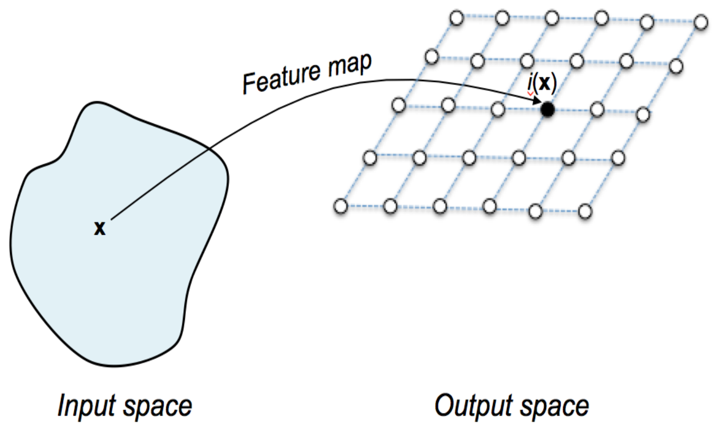

Figure 10.14: Schematic of the SOM mapping from the high-dimensional feature space into the low-dimensional space.

system in which the output layer is organized in rows and columns (for a 2D lattice). This is shown schematically in Figure 10.15.

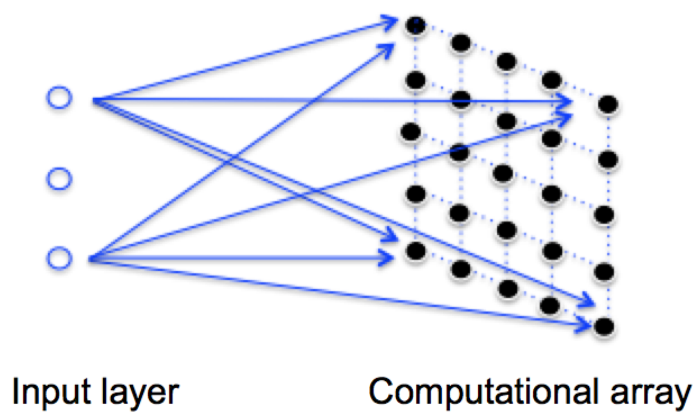

Figure 10.15: Schematic of the Kohonen SOM network.

Components of SOM

Let $\mathbf{w}_j = (w_{j1}, \ldots, w_{jd})^T$ be the synaptic weight corresponding to the j'*th*, $j = 1, \ldots, M$, output neuron, with d being the input space dimension (Fig. 10.15). The index $i(\mathbf{x})$ of the neuron associated with the image of $\mathbf{x}$ (Fig. 10.14) is the one maximizing the scalar product $\mathbf{x}^T \mathbf{w}_j$, i.e.,

$$i(\mathbf{x}) = \underset{j=1,\ldots M}{\operatorname{argmin}} \|\mathbf{x} - \mathbf{w}_j\| \tag{10.11}$$

Equation (13.11) summarizes the competitive process among the M output neurons in which $i(\mathbf{x})$ is the winning neuron. SOM has a number of components that set up SOM algorithm. Next to the competitive process, there are two more processes, namely co-operative and adaptive processes, as detailed below.

The co-operative process concerns the neighborhood structure. The winning neuron determines a topological neighborhood since a firing neuron tends to excite nearby neurons

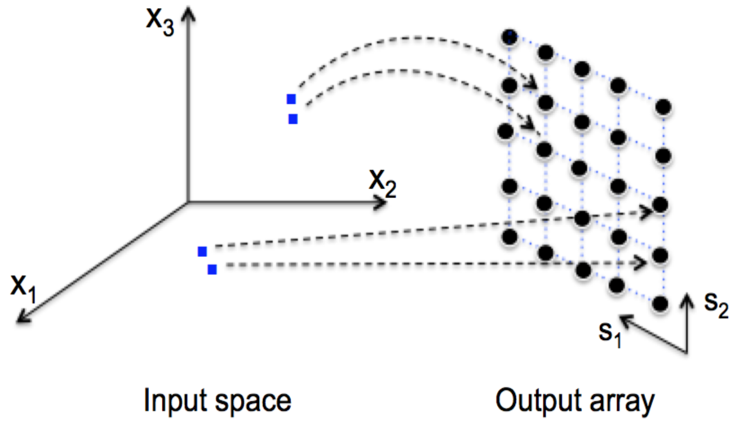

Figure 10.16: Association between the input space and the output array: neighbouring pattern in the input space activate neighbouring winning neurons in the output array

more than those further away. A topology is then defined in the SOM lattice, reflecting the lateral interaction among a set of excited neurons. Let s_{ij} denote the lateral distance between the winning and excited neurons $i(\mathbf{x})$ and j respectively. This then allows to define the topological neighborhood $h_{j,i(\mathbf{x})}$. A typical choice for this topology is given by the Gaussian function:

$$h_{j,i(\mathbf{x})} = \exp\left(-\frac{d^2_{j,i(\mathbf{x})}}{2\sigma^2}\right). \tag{10.12}$$

where d_{kl} denotes the lateral distance between neurons k and l on the SOM grid. Via the similarity metric SOM guarantees that neighbouring patterns in the input space activate neighbouring winning neurons in the output array (Fig. 10.16), and yields an ordered array of output neurons (Mehlig 2022). A characteristic feature of SOM is that the neighborhood size shrinks with time. A typical choice of this shrinking is given by:

$$\sigma(n) = \sigma_0 \exp(-\frac{n}{\tau_0}) \tag{10.13}$$

where n is the timestep, σ_0 is an initial value of σ and τ_0 is a constant.

The adaptive process regards the way of learning process leading to the self-organization of the outputs, and the feature map is formed. The topographic neighborhood mirrors the fact that the weights of the winning and also neighboring neurons get adapted, though not by an equal amount. In practice, the weight update is given by:

$$\Delta\mathbf{w}_j = \mathbf{w}_j(n+1) - \mathbf{w}_j(n) = \eta(n) h_{j,i(\mathbf{x})}\left(\mathbf{x} - \mathbf{w}_j(n)\right), \tag{10.14}$$

which is applied to all neurons in the lattice within the neighborhood of the winning neuron. The discrete time learning rate is given by $\eta(n) = \eta_0 \exp(-\frac{n}{\tau_1})$. The values $\eta_0 = 0.1$ and $\tau_1 = 1000$ are typical examples that can be used in practice (Haykin 2009). Also, in Eq. (13.13), the value $\tau_0 = 1000/log(\sigma_0)$ can be adapted in practice (Haykin 2009).

In order not to end in a metastable state, the adaptive process has two identifiable phases, namely the ordering or self-organizing and the convergence phases. During the former phase the topological ordering of the weight vectors takes place within around 1000 iterations of the SOM algorithm. In this context, the choice of $\eta_0 = 0.1$ and $\tau_1 = 1000$ are satisfactory, and the previous choice of τ_0 in Eq. (13.13) can also be adopted. During the convergence phase, the feature map is fine tuned providing hence an accurate statistical quantification of the input patterns, which takes a number of iterations, around 500 times the number of neurons in the network (Haykin 2009).

Summary of SOM algorithm

The different steps of SOM algorithm can be summarized as follows:

- *(1) Initialization* – Randomly choose initial values of the weights $\mathbf{w}_j(0)$, $j = 1, \dots M$, associated with the M neurons in the lattice. Note that these weights can also be chosen from the input data $\mathbf{x}_t$, $t = 1, \dots N$, where N is the sample size.
- *(2) Sampling* – Draw a sample input vector $\mathbf{x}$ from the input space.
- *(3) Neuron identification* – Find the winning neuron $i(\mathbf{x})$ based on Eq. (13.11).
- *(4) Updating* – Update the synaptic weights $\mathbf{w}_j$, $j = 1, \dots, M$, following Eq. (13.14).
- *(5) Continuation* – Go to step 2 above until the feature map stops changing significantly.

Matlab has several scripts of self organising map. Informaltion can be found by the command *lookfor SOM*. Python also provides clustering via self organising map. Given a data matrix $Y(N, p)$, normalised by removing the mean and scaled by the standard deviation, and the size of the SOM grid, for example 4x4, the following lines obtains SOM:

```
>>> from minisom import MiniSom
>>> import numpy as np
>>> import matplotlib.pyplot as plt
>>> som_shape = (4,4)
>>> som=MiniSom(som_shape[0], som_shape[1], Y[1], sigma=1, learning_rate=0.5)
>>> som.random_weights_init(Y)
>>> som.train_random(Y, 1000) # training with 1000 iterations
>>> # Get the organised map (SOM node indices for each data point)
>>> clustered_maps = np.array ([som.winner(x) for x in Y ])
>>> # Extract i, j indices from the tuples
>>> clusters_i = np.array([x[0] for x in clustered_maps])
>>> clusters_j = np.array([x[1] for x in clustered_maps])
>>> # Optional: flatten or combine the clusters indices into a single identifier
>>> clusters_flat = clusters_i * som_shape[1] + clusters_j # unique cluster ID
>>> # Reshape if needed (e.g., into a time series format for clustering)
>>> clusters_time_series = clusters_flat.reshape (N)
>>> # Plot the SOM weight maps
>>> plt.figure(figsize=(10,10))
>>> for i in range(som_shape[0]):
>>>     for j in range(som_shape[1]):
>>>         plt.subplot(som_shape[0], som_shape[1], i*som_shape[1]+j+1)
>>>         plt.imshow(som.get_weights()[i,j].reshape(la,lo), cmap='coolwarm')
>>>         plt.title(f"Node ({i}, {j})")
```

```
>>>         plt.axis('off')
>>> plt.tight_layout()
>>> plt.savefig('SOM.png')
>>> plt.show()
```

Note that in the above lines, the reshaping is used assuming that the data represent maps with $la \times lo$ grid points, i.e. $p = la \times lo$.

10.5 Radial basis function networks

10.5.1 Introduction

Radial basis functions (RBFs) constitute one of the attractive tools to interpolate and/or smooth scattered data. RBFs have been formally introduced and coined by Powell (1987) in exact multivariate interpolation, although the technique was hanging around before that, see e.g., Hardy (1971), Franke (1982). Given m distinct points[2] $\mathbf{x}_i$, $i = 1, \ldots n$ in $\mathbb{R}^d$, and n real numbers f_i, $i = 1, \ldots n$, these numbers can be regarded as the values at $\mathbf{x}_i$ of a certain unknown function $f(\mathbf{x})$. The problem of RBF is to find a smooth real function $s(\mathbf{x})$ satisfying the interpolation conditions:

$$s(\mathbf{x}_i) = f_i \qquad i = 1, \ldots n \tag{10.15}$$

and that the interpolating function is of the form

$$s(\mathbf{x}) = \sum_{k=1}^{m} \lambda_k \varphi_k(\mathbf{x}) = \sum_{k=1}^{m} \lambda_k \varphi\left(\|\mathbf{x} - \mathbf{x}_k\|\right). \tag{10.16}$$

The functions $\varphi_k(\mathbf{x}) = \varphi\left(\|\mathbf{x} - \mathbf{x}_k\|\right)$ are known as radial basis functions. The real function $\varphi(x)$ is defined over the positive numbers, and $\|.\|$ is any Euclidean norm or Mahalanobis distance. Thus the radial basis function $s(\mathbf{x})$ is a simple linear combination of the shifted radial basis function $\varphi(x)$.

Radial basis function (RBF) networks is a special MLP network used in supervised learning. The method relies first on the approximation of the input data in terms of a set of radial basis functions. Several types of radial basis functions exist in the literature (e.g., Hannachi 2021). One special radial function is the Gaussian given by:

$$\varphi_j(\mathbf{x}) = \exp\left(-\frac{\|\mathbf{x} - \boldsymbol{\mu}_j\|^2}{\sigma_j^2}\right) \tag{10.17}$$

where $\boldsymbol{\mu}_j$ and σ_j are respectively the centre and width of the Gaussian function. These functions have nice properties, but how can they be used in neural networks?

10.5.2 RBF networks

Link between the hidden and output layers

The RBF network is a kind of hybrid network linking the input and output layers in a special way. Let us assume that the input layer used a training set composed of patterns $\mathbf{x}_i \in \mathbb{R}^d$,

[2] generally known as *nodes*, *knots*, or *centres* of interpolation.

$i = 1, \ldots n$, and the output layer has patterns $\mathbf{y}_i \in \mathbb{R}^K$, which are to compare with targets $\mathbf{t}_i$, $i = 1, \ldots n$. The hidden layer is composed of $m = 1, \ldots M$ neurons with 'activation' $\varphi_m(\mathbf{x})$ (Fig. 10.17).

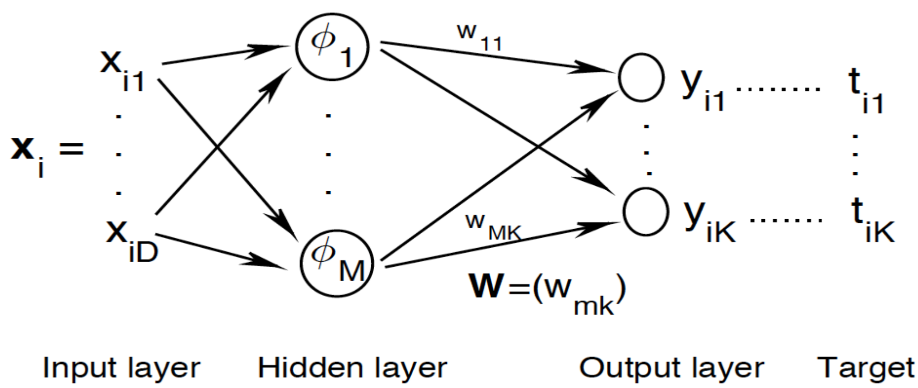

Figure 10.17: Architecture of radial basis function network

The learning pattern in the RBF networks is a bit different from the conventional forward MLP nets in that the output layer is not analysed in the same way as the hidden layer (Mehlig 2022, Haykin 2009, Bishop 2006). The learning process of the weights (w_{mk}) can be obtained simply by solving a linear matrix equation (or gradient method), but the parameters (μ_{mk}), and (σ_m) of the RBF can be estimated independently from the weights (w_{mk}) (Fig. 10.17).

The output $\mathbf{y}_i$, $i = 1, \ldots n$, is given by:

$$y_{ik} = \sum_{m=1}^{M} w_{mk}\varphi_m(\mathbf{x}_i) = \sum_{m=1}^{M} w_{mk} \exp\left(-\frac{\|\mathbf{x}_i - \boldsymbol{\mu}_m\|^2}{\sigma_m^2}\right), \; k = 1, \ldots K. \tag{10.18}$$

The weights are then obtained my minimizing the costfunction E:

$$(w_{mk}) = \text{argmin}\left(E = \frac{1}{2}\sum_{i=1}^{n} \|\mathbf{t}_i - \mathbf{y}_i\|^2\right). \tag{10.19}$$

The parameters of the hidden layers can be obtained using few unsupervised learning tools, including:

- clustering via, e.g., k-means using M clusters where the centres $\boldsymbol{\mu}_m$, $m = 1, \ldots M$ represent the centroids of the clusters of $\mathbf{x}_i$, $i = 1, \ldots n$. The widths σ_m can be approximated by the widths of the corresponding cluster.
- The centres $\boldsymbol{\mu}_m$, $m = 1, \ldots M$, can also be estimated by selecting M patterns randomly from the training set. The widths can be estimated , e.g. by (Mehlig 2022) $M\sigma_m^2 = min(\|\boldsymbol{\mu}_m - \boldsymbol{\mu}_j\|^2)$. Other estimators can also be used such as the average distance $\overline{d}_m$ between $\boldsymbol{\mu}_m$ and the other centres, e.g. $\sigma_m^2 = 8\overline{d}_m$.

Learning the RBF weights

Once the parameters of the basis functions $\varphi_m()$, $m = 1, \ldots M$ are determined one can proceed to estimate the weights (w_{mk}) via supervised learning, see Eq. (19). As it can be seen from Fig. 10.17 there is no 'explicit' weights going into the hidden layer. The link between the input neurons and the neurons of the hidden layer nodes is established via the radial basis functions $\varphi_m()$, $m = 1, \ldots M$. The weights (w_{mk}) are obtained via gradient descent where the gradient of the costfunction is given by:

$$\begin{aligned} \frac{\partial E}{\partial w_{mk}} &= \frac{1}{2}\frac{\partial}{\partial w_{mk}} \sum_i \sum_{k'} (y_{ik'} - t_{ik'})^2 \\ &= \sum_{i,k'} (y_{ik'} - t_{ik'}) \frac{\partial}{\partial w_{mk}} \sum_{m'} w_{m'k'} \varphi_{m'}(\mathbf{x}_i) \end{aligned} \tag{10.20}$$

Letting $\mathbf{T} = (t_{ik})$, $\mathbf{\Phi} = \varphi_{mi}) = (\varphi_m(\mathbf{x}_i)$, and $\mathbf{W} = (w_{mk})$, the optimal weights obtained using the above gradient, Eq. (20), yields the matrix equation:

$$\mathbf{\Phi}\mathbf{\Phi}^T\mathbf{W} = \mathbf{\Phi}\mathbf{T}, \tag{10.21}$$

which can be inverted using the generalized inverse of $\mathbf{\Phi}\mathbf{\Phi}^T$ to yield:

$$\mathbf{W} = (\mathbf{\Phi}\mathbf{\Phi}^T)^{-1}\mathbf{\Phi}\mathbf{T}, \tag{10.22}$$

Exercise

Derive the above equation.

Indication. Equating the above Eq. (20) to zero yields $\sum_i y_{ik}\varphi_m(\mathbf{x}_i) = \sum_i t_{ik}\varphi_m(\mathbf{x}_i)$, that is $\sum_{i,m'} w_{m'k}\varphi_{m'i}\varphi_{mi} = \sum_i t_{ik}\varphi_{mi} = (\mathbf{T}^T\mathbf{\Phi}^T)_{km}$, which can be rewritten as $\sum_{m'} w_{m'k}(\mathbf{\Phi}\mathbf{\Phi}^T)_{m'm} = (\mathbf{W}^T\mathbf{\Phi}\mathbf{\Phi}^T)_{km} = (\mathbf{T}^T\mathbf{\Phi}^T)_{km}$

Remarks

1. The above weights rely on the combination of unsupervised and supervised learning. This hybrid procedure is easy to implement.
2. It is also possible to apply a fully supervised learning procedure by estimating the parameters $(\boldsymbol{\mu}_m)$, (σ_m) and (w_{mk}) using gradient descent and backpropagation. The procedute is more difficult than the hybrid method.
3. It is also possible to use normalized basis functions instead as in the softmax case in logistic regression. The normalized RBF is simple given by $\frac{\varphi_m(\mathbf{x})}{\sum_j \varphi_j(\mathbf{x})}$, see eg. Mehlig (2022) and Hsieh (2023).

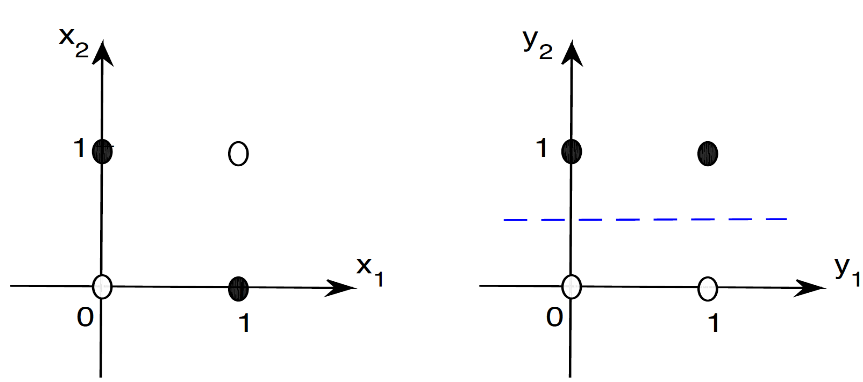

Figure 10.18: Four patterns of the Boolean XOR function (left), and its transformation by the nonlinear map $y_1 = x_1, y_2 = x_1 + x_2 - 2x_1x_2$.

Application

We consider here another interesting Boolean function, namely the exclusive OR, XOR, function: $x_1 XOR x_2 = x_1 \wedge \neg x_2 \vee \neg x_1 \wedge x_2$, where $\neg x$ is the negation of x. Figure 10.18 shows the two classes (truth table 0110) of XOR.

Exercise

Is the Boolean XOR function linearly separable?

Several ways exist to solve the separability problem of XOR function (e.g., Gori et al. 2024). One conventional way is to use perceptrons with hidden layers (see section 13.3.1). One can also use nonlinear mapping. For example, the simple nonlinear transformation $y_1 = x_1$ and $y_2 = x_1 + x_2 - 2x_1x_2$ does the job (Fig. 10.18). Casting the problem nonlinearly in a high-dimensional space, e.g., via kernel methods, can make the problem linearly separable following Cover's theorem[3] (Cover 1965). RBF networks can also be used in a similar way (see exercise below).

Exercise

Use the RBF network to solve the separability problem of the Boolean XOR function.
Indication. Consider the patterns of the XOR function (Fig. 10.18 left), and use two Gaussian basis functions $\varphi_1()$ and $\varphi_2()$ with respective centres $\boldsymbol{\mu}_1 = (1,0)^T$ and $\boldsymbol{\mu}_2 = (0,1)^T$ and $\sigma_{1,2} = 1$, and show that the transformed patterns are linearly separable. For example, the image of $(0,0)$ is $(\varphi_1(0,0) = 0.37, \varphi_2(0,0) = 0.37)$. By equating the outputs (Eq 13.18) to the targets derive the weights of the RBF network. For example, with a bias parameter b the first equation is $0.37w_1 + 0.37w_2 + b = t_1$, where $t_1 = 0$ and represents the target (or class) of $(0,0)$. Note that since the output here is scalar, we do not need to solve an optimization problem as in Eq (13.19), but solve en exact system through the identity $y_i = t_i$, $i = 1, \dots 4$.

10.6 Autoencoders

Linear PCA or EOF analysis belong to unsupervised learning, but the solution is obtained easily using the SVD, which is an exact algorithm. There is no exact algorithm, however, for nonlinear PCA, and this is where machine larning comes in. MLP was initially devised to solve problems in supervised learning, e.g., classification and regression, and was then extended to deal with unsupervised learning as in the case of clustering and signal denoising. Autoencoders constitute a special class of unsupervised feedforward neural net. The main application of autoencoders is towards data compression and dimension reduction such as nonlinear PCA, which is an extension of linear EOF analysis to nonlinear manifold embedding (Brunton and Kutz 2022).

We assume we have input patterns $\mathbf{x}_1, \dots \mathbf{x}_n \in \mathbb{R}^d$, and we want to construct a map onto a generally much lower dimensional space $\mathbf{z} = \mathbf{f}_e(\mathbf{x}) \in \mathbb{R}^r$, with $r \ll d$, with a nonlinear

[3] which states that a complex classification problem may be linearly separable if it is cast nonlinearly into a higher dimensional space.

mapping $\mathbf{f}_e()$. The latent variable $\mathbf{z}$ is then mapped back onto the original space using a different nonlinear mapping $\mathbf{f}_d()$. The objective is then to learn the equation:

$$\mathbf{f}_d(\mathbf{z}) = \mathbf{f}_d(\mathbf{f}_e(\mathbf{x})) = \mathbf{x}, \tag{10.23}$$

that is, the composition function $\mathbf{f}_d o \mathbf{f}_e$ is the identity. The latent variable $\mathbf{z}$ is also known as the *bottleneck* variable. The autoencoder is then composed of two components (Fig. 10.19):
(1) Encoder, which compresses the data via the mapping $\mathbf{z} = \mathbf{f}_e(\mathbf{x})$ onto the bottleneck.
(2) Decoder, which maps the bottleneck back onto the input space, i.e., $\mathbf{f}_d(\mathbf{z}) = \mathbf{x}$.
So the output of the network is to match the input by minimizing the loss-function or error E, possibly with regularization, i.e. a functional of the data $\mathbf{X}$, and the functions $\mathbf{f}_e$ and $\mathbf{f}_d$:

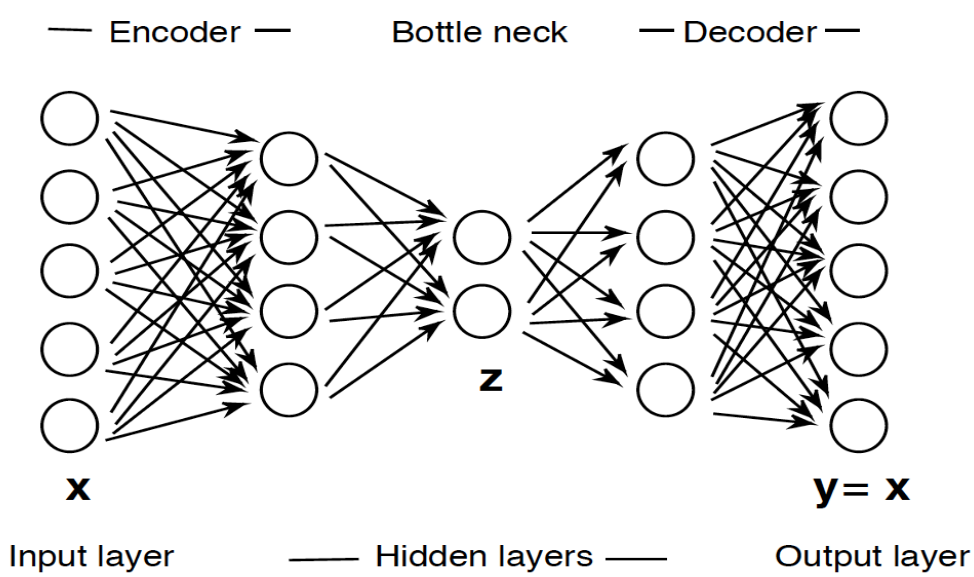

Figure 10.19: Example of an autoencoder neural net with 5 layers showing the encoder (2 layers), bottleneck layer, and the decoder (2 layers)

$$E = \frac{1}{2}\sum_{i=1}^{n} \|\mathbf{x}_i - \mathbf{f}_d(\mathbf{f}_e(\mathbf{x}_i))\|^2 + \sum_{k=1}^{q} \lambda_k g_k(\mathbf{X}, \mathbf{f}_e, \mathbf{f}_d), \tag{10.24}$$

with respect to the weights of the mapping functions. Note that in the illustrative figure 10.19 there are 3 hidden layers and the dimensions of the input and bottleneck spaces are respectively $d = 5$ and $r = 2$. The nonlinear mapping functions $\mathbf{f}_e()$ and $\mathbf{f}_d()$ are referred to as the encoder and decoder respectively. The nonlinearity of the nodes of the hidden layers is normally expressed through a sigmoid activation function. The costfunction E, Eq. (24), can also be regularized by adding a regularization function.

Remark

If the autoencoder is linear then the network yields the leading r EOFs/PCs.

Other variants of autoencoders have been developed, the following are the main ones:
(1) Denoising autoencoders, in which the input and output neurons are no longer assumed

to be identical, as the input data may be noisy and the output have to be filtered. In this case some target clean/filtered data are to be provided.
(2) Variational autoencoders, in which the latent variable is assumed to follow some specific probability distribution $f(\mathbf{z})$, usually taken to be multinormal. The data distribution is then represented as a nonlinear transformation of $f(\mathbf{z})$, which is trained by the decoder as with the MLP where the weights can be estimated using gradient methods by maximizing the data log-likelihood (e.g., Mehlig 2022, Brunton and Kutz 2022).
(3) Sparse autoencoders, in which the bottleneck (and hidden layers) have large number of neurons, perhaps more than the input neurons, but sparse, i.e., only a small number of them are simultaneously active.
(4) Convolutional autoencoder, for which convolutional networks (next section) are used for the encoder and decoder blocs, that is convolution (or spatial smoothing) is used. This is particularly useful for patterns with spatial correlation structure like images.

10.7 Convolutional neural nets

10.7.1 What is convolution and why?

Convolution is a mathematical transformation aiming at highlighting local features. It is widely used in time series and signal processing as a kind of linear filter. Given a kernel $h()$ the convolution of the signal $f()$ is the signal $g()$ according to Eq. (13.25):

$$g(x) = f(x) * h(x) = \int f(\tau)h(x-\tau)d\tau. \tag{10.25}$$

The kernel $h()$ is in general a localised function such as Gaussian or functions with bounded support, and play the role of filters. In the discrete case the convolution is a simple weighted average.

Exercise
Show that the convolution, Eq (13.25), is the area of the overlap between the function $f()$ and the kernel $h()$.
Indication. Simply take $h(z) = 1_{[a,b]}(z)$, then Eq (13.25) becomes $g(x) = \int_{x-b}^{x-a} f(\tau)d\tau$, which is the area of the region delimited by $z = x - b$, $z = x - a$, the (horizontal) z-axis and the curve of $f()$. The convolution at point x is shown by the dotted region in Fig. 10.20.

In many applications, such as image processing, the dimensionality of the input and the number of parameters in a deep neural net can be astronomic. In addition, these objects are chracterised by spatial dependencies. The convolution operation aims at using this spatial dependency, via attempting to preserve the spatial (or gridded) structure of the image, in addition to reducing the problem dimension. The structure of convolutional networks comprises a number of layers, namely the input layer, the convolution layer, which feeds into the activation layer followed by a pooling layer (dimension reduction) and terminating with a fully connected and output layers (Fig. 10.21).

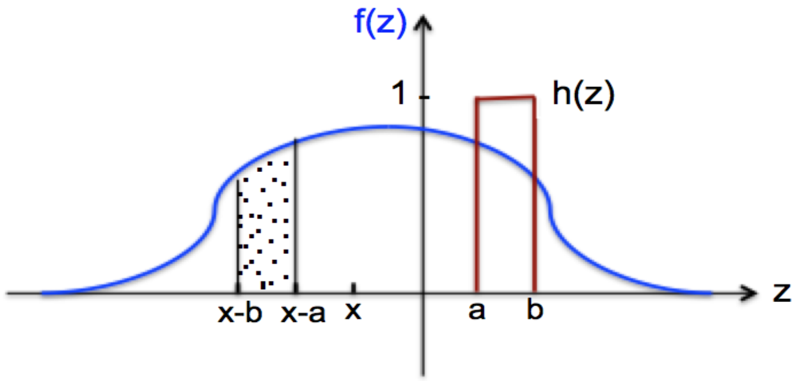

Figure 10.20: Illustration of convolution between $f()$ and $1_{[a,b]}()$. The dotted area represents the convolution of $f()$ and $h()$ at x, $f * h(x)$.

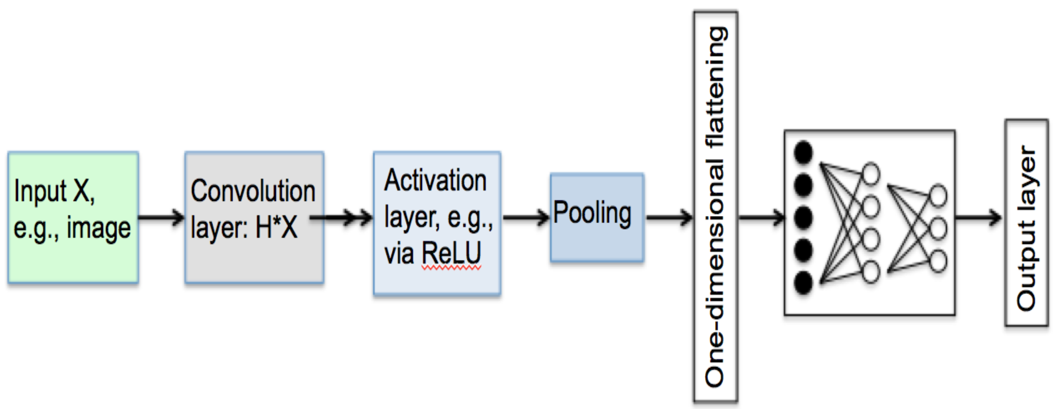

Figure 10.21: Structure of a convolutional neural network

10.7.2 Convolution and activation layer

Given an image (two-dimensional gridded feature) the convolution layer is obtained the filter to the image. Given a two-dimensional filter $H = (h_{kl})$ (set of weights) and an input array $X()$, a feature map $Y()$ is obtained by applying the kernel filter, i.e.,

$$Y(i,j) = (h * X)(i,j) = \sum_{k,l} h_{kl} X(i-k, j-l) \tag{10.26}$$

Example
Consider the following 2×2 filter H and the two-dimensional input array X (Fig. 10.22). The filtering operation is shown in Fig. 10.23. In general one filter is used in convolution, but it is not uncommon that different filters are used for different parts of the input array for example when one wants to focus on specific feature of the image. The obtained feature layer (containing the convolved or filtered array) is followed by a nonlinear activation layer using for example a rectified linear unit (ReLU) $\sigma(x) = \max(0, x)$. Both the binary step function and ReLU have discontinuous derivative at the origin, but one can overcome this, if needed, by approximating them with the sigmoid $\sigma(x) = (1 + e^{-x})^{-1}$ and the SoftPlus

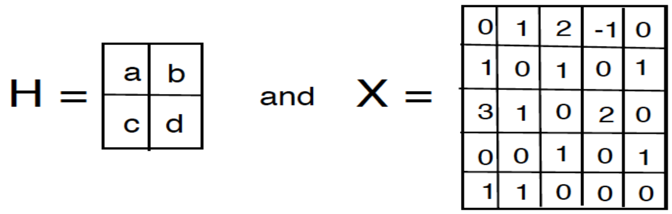

Figure 10.22: Simple example of a two-dimensional filter and input input array X

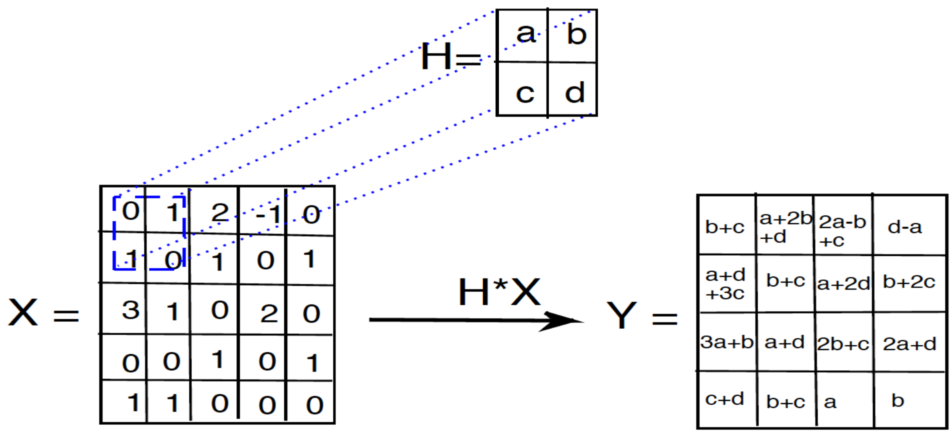

Figure 10.23: Convolution operation of H and X, $H * X$, of Fig. 10.22

$g(x) = log(1 + e^x)$ functions respectively (Fig 10.24).

10.7.3 Pooling layer

Pooling is yet another compression or downsampling operation aiming at further reducing the dimension of the image. Examples of pooling include max or average pooling. The pooling is based on patches of size $m \times m$. For instance, in the previous example with $a = 0, b = 1, c = -1$ and $d = 2$, the filtered matrix and the max pooling are shown in Fig. 10.25. The max pooling (Fig. 10.25) is based on a 2×2 patch. We note here that the convolution and pooling operations can be repeated several times.

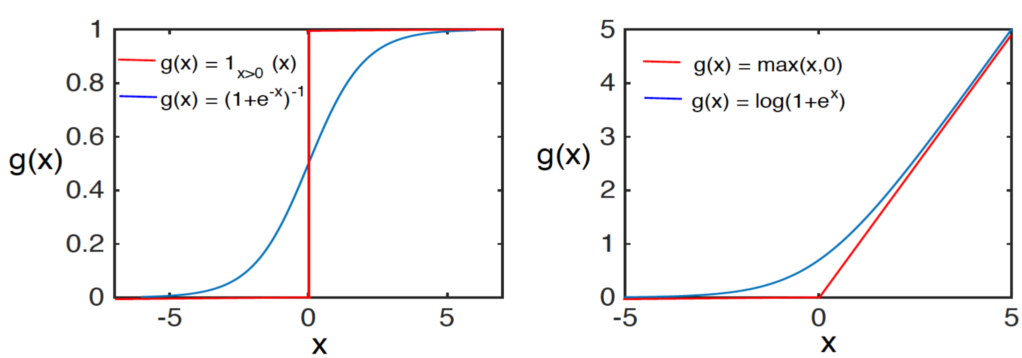

Figure 10.24: Binary step and ReLU functions along with their smooth approximation by sigmoid and SoftPlus functions respectively.

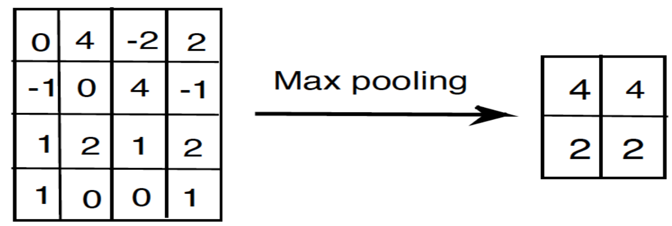

Figure 10.25: An example of max pooling with 2×2 patcg of the convolved example of Fig. 10.23 using $a = 0, b = 1, c = -1$ and $d = 2$.

10.7.4 Flattening and connected layers

After the last pooling the final step includes flattening to a one-dimensional vector, which is fed into a (fully) connected neural net (Fig. 10.21).

10.8 Recurrent neural networks

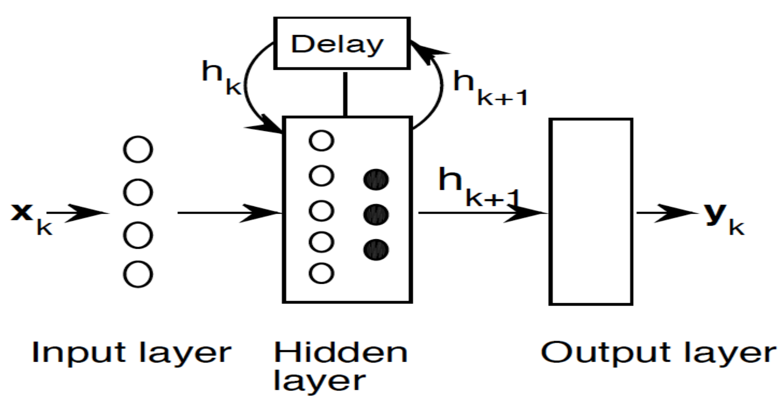

Figure 10.26: General feature of feedback loop

As our last example we discuss another important class of neural nets, namely, recurrent neural networks. In the previous neural networks the inputs are understood to be independent. However, this is not the case in many instances including speech recognition, time series and dynamical systems in which memory (or autocorrelation) is an integral part, and outputs at one step depend on inputs from previous steps. Neural networks using this property are referred to as recurrent neural networks (RNN).

The basic feature of RNN is the existence of feedback connections (or feedback loops) in the hidden layers as illustrated in Fig. 10.26. In brief, RNN behave like dynamical systems and therefore can have stability and controllability problems. The feedback loop in the hidden layer behaves precisely like a latent representation in a latent space. The structure in RNN flow looks like that of dynamical systems with a recurrent map:

$$\mathbf{x}_{k+1} = f_{\mathbf{w}}(\mathbf{x}_k), \tag{10.27}$$

where $\mathbf{w}$ is a set of parameters of the dynamical system or a set of weights for the neural network. In RNN training is normally done over a sequence of consecutive snapshots, i.e.,

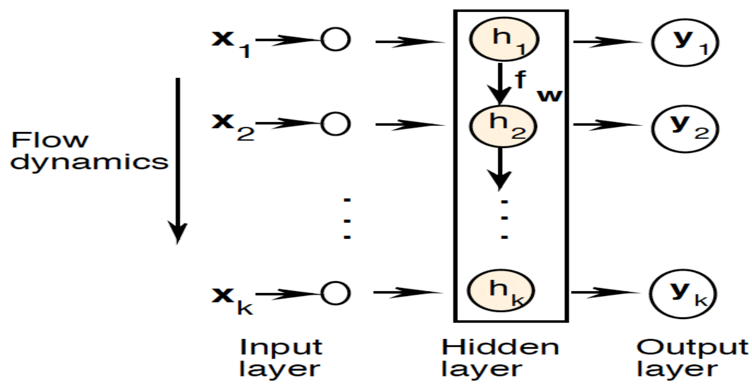

Figure 10.27: Structure of a RNN with inputs $\mathbf{x}_k$, outputs $\mathbf{y}_k$, $k = 1, 2, \ldots$, and feedback loops. The latent space is represented by the box.

$\mathbf{x}_{k+p} = f_{\mathbf{w}}^{(p)}(\mathbf{x}_k)$. As is mentioned above, the mapping in RNN uses the concept of latent space with latent representation (Fig. 10.27):

$$h_{k+1} = f_{\mathbf{w}}(h_k, \mathbf{x}_{k+1}), \tag{10.28}$$

so that the output $\mathbf{y}_k$, at step (or time) k depends not only on the input $\mathbf{x}_k$ but also on the previous input $\mathbf{x}_{k-1}$ via a feedback $h_k = f_{\mathbf{w}}(h_{k-1}, \mathbf{x}_k)$ (Fig. 10.27).

One particular algorithm of RNN is the long short-term memory (LSTM). LSTM is a kind of regularization and has feedback loops allowing processing an entire sequence, and avoids vanishing gradient a problem that is often encountered in training networks. Basically, with LSTM the network generates the output using the previous input (short-term) while considering the long-term memory, see Eq. (28), hence the name LSTM. Several programming languages consider LSTM in RNN, including Matlab and Python (e.g., Brunton and Kutz 2022, Shah 2023).

Chapter 11

Application to Forecast Verification

11.1 Background

11.1.1 What is forecast verification and why?

Weather forecasting is a complex process that involves theorists, experimentalists and modellers. Observations are collected, processed and used via data assimilation along with background, usually previous, forecasts to determine the initial conditions. These initial conditions are used by the numerical weather prediction (NWP) model to issue forecasts. Once a forecast is issued, it is then used by customers and concerned agencies. But is that all? Are these forecasts thrown away?

Numerical weather forecasts are produced by numerical weather agencies such as the European Centre for Medium Weather Forecasts (ECMWF), the National Center for Environmental Prediction (NCEP) and the Japan Meterorological Agency (JMA). These products, like many others, are subjected to quality check allowing thus for an evaluation of the forecast quality. This is known as *forecast verification* and sometimes *forecast validation* since we know that all forecasts are not exact or perfect[1]. It is seen therefore that forecast verification is the process of assessing the quality of a forecast.

Forecast verification is an essential and necessary component of the forecasting system as a whole (Murphy and Winkler 1987; Doswell et al. 1990). Forecast verification provides in fact a "measure" of the quality and the value of a forecast. The latter indicates the economic benefit provided by a "correct" forecast whereas the former refers to the model producing the forecast. According to Brier and Allen (1951) there are at least three main purposes behind forecast verification, namely:

1. Administrative – this relates to the performance of the operational NWP model, where the aim is to evaluate the continuous improvement of the forecasting model.
2. Economic – this relates to the value of the forecast and embeds the economic benefit of a correct forecast to the decision making bodies.

[1] "It is very hard to predict, especially the future", a quote attributed to Niels Bohr by the Polish-born mathematician Stanislaw Ulam in page 286 of his book (Ulam 1976).

3. Scientific – and refers to the relationship between forecast and observations in order to stress on the ability of the model to provide correct forecasts.

So it is clear that forecast verification is an important step in closing the loop of the forecast system. This step is important since it allows:

- following and measuring NWP model improvement.
- using the model as surrogate to observations, in areas where observations are scarce or not available, for various physical-process studies. This is possibel only when the forecasts have been verified and are correct.
- decision to be taken regarding the choice between the different physical parametrisation schemes.
- objective decision-making.

In addition, in model intercomparison projects, where different models from different weather and climate centres are compared. It is important that forecasts have been verified. This allows for an objective competition between models.

11.1.2 Terminology

In forecasting literature terms like accuracy, bias, skill and reference forecast are frequently used. Below is a brief description of these terms, see also Wilks (2006).

- *Accuracy* – provides an overall measure of correspondence between observed and forecast values.
- *Bias* – this is a measure of the correspondence between the average of forecast and the average of observations.
- *Skill* – it is a measure of the accuracy of a forecast relative to a reference forecast. This reference often has low or no skill.
- *Reference forecast* – this refers to an easily available, non-model based, data often with low or no skill.

The first three correspond to what Murphy (1993) describes as attributes of a forecast. In fact, besides the above three basic attributes Allan Murphy describes six more attributes, namely; association, reliability, resolution, sharpness, discrimination and uncertainty, see Murphy (1993) for more details and definitions.

11.2 Some basic metrics for forecast verification

Forecast products can be of different nature. One can have forecasts for continuous variables such as pressure, temperature and winds. One can also have forecasts for discrete variables such as rain or no rain. Note that for this latter case one talks specifically of categorical or binary variables. These variables can be measured at a given point in space in which case single numbers (observed and forecast) are compared. Quite often these have a spatial distribution, in which case we have to compare maps. One of the oldest and very basic ways of comparing maps is through visual inspection or "*eyeball*" method. Following this method the forecaster compares the forecast and observations side by side using the human skill of patterns recognition to make his/her own judgement. This is a good way when we have a few forecasts but remains, never the less, subjective and is no-quantitative. In the sequel we let $\mathbf{x}_k^f$ and $\mathbf{x}_k^o$, $k = 1, \ldots n$, designate respectively the k'*th* forecast and observed variables. The bold face means that we are dealing in general with maps. When we deal with single variables we use lower case instead.

11.2.1 Accuracy measure

Single continuous variables

In the case of single variables several accuracy measures exist:
• Mean absolute error (MAE) – It is defined as the arithmetic mean of the absolute difference between forecast and observed variables:

$$MAE = \frac{1}{n}\sum_{k=1}^{n} |x_k^f - x_k^o|. \tag{11.1}$$

A perfect forecast corresponds to $MAE = 0$.
• Mean square error (MSE) – It is the average squared difference between observations and forecast:

$$MSE = \frac{1}{n}\sum_{k=1}^{n} \left(x_k^f - x_k^o\right)^2. \tag{11.2}$$

For a perfect forecast MSE is also zero. Notice, however, that unlike the MAE, the MSE is sensitive to large errors.
• Root means quare error (RMSE) – it is the square root of the MSE:

$$MSE = \sqrt{\frac{1}{n}\sum_{k=1}^{n} \left(x_k^f - x_k^o\right)^2} \tag{11.3}$$

• Correlation coefficient – it is the correlation coefficient between the forecast and observations:

$$r = \frac{\sum_{k=1}^{n}\left(x_k^f - \overline{x^f}\right)\left(x_k^o - \overline{x^o}\right)}{\sqrt{\sum_{k=1}^{n}\left(x_k^f - \overline{x^f}\right)^2}\sqrt{\sum_{k=1}^{n}\left(x_k^o - \overline{x^o}\right)^2}} \tag{11.4}$$

where $\overline{x}$ is the average of x_k, $k = 1, \ldots n$. The correlation coefficient satisfies $-1 \leq r \leq 1$. A perfect forecast corresponds to $r = 1$.
• Anomaly correlation (AC) – it is defined as the correlation coefficients between the anomalies with respect to the climatology.

Multivariate continuous case

• Anomaly correlation (AC) – It is a measure of the correlation between anomalies, with respect to climatological means, of observations and forecasts.

• Spatial correlation (SC) – it is the spatial correlation between observed and forecast patterns, defined as:

$$SC_k = \frac{\mathbf{x}_k^f \bullet \mathbf{x}_k^o}{\|\mathbf{x}_k^f\| \|\mathbf{x}_k^o\|} = \cos\left(\mathbf{x}_k^f, \mathbf{x}_k^o\right) \tag{11.5}$$

and measures the cosine of the angle between observation and forecast fields considered as vectors. A perfect forecast corresponds to $SC_k = 1$.

In addition to those measures we can also use the MSE or the RMSE or even the MAE by spatially averaging those mesures taken at individual grid points.

Discrete or categorical variables

We now consider a categorical variable where the outcome is "yes" or "no" (or 1 and 0). If n forecasts and observations, x_k^f and x_k^o, $k = 1, \dots n$ are available we denote by:

- a : the number of hits, that is the number of times an observed event was correctly forecast. When this happens we say we have a hit.
- b: the number of false alarm, that is the number of times the forecast was positive or "yes" (also "1") but the event was not observed (i.e. "no" or "0").
- c: number of misses, that is the number of times the observed events were forecast not to occur, i.e. forecast is negative but the event did occur.
- d: number of correct negatives, which represents the number of times that the forecast was negative for an event that did not occur.

These numbers can be brought together in a contingency table as shown in table 11.1. The

	Observed variable: YES	Observed variable: NO	Marginal
Forecast variable: YES	hit a	false alarm b	a+b
Forecast variable: NO	c miss	d correct –ve	c+d
Marginal	a+c	b+d	n= a+b+c+d

Table 11.1: 2×2 contingency table

contingency table is used as a basis for defining the accuracy of forecasts for binary (or categorical) variables. The following measures are often used in categorical forecast evaluation:

• *Proportion of correct* – PC – It represents the proportion of correct forecasts, i.e. hits and correct negatives, and is given by:

$$PC = \frac{a+d}{n} \tag{11.6}$$

The PC gives equal weighting to the hits and correct negatives. This can have some disadvantages when, for example, the variable to forecast is quite 'predictable' such as rain in Oslo in winter or snow in Moscow also in winter. An alternative is given by:

• *Threat score* – TS – It is also referred to as *critical success index* (CSI), and is given by the ratio:

$$TS = \frac{a}{a+b+c} \tag{11.7}$$

• *False alarm ratio* – FAR – It is defined as the fraction of false alarm to the total number of non-occurrence of the event:

$$FAR = \frac{b}{b+d} \tag{11.8}$$

• *Hit ratio* – HR – It is also known as the probability of detection (POD), and is defined as the fraction of hits to the total number of occurrences of event:

$$HR = POD = \frac{a}{a+c} \tag{11.9}$$

11.2.2 Bias

Continuous case

For continuous variables, the bias is simply measured by the mean error (ME):

$$ME = \frac{1}{n}\sum_{k=1}^{n}\left(x_k^f - x_k^o\right) = \overline{x^f} - \overline{x^o}. \tag{11.10}$$

This bias is also referred to as the systematic error since it is given by the difference between climatologies. The systematic error, or bias, is computed for both the univariate and the multivariate cases.

Discrete case

For the categorical forecast the bias is given by comparing the total "Yes" forecast to the total occurrence of events:

$$B = \frac{a+b}{a+c} \tag{11.11}$$

11.2.3 Skill score

The skill of a forecast always requires a reference forecast, which will be discussed later. We let A designate one of the above accuracy measures, e.g. MAE for continuous variables or POD for categorical variables. We also let A_{ref} represent the same accuracy measure obtained from a reference forecast. The skill score is given in percentage by:

$$SS = \frac{A - A_{ref}}{A_{perf} - A_{ref}} \times 100\%. \tag{11.12}$$

where A_{perf} is the same accuracy measure but for a perfect score. The SS is 100% when the forecast is perfect. Note also that the SS can be negative when the forecast is worse than the reference forecast. The above skill score is valid for both the continuous and discrete cases. When we specify the type of variables to be forecast then we can provide more specific skill scores.

Categorical case

In this particular case there are particular measures of skill scores. One of the most-frequently used is the Heidke skill score (HSS). The HSS is based on the proportion of correct forecast, PC, see eq (11.6) as a measure of accuracy. If the reference forecast is the random forecast, independent of observations, then the HSS is given by (e.g. Wilks 2006):

$$HSS = \frac{2(ad - bc)}{(a+c)(c+d) + (a+b)(b+d)} \tag{11.13}$$

If the threat score (TS), eq (11.7), is used as the accuracy measure instead, and the reference forecast is again random, independent of observations, we obatain the so-called Gilbert skill score (GSS), known also as the equitable threat score (ETS), given by:

$$GSS = \frac{a - a_{ref}}{a - a_{ref} + b + c} \tag{11.14}$$

where a_{ref} is given by:

$$a_{ref} = \frac{(a+b)(a+c)}{n} \tag{11.15}$$

Continuous case

For the continuous case the persistence or the climatology are frequently used as reference forecasts. Let us take, for example, the mean square error (MSE) as a measure of accuracy, then the reference MSE for the climatology and persistence are respectively:

$$MSE_{clim} = \frac{1}{n}\sum_{k=1}^{n}\left(\overline{x^o} - x_k^o\right)^2 = \frac{1}{n}\sum_{k=1}^{n}(x_k^o)^2 - (\overline{x^o})^2 \tag{11.16}$$

$$MSE_{pers} = \frac{1}{n}\sum_{k=1}^{n}\left(x_{k-1}^o - x_k^o\right)^2 \tag{11.17}$$

The skill score is then given by:

$$SS = -\frac{MSE - MSE_{ref}}{MSE_{ref}} = 1 - \frac{MSE}{MSE_{ref}} \tag{11.18}$$

There are a number of other measures of skill score in the literature, but we do not discuss them here, see e.g. Wilks (2006) or Gilleland et al. (2009).

Remarks

1. There are mainly three types of reference forecasts:
   - climatology (for continuous variables)
   - persistence
   - Random forecast

The climatology is based on the average over, e.g. a season. The persistence forecast is based on using the precedent (in time) observed variable, i.e. x^o_{t-1}, as the next forecast x^f_t, i.e.

$$x^f_t = x^o_{t-1} \tag{11.19}$$

The random forecast is based on drawing randomly and repeatedly samples from the available observation record over the study period. The forecast at any time t is then one such random sample. The statistics of the obtained verification scores are then computed from a large sample of forecast−observation pairs.

2. Observations are not restricted to in-situ data, e.g, rain gauges, radio-sondes and surface stations but include also remotely sensed observations, such as satellite radiances. Re-analyses are also used as surrogates for observations when no observations are available. Differences between the forecast verification and reanalyses have to be taken care of. For instance, the forecast model may not have the same resolution as the analyses or reanalyses. This is also the case, in general, for (in-situ and remote) observations. It is recommended in this case to interpolate the forecast to the locations of the observations.

11.3 Some advanced verification methods for the 1D case

11.3.1 Multicategory forecast

In the case when we have more than two categories (multi-category forecasts) we use a similar approach to that of the categorical (”yes” and ”No”) forecast based on the (multicategory) contingency table shown in table 11.2. In this table we have K categorical variables (observed and forecast). The number N_{ij}, for each $i, j = 1, \dots K$, denotes the number of forecasts in the i'*th* category that has been observed in the the j'*th* category. The number N^o_i, for each i, $i = 1, \dots n$ represents the total number of observations in category i, and N^f_i is the total number of forecasts in the same category.

Multicategory forecasts arise in many applications. For instance, when we are inrterested in forecasting different bins of a given variable we use the muticategory contingency table. Let us take the example of total daily precipitation. Instead of considering the dichotomy wet or dry, we can extend this to ternary case by considering the three categorical variables: dry, moderately wet (say precipitation greater than zero less than 10 mm) and very wet (greater than 10 mm). The above multicategory contingency table can also be represented as a two-dimensional histogram. In this case the accuracy measures for discrete variables (see section 11.2) can be used here. For example for the PC (11.6) and HSS (11.13) we get:

$$PC = \frac{1}{n}\sum_{i=1}^{K} N_{ii} \tag{11.20}$$

$$HSS = \frac{\frac{1}{n}\sum_{i=1}^{K} N_{ii} - \frac{1}{n^2}\sum_{i=1}^{K} N^o_i N^f_i}{1 - \frac{1}{n^2}\sum_{i=1}^{K} N^o_i N^f_i} \tag{11.21}$$

Forecast variables \ Observed variables	1	2	. . .	K	Marginal
1	N_{11}	N_{12}	. . .	N_{1K}	N^f_1
2	N_{21}	N_{22}	. . .	N_{2K}	N^f_2
⋮	⋮	⋮	⋱	⋮	⋮
K	N_{K1}	N_{K2}	. . .	N_{KK}	N^f_K
Marginal	N^o_1	N^o_2	. . .	N^o_K	n

Table 11.2: Multicategory contingency table

Note that a 'good' forecast should show high values at and around the diagonal. Perfect forecast will only have non-zero values on the diagonal.

11.3.2 Methods for probabilistic forecasts

Probabilistic forecasts are used to issue probabilities for the occurrence of an event. Since in general single probabilistic forecasts are difficult to verify, it is often recommended to look at a set of probabilistic forecasts and verify them with observations using the categorical contingency table. Probabilistic forecasts are often evaluated through their attributes, namely *reliability, sharpness* and *resolution*:

• *Reliability* – it is a measure of the agreement between forecast probability and mean observed frequency.

• *Sharpness* – it is a measure of the tendency to forecast probabilities near 0 or 1 versus values clustered around the mean.

• *Resolution* – it reflects the ability of the forecast to resolve the different chraracteristics of the sample.

One of the most useful diagrams used to analyse the above attributes is the *reliability diagram*, which plots the observed frequencies, O_k, versus the forecast probabilities, p_k^f, $k = 1, \dots n$, binned into a number of bins. The sample size in the bins is normally plotted as a histogram next to the frequency diagram. The reliability is indicated by the proximity to the diagonal, whereas the resolution is reflected by the flatness of the diagram. The sharpness of the forecast is provided by the forecast frequency in each bin.

For the skill measures one defines the Brier skill and skill scores in addition to the relative operating characteristics.

• *Brier score* (BS) – it is given by:

$$BS = \frac{1}{n}\sum_{i=1}^{K}\left(p_i^f - O_i\right)^2 = \frac{1}{n}\sum_{k=1}^{K} n_k\left(p_k^f - \overline{o_k}\right)^2 - \frac{1}{n}\sum_{k=1}^{K} n_k\left(\overline{o_k} - \overline{o}\right)^2 + \overline{o}\left(1-\overline{o}\right) \tag{11.22}$$

• *Brier skill score* (BSS) – It is a measure of skill with respect to a reference forecast, and is given by:

$$BSS = \frac{BS - BS_{ref}}{0 - BS_{ref}} = 1 - \frac{BS}{BS_{ref}} \tag{11.23}$$

• *Relative operating characteristics* (ROC) – It is based on plotting the hit rate POD, eq (11.9), versus the false alarm rate, FAR, eq (11.8), based on a set of increasing probability thresholds (figure 11.1). The area under the ROC curve (dotted in figure 11.1) is frequently used as a measure of score. Forecat verification is not limited to single or dichotomic variables

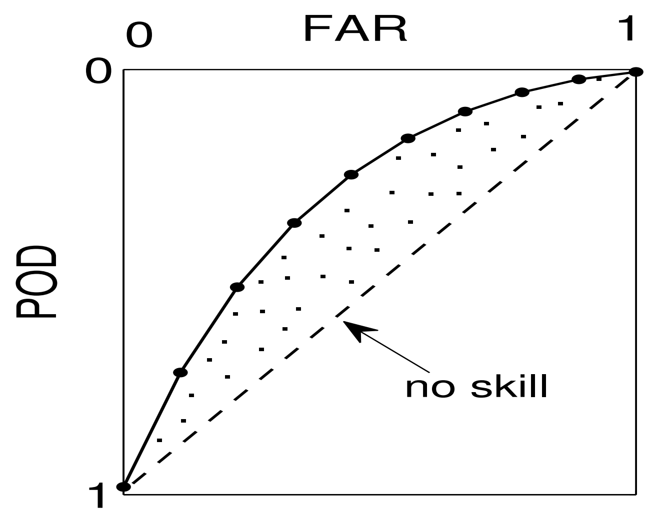

Figure 11.1: Diagram showing the relative operating characteristics

but extends to conventional continuous gridded or even object-based data. Various methods exist to evaluate forecast verification for this kind of data. We have mentioned above some basic methods such spatial correlations, but other metrics can also be used as given below.

11.3.3 Scale-based methods

These methods attempt to compare forecasts and observations at different scale levels. Basically, the forecasts and observations are decomposed using various scales and verification is performed between the associated scales. The following provide examples of scale decomposition:

• *Spectral transforms* – These can be obtained using Fourier or cosine transforms (Denis et al. 2002), and wavelet decomposition (Briggs and Levine 1997). Another method based on the intensity-scale relationship was developed Casati et al. (2004). The method examines how the skill of spatial precipitation forecasts depends on both the scale of the forecast error and the intensity of rainfall events. Forecast skills are also evaluated based on different spatial scales using, e.g. spatial spectra.

• *Variance*

• *Temporal spectra.*

11.3.4 Feature- and object-based methods

In various cases, such as convective precipitation, forecast verification is more suitably performed using feature-based methods (e.g. Davis et al. 2006). The verification approach is based on identifying and describing the different (mostly geometric) features in the observed and forecast fields, then associating the features between observations and forecasts. For precipitation, as an example, the features can be defined based on various precipitation threshols.

The feature- and object-based methods also attempt to identify the location error of the forecast. The total error is then decomposed into contributions due to incorrect location and/or fine scale features. The object-oriented methods (Ebert and McBride 2000) normally look at how objects from observations and forecasts differ. In most cases objects are defined by regions or quantities delimited by closed countours, such as contiguous rain areas (CRAs), low pressure minima, etc. Pattern matching techniques are then applied to identify locations, intensity and shapes, etc.

11.4 Various other methods

11.4.1 Methods for rare events

Methods for rare events (e.g. Hewson 2007) are crucial to evaluate the model performance in forecasting extremes. An example of score measure for rare events is the extreme dependency score given by:

$$EDS = \frac{2\log\left(\frac{\text{hits+misses}}{\text{total}}\right)}{\log\left(\frac{\text{hits}}{\text{total}}\right)} \tag{11.24}$$

Ferro (2007) also proposed a probability-based method for mesuring skill scores for extreme events.

11.4.2 Taylor diagram

The Taylor diagram (Taylor 2001) represents various measures such as correlation coefficients, RMSE, and standard deviation in the same plot. An example of Taylor diagram is shown in Fig. 11.2. In this sample diagram spatial mean annual precipitation is compared between observations and eight global climate model simulations (A, B ... H). The comparison is based on three measures, namely Pearson correlation coefficient, root mean square error and standard deviation. The observation is shown on the x-axis. The different curves in the plot refer to the different measures. Correlation is represented by the blue lines, the standard deviation is represented by the excentric circles centered at the origine, and the centered RMS error is represented by the (blue) eccentric circles centered at the observation providing the distance from it. Note that following this representation the correlation between the observation and the model is given by the cosine of the angle between the x-axis and the line linking the model and the origin. Note also, that one needs only two parameters to find the model position, e.g. correlation and RMS error. This is because, in vector form and denoting

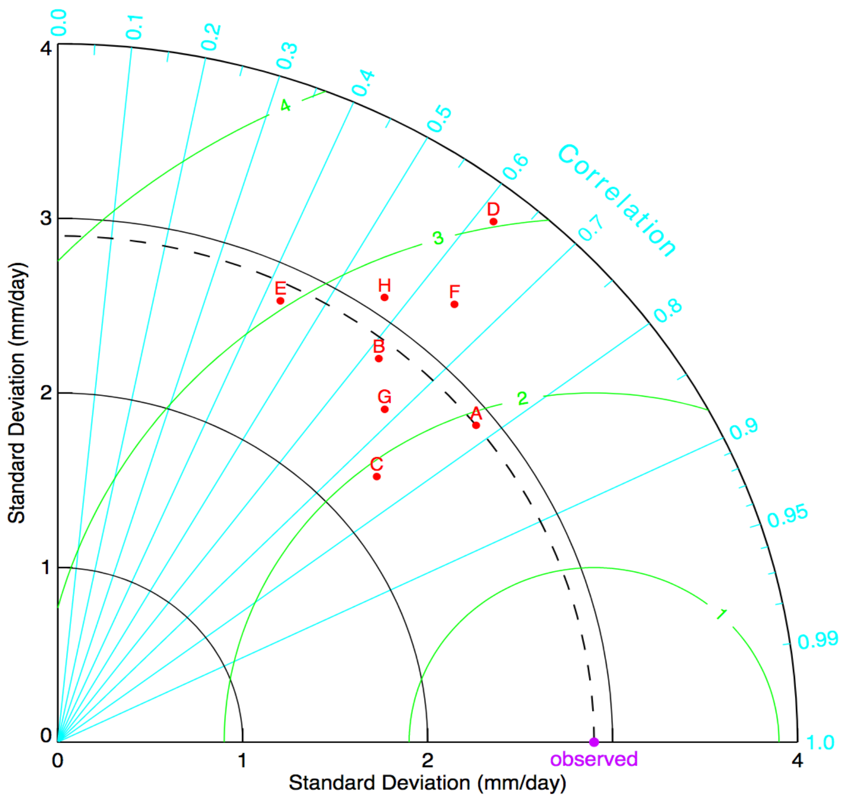

Figure 11.2: Example of Taylor diagram comparing precipitation between observations and eight climate model simulations represented by A, B ... H. Adapted from https://en.wikipedia.org/wiki/Taylor_diagram.

respectively by $\mathbf{r}$, $\mathbf{s}$ and $\mathbf{o}$ the model RMS, model standard error, and observation standard error, we have: $\mathbf{r} + \mathbf{o} = \mathbf{s}$, that is $|\mathbf{r}|^2 = |\mathbf{s}|^2 + |\mathbf{o}|^2 - 2\mathbf{s}.\mathbf{o} = |\mathbf{s}|^2 + |\mathbf{o}|^2 - 2|\mathbf{s}|.|\mathbf{o}|\cos(\mathbf{s}, \mathbf{o})$, and $\cos(\mathbf{s}, \mathbf{o})$ is precisely the correlation coefficient.

In the diagram shown in in Fig. 11.2, the correlation goes from 0 to 1, but it can be extended to include negative values. The lowest correlations (with the observations) is 0.42 and is obtained with model E. Models E and D have the largest RMS error, around 3 mm/day. Among all the models shown on the diagram, models C and D have, respectively, the smallest and largest spatial variability with respective standard deviations around 2.3 mm/day and 3.8 mm/day. One can see also that models A and C generally perform well whereas models D and E have relatively low performance. Examples of Matlab and Python codes can be found here[2]. A similar diagram, the BLT diagram, was also proposed by Boer and Lambert (2001) for the relative climate mean squared difference, variance ratio and effective correlation. The categorical performance diagram (CPD) proposed by Roebber (2009) is also similar to the Taylor diagram and is used for dichotomic forecast verification.

[2]https://github.com/PeterRochford/SkillMetricsToolbox/wiki
https://www.mathworks.com/matlabcentral/fileexchange/20559-taylor-diagram
https://cdat.llnl.gov/Jupyter-notebooks/vcs/Taylor_Diagrams/Taylor_Diagrams.html

11.4.3 Exploratory methods

There are a number of other exploratory tools for forecast verification. These are basic tools that can visually provide a first hand information on the forecast quality. The following examples are most widely used:

- Scater plots of observations versus forecast.
- Scatter plots between the forecast values and the errors.
- Box plots.
- Linear error in probability space (LEPS) − it is defined by:

$$LEPS = \frac{1}{N}\sum_{i=1}^{N}\left(F^o(x_i^f) - F^o(x_i^o)\right) \tag{11.25}$$

where $F^o()$ is the cumulative distribution function of the observations. The LEPS measures the forecast error in the probability space.

11.5 Elements of ensemble prediction

11.5.1 Background

Weather forecasting relies on the forecasting model and the initial conditions, and the quality of the forecast depends closely on the quality of both the model and the initial conditions. The forecast will never be 'exact' due to imperfect prediction model and the initial conditions, and uncertainties will always be present. The real problem, however, is to get a good estimate of these uncertainties.

One way to obtain estimates of those uncertainties is to use probabilistic forecasts by computing the pdf of weather states and propagate it forward (e.g. Epstein 1969). This operation is, however, difficult to apply as we do not know the pdf of the system, in addition to its very high dimension. A more practical way is to obtain an ensemble, with typical size of the order $O(20-100)$ ensemble members, of forecasts by running the prediction model forward using different initial conditions obtained by perturbing the optimal initial condition or reanalysis (e.g. Leith 1974; Leutbecher and Palmer 2008; Vannitsem et al. 2018). If the forecasting system is consistent then the spread of the ensemble will reflect the forecast uncertainty and provides hence information on the system predictability. For example, if the ensemble members are coherent then one would expect the system to be predictable, and the forecast to be 'good', and vice-versa.

Differenr weather services use different methods of initial condition perturbations, such as breeding vectors at NCEP, ensemble Kalman filters at the Canadian Meteorological Service, and a combination of singular vectors and ensemble data assimilation at ECMWF. Model uncertainties related to physical parametrization, such as turbulent fluxes, convection schemes and clouds, are studied via stochastic physics in the tendency equations.

11.5.2 Ensemble forecast verification

In ensemble prediction the forecast skill is normally measured using the ensemble mean, and the uncertainties are measured by the ensemble spread. An ensemble forecast is reliable when the ensemble mean error balances, on average, the ensemble spread. In this case the forecasting system is also known to be well-balanced (or tuned). When the ensemble members are tight together or that the spread is small, then the ensemble forecast is accurate. Below we list a few examples of traditional ensemble forecasting verification metrics.

Spread-skill relation

Consider M ensemble forecasts $z_{i,t}$, $i = 1, \ldots N$, $t = 1, \ldots M$, each of size N (i.e., N ensemble members), $\sigma^2_{z,t}$ the variance of forecasts around the ensemble mean, and $\varepsilon_t = y_t - \overline{z}_t$ the bias of the ensemble mean ($\overline{z}_t = \sum_i z_{i,t}/M$), and y_t is the observation, for $t = 1, \ldots N$. Then the ensemble forecast, in case of no observation errors, is reliable when (Leutbecher and Palmer 2008):

$$\lim_{M\to\infty} \frac{1}{M} \sum_{t=1}^{M} \left[\varepsilon_t^2 - \frac{N+1}{N-1} \sigma^2_{z,t} \right] = 0, \tag{11.26}$$

that is, the ensemble mean error (or bias) balances, on average, the ensemble spread, and provides therefore the right forecast uncertainty. When observation errors ($\sigma_{o,t}$) are present, the balance equation becomes (Yamaguchi et al. 2016):

$$\lim_{M\to\infty} \frac{1}{M} \sum_{t=1}^{M} \left[\varepsilon_t^2 - \sigma^2_{o,t} - \frac{N+1}{N-1} \sigma^2_{z,t} \right] = 0, \tag{11.27}$$

Rank histogram

The rank histogram (Hamill 2001) is obtained as follows. The ensemble forecasts (containing

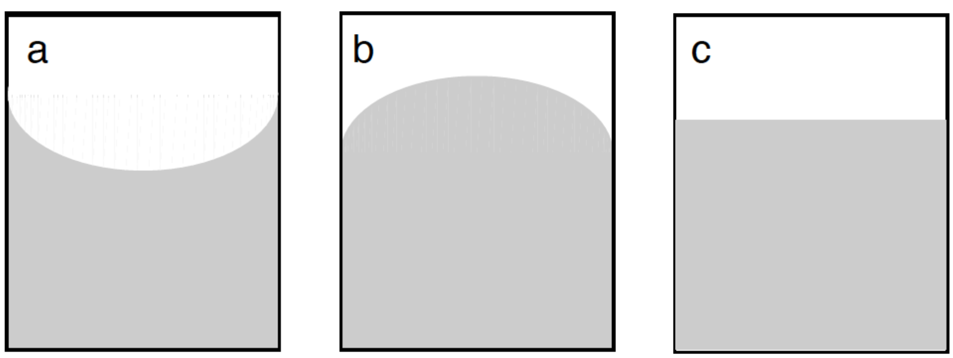

Figure 11.3: Schematic showing a under-dispersed (a), over-dispersed (b) and well-tuned cases.

N ensemble memebers) are first ordered, then the observation is ranked within the set of ordered forecasts. The set of all ranks of the observations are then considered and their histogram obtained. a A-shape of the histogram means that the forecasting system has much spread, i.e. the observations are most of the time in the middle (Fig. 11.3b). A well-balanced system provides a flat histogram (Fig. 11.3c) while a system lacking spread yields a U-shape histogram as shown in Fig. 11.3a.

Ignorance score

This is given by the opposite of the logarithm of the pdf of the forecast error. The Dawid-Sebastiani score is an example. It is obtained based on a Gaussian distribution of the ensemble forecast forecast bias $y - \overline{z}$. In the multivariate case it is given by (Vannitsem et al. 2018):

$$DSS = (\mathbf{y} - \overline{\mathbf{z}})\mathbf{\Sigma}_z^{-1}(\mathbf{y} - \overline{\mathbf{z}}) + \log|\mathbf{\Sigma}_z| \quad (11.28)$$

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